Cyclicity three for the quadratic hemicycle H_{14}^3

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Abstract

We determine the cyclicity of the hemicycle H_{14}^3 bounding the period annulus of the quadratic center $\dot{x} = -y$, $\dot{y} = x(1 + y)$. For all fields in a neighborhood of this center in the full twelve-dimensional quadratic family, the isolated periodic orbits contained in one fixed spherical neighborhood of the hemicycle have total Poincaré multiplicity at most three. Arbitrarily small perturbations produce three simple cycles, each converging to the entire hemicycle. The upper bound compares two actual return paths in a common central-height coordinate. Relative contraction estimates control their nonlinear terms while retaining the divergent slow-passage actions. The signs at the entrances then join the local critical-point bounds into a single count, including simultaneous passage losses and unrestricted separation of two parameter scales. A separate variation at a reversible center realizes the three cycles in the original five-parameter subfamily. Together these arguments give the sharp bound at this graphic with semi-hyperbolic and nilpotent singularities at infinity.

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1 The hemicycle and the main result

The quadratic field

$$\dot{x} = -y, \qquad \dot{y} = x(1+y)$$

has a center whose period annulus fills the half-plane $y > -1$. On the Poincaré sphere its outer boundary is a hemicycle: the compactified line $y = -1$ joins two semi-hyperbolic points at infinity, and the upper equatorial semicircle joins them through a nilpotent point. We denote this graphic by $\Gamma = H_{14}^3$. The question is how many isolated periodic orbits can appear near this entire boundary under arbitrary quadratic perturbations.

The reduction of the quadratic finiteness problem to cyclicity questions for a finite collection of graphics was developed by Dumortier, Roussarie and Rousseau [2]. For center graphics through a nilpotent point, Roussarie and Rousseau [3] prove finite cyclicity of the boundary limit periodic sets for several neighboring configurations. Their Theorem 1.1 explicitly leaves H_{14}^3 aside. Their family $\dot{x} = -y + B_0x^2$, $\dot{y} = x + xy$ is treated for $B_0 > 0$; the field above has $B_0 = 0$. At the end of their introduction they identify the two semi-hyperbolic equatorial points as the additional difficulty in this hemicycle. A bound near a fixed positive B_0 does not give a bound uniform as these points degenerate at $B_0 = 0$. The hemicycles studied by Marín and Villadelprat [4] have two hyperbolic saddles at infinity; the semi-hyperbolic and nilpotent points here require additional slow-passage estimates.

We use the ordinary Euclidean topology on the twelve coefficients of a planar quadratic vector field. All neighborhoods of Γ are taken on the closed northern hemisphere. A cycle is counted only when its entire affine periodic orbit lies in that neighborhood. Its multiplicity is the order of the isolated zero of a local Poincaré displacement; a period annulus contributes no isolated cycles.

Theorem 1.1 (Cyclicity three)**.** *Let $X_0 = (-y, x(1+y))$ and let Γ be its hemicycle described above.*

(i) *There are an open neighborhood $\mathscr{U}$ of Γ and an open coefficient neighborhood $\mathscr{V}$ of X_0 such that every quadratic field in $\mathscr{V}$ has total multiplicity at most three among its isolated periodic orbits contained in $\mathscr{U}$.*

(ii) *For every open neighborhood $\mathscr{U}_1$ of Γ and every open coefficient neighborhood $\mathscr{V}_1$ of X_0, some quadratic field in $\mathscr{V}_1$ has three distinct simple limit cycles contained in $\mathscr{U}_1$. These fields can be chosen in a sequence tending to X_0 such that each of the three cycles converges to the entire Γ in spherical Hausdorff distance.*

Consequently the cyclicity of H_{14}^3 in the full quadratic family is exactly three.

The upper bound controls all isolated cycles whose entire orbits lie in the fixed neighborhood, with their full return multiplicities. The lower bound also gives cyclicity three in the usual Hausdorff sense of [3, Definition 2.1]. It is realized in the five-parameter family

$$\dot{x} = -y + Bx^2 + my^2 + ax, \qquad \dot{y} = x(1+y) + cy + dy^2.$$

Near-identity affine transformations and positive time changes make this family a local slice for every quadratic perturbation. The two parts of the theorem are proved independently in Theorems 12.10 and 13.3.

1.1 The mechanism of the proof

Two difficulties intervene between local passage estimates and a sharp bound. Several passages can lose their exit simultaneously, making their linear contraction actions arbitrarily large. Near the nilpotent end, a coefficient direction can also reveal a second scale with unrestricted

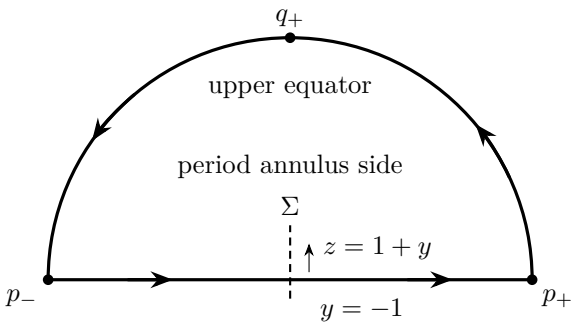


Figure 1: The base hemicycle, as a topological schematic of the northern compactification. Solid curves are the invariant finite line and the upper equatorial arcs; arrows on these curves give the base field's time direction. The thin arrow labels the central height, and the dashed segment is the central transverse section. The vertices $p_\pm$ are semi-hyperbolic and q_+ is nilpotent. Flattening the compactified finite line to a diameter is schematic; distances and curvature have no metric meaning.

separation from the first. These phenomena require estimates uniform in the complete passing domains. Even with such estimates, adding the counts of neighboring charts would exceed three. We instead compare the same two actual paths on one interval and keep the direction in which their critical values cross zero.

The two paths start at the same outgoing point on $y = 0$: one follows the lower arc backward to the central section, and the other follows the upper arc and the incoming lower arc forward. We compare their outputs in the original central height $z = 1 + y$. They have outputs $s_+ + U$ and $s_- + V$, where the reference difference $s_- - s_+$ depends on the field and is independent of the section input. With $\epsilon > 0$ the primary scale of Definition 2.6, we study

$$F = U - V - (s_- - s_+), \qquad K = \epsilon \log(V'/U').$$

The derivatives U', V' have the same nonzero sign, and

$$F' = U'(1 - e^{K/\epsilon}).$$

Hence F' and K have the same zero multiplicities. Once K has at most two zeros on that interval, Rolle's theorem gives at most three closing zeros. If a sufficiently small cycle occurs before the interval, it is unique and simple. It supplies a signed entrance at which the bound for K improves to one, leaving at most two further closing zeros. Section 3 states the precise thresholds and proves this counting implication.

Two additional facts make this implication applicable to the whole hemicycle. The bounded-energy orbits and the large-energy orbits on which the coefficients remain small are counted together by one source return. Whenever one monotone comparison does not already settle the outer cycles, an outer cycle supplies a coefficient inequality and bounds three equilibrium-free returning strips. Every intermediate input, including a nonclosing one, then has both complete central outputs. This is the physical interval proved in Proposition 11.6. The source bound is not added to a count on this interval when it already includes those source cycles.

Proposition 11.8 arranges the physical interval as a regular prefix followed by at most one terminal union. The regular crossings have one direction and supply the signs at its entrance. The hardest endpoint charts initially allow four closing zeros. Their derivative estimates alone therefore do not prove the theorem. On a fixed coefficient fibre, Proposition 11.9 transports the outer-cycle inequality to each actual open overlap. The three-saddle and weak-end arguments then use the already fixed entrance to obtain a two-zero bound for K on the whole union,

with a one-zero bound under the small-cycle entrance. These are the exceptional steps in Propositions 12.5 and 12.6; Proposition 12.9 joins them to the other endpoints.

At secondary scale the entrance is carried through the primary–wedge–secondary regular chain; the actual saddle ratio and the increasing weak numerator give the terminal bounds in Propositions 12.7 and 12.8.

Sections 4–5 construct the original passages and their real center comparison domains before coefficient division. Relative amplitudes are estimated separately from possibly divergent contraction actions. This distinction permits simultaneous small gaps, moving weak roots and exact cancellation of common powers. Sections 7–10 then classify the finite, infinite and secondary endpoints on their whole passing domains. The long secondary passage retains its exact scale exponent even when the product of the primary scale and the logarithm of the secondary scale is unbounded.

The lower bound starts from the reversible face $B = 0$, $m = \epsilon$. Section 5.3 describes its three-root variation after the basic passages and real centers are available. Section 13 constructs the coefficient curve, preserves all three real return paths for one field, and proves that each resulting cycle approaches the entire hemicycle.

The appendices provide the independent Abelian-integral threshold, the common-domain calculus, the endpoint algebra and uniform relative estimates. The domains and derivative orders are stated with the transitions and endpoint comparisons that use them.

2 Geometry and the return problem

We count cycles on one transverse section in a fixed neighborhood of the hemicycle. The geometric construction ensures that every counted cycle crosses this section exactly once. We then define the return on the inputs whose finite return paths stay in the neighborhood; the later changes of scale estimate this same return problem.

The base field is

$$X_0 : \qquad \dot{x} = -y, \qquad \dot{y} = x(1+y). \tag{2.1}$$

Put $z = 1 + y$. On $z > 0$ it has the first integral

$$H_0(x, z) = \frac{x^2}{2} + z - 1 - \log z. \tag{2.2}$$

Its Hessian is $\operatorname{diag}(1, z^{-2})$, its unique critical point is $(0, 1)$, and its sublevel sets are compact in $z > 0$. Consequently every positive level is a smooth strictly convex oval. The field is nonzero on that oval and tangent to it, so the oval is a periodic orbit.

We use the northern embedding

$$\iota(x, y) = \frac{(x, y, 1)}{\sqrt{1 + x^2 + y^2}}.$$

The boundary of this period annulus on the sphere is the Jordan curve Γ consisting of the closure of $y = -1$ and the upper equatorial semicircle. Indeed, at bounded x the lower branch of a level tending to infinity approaches $y = -1$. Every unbounded sequence on these ovals has $y > -1$ and hence can accumulate only on the upper semicircle; intersections with each fixed ray $y = k|x|$, $k \geq 0$, tend to infinity, as does the upper turning point. Write p_-, p_+ for the negative and positive horizontal points at infinity, and q_+ for the positive vertical point. The direction on $y = -1$ is from p_- to p_+. In the chart $y = 1/r$, $x = w/r$, multiplication by $r > 0$ gives

$$r' = -rw(1 + r), \qquad w' = -w^2 - r(1 + w^2).$$

Thus the other half of Γ runs from p_+ through q_+ to p_-.

2.1 A coefficient family for all quadratic perturbations

Consider the five-parameter family

$$\begin{aligned} X_\alpha = (P_\alpha, Q_\alpha), \qquad & P_\alpha = -y + Bx^2 + my^2 + ax, \\ & Q_\alpha = x(1+y) + cy + dy^2, \qquad \alpha = (B, m, a, c, d). \end{aligned} \tag{2.3}$$

The coefficient which breaks the invariant line is

$$\ell = d - c, \qquad \dot{z}\big|_{z=0} = \ell. \tag{2.4}$$

The origin remains an equilibrium; its linearization has determinant $1 + ac > 0$ near $\alpha = 0$.

Proposition 2.1. *The family (2.3), near-identity affine coordinate changes, and positive constant time changes form analytic local coordinates on the twelve-dimensional space of real quadratic planar vector fields near X_0. A uniform bound for isolated cycles of (2.3) in a fixed spherical collar of Γ, counted with return-germ multiplicity, therefore gives the same bound on a smaller fixed collar for every quadratic field in an open coefficient neighborhood of X_0.*

Proof. For A near the identity matrix, $b \in \mathbb{R}^2$ near zero, and $r \in \mathbb{R}$ near zero, define

$$\mathcal{S}(A, b, r, \alpha)(v) = e^r A^{-1} X_\alpha(Av + b).$$

There are four matrix entries, two translation parameters, one time parameter, and five parameters in α. For an affine infinitesimal change h, its column in the differential at the base point is $DX_0 h - Dh X_0$. The changes $(1,0), (0,1), (x,0), (y,0), (0,x), (0,y)$ give, respectively,

$$\begin{aligned} &v_1 = (0, 1+y), && v_2 = (-1, x), && v_3 = (y, x+xy), \\ &v_4 = (-x-xy, y+y^2), && v_5 = (-x, x^2+y), && v_6 = (-y, -x). \end{aligned}$$

The time column is $v_7 = (-y, x + xy)$. The remaining columns, in the parameter order of (2.3), are

$$(x^2, 0), \quad (y^2, 0), \quad (x, 0), \quad (0, y), \quad (0, y^2).$$

In a linear relation, the constant, first-component xy and second-component x^2 coefficients eliminate v_1, v_2, v_4, v_5. The remaining coefficients u, v, w of v_3, v_6, v_7 satisfy

$$\begin{pmatrix} 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = 0.$$

Its determinant is -2, and the five family columns occupy distinct monomials. Thus $D\mathcal{S}$ is invertible; the analytic inverse function theorem proves the local-coordinate assertion.

If $Y = \mathcal{S}(A, b, r, \alpha)$ and $\Phi(v) = Av + b$, then $D\Phi\, Y = e^r X_\alpha \circ \Phi$. Thus Φ sends trajectories to trajectories preserving time orientation. On transverse sections it conjugates the corresponding return germs. If ψ is the section coordinate change and P a return, then

$$\psi(P(u)) - \psi(u) = (P(u) - u) \int_0^1 \psi'(u + t(P(u) - u))\, \mathrm{d}t. \tag{2.5}$$

At a fixed point the last factor is nonzero. Isolation and the order of the displacement zero are preserved.

In homogeneous coordinates the spherical extension of Φ is

$$\widehat{\Phi}(\xi_1, \xi_2, \xi_3) = \frac{(A(\xi_1, \xi_2)^{\mathsf{T}} + b\xi_3, \xi_3)}{\|(A(\xi_1, \xi_2)^{\mathsf{T}} + b\xi_3, \xi_3)\|}.$$

Its homogeneous matrix is invertible, so it is a sphere diffeomorphism preserving the closed northern hemisphere. It converges uniformly to the identity as (A, b) tends to $(I, 0)$. Given a collar U_* for the normalized family, choose U with $\overline{U} \subset U_*$. One sufficiently small affine neighborhood then satisfies $\widehat{\Phi}(\overline{U}) \subset U_*$ for every such Φ. Intersecting the corresponding product neighborhood with the local-coordinate neighborhood of $\mathcal{S}$ gives an open twelve-dimensional coefficient neighborhood. Cycles in U map injectively to cycles in U_* with their multiplicities unchanged. □

2.2 A fixed collar and its transverse section

The initial energy count uses the base-field Chebyshev estimate of Proposition A.1. Fix its threshold H_{ECT} and choose the collar so that every point on its positive lower section has energy above that threshold.

Proposition 2.2. *There are a closed spherical annulus $\mathcal{A}$ containing Γ in its interior, a number $0 < z_L < 1/8$, and a coefficient neighborhood $\mathcal{V}$ of zero with the following properties. The annulus excludes the origin and the negative vertical point at infinity, and*

$$h(z_L) := z_L - 1 - \log z_L > H_{\mathrm{ECT}} + 2, \qquad \Sigma = \{x = 0,\ |z| < z_L\}. \tag{2.6}$$

Its lower part over $-1 \le x \le 1$ is the rectangle $[-1, 1] \times [-z_L, z_L]$ in (x, z) coordinates. Every affine periodic orbit contained in $U = \operatorname{int} \mathcal{A}$ is essential in $\mathcal{A}$ and crosses Σ exactly once, with $\dot{x} > 0$. Both $\mathcal{A}$ and $\mathcal{V}$ can be chosen invariant under the spatial and coefficient reflections defined below.

Proof. We first choose three vertex boxes in which no periodic orbit can be contained. Throughout this construction let all five coefficient moduli be at most $p_* = 1/64$. At a horizontal vertex put $x = \sigma/r$, $y = v/r$, where $\sigma \in \{-1, 1\}$, and multiply the field by $r > 0$. Its equations are

$$\begin{aligned} r' &= -\sigma B r - \sigma m r v^2 + \sigma r^2 v - a r^2, \\ v' &= \sigma r + \sigma(1 - B) v + d v^2 - \sigma m v^3 + \sigma r v^2 + (c - a) r v. \end{aligned}$$

The divergence of this polynomial field is

$$\sigma(1 - 2B) + 2dv - 4\sigma m v^2 + 4\sigma r v + (c - 3a) r.$$

On the square $|r|, |v| \le 1/8$ its difference from σ is at most $109/1024 < 1/8$. If a periodic orbit were contained in its northern part $r > 0$, its Jordan disk would lie in the same square and half-plane. Green's theorem would give zero for the integral of the divergence over that disk, contrary to its strict sign.

At the upper vertex use $y = 1/r$, $x = w/r$. The positively rescaled field is $r' = -rF$, $w' = G$, where

$$\begin{aligned} F &= (1 + r) w + d + r c, \\ G &= m - d w + (B - 1) w^2 + r[-1 + (a - c) w - w^2]. \end{aligned}$$

In $0 < r < 1/4$, $|w| < 1/4$, an equilibrium lies on

$$w = w_s(r) = -\frac{d + rc}{1 + r}, \qquad |w_s| \le p_*, \quad |w_s'| \le 2p_*.$$

On this graph,

$$G_r \le -1 + 2p_*^2, \qquad |G_w| \le 4p_* + 2p_*^2,$$

so

$$\frac{\mathrm{d}}{\mathrm{d}r} G(r, w_s(r)) \le -1 + 10p_*^2 + 4p_*^3 < -\frac{1}{2}.$$

There is at most one equilibrium in the box, and its determinant is

$$\det D(-rF, G) = r(1+r)\frac{\mathrm{d}}{\mathrm{d}r}G(r, w_s(r)) < 0.$$

It is a saddle. A periodic orbit contained in this rectangle would enclose either no equilibrium or one saddle, whereas its index is one. This excludes such an orbit.

Cut the regular parts of Γ at finitely many points, with two cuts inside each vertex box. The remaining compact regular arcs have flow-box neighborhoods. A sufficiently thin bicollar can be divided by radial crosscuts into strips, each contained in one of these flow boxes or vertex boxes. Make the crosscuts transverse at the chosen regular points, with their positive normal pointing in the direction of the base flow. Include the lines $x = -1, 0, 1$ on the finite arc and choose the stated rectangular part there. Its width can be made arbitrarily small; choose z_L to satisfy (2.6). The other widths can then be decreased so that the annulus excludes the two points specified in the statement. The construction can be made symmetric under $x \mapsto -x$ by choosing the two horizontal boxes and the regular crosscuts in reflected pairs.

On the central rectangle, $P_\alpha > 3/4$ after shrinking the coefficient neighborhood. On every other crosscut the normal component also has a positive minimum on its compact length. The finitely many strict inequalities persist on one common coefficient neighborhood, which we intersect with its reflected image.

Any periodic orbit in U must meet a crosscut. Otherwise it is contained in a single strip, where a flow-box coordinate or one of the vertex arguments excludes it. All crosscut intersections are transverse and positive. There are finitely many, since the orbit is compact and a transverse intersection is isolated. Each crosscut represents the same relative homology class in the annulus, so its signed intersection number with the orbit is the orbit's winding number. A simple closed curve in an annulus has winding number 0 or ± 1; positivity forces the value one. Hence the orbit is essential and meets every crosscut, in particular Σ, exactly once. □

2.3 Actual returns and periodic points

Only isolated affine periodic orbits whose northern images lie in U are counted. We first locate these orbits relative to the line $z = 0$ and bound the line-breaking coefficient by their section height. The return domain will then be defined by the complete trajectory.

Lemma 2.3. *Every periodic orbit contained in U encloses the origin and lies in $z > 0$. If s is its intersection with Σ, then*

$$0 < s < z_L, \qquad |\ell| < 3s. \tag{2.7}$$

Proof. An essential curve in $\mathcal{A}$ separates its two complementary components. The origin is on the bounded side of Γ, while the southern hemisphere is on the other side. Thus the affine Jordan disk of each such periodic orbit contains the origin.

For $\ell \neq 0$, every intersection with $z = 0$ has the same nonzero normal velocity ℓ. A periodic scalar function cannot have only crossings of one sign, so a periodic orbit cannot meet this line. For $\ell = 0$, invariance and uniqueness give the same exclusion. Choose one of the regular equatorial crosscuts of the preceding construction with its northern part in $y > 0$. Every counted orbit meets it, and hence is on the side $z > 0$ throughout.

Its lower segment is a graph over $[-1, 1]$, since $P_\alpha > 3/4$ on the central rectangle. On that graph the exact equation is

$$z_x = \ell A(x, z) + b(x, z)z, \quad A = \frac{(1-z)^2}{D}, \quad b = \frac{x + c(z-1)}{D},$$

where

$$D = 1 - z + Bx^2 + m(1-z)^2 + ax.$$

For $0 \le z \le z_L < 1/8$ and the coefficient bound used above, $3/4 < D < 9/8$, $A > 2/3$, and $|b| < 3/2$. If $\ell < 0$, integrate from 0 to 1 and use $z(1) > 0$ to obtain

$$s > |\ell| \int_0^1 A(t, z(t)) e^{-\int_0^t b(u,z(u))\,\mathrm{d}u}\,\mathrm{d}t \ge \frac{4}{9}(1 - e^{-3/2})|\ell| > \frac{|\ell|}{3}.$$

If $\ell > 0$, variation of constants from -1 to 0 gives the same lower bound for s after discarding its positive homogeneous term. The case $\ell = 0$ is immediate. □

Let $\Sigma_+ = \{x = 0,\ 0 < z < z_L\}$, identified with $(0, z_L)$. For a fixed α, define $\mathcal{D}_\alpha$ to be the set of $s \in \Sigma_+$ whose trajectory has a finite first positive-time return to Σ_+, with the entire intervening path affine, contained in U, and contained in $z > 0$. Write this return as $\mathcal{P}_\alpha(s)$.

Lemma 2.4. *The domain $\mathcal{D}_\alpha$ is open. The return is locally jointly analytic in its input and coefficients and is a local increasing diffeomorphism. Its isolated fixed points correspond bijectively to the cycles counted in U.*

Proof. A finite return path is compact in the prescribed open region and ends transversely. Analytic flow dependence and the implicit function theorem preserve its return under small input and coefficient changes. Compactness away from the endpoint flow boxes, and transversality inside them, exclude an earlier intersection. This proves openness and local joint analyticity. The variational determinant formula for a transverse return gives

$$\mathcal{P}'_\alpha(s) = \frac{P_\alpha(0, s-1)}{P_\alpha(0, \mathcal{P}_\alpha(s) - 1)} \exp\left(\int_0^{T(s)} \operatorname{div} X_\alpha(\Phi_t^\alpha(0, s-1))\,\mathrm{d}t \right) > 0.$$

Here P_α is the first component of the vector field in (2.3). Both normal velocities are positive. Finally, a fixed point closes the prescribed trajectory, and a counted cycle has exactly one section intersection by Proposition 2.2 and belongs to this domain by Lemma 2.3. □

Definition 2.5. The Poincaré multiplicity of an isolated cycle is the order of the zero of $\mathcal{P}_\alpha(s) - s$ at its section intersection. A return germ identically equal to the identity contributes no isolated cycle.

This definition is independent of the analytic section coordinate by (2.5). Multiplication of a displacement by an analytic nonvanishing factor also preserves its zero orders. Derivative zeros require a separate comparison, since differentiating a unit multiple also differentiates the unit. The domain $\mathcal{D}_\alpha$ may be disconnected. Thus a periodic orbit supplies a local return, while a comparison with more distant section points requires the intervening paths to be constructed.

2.4 One direction for each fixed field

The reflection used throughout the paper reverses time as well as space. Let

$$R(x, y) = (-x, y), \qquad \alpha^R = (B, m, -a, -c, -d). \tag{2.8}$$

Direct substitution in (2.3) gives

$$X_{\alpha^R}(Rv) = -DR\,X_\alpha(v), \qquad \Phi_t^{\alpha^R}(Rv) = R\Phi_{-t}^\alpha(v). \tag{2.9}$$

The collar and coefficient neighborhood are reflection invariant, and R fixes every point of Σ_+. Thus it preserves cycles, their central height, and isolation.

Let s_* be a periodic point and restrict its original forward return to an analytic increasing diffeomorphism $P : I \to P(I)$ on a small positive section interval. The reflected forward return near that point is $Q = P^{-1} : P(I) \to I$, and

$$Q(s) - s = -\big(P(Q(s)) - Q(s)\big). \tag{2.10}$$

Since $Q'(s_*) > 0$, composition with Q preserves the order of the displacement zero. The multiplicity is therefore unchanged by (2.8). This is an identity of germs at a periodic point. The reflected positive return follows the original trajectory in reverse time, so its domain is determined by that reverse passage.

We may consequently reflect a fixed field with $\ell < 0$ and carry out its count with $\ell \geq 0$. All section labels and comparison paths for that field are thereafter taken in this selected direction. Numerical cutoffs used uniformly in the family are chosen once on a reflection-invariant coefficient neighborhood, not chosen anew for each field. This sign-dependent choice is made only after the field is fixed. Coefficient differentiation and division across $\ell = 0$ use the original analytic auxiliary maps on their stated common domains.

2.5 Energy ranges and the primary normalization

The lower section has the analytic coordinate

$$h = h(s) = s - 1 - \log s > H_{\mathrm{ECT}} + 2, \qquad h'(s) = 1 - s^{-1} < 0.$$

For $0 < \eta < 1$ and a fixed $H_* > h(z_L) + 1$, set

$$\mathscr{M}(h) = \max\{|B|h, |m|h, |a|\sqrt{h}, |c|\sqrt{h}\}.$$

The three energy classes are

$$\begin{array}{lll} h \leq H_*, & & \text{bounded energy,} \\ h > H_*, & \mathscr{M}(h) \leq \eta, & \text{small coefficients at the energy scale,} \\ h > H_*, & \mathscr{M}(h) > \eta, & \text{parameter-dominant energy.} \end{array} \tag{2.11}$$

The first two classes will be counted together. Their thresholds are fixed before the final coefficient shrink, and the bounded-energy interval lies entirely above H_{ECT}. This classification applies to actual periodic points and extends no return domain.

Definition 2.6. The primary scale and normalized coefficients, for $\rho > 0$, are

$$\begin{aligned} \rho = \max\{\sqrt{|B|}, \sqrt{|m|}, |a|, |c|\}, \qquad \epsilon = \rho^2, \\ (B, m, a, c) = (\rho^2 \bar{B}, \rho^2 \bar{m}, \rho \bar{a}, \rho \bar{c}), \qquad \nu = \ell/\rho, \quad \bar{d} = d/\rho = \bar{c} + \nu. \end{aligned} \tag{2.12}$$

Thus $\max(|\bar{B}|, |\bar{m}|, |\bar{a}|, |\bar{c}|) = 1$. The generators in original and primary coefficients are

$$\begin{aligned} \tau_0 = a + c, \qquad \sigma_0 = c(B + m), \\ \tau_1 = \bar{a} + \bar{c} = \tau_0/\rho, \qquad \sigma_1 = \bar{c}(\bar{B} + \bar{m}) = \sigma_0/\rho^3. \end{aligned} \tag{2.13}$$

The normalized vector lies on the union of the signed faces of $[-1, 1]^4$. Smooth parameter arguments are made in the indicated local coefficient charts, not by differentiating the maximum across its corners. For $\rho > 0$, the two products $\tau_0\sigma_0$ and $\tau_1\sigma_1$ have the same sign, but

$$\frac{\tau_1}{\sigma_1} = \epsilon \frac{\tau_0}{\sigma_0} \qquad (\sigma_0 \neq 0). \tag{2.14}$$

If $\rho = 0$, then $B = m = a = c = 0$, while $d = \ell$ may be nonzero. All energies belong to the first two classes because $\mathscr{M} = 0$. No barred coefficients, ν, or quotient by ϵ is used in that case.

For later calculations we record the exact scaled equations. With $\xi = t/\rho$, $X = \rho x$, and $z = 1 + y$, the lower equations are

$$\begin{aligned} X_\xi &= \epsilon\{1 - z + \bar{B}X^2 + \bar{a}X + \epsilon\bar{m}(1-z)^2\}, \\ z_\xi &= Xz + \epsilon\bar{c}\,z(z-1) + \rho\ell(1-z)^2. \end{aligned} \tag{2.15}$$

With $X = \rho x$ and $Y = \rho^2 y$, the same field and time give

$$\begin{aligned} X_\xi &= Y(\bar{m}Y - 1) + \epsilon(\bar{B}X^2 + \bar{a}X), \\ Y_\xi &= (Y + \epsilon)(X + \bar{c}Y) + \nu Y^2. \end{aligned} \tag{2.16}$$

These identities do not assume that ν is bounded.

Lemma 2.7. *A parameter-dominant cycle has $\rho > 0$ and satisfies*

$$\rho\sqrt{h} > \eta, \qquad |\ell| < 3e^{-\eta^2/\rho^2}, \qquad \frac{|\ell|}{\rho^k} \le C_{k,\eta} e^{-\eta^2/(2\rho^2)} \quad (k \ge 0).$$

Proof. If $\rho\sqrt{h} \le \eta$, the four terms in $\mathscr{M}(h)$ are bounded by $\eta^2, \eta^2, \eta, \eta$, respectively, contradicting the third class. Since $0 < s < 1$, the exact identity $s = e^{-h-1+s}$ gives $s < e^{-h}$. Apply (2.7) and retain half of the exponential to absorb any fixed power of ρ^{-1}. □

The estimate uses the height of an actual cycle. The ratio ℓ/s still has only the bound in (2.7); derivatives along parameter curves require the common-domain estimates below.

Lemma 2.8. *Every periodic orbit counted in U meets $y = 0$ exactly once on each half-axis. For $\rho > 0$, write these intersections, in the selected direction, as $(p/\rho, 0)$ and $(-q/\rho, 0)$ with $p, q > 0$. The upper connecting arc lies in $y > 0$. Nested cycles have increasing p and q towards the outer cycle. Reflection exchanges the labels: $p_{\mathrm{ref}} = q_{\mathrm{orig}}$ and $q_{\mathrm{ref}} = p_{\mathrm{orig}}$.*

Proof. By Lemma 2.3, the Jordan disk contains the origin. Each horizontal half-axis therefore meets its boundary. On $y = 0$, the normal velocity is $\dot{y} = x$, so all crossings of the positive ray are upward and all crossings of the negative ray are downward. Their signed intersection number is the winding number one; consequently each ray is crossed exactly once. The arc between the upward and downward crossings is in $y > 0$. Inclusion of the Jordan disks orders their two ray intersections and proves the nesting assertion. Equation (2.9) exchanges the two crossings while reversing time, giving the last assertion. □

In particular a cutoff imposed on the reflected outgoing label is a cutoff on the original incoming label, not on the original outgoing label. All later endpoint cutoffs refer to the direction already selected for the fixed field. Varying a section input always keeps the original coefficient vector fixed; any induced changes of scaled coordinates must be included in the chain rule.

3 A conditional reduction to three closing zeros

The sharp count uses one passing interval for the selected outer cycles. Four statements about a fixed field provide this interval and its zero bounds: the initial energy classes have a joint count, an outer cycle supplies the complete comparison paths, the logarithmic derivative ratio has at most two zeros there, and a small cycle forces a stronger entrance sign. We prove the resulting count here. Sections 6, 11 and 12 establish these hypotheses on the same interval.

3.1 Multiplicity and the three counting levels

On a regular section interval, all derivatives in this section are taken with the original field fixed. If f is real analytic on a neighborhood of an interval J, write $Z(f;J)$ for the sum of the orders of its isolated zeros in J. The identically zero function contributes no isolated zeros. At an included finite endpoint we count the full order of the analytic germ, not a one-sided order. For an open or unbounded interval a bound for Z means the same bound for every finite selection of zeros with their full multiplicities. No regular extension at an excluded singular endpoint is implicit in this convention.

Lemma 3.1 (Rolle's inequalities with multiplicity)**.** *For a real analytic function on a neighborhood of an interval J,*

$$Z(f;J) \le Z(f';J) + 1 \tag{3.1}$$

whenever the right-hand side is finite. Suppose in addition that a is the left endpoint of J, f extends analytically across a, and $f(a)f'(a) > 0$. On the part of J to the right of a,

$$Z(f;J) \le Z(f';J). \tag{3.2}$$

For a right entrance the corresponding condition is $f(a)f'(a) < 0$. Equivalently, in either case the derivative in the direction from the entrance into the interval must have the same sign as $f(a)$.

Proof. For any finite selection of zeros, differentiation lowers each order by one and Rolle gives an additional zero between consecutive zeros; this proves (3.1). Under the left-entrance condition, after changing signs, $f(a), f'(a) > 0$. Before the first selected zero, the derivative must change from positive to negative, supplying one additional derivative zero. This proves (3.2); reverse the coordinate at a right entrance. Constant functions have no isolated zeros. □

Now assume $\rho > 0$ and put $\epsilon = \rho^2$. Let two comparison paths be defined on one open passing interval I, with analytic central outputs $s_+ + U(t)$ and $s_- + V(t)$. The central coordinate is the original height $z = 1 + y$. The reference values $s_\pm$ are fixed while t varies; in the applications their difference is

$$s_- - s_+ = \rho\nu H = \ell H, \qquad H > 0,$$

where H may depend on the field but not on the input. Assume that U' and V' are nonzero and have the same sign throughout I. Define

$$F = U - V - \rho\nu H, \qquad K = \epsilon \log \frac{V'}{U'}, \qquad M = wK', \qquad w > 0. \tag{3.3}$$

The multiplier w is real analytic and positive on I. It may tend to zero or infinity at an excluded singular endpoint. Only the positivity of V'/U' is needed to define K; neither derivative is required to be positive individually.

Lemma 3.2 (Closing differences and logarithmic ratios)**.** *Under these hypotheses, on every interval $J \subset I$,*

$$\begin{aligned} Z(F';J) &= Z(K;J), \\ Z(K';J) &= Z(M;J), \\ Z(F;J) &\le 1 + Z(K;J) \le 2 + Z(M;J), \end{aligned} \tag{3.4}$$

provided the indicated bounds are finite. The statements include multiplicities. If an outward entrance for K is available, the second Rolle step improves to $Z(K;J) \le Z(M;J)$ on the interval beyond that entrance.

Proof. The input-independent reference difference disappears upon differentiation, giving the exact identity

$$F' = U'(1 - e^{K/\epsilon}).$$

The function $(1 - e^{v/\epsilon})/v$ extends analytically and is nonzero at $v = 0$, with value $-1/\epsilon$. Consequently F' and K have precisely the same zero orders. Multiplication by w likewise preserves the zero orders of K'. Apply Lemma 3.1 first to F, then to K. If K is constant, F is constant or strictly monotone; this also settles the constant cases. The entrance improvement is (3.2) applied to K. □

The passage from M to closing zeros therefore uses two Rolle steps, through K and then F. A signed entrance for K improves the first step; the step from K to F still costs one zero. The functions U, V include every regular connection along their respective paths, so those connections also enter the derivative ratio.

The following elementary observation will join local estimates without adding their bounds.

Lemma 3.3 (Joining crossings of one sign)**.** *Let I_1, I_2 be intervals with a nonempty open overlap, with the part of I_1 outside the overlap to its left and the part of I_2 outside the overlap to its right. Let g be C^1 on an open neighborhood of their union. Suppose that each interval contains at most one zero, that every zero is simple, and that all zeros cross from the same sign $s_0 \in \{-1, 1\}$ to $-s_0$ as the common coordinate increases. Then their union contains at most one zero, with multiplicity. The same conclusion holds for a finite ordered chain of such intervals.*

Proof. Two zeros would lie on opposite sides of the overlap, forcing signs $-s_0$ and s_0, respectively, throughout that overlap. This contradiction proves the assertion; induction gives the finite chain. □

The joining argument uses the sign of one function throughout an open overlap of actual passing inputs. Endpoint asymptotics will therefore be used together with such an overlap.

3.2 Changes of representation and reflection

An analytic unit multiplying F preserves its isolated zeros and their orders. It does not preserve derivative zeros: for a nonconstant positive unit $a(t)$, $(aF)' = aF' + a'F$. Thus a source closing function and F may count the same fixed points with the same multiplicities, while their derivative zeros must be estimated in the chosen representation.

A common change of input has a different effect. If $t = \phi(v)$ is an analytic local diffeomorphism, then

$$\epsilon \log \frac{(V \circ \phi)'(v)}{(U \circ \phi)'(v)} = K(\phi(v)). \tag{3.5}$$

It preserves the zero orders of K and, since $(K \circ \phi)' = (K' \circ \phi)\phi'$, those of its derivative. When $\phi' < 0$, the entrance endpoint and the direction of departure from it must both be reversed. These identities retain the same central output coordinate. A nonlinear change $z \mapsto \psi(z)$ of that output would multiply the derivative ratio by $\psi'(s_- + V)/\psi'(s_+ + U)$, which is not generally one away from a closing zero.

The reflection (2.8) reverses time and fixes the central height. It preserves ρ, ϵ and changes the signs of $\ell, \tau_0, \sigma_0, \tau_1, \sigma_1$.

The transformation of K follows from the actual comparison paths. Write the upper transition as $q = \mathcal{U}(p)$, with $\mathcal{U}'(p) > 0$, and the two lower central outputs as $L_+(p)$ and $L_-(q)$. On their common passing domain,

$$K(p) = \epsilon \log \frac{L'_-(\mathcal{U}(p))\mathcal{U}'(p)}{L'_+(p)}.$$

Reflection exchanges the lower paths and replaces the upper transition by its inverse. Its outgoing coordinate is the original q, not the original p. The chain rule therefore gives

$$\begin{aligned} K_{\mathrm{ref}}(q) &= -K(p), \\ K_{\mathrm{ref}}(q)\,\partial_q K_{\mathrm{ref}}(q) &= \frac{K(p)K'(p)}{\mathcal{U}'(p)}, \qquad q = \mathcal{U}(p). \end{aligned} \tag{3.6}$$

The second line preserves $KK' > 0$ at corresponding entrances. Since σ_1 changes sign, the condition $\operatorname{sgn} K = -\operatorname{sgn}\sigma_1$ is preserved at corresponding points, as is $\tau_1\sigma_1 < 0$.

The fixed-field direction, exchanged p, q labels and coefficient division follow the conventions of Section 2; (2.10) preserves the return multiplicity.

3.3 The four hypotheses

Fix the collar and the return convention of 2.2. A counted cycle means an isolated periodic orbit in that collar, with its Poincaré multiplicity. For $\rho > 0$, label it by $p = \rho x_{\mathrm{out}}$, where $(x_{\mathrm{out}}, 0)$ is its outgoing intersection with the positive horizontal half-axis in the chosen orientation. Distinct nested cycles have distinct ordered labels. The normalization and generator ratios are (2.13)–(2.14), with the zero-scale convention of Definition 2.6.

The initial block $\mathcal{I}_0$ combines the bounded-energy and small-coefficient classes of (2.11); the remaining class is parameter-dominant. The source representative F_{src} counts these two initial classes together, even when it differs from (3.3).

Fix $p_\sharp$ with $0 < p_\sharp \le \eta/8$. The constants, collar, and coefficient neighborhood in the following hypotheses are fixed before a field and a finite selection of its cycles are chosen. Three distinct thresholds will be used: $p_\sharp$ for an outer witness, $p_\sharp/8$ for the small-cycle trigger, and $p_\sharp/4$ for uniqueness and the strengthened entrance. The hypotheses concern actual cycles and actual passing intervals. Hypotheses (II)–(IV) concern $\rho > 0$; the zero-scale case is included in (I).

(I) *The initial count and the outer threshold.* The total multiplicity in $\mathcal{I}_0$ is at most two. Every parameter-dominant cycle has $p > \eta/8 \ge p_\sharp$. When $\rho = 0$, all counted cycles belong to $\mathcal{I}_0$.

(II) *An outer witness and a common physical interval.* In either alternative below, the closing zeros on a comparison interval represent all selected cycles contained in it, with their full Poincaré multiplicities. Suppose a finite selection contains a cycle with $p \ge p_\sharp$, and let x be its first such member in increasing p. One of the following alternatives holds. Either a single physical comparison interval contains x and every selected cycle at or beyond it, with F' of strict constant sign on that interval; or the original coefficients satisfy

$$|\ell| \le \exp\left(-\frac{p_\sharp^2}{20\epsilon}\right)(|\tau_0| + |\sigma_0|). \tag{3.7}$$

In the latter alternative, let γ be any outermost member of a finite selection with $p_\gamma \ge p_\sharp$. For either $p_0 \in \{p_\sharp/8, p_\sharp/4\}$, the entire segment

$$J(p_0, \gamma) = [p_0, p_\gamma]$$

is a physical passing interval for the two comparisons in (3.3). It includes every intermediate nonperiodic comparison point and every selected cycle with label in this range, including cycles in $\mathcal{I}_0$. At the two endpoints the comparison maps extend to open neighborhoods.

(III) *The logarithmic-ratio count and its signed entrance.* Under (3.7), on each segment supplied by (II),

$$Z(K; J(p_0, \gamma)) \le 2.$$

If also $\tau_1\sigma_1 < 0$ and $\operatorname{sgn} K(p_0) = -\operatorname{sgn}\sigma_1$, then

$$Z(K; J(p_0, \gamma)) \le 1.$$

This strengthened conclusion is an assumption about the same physical interval and the specified coefficient and entrance conditions. It is not deduced from a sign condition for FF'.

(IV) *The information supplied by a small cycle.* Under (3.7), at most one actual counted cycle has $p < p_\sharp/4$, and any such cycle is simple. If an actual cycle has $p < p_\sharp/8$, the regular comparison is defined at $p = p_\sharp/4$ and satisfies

$$\tau_1\sigma_1 < 0, \qquad 0 < -\frac{\tau_1}{\sigma_1} \le \frac{p_\sharp^2}{300}, \qquad \operatorname{sgn} K(p_\sharp/4) = -\operatorname{sgn}\sigma_1 \tag{3.8}$$

Thus a segment supplied by (II) with entrance $p_\sharp/4$ has the entrance required by the second clause of (III).

Hypothesis (II) constructs the two paths at every intermediate input, so (III) can bound one logarithmic ratio on the entire segment. The stronger bound in (III) will follow by joining the endpoint estimates with the stated coefficient signs and entrance. The entrance form of Rolle's inequality then gives the required zero count.

Proposition 3.4 (Conditional bound of three)**.** *Assume* (I)–(IV). *Then every finite selection of counted cycles of a fixed field has total Poincaré multiplicity at most three. Consequently the complete set of isolated counted cycles satisfies the same bound. Moreover, if* $\rho > 0$, (3.7) *holds, and there is an actual cycle with* $p < p_\sharp/8$, *the total multiplicity of cycles with* $p \ge p_\sharp/8$ *is at most two.*

Proof. The case $\rho = 0$ follows from (I). For $\rho > 0$, take a finite selection, retaining full multiplicities, and adjoin an actual outer witness $p \ge p_\sharp$ and an actual small cycle $p < p_\sharp/8$ whenever such cycles exist but are absent from the selection. Bounds for this enlarged selection imply bounds for the original one.

If no outer witness exists, (I) puts every selected cycle in $\mathcal{I}_0$, giving the bound two. Otherwise let x be the first selected witness. The selected cycles preceding x belong to $\mathcal{I}_0$ and contribute at most two. In the monotone alternative of (II), the remaining cycles contribute at most one simple zero, giving the bound three.

In the other alternative, (3.7) holds. Let γ have the largest label in the enlarged selection; no outermost cycle among all cycles of the field is required. If no small cycle exists, all selected cycles lie in $J = J(p_\sharp/8, \gamma)$. Hypotheses (II)–(III) and Lemma 3.2 give

$$Z(F; J) \le 1 + Z(K; J) \le 3.$$

This already counts any cycles of $\mathcal{I}_0$ in the selection. If a small cycle exists, (IV) makes it simple and unique below $p_\sharp/4$. All remaining selected cycles lie in $J = J(p_\sharp/4, \gamma)$, where (3.8) supplies the strengthened clause of (III). Hence

$$Z(F; J) \le 1 + Z(K; J) \le 2.$$

Restoring the small cycle gives three; the entrance improvement is used only at the K level.

For the additional assertion, adjoin an actual small cycle to any finite selection above $p_\sharp/8$, and adjoin an outer witness if one exists. The last case directly gives the bound two above the threshold. Without an outer witness, (I) allows at most one additional multiplicity; in the monotone case the small cycle uses one of the two initial multiplicities, leaving at most one there and one in the monotone remainder. This again gives two above the threshold.

The bounds for arbitrary finite selections bound the complete set of isolated cycles. Equation (2.10) transfers them to the opposite orientation; identity returns contribute no isolated cycles. □

Section 12 proves (III) on the interval supplied by Propositions 11.8 and 11.9. The regular chain supplies the entrance signs. The primary three-saddle and weak ends use the fixed seam, while the secondary ends carry the entrance through the wedge. Each argument controls the same K on the whole union, which is what the count above requires.

4 Uniform transitions from the original field

When a slow root blocks passage between separated sections, the transit time diverges and the normal output tends to zero. To compare such passages uniformly, we separate the exact scalar phase from the contracting normal motion and estimate its relative amplitude before connecting to the original central height. The relative amplitude has bounded finite jets at a lost exit, while the absolute output and its jets vanish. These are the estimates needed for center division and zero counting at the simple, repeated and weak ends of the primary atlas.

Throughout this section $\mathbf{b} = (\bar{B}, \bar{m}, \bar{a}, \bar{c}, \nu)$ ranges in an ordinary bounded coefficient box, slightly larger than the compact set being used. Coefficient variations are taken in this fixed box, with its normalization held fixed. The scale $\epsilon > 0$ is independent on the auxiliary box. The physical coefficients are recovered by Definition 2.6. Put $E_\epsilon = \epsilon\partial_\epsilon$. For a section input s, a bound of total order N means a bound for

$$\partial_s^i \partial_{\mathbf{b}}^\alpha E_\epsilon^j, \qquad i + |\alpha| + j \le N.$$

At a singular power endpoint, $s\partial_s$ replaces ∂_s when explicitly stated; several inputs contribute the sum of their orders. Graph construction uses ordinary scale derivatives on a signed scale neighborhood, where they can remove an explicit ϵ factor.

Choose the finite derivative order first, including the orders spent by factorization and coordinate changes on larger domains. Then choose the normal neighborhoods and scale cutoff. With these choices fixed, the constants are uniform in the transit time and minimum positive slow gap. Each physical passage follows an original-field trajectory between the stated sections.

4.1 An exact phase on the lower strip

The lower strip contracts in the normal direction while its slow phase moves at order ϵ. The horizontal coordinate of the compactified chart also moves during fast contraction, so it cannot serve as this phase. We first project along the zero-scale fast fibres, then correct the projection at positive scale to obtain an exact scalar equation. In the incoming chart put

$$T = (-X)^{-1}, \qquad v = Y/(-X), \qquad w = v + \epsilon T = \epsilon z T, \qquad \frac{\mathrm{d}\zeta}{\mathrm{d}\xi} = -X > 0.$$

Direct substitution in the scaled original field gives

$$\begin{aligned} \dot{T} &= \epsilon T[\bar{B} - \bar{a}T + (1 + \epsilon\bar{m})T^2] - (1 + 2\epsilon\bar{m})T^2 w + \bar{m}Tw^2, \\ \dot{w} &= \epsilon^2 \nu T^2 + [-1 + \epsilon\bar{B} + \epsilon(1 + \epsilon\bar{m})T^2 - \epsilon(\bar{a} + \bar{c} + 2\nu)T]w \\ &\quad + [\bar{c} + \nu - (1 + 2\epsilon\bar{m})T]w^2 + \bar{m}w^3. \end{aligned} \tag{4.1}$$

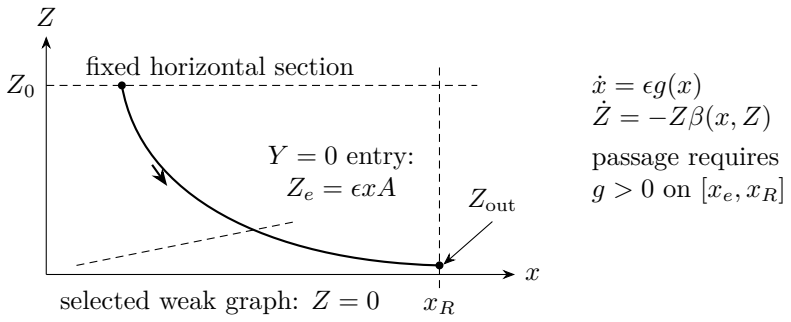


Figure 2: Section conventions in the straightened lower phase chart (schematic). The solid curve illustrates a forward normally contracting passage on its positive slow component; dashed lines are transverse sections, not invariant manifolds. A fixed horizontal entry and the finite $Y = 0$ entry have different normal sizes. Their labels are retained in the estimates. The drawing assumes separated entry and exit; it does not represent a passage through a scalar root or a collision of sections.

Dots in this subsection denote ζ-time. The outgoing chart is obtained by reversing time and replacing $(\bar{a}, \bar{c}, \nu)$ by $(-\bar{a}, -\bar{c}, -\nu)$.

Fix a small $R > 0$ such that $1 \pm \bar{a}X + \bar{B}X^2 > 3/4$ for $0 \le X \le 2R$ throughout the chosen bounded coefficient box, and put $T_R = 1/R$. All constructions below take place first on a slightly larger T-interval than $[0, T_R]$.

Proposition 4.1 (Lower phase)**.** *For each prescribed finite order N, there are a normal width and a scale cutoff for which the following statements hold on common smaller state and coefficient domains. There is an invariant graph*

$$w_g(T) = \epsilon^2 \nu T^2 H(T; \mathbf{b}, \epsilon), \qquad H(0; \mathbf{b}, \epsilon) = \frac{1}{1 + \epsilon \bar{B}}, \tag{4.2}$$

and coordinates (x, Z), with $Z = w - w_g(T)$ and $x = T$ on that graph, in which the field is exactly

$$\dot{x} = \epsilon g_I(x), \qquad g_I(x) = xF_I(x), \qquad \dot{Z} = -Z\beta_I(x, Z). \tag{4.3}$$

Their finite mixed derivatives, including the ordinary scale derivatives used in their construction, are bounded. The normal rates satisfy

$$\beta_I > \beta_*, \qquad \beta_I + Z\partial_Z \beta_I > \beta_*, \qquad \beta_I(0, 0) = 1 - \epsilon \bar{B}.$$

Moreover,

$$\begin{aligned} &F_I(x) = \bar{B} - \bar{a}x + (1 + \epsilon \bar{m})x^2 - \epsilon\nu(1 + 2\epsilon\bar{m})x^3 H(x) + \epsilon^3 \nu^2 \bar{m} x^4 H(x)^2, \\ &F_I(0) = \bar{B}, \qquad F_I'(0) = -\bar{a}. \end{aligned} \tag{4.4}$$

Thus $F_I'' > 1$ after the scale cutoff is reduced, and F_I has a positive margin near the fixed exit $x_R = T_R$. In the derivative convention of this section, $\beta_I(x, 0) = 1 + O_{C^N}(\epsilon)$. Ordinary differentiation with respect to ϵ may remove the factor ϵ in this last estimate.

Proof. At $\epsilon = 0$ the field is

$$\dot{T} = Tw(\bar{m}w - T), \qquad \dot{w} = w[-1 + (\bar{c} + \nu - T)w + \bar{m}w^2].$$

Choose the normal width small enough that the bracket in the second equation is less than $-2\beta_*$, and that the variation of T before normal contraction is contained in the larger T-interval. Extend the field smoothly outside this larger interval, preserving $T = 0$, $w = 0$ and

this contraction. One can do this by bounded cutoffs in the coefficient functions multiplying their displayed factors. The extension is used only to define a complete auxiliary flow.

Define its fast projection by

$$\pi_0(T,w)=\lim_{t\to\infty}T(t).$$

The decay $|w(t)|\le C|w|e^{-2\beta_* t}$ makes $\dot T$ integrable. The variational system has a negative normal diagonal, an integrable horizontal diagonal, and horizontal and coefficient couplings containing $w(t)$. Variation of constants bounds horizontal variations and gives normal decay. Higher normal forcings and derivatives of $\dot T$ retain a normal state or lower normal variation; induction bounds them by $C(1+t)^M e^{-\beta_* t}$, including frozen coefficient derivatives. Hence

$$\pi_0(T,0)=T,\qquad \pi_0(0,w)=0,\qquad \pi_0-T=O\big(Tw(|T|+|w|)\big),\qquad \partial_T\pi_0>1/2. \tag{4.5}$$

The order follows by integrating $Tw(\bar m w-T)$, whose factors the flow preserves. A smaller normal width gives the derivative margin.

Use $\chi=\pi_0(T,w)$ as base coordinate. Its velocity vanishes identically when $\epsilon=0$, and on $\chi=0$. Successive Hadamard integration therefore writes it as $\epsilon\chi A$. Its time-one state derivative in the base direction is $1+O(\epsilon)$, whereas the normal multiplier is at most $e^{-\beta_*}+O(\epsilon)$. The time-one graph transform is consequently a contraction on a small set of graphs over the extended base. For its kth state derivative the coefficient of the highest graph derivative is bounded by

$$e^{-\beta_*}(1+C|\epsilon|)^k+C|\epsilon|<1$$

once k is fixed and the scale cutoff is small enough. The lower terms depend on lower graph derivatives. This proves convergence in C^k by induction, applied also to differences of consecutive iterates.

Ordinary ϵ differentiation introduces an $O(1)$ base shear. Order the induction first by total order, then by the number of coefficient derivatives. The highest unknown retains the same contracting multiplier, and forcing of the same total order involves fewer coefficient derivatives. The scalar variational equations give

$$|D^\alpha\phi_t|\le C_\alpha(1+|t|)^{m_\alpha}e^{C_\alpha|\epsilon||t|}. \tag{4.6}$$

An ordinary scale derivative adds the time factor in the variation-of-constants integral; iteration gives the displayed bound. Normal contraction absorbs all finite polynomial factors. The graph transform therefore converges with the prescribed mixed smoothness, and uniqueness makes its graph invariant under the whole flow.

Return to the original T parametrization. The graph is zero at $\nu=0$ and at $\epsilon=0$. Its first ordinary ϵ derivative is also zero at $\epsilon=0$: differentiation of the invariance equation has zero forcing there, because the independent forcing in (4.1) begins with $\epsilon^2\nu T^2$. Finally its value and first T derivative at $T=0$ vanish. The point $(0,0)$ is invariant, and its weak and normal linear eigendirections are distinct. These statements give the five factors $\epsilon^2\nu T^2$ by finite-order Hadamard integration. To retain order $N+2$ afterwards, construct the graph with at least $N+7$ joint derivatives. Comparing the T^2 coefficient in its invariance equation gives

$$2\epsilon\bar B\, w_{g,2}=\epsilon^2\nu+(-1+\epsilon\bar B)w_{g,2},$$

which proves the exact value of $H(0)$.

Subtract the graph and reparametrize the base so that its coordinate x_0 agrees with T on the graph. In these coordinates

$$\dot x_0=\epsilon a(x_0)+\epsilon Z\Gamma(x_0,Z),\qquad \dot Z=-Z\widetilde\beta(x_0,Z).$$

The factor ϵ survives because the zero-scale coordinate is π_0; invariance of $T = 0$ also makes $\Gamma(0, Z) = 0$. Let ϕ_t be the complete scalar flow of $\dot{x}_0 = \epsilon a(x_0)$. Set

$$x = \lim_{t\to\infty} \phi_{-t}(x_0(t)).$$

Indeed,

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi_{-t}(x_0(t)) = \epsilon D\phi_{-t}(x_0(t))Z(t)\Gamma(x_0(t), Z(t)),$$

and each selected derivative is bounded by $C(1+t)^M e^{-(\beta_* - C_N|\epsilon|)t}$, by (4.6). Reduce the scale cutoff so that $C_N|\epsilon| < \beta_*/8$ for all the finitely many variational products used below; the majorants are then integrable. The limit is a coordinate with the prescribed finite smoothness, equals x_0 on the graph, and differs from x_0 by $\epsilon x_0 Z O_{C^N}(1)$. The flow identity for the limit gives $\dot{x} = \epsilon a(x)$ exactly. Restriction of the original T equation to the graph now yields (4.4).

The normal linear eigenvalue at $T = w = 0$ is exactly $-1 + \epsilon\bar{B}$. At zero scale its value along $w = 0$ is -1. Continuity, on the common compact domains and with the stated finite jets, proves the normal estimates. Convexity follows from (4.4). Finally $x_R = 1/R$ and the choice of R give $\bar{B} - \bar{a}x_R + x_R^2 > 3x_R^2/4$; reducing the scale cutoff preserves a fixed positive margin there. □

Lemma 4.2 (Sections and the actual exit)**.** *On the fixed horizontal section $v = v_h > 0$, with v_h sufficiently small, the phase input is $t = x(T, v_h + \epsilon T) = Tk(T)$, where $k > c > 0$ and $t_T > c > 0$. The normal input has a positive lower bound. On the section $Y = 0$ with T in a fixed compact subset of $(0, T_R)$, it is instead $w = \epsilon T$, the normal input is ϵ times a positive bounded unit, and $x = T + O_{C^N}(\epsilon)$.*

Restrict either entry to a fixed positive phase distance from x_R. Its forward orbit reaches the fixed phase exit $x = x_R$ if and only if

$$g_I(s) > 0 \qquad (t \le s \le x_R). \tag{4.7}$$

This criterion concerns the phase exit in the signed tube. Positivity of the original central height is imposed separately after composition. The entire transit remains in the chosen strip, and its duration T_f is at least c/ϵ. Its selected finite time jets obey

$$|D^\alpha T_f| \le C\epsilon^{-M}(1 + T_f)^M e^{C_N \epsilon T_f}.$$

Proof. Equation (4.5) and $w_g = O(\epsilon^2 T^2)$ give $Z > v_h/2$ on the horizontal entry and positive-unit Z/ϵ on the finite entry. They also give the phase margins, reversed in $q = 1/T$ on fixed finite compact intervals.

The normal coordinate contracts, so it stays in the selected tube. The scalar phase crosses exactly under (4.7): an equilibrium blocks finite-time crossing, whereas positivity on the closed interval gives a finite transit integral. The larger domain keeps the original orbit in the unchanged field. Bounded g_I and separated sections give $T_f \ge c/\epsilon$. For the time jets apply the implicit hitting calculation of Appendix D.1 to $\phi_{T_f}(t) = x_R$, using (4.6) and the exit speed $\epsilon g_I(x_R) \ge c\epsilon$. Thus no minimum-gap denominator is introduced. □

4.2 Relative contraction and the original central height

The exact phase reduces a passage to scalar transit and normal contraction. Its linear normal factor may have a divergent action, so we first estimate the nonlinear amplitude relative to that factor. These relative estimates remain uniform as the transit time diverges. We then compute the rates for a moving entry and connect the output to $X = 0$, where the closing difference is measured.

Proposition 4.3 (Relative normal contraction)**.** *Fix $N \geq 1$ and a slightly larger state and coefficient domain on which*

$$\dot{w} = qf(w; \mathbf{b}), \qquad \dot{Z} = -Z\beta(w, Z; \mathbf{b}), \qquad \beta,\ \beta + Z\beta_Z \geq b_* > 0. \tag{4.8}$$

Here $0 < q \leq 1$, the coefficients and the field have bounded selected jets through order $N+2$. The triangular field has a complete auxiliary extension preserving these bounds and the normal margins. The selected derivatives are ordinary normal-input, slow-entry and coefficient derivatives and, when $q = \epsilon$, Euler scale derivatives. Suppose that their scalar flow variations through the required order are bounded by

$$C_N(1+t)^{M_N} e^{c_N \varpi t}. \tag{4.9}$$

Choose the normal neighborhood and slow width so that $c_N \varpi < b_*/8$, increasing c_N first to cover all products of variations through this order. At the fixed slow exit suppose that its velocity has magnitude at least cq and that its hitting time satisfies

$$|D^\alpha T| \leq C_N q^{-M_N} (1+T)^{M_N} e^{c_N \varpi T}, \qquad 1 \leq |\alpha| \leq N+1. \tag{4.10}$$

If q tends to zero, also require $T \geq c/q$.

For every actual passing orbit and signed normal input z in a fixed small interval, its normal output is

$$G(z) = Lze^{j(z)}, \qquad L = \exp\left[-\int_0^T \beta(w(t), 0; \mathbf{b})\,\mathrm{d}t\right] > 0, \qquad j(0) = 0. \tag{4.11}$$

The selected jets of j through order N are bounded uniformly in T and the minimum positive slow gap. The jets of L and G have absolute bounds $C_N e^{-b_* T/4}$, after increasing C_N; this assertion includes mixed normal-input derivatives of G. In particular, normal input zero is included in the relative statement.

If $z_*(\mathbf{b})$ has bounded selected jets and stays in the smaller normal interval, then

$$G(z_* + u) - G(z_*) = G'(z_*)ue^{r(u)}, \qquad r(0) = 0, \qquad \|r\|_N \leq C_N, \tag{4.12}$$

on a fixed smaller u interval. At a slow-root loss of the exit, with entry and exit sections remaining separated and $T \to \infty$, define the relative integral along the complete auxiliary flow up to infinite time and extend L and the absolute output by zero. These definitions give extensions of class C^N in the selected ordinary variables through each such boundary, with the stated Euler bounds. At $\epsilon = 0$ this assertion uses the Euler scale class just specified, not ordinary scale derivatives. They do not assign a physical passage to a nonpassing parameter.

Proof. Write $g(w, Z) = Z\beta(w, Z)$. The normal equation and its first variation give

$$|Z(t; z)| \leq |z|e^{-b_* t}, \qquad G_t'(z) = \exp\left[-\int_0^t g_Z(w(s), Z(s; z))\,\mathrm{d}s\right] > 0.$$

Subtracting the invariant-graph rate leaves the nonlinear defect

$$j(z) = -\int_0^T \mathcal{I}(t, z)\,\mathrm{d}t, \qquad \mathcal{I} = \beta(w(t), Z(t; z)) - \beta(w(t), 0) = Z(t; z)\int_0^1 \beta_Z(w(t), sZ(t; z))\,\mathrm{d}s. \tag{4.13}$$

The mixed normal induction in Appendix D.1 gives

$$|D^\alpha \mathcal{I}(t, z)| \leq C_N(1+t)^{M_N} e^{-(b_* - c_N\varpi)t}, \qquad |\alpha| \leq N. \tag{4.14}$$

Its endpoint calculation uses (4.10) and, for $q \to 0$, $q^{-1} \le T/c$. It bounds the relative jets and proves the absolute estimates for G, L separately.

For the difference formula, use instead

$$\log \frac{G'(z)}{L} = -\int_0^T [g_Z(w(t), Z(t;z)) - \beta(w(t), 0)]\, \mathrm{d}t.$$

The same majorant makes $G'(z_* + \theta u)/G'(z_*)$ a positive unit with bounded selected jets. Integrate it for $0 \le \theta \le 1$ and take its logarithm to obtain (4.12).

Persistence of a finite transverse exit forces $T \to \infty$ at the specified separated-section slow-root boundary. The appendix proves the relative extension by differentiated tails and the absolute zero extension. Entry–exit collision, where $T \to 0$, is excluded. □

For the lower phase, take $q = \epsilon$ and $\varpi = O(\epsilon)$; separated sections give $T \ge c/\epsilon$. For an upper angular passage, take $q = 1$ and use a small angular width to obtain normal domination. In both cases the estimates allow independently vanishing slow gaps. The action $-\log L$ remains in the linear factor, with no bound required for it or its gap derivatives.

Corollary 4.4 (Continuity of the relative passage)**.** *Consider a family satisfying Proposition 4.3 with common domains, margins and finite-order bounds. Suppose its auxiliary trajectories and reserved mixed jets converge on bounded time intervals, and its exit times converge to the corresponding finite exit or to infinity. Then the relative amplitude and the logarithm of the normal derivative divided by its linear factor converge in the selected finite norms, including at a lost exit. No derivative of the divergent linear action is included.*

Proof. Use (4.14) and the differentiated endpoint bound (D.3): pass to the limit on a fixed time interval, then send its endpoint to infinity. At a finite exit, transversality also gives convergence of the time jets. □

Lemma 4.5 (A moving slow entry)**.** *In Proposition 4.3, fix a normal input $Z_0 > 0$ and the exit v_R. Let the slow entry v vary with $f(s) > 0$ on the whole interval $[v, v_R]$. With $\beta_0(s) = \beta(s, 0)$, the actual output is*

$$G(v) = Z_0 \exp\left[-q^{-1} \int_v^{v_R} \frac{\beta_0(s)}{f(s)}\, \mathrm{d}s + j(v)\right]. \tag{4.15}$$

The relative function j has the selected jets of the preceding proposition, with v counted as an ordinary input. Put

$$N_0 = \beta_0 + qfj', \qquad A_0 = N_0 - qf' + qf\frac{N_0'}{N_0}. \tag{4.16}$$

Where $N_0 \ge n_* > 0$,

$$\frac{G'}{G} = \frac{N_0}{qf}, \qquad \frac{G''}{G'} = \frac{A_0}{qf}.$$

For a regular change $v = v(t)$, $v' \ne 0$, the latter formula becomes

$$\frac{(G \circ v)''}{(G \circ v)'} = \frac{A_0 v' + qf\, v''/v'}{qf}. \tag{4.17}$$

Thus bounds through order k for A_0 use $k+2$ entry derivatives of j, and the numerator after a change of input uses $k+2$ derivatives of that change and a bound for $1/v'$.

Proof. Separate the linear rate in (4.13) and logarithmically differentiate; the chain rule gives (4.17). The denominators N_0, v' require the stated margins and derivative orders. The transit integral uses $f > 0$ along the whole path. □

Lemma 4.6 (The fixed phase exit in original coordinates)**.** *Choose a fixed lower phase exit away from $T = 0$. Use the compact lower coordinates and their inverse from Proposition 4.1. At this exit write $Z = Z_{\mathrm{out}}$ and set $\eta = Z_{\mathrm{out}}/\epsilon$. On a fixed sufficiently small signed η interval, passage from this curved exit to a fixed original section $|X| = R$ has bounded ordinary η, coefficient and Euler scale jets of every prescribed finite order, provided the phase construction retains these orders. After orienting the normal coordinate positively, the deviation from the same weak reference on $|X| = R$ is*

$$W = \eta u_c(\eta, \mathbf{b}, \epsilon), \qquad u_c \geq c_* > 0, \tag{4.18}$$

with bounded jets of $\log u_c$. The original horizontal displacement at the curved exit is $O_{C^N}(\epsilon\eta)$.

Proof. At the fixed phase exit, inversion gives $T = T_b + ZO_{C^N}(1)$, with $T_b > c > 0$; hence $X = -1/T$ moves by $ZO_{C^N}(1)$. The exact compact relation is $w = \epsilon z T$ and $w = \epsilon^2 \nu T^2 H(T) + Z$. Hence

$$z = \epsilon \nu T H(T) + \frac{Z}{\epsilon T}.$$

Subtract the same weak reference at the shifted T and substitute $Z = \epsilon\eta$ before differentiation. The deviation is η/T, a positive bounded unit times η.

Put $|X| = R + \epsilon\xi$. Its graph equation has bounded finite jets on a fixed (ξ, η) box, since $R > 0$ and the horizontal denominator has a positive margin. The section moves by $\xi = O(\eta)$; finite ODE dependence and the positive normal derivative give (4.18) and the inverse bounds.

For an actual lower slow passage, retain a fixed final piece with time at least c/ϵ. Proposition 4.3 then implies that $Z_{\mathrm{out}}/\epsilon$ and all selected jets are exponentially small. Thus the actual output lies in the signed box used for this construction. □

Proposition 4.7 (The Gaussian central connection)**.** *Fix $0 < R_0 < R_1$ and a compact coefficient box such that*

$$F_+(X) = 1 + \bar{a}X + \bar{B}X^2 \geq f_* > 0 \quad (0 \leq X \leq R_1).$$

Let $R \in [R_0, R_1]$ be fixed independently of the section input, with bounded selected coefficient and Euler jets if it varies with parameters. Continue the chosen weak reference from $z_b(R) = \epsilon\nu h_b$, where h_b has bounded selected jets. For sufficiently small $\epsilon > 0$ and $|\nu|$, this continuation exists on $0 \leq X \leq R$ and satisfies

$$|z_b(X)| \leq \frac{C\epsilon|\nu|}{X + \sqrt{\epsilon}}, \qquad |D^\alpha z_b(Ry)| \leq \frac{C_N \epsilon}{Ry + \sqrt{\epsilon}}, \quad |\alpha| \leq N. \tag{4.19}$$

Here $0 \leq y \leq 1$ is fixed when taking coefficient or Euler derivatives. The factor ν remains whenever no ν derivative is taken.

For signed input $z(R) = z_b(R) + W$ in a fixed small interval, its central height is

$$z(0) = s_b + e^{-S_c/\epsilon} a_c W e^{j_c(W)}, \qquad s_b = z_b(0), \qquad S_c = \int_0^R \frac{X}{F_+(X)}\,\mathrm{d}X > 0. \tag{4.20}$$

The positive factors a_c, a_c^{-1} and the selected jets of $\log a_c, j_c$ through order N are bounded, and $j_c(0) = 0$. Ordinary W derivatives are included. Composing with (4.18) gives the same statement with W replaced by η and with adjusted positive units. All reference choices are independent of the moving section input.

Proof. The exact original graph equation is

$$\frac{\mathrm{d}z}{\mathrm{d}X} = \mathfrak{f}(X, z) = \frac{(X/\epsilon)z + \bar{c}z(z-1) + \nu(1-z)^2}{D(X, z)}, \quad D = F_+ + \epsilon\bar{m} - (1 + 2\epsilon\bar{m})z + \epsilon\bar{m}z^2. \tag{4.21}$$

Its denominator stays positive on a fixed small signed tube. The weighted Gaussian kernel and Volterra estimates in Appendix D.2 give (4.19), including its selected mixed jets.

Put $v = z - z_b$. Exact Taylor division gives

$$v_X = a(X)v + B_c(X, v)v^2, \quad a(X) = \frac{X}{\epsilon F_+(X)} + a_r(X), \quad |a_r| \le C, \quad |\partial_v^j B_c| \le C_j(X/\epsilon + 1).$$

These bounds include selected coefficient and Euler jets because (4.19) controls $(X/\epsilon)z_b$. Set $E(X) = \exp[-\int_X^R a(s)\,\mathrm{d}s]$ and solve directly for the relative logarithm:

$$Q(X) = -W \int_X^R B_c(s, WE(s)e^{Q(s)})E(s)e^{Q(s)}\,\mathrm{d}s. \tag{4.22}$$

The weighted kernel bound (D.6) and the differentiated contraction proof in Appendix D.2 give $\|Q\|_\infty \le C|W|$ and bounded selected jets on a fixed small W interval, including zero. Thus $v(X) = WE(X)e^{Q(X)}$. Set $a_c = \exp[-\int_0^R a_r]$ and $j_c(W) = Q(0, W)$. This proves (4.20), retaining the exponentially small factor in E throughout. □

The Gaussian layer is controlled by the weighted reference estimate; ordinary high-order X derivatives there may be unbounded. All derivatives of the finitely smooth phase and reference are taken on their real parameter domains. The signed construction supplies the estimates and coefficient comparisons, and its physical restriction consists of positive-height original trajectories with finite transverse passages. For the incoming core, apply the same construction after reflection with $(\bar{a}, \bar{c}, \nu)$ replaced by $(-\bar{a}, -\bar{c}, -\nu)$.

4.3 Simple roots, quadratic passages and the cubic boundary

The form of the lower transition depends on which root controls its transit. A separated simple root gives a power law, while a quadratic minimum is described by its full transit integral. At the horizontal end a positive $\bar{B}$ gives an exact invariant-axis exponent; as $\bar{B} = \bar{a} = 0$ is approached, we retain the whole slow factor $xF_I(x)$ to avoid division by the vanishing coefficients. Each representation ends at the fixed phase section of Lemma 4.2 and then uses the same central connection.

Proposition 4.8 (Lower preparations)**.** *Fix a finite derivative order and a compact set of the indicated data. The following representations hold on common auxiliary domains and on the actual passing portions specified below.*

(i) Suppose the first positive root q_0 of $1 - \bar{a}q + \bar{B}q^2$ is simple, with $\Delta = -\partial_q(1 - \bar{a}q + \bar{B}q^2)|_{q=q_0} > 0$. Assume q_0 and Δ have positive upper and lower bounds. There is an actual strong intersection $\widehat{q} = q_0 + O(\epsilon)$ on $Y = 0$ and a regular power coordinate $t = \chi(\widehat{q} - q)$, where $\chi(0) = 0$ and $\chi' > c > 0$. On its passing side $t > 0$ the central output is

$$L_-(q) = s_- + D_-(t), \qquad D_-(t) = K_- t^{h_-/\epsilon} e^{j_-(t)}, \qquad j_-(0) = 0. \tag{4.23}$$

Here $h_- > c > 0$, $h_- = q_0/\Delta + O(\epsilon)$, $K_- > 0$, and j_- has bounded ordinary input and mixed coefficient and Euler-scale derivatives through the prescribed order. On this compact simple-root block the derivatives of $\epsilon \log K_-$ in the coefficients and Euler scale are bounded. The reflected outgoing preparation has the same properties.

(ii) Near an interior double root of F_I, there are an increasing regular coordinate u and a positive bounded time change for which the weak equation is

$$u' = \epsilon(\mu + u^2). \tag{4.24}$$

The complete interval to the selected right exit must satisfy $\mu + s^2 > 0$. The normal equation retains a fixed contracting margin. The relative amplitude and the absolute output have the finite-order extensions of Proposition 4.3, without any bound on ϵ/μ.

(iii) On a compact block $\bar{B} \geq B_* > 0$, the fixed horizontal entry of Lemma 4.2 has

$$D_-(t) = K_- t^{h_I/\epsilon} e^{P_-(t)/\epsilon}, \qquad h_I = \frac{1 - \epsilon \bar{B}}{\bar{B}}, \qquad P_-(0) = 0. \tag{4.25}$$

The selected finite jets of P_- are bounded, including at $t = 0$. Its domain is the positive component that connects to the fixed exit. The reflected exponent is exactly the same at each positive scale. All action accumulated after a fixed small horizontal section belongs to K_-; its logarithm need not have bounded gap derivatives.

(iv) On a neighborhood of $\bar{B} = \bar{a} = 0$, including both signs of $\bar{B}$, write $g = g_I = xF_I(x)$. The horizontal deviation admits

$$D(x) = C \exp\left[-\frac{1}{\epsilon} \int_x^{x_R} \frac{\beta_I(s,0)}{g(s)}\, \mathrm{d}s\right] e^{\widetilde{j}(x)}, \qquad C > 0, \tag{4.26}$$

where the normal entry and central relative units are included in $\widetilde{j}$. Its selected ordinary input and mixed jets are bounded on the common auxiliary domain. On the actual passing component put

$$N_I = \beta_I(x,0) + \epsilon g \widetilde{j}', \qquad A_I = N_I - \epsilon g' + \epsilon g \frac{N_I'}{N_I}. \tag{4.27}$$

Then

$$D' > 0, \qquad \frac{D''}{D'} = \frac{A_I}{\epsilon g}, \qquad A_I = 1 + O_{C^N}(\epsilon), \qquad A_I(0) = 1 - 2\epsilon \bar{B}.$$

The rate coefficients are defined and bounded across the auxiliary nonpassing parameters; the quotient is used only where $g > 0$.

The reference value s_- in these formulas is the central value of the one fixed weak graph, independent of the input. The outgoing reference is obtained by the fixed-field reflection convention. No assertion about the number of closing zeros is part of this proposition.

Proof. For a simple root, the first task is to locate the actual strong separatrix and normalize its input. In the phase coordinate the root r is close to $1/q_0$; $g_I'(r) > c > 0$. The strong fibre is $x = r$. On a fixed finite root block use the original normal height coordinate $U_0 = Z/\epsilon = Tz - \epsilon\nu T^2 H(T)$, which extends smoothly through $\epsilon = 0$ after substitution of $w = \epsilon T z$. The phase on these states satisfies $x = T + O_{C^N}(\epsilon)$. Thus $x = r$ is a graph over z, with horizontal slope $O(\epsilon)$, and intersects $z = 1$ at $q = \widehat{q}$ with $\widehat{q} = q_0 + O(\epsilon)$. Its normal rate has a fixed nonzero margin outside a smaller saddle box. It is the strong separatrix throughout $0 < z < 1$ and is a barrier to the incoming passage. The sign in (4.7) selects $q < \widehat{q}$.

Locally put $u = U_0$ and $v = x - r$. The invariant axes give

$$\dot{u} = -uA(u,v), \qquad \dot{v} = \epsilon v B(u,v), \qquad A, B > c > 0.$$

At the equilibrium their ratio is $h = A(0,0)/B(0,0) = q_0/\Delta + O(\epsilon)$. Let $H_o = A/B$. Normalize the axes by the regular changes

$$\frac{\mathrm{d}\log U}{\mathrm{d}\log u} = \frac{h}{H_o(u,0)}, \qquad \frac{\mathrm{d}\log V}{\mathrm{d}\log v} = \frac{H_o(0,v)}{h}. \tag{4.28}$$

The right sides equal one at zero. Subtracting one before integration shows that U/u and V/v are positive regular units with bounded jets and inverses. In the new coordinates the orbital coefficient is $h^2 H_o(u,v)/(H_o(u,0)H_o(0,v))$. It equals h on both axes. Two Hadamard divisions therefore give

$$\frac{\mathrm{d}U}{\mathrm{d}V} = -\frac{h}{\epsilon}\frac{U}{V} - \frac{U^2}{\epsilon}R(U,V). \tag{4.29}$$

The finite jets of R are bounded uniformly. To obtain final order N, use the phase and axis constructions two orders higher for these two divisions, and retain another two orders for the connections and inverses. Together with Proposition 4.1, a graph order $N+11$ suffices for this argument. Only finitely many orders are used.

Fix small positive sections $U = U_*$ and $V = V_*$ and take $0 < t < t_0 < V_*/2$. The reciprocal equation gives the exact passage

$$\mathscr{D}(t) = \frac{U_*(t/V_*)^{h/\epsilon}}{1 + (U_*/\epsilon)t^{h/\epsilon}\int_t^{V_*} s^{-h/\epsilon}R(U(s),s)\,\mathrm{d}s}. \tag{4.30}$$

The integral retains the dependence of R on the solution U. Set $V = te^{\epsilon\theta}$ and $T_f = \epsilon^{-1}\log(V_*/t)$. The denominator is $1 + U_* t I$, where

$$I = \int_0^{T_f} e^{-(h-\epsilon)\theta}R(U(\theta), te^{\epsilon\theta})\,\mathrm{d}\theta, \qquad U_\theta = -U[h + UVR(U,V)].$$

Here U contracts uniformly and the explicit scalar flow $V = te^{\epsilon\theta}$ satisfies (4.9) with $\varpi = O(\epsilon)$. The mixed normal induction of Appendix D.1 bounds the differentiated integrands, with their additional weight $e^{-(h-\epsilon)\theta}$, by $C(1+\theta)^M e^{-(h_*-C_N\epsilon)\theta}$. Extend R boundedly beyond V_* while retaining contraction. The infinite integral has bounded ordinary and mixed jets. The finite tail and its endpoint terms are bounded by finite sums $C\epsilon^{-a}t^{-b}(1+|\log(t/V_*)|)^M(t/V_*)^{h/\epsilon-1}$. Half the exponent absorbs the input powers and logarithms, and $t/V_* < 1/2$ absorbs the inverse scale powers. Thus the selected jets extend at $t = 0$. Reducing t_0 makes the denominator positive and $j = -\log(1 + U_* t I)$ has bounded jets with $j(0) = 0$.

The transverse connection from $z = 1$ to the local fast section has bounded jets and inverse jets in the original finite-height coordinates; composing its input with V gives the regular deficit t. The remaining leg stays away from the simple root and reaches the fixed phase exit. Its input-independent linear contraction belongs to K_-; its relative unit is controlled by Proposition 4.3 and Lemma 4.6, followed by Proposition 4.7. The large power absorbs the finite ordinary derivative losses, so composition preserves the relative jets. Absorb the unit's value at zero into K_-. On this simple-root compact block the remaining linear action is an integral over intervals whose slow denominators have positive margins. It follows that $\epsilon\log K_-$ has bounded coefficient and Euler-scale jets. This proves (i), including the fixed central reference.

For (ii), let x_0 be the minimum of F_I near the double root. It is interior and bounded away from zero. Write

$$F_I(x) = \mu + (x - x_0)^2 a(x), \qquad a(x) = \int_0^1 (1-s)F_I''(x_0 + s(x-x_0))\,\mathrm{d}s > c.$$

The increasing Morse coordinate $u = (x - x_0)\sqrt{a(x)}$ is regular. Its velocity is $\epsilon(\mu + u^2)$ times the bounded positive unit $x\partial_x u$. Dividing time by this unit gives (4.24). The normal contraction remains uniformly positive. Apply Proposition 4.3 on the whole passing interval, with the fixed exit margin. Crossing the minimum requires $\mu > 0$; for $\mu \le 0$ a right-hand entry must exceed $\sqrt{-\mu}$. The same fixed phase and central connections complete the passage.

For (iii), choose $x_h > 0$ small enough that $F_I > c > 0$ on $[0, x_h]$ throughout the positive-$\bar{B}$ block. Since $\beta_I(0,0) = 1 - \epsilon\bar{B}$ and $F_I(0) = \bar{B}$,

$$\frac{\beta_I(x,0)}{xF_I(x)} = \frac{h_I}{x} + q_0(x) \tag{4.31}$$

with q_0 having bounded finite jets at zero; subtraction of the constant value and one Hadamard division prove this statement. The exact normal integral from the fixed horizontal entry is

$$Z_{\text{out}} = Z_e(t) \exp\left[-\frac{1}{\epsilon}\int_t^{x_R} \frac{\beta_I(s,0)}{sF_I(s)}\,\mathrm{d}s\right] e^{j(t)}.$$

The factor Z_e is bounded away from zero for this section. Separate the integral at x_h and use (4.31). After the central relative unit is included, its input-dependent logarithm multiplied by ϵ is

$$P_-(t) = \int_0^t q_0(s)\,\mathrm{d}s + \epsilon \log \frac{Z_e(t)}{Z_e(0)} + \epsilon(\widetilde{j}(t) - \widetilde{j}(0)).$$

All its required jets are bounded by the relative estimates, even when a later slow exit is lost. For each passing parameter the input-independent action on $[x_h, x_R]$ belongs to K_-. A later loss of passage gives the absolute zero extension of Proposition 4.3; $\log K_-$ may still have unbounded gap derivatives there. Reflection fixes $\bar{B}$ and ϵ, so h_I agrees exactly on the two sides. If a separately normalized power input is needed, $t\exp(h_I^{-1}\int_0^t q_0)$ absorbs the pure weak-axis action; it and its inverse have bounded jets. In a common actual input the scaled remainder P_- is the appropriate representation.

For (iv), retain the undistributed slow factor $g = xF_I$. The exact linear normal integral and the relative amplitudes give (4.26); the fixed horizontal normal input is a positive unit, so its logarithm can be included in $\widetilde{j}$. Use (4.16) with $(q, f, j) = (\epsilon, g, \widetilde{j})$ and $N_I = 1 + O(\epsilon) > 0$ to obtain the rate formula. At zero, $g(0) = 0$ and $g'(0) = \bar{B}$; hence $N_I(0) = 1 - \epsilon\bar{B}$ and $A_I(0) = 1 - 2\epsilon\bar{B}$ exactly. The finite jets of the relative amplitude and of g, β_I give those of N_I, A_I. To retain a rate coefficient through order N, retain the primitive through order $N + 2$ before differentiating. The construction remains regular when $\bar{B}$, $\bar{a}$ or a slow gap vanishes, since these factors have not been divided out. The values and logarithmic derivatives of D are used on the passing component (4.7); its relative coefficients and absolute zero extension are defined on the larger auxiliary domain. □

Remark 4.9 (The choice of section matters). The exponent in (4.25) belongs to a fixed positive horizontal fast section. The finite section $Y = 0$ has normal input ϵ times a unit on finite T-intervals; at $T = 0$ it has an additional vanishing T factor. Thus extending that section to the horizontal boundary does not preserve the displayed power without accounting for its input factor. Likewise an arbitrary regular substitution $t \mapsto tk(t)$ contributes $(h/\epsilon)\log k(t)$ to an unscaled power logarithm. The normalized input of (i) absorbs this action; the common-input formula (iii) records it in P/ϵ.

4.4 A finite upper saddle near ratio one

We first treat the finite hyperbolic saddle, then the repeated and weak ends at infinity. Near ratio one, expanding a power in the ratio would lose uniformity when its logarithmic input is large. Instead we keep the exact compensator. The original field is affine at zero scale, so its nonlinear orbital remainder has a factor ϵ, which persists in the compensator and curvature estimates. Fix a compact interval $0 < m_* \le \bar{m} \le m^*$, bounded remaining coefficients and a small neighborhood on which its eigenvalue ratio satisfies $|r - 1| \le 1/64$. The ratio used below is the actual ratio of the original field. Dividing (2.16) by $Y + \epsilon > 0$ gives

$$\dot{X} = \frac{Y(\bar{m}Y - 1) + \epsilon(\bar{B}X^2 + \bar{a}X)}{Y + \epsilon}, \qquad \dot{Y} = X + \bar{c}Y + \frac{\nu Y^2}{Y + \epsilon}. \tag{4.32}$$

At $\epsilon = 0$ this is an affine field. Its saddle is $(-\bar{d}/\bar{m}, 1/\bar{m})$, where $\bar{d} = \bar{c} + \nu$, and its two eigenvalues have opposite signs separated from zero. The implicit function theorem gives the perturbed saddle and its eigendirections on a common analytic coefficient neighborhood. Write its eigenvalues as $\lambda_u > 0 > \lambda_s$ and set $r = -\lambda_s/\lambda_u$.

To bound the analytic jets near this finite saddle, choose a complex neighborhood whose real part lies in $Y > Y_s/2 > 0$. On a sufficiently small such neighborhood, $Y + \epsilon$ stays away from zero. In linear eigenvector coordinates the stable and unstable graph equations have coefficient denominators $n\lambda_u - \lambda_s$ and $\lambda_u - n\lambda_s$, respectively. Their absolute values are at least $c(n+1)$ for $n \geq 1$ on a smaller complex coefficient disk. The nonlinear part of (4.32) is ϵ times an analytic function. On a sufficiently small state disk the coefficient recursion is therefore a contraction in the norm $\sum |a_n|\delta^n$; the factor $(n+1)^{-1}$ absorbs the derivative index in the graph equation. This constructs both axes on one disk, with their nonlinear parts divisible by ϵ. Cauchy estimates on smaller disks give any prescribed finite state and coefficient jets, including Euler scale jets of the quotient by ϵ.

Straighten these axes, orient their coordinates U, V positively in the quadrant of the local returning arc, and write the orbital equation as $V_U = -(V/U)H(U,V)$, with $H(0,0) = r > 0$. Separate positive changes on the two axes, obtained by integrating $d\log\widetilde{U}/d\log U = H(U,0)/r$ and $d\log\widetilde{V}/d\log V = r/H(0,V)$, make the orbital coefficient constant on each axis. Its difference from r is then divisible by UV. Dropping the tildes gives

$$\frac{dV}{dU} = -r\frac{V}{U} - V^2 R(U,V), \qquad R = \epsilon\widetilde{R}, \tag{4.33}$$

where $\widetilde{R}$ has common analytic bounds. The local maps below follow this positive quadrant. To obtain the complete upper path, they will be composed with its two regular connecting arcs.

The coordinates can be chosen to preserve the reversible component $\bar{a} = \bar{c} = \nu = 0$ exactly. Indeed, with $Y_s = 1/\bar{m}$, put

$$W(Y) = \left(\frac{Y+\epsilon}{Y_s+\epsilon}\right)^{-2\epsilon\bar{B}}, \qquad I(Y) = \int_{Y_s}^{Y} \frac{2s(\bar{m}s-1)}{s+\epsilon} W(s)\, ds.$$

Direct differentiation gives the first integral $WX^2 - I$. Writing $I = (Y - Y_s)^2 J$ yields $J(Y_s) = \bar{m}/(1+\epsilon\bar{m}) > 0$. Thus $\eta = (Y - Y_s)\sqrt{J/\bar{m}}$, $\zeta = \sqrt{W/\bar{m}}X$ and $U = -(\eta+\zeta)/2$, $V = -(\eta-\zeta)/2$ transform that integral to $-4\bar{m}UV$. Extend these analytic coordinates to the coefficient neighborhood before straightening the perturbed axes. The subsequent axis changes are the identity on this component; there $r = 1$ and $R = 0$ exactly. This local first integral supplies the normalization on the reversible component.

Proposition 4.10 (The uniform resonant passage)**.** *After choosing common small local sections and positive coordinates on them, the passage from $(t,1)$ to $(1,v)$ has the form*

$$v = t^r e^{w(t)}, \qquad w = -\sigma t\omega(t; r-1) + R_w(t), \qquad \omega(t;\kappa) = \int_0^{\log(1/t)} e^{-\kappa s}\, ds. \tag{4.34}$$

For each prescribed finite integer N, the neighborhoods can be chosen so that, with $D = t\partial_t$,

$$|D^i \partial_{\mathfrak{b}}^{\alpha} E_\epsilon^j R_w| \leq C_N \epsilon t^{7/4}, \qquad i + |\alpha| + j \leq N, \qquad 0 < t < e^{-6}. \tag{4.35}$$

Ratio derivatives may also be included, counting towards the same total order. The coefficients satisfy $\sigma = O(\epsilon)$ in these norms. In particular $N = 6$ supplies four input Euler derivatives and two coefficient derivatives of R_w. The estimate places no bound on $|(r-1)\log t|$.

Proof. First remove the nonconstant restrictions of R to its axes without dividing by $r - 1$. Put $s_0 = R(0,0)$ and, writing $R(U,0) = s_0 + \sum_{n\geq 1} R_{n0}U^n$, set

$$f(U) = -\sum_{n\geq 1} \frac{R_{n0}U^{n+1}}{n+1-r}, \qquad V_1 = \frac{V}{1+f(U)V}.$$

A direct differentiation gives

$$R_1(U,V_1) = R\left(U, \frac{V_1}{1-f(U)V_1}\right) + f'(U) - rf(U)/U, \qquad R_1(U,0) = s_0.$$

Next set $U_1 = U/(1+g(V_1)U)$, where $g(0) = g'(0) = 0$ solves

$$g'(V) - \frac{g(V)}{rV} = \frac{R_1(0,V) - s_0}{r^2}.$$

The new axis restriction is $R_2(0,V) = R_1(0,V) - r^2 g'(V) + rg(V)/V = s_0$. Since $g = O(V^2)$, the previously normalized axis is unchanged. The denominators of the two series are $n+1-r$ and $n+1-1/r$, $n \geq 1$. On $|r-1| \leq 1/64$ they have a fixed positive margin and grow linearly with n. The changes and their inverses therefore have uniform analytic bounds on smaller disks, and f, g retain their factor ϵ. Exact division of $R_2 - s_0$ by U_1V_1, followed by one common positive rescaling of both sections, gives

$$V_U = -rV/U - \sigma V^2 - UV^3\mathcal{R}(U,V), \qquad |\sigma| + \|\mathcal{R}\| \leq C\epsilon. \tag{4.36}$$

The bound includes whatever finite jets are reserved before this construction. The two shears are also the identity on the reversible component considered above.

The reciprocal equation before the shears has the exact integral representation

$$V(U) = \frac{(t/U)^r}{1 + t^r \int_t^U s^{-r} R(s, V(s))\, ds}.$$

It proves positivity and existence on the full rectangle for small ϵ. In (4.36) put $U = te^\theta$, $V = e^{-r\theta}\zeta$, and $L = \log(1/t)$. Then, exactly,

$$\zeta_\theta = -\sigma t e^{-(r-1)\theta}\zeta^2 - t^2 e^{-2(r-1)\theta}\zeta^3 \mathcal{R}(te^\theta, e^{-r\theta}\zeta), \qquad \zeta(0) = 1. \tag{4.37}$$

With $\delta = 1/64$ the two kernel integrals are bounded respectively by $t^{1-\delta}L$ and $t^{2-2\delta}L$. A bootstrap on the integral equation gives $1/2 < \zeta < 2$ and $\zeta - 1 = O(\epsilon t^{1-\delta}(1+L))$ on $[0, L]$. Integrating its logarithmic derivative now yields

$$\begin{aligned} R_w = & -\sigma t \int_0^L e^{-(r-1)\theta}(\zeta - 1)\, d\theta \\ & - t^2 \int_0^L e^{-2(r-1)\theta}\zeta^2 \mathcal{R}(te^\theta, e^{-r\theta}\zeta)\, d\theta. \end{aligned}$$

Consequently $|R_w| \leq C\epsilon t^{2-2\delta}(1+L)^2$.

Differentiate (4.37) at fixed θ. Ratio derivatives add polynomial factors in θ; input Euler derivatives preserve te^θ. The highest variation has the same small-L^1 linear kernel, so induction bounds it by the displayed powers of t times a fixed power of $1+L$. Endpoint rows use $DL = -1$ and retain those powers. In the first integral for R_w, all variations of $\zeta - 1$ retain their first-order smallness. Therefore

$$|D^i \partial_{\mathfrak{b}}^\alpha E_\epsilon^j R_w| \leq C_N \epsilon t^{2-2/64}(1+L)^{M_N}.$$

The strict exponent margin $2 - 2/64 - 7/4 = 7/32$ absorbs every such fixed logarithmic power and proves (4.35). Ordinary ϵ derivatives can differentiate away the front factor ϵ and are not covered by that displayed conclusion. □

The curvature formula follows from the exact identity

$$E := t\frac{v''}{v'} - (r-1) = Dw + \frac{D^2w}{r+Dw}.$$

Inserting (4.34) gives

$$E = -\sigma Q + R_E, \qquad Q = 2rt\omega(t; r-1) - (2+1/r)t, \qquad Q'' = -2r^2t^{r-2}. \tag{4.38}$$

Here $|D^i\partial_{\mathfrak{b}}^{\alpha}E_{\epsilon}^{j}R_E| \le C\epsilon t^{7/4}$ for $i+|\alpha|+j \le N-2$ if N orders were retained in (4.35). Indeed the error in replacing $(r+Dw)^{-1}$ by $1/r$ is a product of two first corrections, bounded by the same $t^{2-2/64}(1+L)^M$ estimate. Thus a curvature use requiring four Euler and two coefficient derivatives reserves $N=8$, whereas four Euler and two coefficient derivatives of w itself require $N=6$. Since $\omega \ge \int_0^6 e^{-s/64}\,ds > 5$ and $Q' = 2r^2\omega - 2r - 2 - 1/r$, both Q and Q' are positive on $0 < t < e^{-6}$ for the stated ratio interval. Both estimates retain t^{r-1} exactly and therefore remain valid when $|(r-1)\log t|$ is large.

The finite connecting arcs contribute a separate regular factor. On a small low-height segment where $|X| \ge c > 0$, the denominator of dX/dY obtained from (4.32) is bounded below in modulus by $c(Y+\epsilon)$. The numerator and all its fixed X and coefficient derivatives are $O(Y+\epsilon)$; Euler scale differentiation preserves this property. The equation in Y and its finite variational equations consequently have bounded coefficients. Comparing it with its zero-scale equation leaves only the integrable layer carrier $\epsilon/(Y+\epsilon)$, whose integral is $\epsilon\log(1+\delta/\epsilon)$. Beyond this segment the vector field is regular on a compact arc. These arguments give the required finite section and coefficient jets, with zero-scale error $O(\epsilon(1+|\log\epsilon|)+|\nu|)$ when the comparison also sets $\nu = 0$. The scale derivatives in this estimate are Euler derivatives; ordinary ϵ derivatives of the error may be unbounded.

4.5 Repeated and weak upper passages

As an upper root becomes repeated or weak, the preceding hyperbolic preparation ceases to be uniform. We instead use a small angular width to make the slow variational growth weaker than the normal contraction. In projective coordinates $R = 1/Y$, $\varphi = X/Y$, put $b = 1-\epsilon\bar{B}$ and $\bar{d} = \bar{c} + \nu$. A positive projective time change gives the exact field

$$\begin{aligned} R' &= -R(\varphi + \bar{d}) - \epsilon R^2(\varphi + \bar{c}), \\ \varphi' &= \bar{m} - b\varphi^2 - \bar{d}\varphi + R[-1 + \epsilon\varphi(\bar{a} - \bar{c} - \varphi)]. \end{aligned} \tag{4.39}$$

We describe separately the quadratic angular minimum and the weak radial factor. In both constructions all finite derivative orders are fixed before shrinking the strip.

The quadratic angular factor. Fix a limiting repeated field $\bar{c} = 2a_* > 0$, $\bar{m} = -a_*^2$, $\epsilon = \nu = 0$, with bounded $\bar{B}, \bar{a}$. The symbol a_* here is a positive constant, unrelated to $\bar{a}$. On $R = 0$ set

$$v_0 = -\varphi - \frac{\bar{d}}{2b}, \qquad \mu_U = -\frac{\bar{d}^2 + 4b\bar{m}}{4b^2}, \qquad \Xi = \frac{\bar{d}(1-2\epsilon\bar{B})}{2b}.$$

Dividing the vector field by the positive factor b makes its angular restriction exactly $v_0' = \mu_U + v_0^2$, and its normal rate is $(\Xi - v_0)/b$. Near the chosen field Ξ/b has a fixed positive margin. The line $R = 0$ is exactly invariant.

Choose an angular outer width $\varpi > 0$. Extend v_0^2 outside $|v_0| < \varpi$ to a smooth bounded function with first derivative $O(\varpi)$, and extend the normal coefficients preserving positive margins for β and $\beta + R\beta_R$. The extension is made on a larger real rectangle than the one used for the original passage. Since its scalar derivative is $O(\varpi)$, the flow induction (4.6),

now with $q = 1$, gives $C(1+t)^M e^{C_N \varpi |t|}$ through order $N+2$, also for the inverse flow and parameter derivatives of μ_U. Reserve all products of these and the exit-time jets in C_N, then choose $C_N \varpi < \beta_*/16$, followed by radial width and coefficient neighborhood. The angular equation has no small factor ϵ.

Let ϕ_t denote the complete scalar flow on $R = 0$ and let $(v_0(t), R(t))$ be a trajectory of the complete auxiliary field. The phase integral

$$v = \lim_{t \longrightarrow \infty} \phi_{-t}(v_0(t))$$

exists by the inverse-flow construction of Proposition 4.1: the forcing $RO(1)$ gives the majorant $C(1+t)^M e^{-(\beta_* - C_N \varpi)t}$. The limit equals v_0 on the axis and obeys its scalar flow identity; hence $v = v_0 + RO_{C^N}(1)$ is a coordinate. Reusing R for a positive normal coordinate, the original field, on the smaller rectangle, is exactly

$$v' = \mu_U + v^2, \qquad R' = -R\beta_U(v, R), \qquad \beta_U(v, 0) = \frac{\Xi - v}{b}, \qquad \beta_U,\ \beta_U + R\beta_{U,R} > \beta_* > 0. \tag{4.40}$$

Shrinking the radial rectangle retains the normal margins. Together with the angular variation bound and the fixed transverse exit below, these estimates give the hypotheses of Proposition 4.3 with $q = 1$. The small angular width provides the required domination in this upper passage.

For a fixed small radial input $R_0 > 0$ and a fixed positive angular exit $v_R < \varpi/2$, restrict $|\mu_U| < v_R^2/2$. The exit speed is then at least $v_R^2/2$. The actual outgoing normal value is

$$G(v) = R_0 \exp\left[-\int_v^{v_R} \frac{\beta_U(s, 0)}{\mu_U + s^2}\, ds\right] e^{j_U(v)}. \tag{4.41}$$

Its relative amplitude j_U has the finite mixed bounds of Proposition 4.3 and Lemma 4.5, on the stated auxiliary domain. The physical domain is the whole-path condition

$$\mu_U + s^2 > 0 \quad (v \le s \le v_R), \quad \text{equivalently} \quad \mu_U > 0 \text{ or } [\mu_U \le 0 \text{ and } v > \sqrt{-\mu_U}], \qquad v < v_R. \tag{4.42}$$

A scalar root cannot be crossed in finite time. The exact phase coordinate carries this obstruction to every stable fibre of the original field. A fixed passage from $-v_R$ to v_R therefore requires $\mu_U > 0$. The full-interval condition is needed even when the slow factor is positive at both endpoints.

To apply this preparation to the upper comparison, we next locate the entry reached from the finite section, on both sides of its limiting threshold. At the limiting repeated field the affine upper system in (4.32) has the exact coordinates

$$V = Y + a_*^{-2}, \qquad Z = X + a_* Y - a_*^{-1}, \qquad r_0 = Z/V,$$

which satisfy $V' = a_* V + Z$, $Z' = a_* Z$, and $r_0' = -r_0^2$. For an orbit starting at $(p, 0)$, put $P = 1/a_*$ and $Z_0 = p - P$. At a fixed large $V = V_e$ reached at time T,

$$V_e = e^{a_* T}(a_*^{-2} + Z_0 T), \qquad \frac{\partial r_{0,e}}{\partial p} = \frac{e^{a_* T}}{V_e} \frac{p}{a_* Z_0 T + Z_0 + a_*^{-1}} > 0.$$

In particular the derivative at $p = P$ is a_*^2. Choose $v = -r_0$ and normal coordinate $1/V$ at this limiting field. Then (4.40) is exact with $\beta_U = a_* - v$. Choose the auxiliary extension in these exact limiting coordinates, and extend this choice smoothly in the parameters; thus $j_U = 0$ there. On each fixed compact finite entry interval about P on which the connecting orbit reaches this section transversally, persistence and the low-layer estimate above give $v_e'(p) < -c < 0$ and bounded finite section and coefficient jets. The limiting entry is zero at P and negative for $p > P$; such negative entries are retained and pass precisely when (4.42) permits them. By the limiting normalization and continuity, j_U tends to zero in the reserved norms as the coefficients tend to this limiting field.

The weak radial factor. Fix $\bar{c}_0 > 0$, $\bar{m} = 0$, and $\epsilon = \nu = 0$. On a sufficiently small coefficient neighborhood define the actual quantities

$$\begin{aligned} \Delta &= \sqrt{\bar{d}^2 + 4b\bar{m}}, \qquad \varphi_s = \frac{-\bar{d} - \Delta}{2b}, \\ \gamma &= \varphi_s + \bar{d} = -\frac{2(\bar{m} + \epsilon\bar{B}\bar{d}^2)}{\bar{d}(1 - 2\epsilon\bar{B}) + \Delta}. \end{aligned} \tag{4.43}$$

The denominator is positive and bounded away from zero. In particular the finite-scale weak face is $\bar{m} + \epsilon\bar{B}\bar{d}^2 = 0$; replacing it by $\bar{m} = 0$ would move the radial root. Reverse the projective time in (4.39) and put $y = \varphi - \varphi_s$. The exact system is

$$\begin{aligned} \dot{R} &= R(\gamma + y) + \epsilon R^2(\varphi_s + y + \bar{c}), \\ \dot{y} &= -\Delta y + by^2 + R\{1 - \epsilon(\varphi_s + y)(\bar{a} - \bar{c} - \varphi_s - y)\}. \end{aligned} \tag{4.44}$$

Its normal eigenvalue has a fixed negative margin and its slow radial derivative is small when $|\gamma|$, the radial width and the normal width are small.

We construct the weak graph with the prescribed finite smoothness on a common real domain. Extend (4.44) smoothly outside a slightly larger rectangle, preserving $R = 0$, normal contraction and the small radial derivative. The time-one graph transform has base variation at most $e^{C\varpi}$ and normal variation at most $e^{-\Delta_*/2}$ after reducing the rectangle. Choose its width so that $e^{-\Delta_*/2}e^{(N+2)C\varpi} < 1$ and the bound $C_N\varpi < \beta_*/16$ holds for all the finite products reserved above. The mixed graph-transform induction of Proposition 4.1, with this contracting highest-jet multiplier, constructs $y = h(R)$, $h(0) = 0$, with the reserved mixed bounds on the common auxiliary domain.

The linear graph equation gives

$$h'(0) = \alpha = \frac{1 - \epsilon\varphi_s(\bar{a} - \bar{c} - \varphi_s)}{\Delta + \gamma}.$$

The reduced equation is $\dot{R} = RG(R)$, where

$$G(R) = \gamma + h(R) + \epsilon R(\varphi_s + h(R) + \bar{c}), \qquad G'(0) = \alpha + \epsilon(\varphi_s + \bar{c}) \longrightarrow 1/\bar{c}_0 > 0. \tag{4.45}$$

The same inverse-flow phase integral just used for the repeated strip, now relative to this reduced equation, gives a coordinate $\widetilde{R}$ constant on its strong fibres and equal to R on the weak graph. Its normal forcing is divisible by R, since $R = 0$ is invariant in (4.44). Thus $\widetilde{R} = R$ times a positive unit on the smaller rectangle. In particular physical $R > 0$ corresponds exactly to $\widetilde{R} > 0$. The mixed convergence follows from the normal margin and the chosen small slow width, as above.

Define

$$u = G(\widetilde{R}) - \gamma = \widetilde{R}\widetilde{G}(\widetilde{R}), \qquad \widetilde{G}(R) = \int_0^1 G'(sR)\,ds > c > 0.$$

The positive time factor $G'/\widetilde{G}$ has bounded finite jets and is equal to one at $R = 0$. Dividing the vector field by this factor, and straightening the weak graph, gives exactly

$$\dot{u} = u(\gamma + u), \qquad \dot{Z} = -Z\beta(u, Z), \qquad \beta(0, 0) = \Delta, \qquad \beta,\ \beta + Z\beta_Z > \beta_* > 0. \tag{4.46}$$

Only one derivative of G was spent here; it was reserved before the graph construction. The change is regular across $\gamma = 0$. Its slow derivative $\gamma + 2u$ is $O(\varpi)$ on the extended small strip, while its normal rate has the stated positive margin. All hypotheses of Proposition 4.3 are therefore again verified for $q = 1$.

Fix a positive weak exit u_R and count inputs below $u_0 < u_R/2$, taking $|\gamma| < u_R/2$. The actual passage domain is

$$\max(0, -\gamma) < u < u_0. \tag{4.47}$$

This is precisely positivity of $u(\gamma + u)$ on the entire path to u_R. At $\gamma < 0$ its positive root $-\gamma$ blocks every entry on its left. The phase projection and normal contraction give the same obstruction for the original two-dimensional field.

The finite outgoing normal value, followed by its regular connector to $Y = 0$, is the inverse upper map. Choose its positive displacement t from the weak intersection $\hat{p}$ so that $p = \hat{p} - t(u)$. The limiting weak graph is $X + \bar{c}_0 Y = 1/\bar{c}_0$; hence $\hat{p} \to 1/\bar{c}_0$, and its finite continuation is transverse. The low-layer equation above bounds all the required section and coefficient jets of this continuation. With $f = u(\gamma + u)$, the relative normal formula and this connector give

$$t, t' > 0, \qquad \frac{t'}{t} = \frac{N_U}{f}, \qquad \frac{t''}{t'} = \frac{A_U}{f}, \qquad N_U = \beta(u, 0) + f\widetilde{j}'(u), \qquad A_U = N_U - f' + f\frac{N_U'}{N_U}. \tag{4.48}$$

Here $\widetilde{j}$ includes the logarithm of the positive regular connector unit and has the finite bounds from Lemma 4.5. Since f and f' are uniformly small, $N_U, A_U > c > 0$ on a sufficiently small input and coefficient neighborhood. The absolute output and all its selected finite derivatives tend to zero uniformly as these neighborhoods shrink: the travel time diverges, and normal decay dominates the finite slow variational growth.

Moving inputs and lost exits. The phase maps were constructed on larger domains. Their bounded mixed jets therefore also control their restrictions to moving transverse sections by composition. For a full repeated strip, $T = \int_{-v_R}^{v_R} (\mu_U + s^2)^{-1}\, ds$ and $|\partial_{\mu_U}^j T| \le C_j(1+T)^{2j+1}$. For moving entries and the weak factor, apply the fixed-exit calculation of Appendix D.1 with $q = 1$, the stated positive exit speeds and slow exponent $C_N\varpi$. This verifies Proposition 4.3; its extensions and Corollary 4.4 apply in the reserved mixed orders. The physical domains remain (4.42) and (4.47), with the stated regular connections.

4.6 Derivative comparison at a weak passage

The weak inverse map has a uniformly flat output, but its logarithmic rates divide by the vanishing scalar speed. Multiplying the derivative comparison by that speed gives a regular numerator. The positive weight below also retains the estimate when the incoming lower factor approaches a root.

Lemma 4.11 (Weighted weak rates)**.** *On one passing interval let $f = u(\gamma + u) > 0$, $t \in C^3$, $t, t' > 0$, and*

$$\frac{t'}{t} = \frac{N}{f}, \qquad \frac{t''}{t'} = \frac{A_U}{f}.$$

Let $J(u), O(t) \in C^3$ be central maps with same-sign nonzero derivatives after composition, and let $h \in C^1$, $h(u) > 0$. Set

$$K = \epsilon \log \frac{J'}{(O \circ t)'}, \qquad h\epsilon\frac{J''}{J'} = \frac{A}{u} + \epsilon R, \qquad E = hN\left(\epsilon\frac{O''}{O'}\right) \circ t.$$

Then the exact numerator is

$$M = hfK' = A(\gamma + u) + \epsilon f R - tE - \epsilon h A_U. \tag{4.49}$$

Suppose $A \ge a_* > 0$ and A, R, E, hA_U have common bounded ordinary C^1 norms. For sufficiently small $|\gamma| + u + \epsilon + \|t\|_{C^1}$, depending only on these bounds, $M' > a_*/2$. Hence K'

has at most one zero, which is simple, and K has at most two zeros counted with their orders. Every such zero has order at most two. A regular C^3 input change preserves these orders in the original analytic comparison. There is no lower bound on h at an excluded endpoint.

Proof. The chain rule gives $\epsilon(O\circ t)''/(O\circ t)' = (\epsilon O''/O')t' + \epsilon t''/t'$. Substituting the stated rates proves (4.49). Its derivative is

$$M' = A + (\gamma + u)A' + \epsilon(fR)' - (tE)' - \epsilon(hA_U)'.$$

The last four terms are uniformly small under the stated bounds. Since $hf > 0$, a zero of K' is simple, and a zero of K at that point has $K'' = M'/(hf) \neq 0$. All other zeros of K are simple. Rolle's theorem, counting a double zero once also in K', gives the asserted bounds for every finite selection. A regular C^3 change of input preserves the first nonzero derivative and hence these original analytic zero orders. □

4.7 Exact common powers and the joint derivative budget

The finite and radial preparations label the same outgoing orbits in different ways: by p on $Y = 0$ and by the horizontal phase. We identify these labels through a regular connector before applying an upper slow passage. This gives a common input for the two reflected lower powers. Their exact exponents agree for the same field, so their singular power factors cancel in the logarithmic derivative ratio.

Lemma 4.12 (A common actual input)**.** *Fix a compact coefficient box and a fixed annulus $s_1 \le \sigma \le 2s_1$, where $\sigma = 1/p$ and p labels the outgoing intersection with $Y = 0$. Suppose this annulus and a slightly larger one lie in the common low-layer connector to the selected horizontal section, before every upper slow block under consideration. Let t be the reflected lower phase on that horizontal section. For each fixed finite order N,*

$$t = t_\epsilon(\sigma), \qquad \|t_\epsilon - \sigma\|_{C^N_{\sigma,\mathbf{b}},E_\epsilon} \le C_N\epsilon(1 + |\log\epsilon|)^{M_N} = o(1).$$

There is consequently a common coordinate on these outgoing orbits which agrees with t near the radial end and with $1/p$ on the finite side. On the overlap its selected jets and inverse jets are bounded and its derivative is positive. The choice is uniform in subsequent slow gaps, provided the stated connector remains in the fixed common domain.

Proof. Use the reflected version of the lower phase and reverse its time. On $Y = 0$ its phase is $\sigma + O_{C^N}(\epsilon)$ and its normal input is $Z_e = \epsilon A$, where A is a positive bounded unit on this fixed annulus. Its backward normal velocity is $Z\beta_I$. Follow it to the fixed positive horizontal section, retaining the slightly larger annulus. In the normal logarithm $r = \log(Z/Z_e)$, the phase equation is

$$\frac{\mathrm{d}x}{\mathrm{d}r} = -\frac{\epsilon g_I(x)}{\beta_I(x, Z_e e^r)}.$$

The terminal value of r is $\log(1/\epsilon) + O(1)$, since the horizontal section has a positive normal height. Its transversality has a fixed margin. Throughout this connector $Z_e e^r$ stays in the fixed normal tube. The phase moves by $O(\epsilon(1 + |\log\epsilon|))$, so the larger annulus indeed contains it when the scale is small.

For the mixed rows, the highest phase variation has coefficient $O(\epsilon)$ and its forcing is ϵ times bounded lower variations. The selected derivatives of $\log Z_e$ are bounded and $Z_e e^r$ stays in the tube. The scalar variational induction and Gronwall on this $O(1 + |\log\epsilon|)$ interval therefore give the claimed $C_N\epsilon(1+|\log\epsilon|)^{M_N}$ bound. The fixed terminal transversality controls its endpoint rows; no slow denominator is inverted.

On the fixed overlap choose a smooth cutoff χ, equal to one at its radial side and zero at its finite side, and use

$$\sigma + \chi(\sigma)(t_\epsilon(\sigma) - \sigma).$$

Its derivative is $1 + o(1)$, and finite-order inverse differentiation gives the inverse bounds. Continue it by t toward the radial end and by σ toward the finite side. These are coordinates on the same actual outgoing orbits, obtained entirely within the regular connector. Subsequent coefficient paths and passing conditions are chosen inside these common charts. □

We now check the exponent equality needed for this cancellation. In the horizontal chart used above, the weak and normal rates at its origin are $\epsilon\bar{B}$ and $-(1 - \epsilon\bar{B})$. Reflection fixes both ϵ and $\bar{B}$. Thus, on a compact set $\bar{B} \geq B_* > 0$, the two horizontal passages have the same exact exponent

$$\mu = \frac{1 - \epsilon\bar{B}}{\epsilon\bar{B}}, \qquad h = \epsilon\mu = \frac{1 - \epsilon\bar{B}}{\bar{B}} > h_* > 0. \tag{4.50}$$

The weak boundary $\bar{B} = 0$ is treated by the unnormalized slow factor of Proposition 4.8.

Lemma 4.13 (Cancellation before differentiation)**.** *Let $D_\pm(s) = K_\pm s^{h/\epsilon} e^{P_\pm(s)/\epsilon}$, $K_\pm > 0$, on $s > 0$, with the same h in both maps. Let $f(s) = sk(s)$, $k > 0$, and suppose the following three factors have positive lower bounds:*

$$1 + \frac{sk'}{k}, \qquad h + sP'_+(s), \qquad h + f(s)P'_-(f(s)).$$

Assume bounded total order $N + 1$ for $k, P_\pm$ and order N for $h, 1/k$, on a common input and coefficient neighborhood. Then

$$\begin{aligned} \epsilon \log \frac{(D_- \circ f)'(s)}{D'_+(s)} &= C(\mathbf{b}, \epsilon) + h \log k(s) + P_-(f(s)) - P_+(s) \\ &\quad + \epsilon \log\left(1 + \frac{sk'(s)}{k(s)}\right) + \epsilon \log \frac{h + f(s)P'_-(f(s))}{h + sP'_+(s)}, \\ C(\mathbf{b}, \epsilon) &= \epsilon \log(K_-/K_+). \end{aligned} \tag{4.51}$$

After subtracting C, the right side has bounded total order N. No boundedness assertion for C follows from this cancellation.

Proof. Differentiate both amplitudes before cancelling factors. The power part of the ratio is

$$(f/s)^{h/\epsilon - 1} f' = k^{h/\epsilon}\left(1 + \frac{sk'}{k}\right).$$

The remaining factors are $h + fP'_-(f)$ and $h + sP'_+(s)$; logarithms give the identity. Their positive margins bound the reciprocals and logarithms. Each differentiated primitive costs one order, so the total-order chain rule gives the stated budget, including coefficient dependence of f and Euler scale derivatives. □

The identity makes the derivative requirement explicit. Primitive total order five leaves total order four for the nonconstant $R = K - K(0)$. For $H = \partial_s R$, the orders $s^3\mathbf{b}^0$ and $s^2\mathbf{b}^1$ give two input derivatives and coefficient Lipschitz control of its first derivative. A Hadamard column additionally uses $s^3\mathbf{b}^1$ and $s^2\mathbf{b}^2$, still total order four. To use that column, one must also establish its center identity and the common domain along the whole coefficient segment; these are proved separately.

For example $k = \alpha e^\psi$, with an input-independent $\alpha > 0$ that may be small, gives

$$h \log k = h \log \alpha + h\psi, \qquad 1 + sk'/k = 1 + s\psi'.$$

Group $h \log \alpha$ with C; the remaining compositions use bounded jets of f, ψ. Derivatives of the coefficient-only constant require separate action bounds. Exact exponent equality is essential: an error $\delta\mu$ leaves the unbounded term $\epsilon\delta\mu \log f(s)$ at $s = 0$.

The constructions must therefore reserve the derivative orders before their domains are reduced. The following allocation records the orders used later.

Use	Primitive order and operations	Result used later
Three lower connections	total order at least ten, including six input, two coefficient and two Euler orders	bounded relative units and their compositions on the stated common domains
Separated joint columns	total order five; one derivative in (4.51)	total order four for the nonconstant comparison, including $s^3\mathbf{b}^1$ and $s^2\mathbf{b}^2$
Contact comparison	total order eight if three input and three coefficient derivatives of its numerator are retained	at most two orders spent on logarithmic rates and connection curvature
Weak or cubic rate	total order eight; two rate derivatives and one later Hadamard division	three input and two coefficient orders of the divided numerator, once that division is justified
Upper saddle	primitive total order six for the relative logarithm; total order eight for its curvature	four Euler and two coefficient orders of the respective object; curvature consumes two extra Euler orders

The exact long-scale maps and their ordinary terminal-height derivatives are constructed separately.

5 Centers, common domains and division

A center identity can be used for coefficient division only when its return maps are defined on the input interval being compared. We first establish the real period domains of the two center branches, then use actual gaps and splittings to construct comparison centers valid on a whole input interval. Integral division expresses closing differences in the center-breaking coefficients while preserving the absolute contraction estimates. At the intersection of the branches, the divisor includes their product generator.

5.1 The two center branches and their real period domains

The center identities used below are identities of actual return maps on specified real domains. In original coefficients the two branches are

$$\mathcal{C}_r = \{a = c = d = 0\}, \qquad \mathcal{C}_c = \{m = -B,\ a = -c,\ d = c\}.$$

For $\rho > 0$ these become, respectively,

$$(\bar{a}, \bar{c}, \nu) = (0, 0, 0), \qquad (\bar{m}, \bar{a}, \nu) = (-\bar{B}, -\bar{c}, 0).$$

Their union is exactly the algebraic set $\nu = \tau_1 = \sigma_1 = 0$, because $\tau_1 = \bar{a} + \bar{c}$ and $\sigma_1 = \bar{c}(\bar{B} + \bar{m})$. We construct their real period domains by explicit first integrals with compact level sets, fixing $\epsilon > 0$ throughout this subsection. These domains supply the actual center returns used in division. The transition estimates of Section 4 supply the uniform finite jets as a gap or the scale vanishes.

Proposition 5.1 (Real separated center)**.** *On $\mathcal{C}_c$, assume*

$$b = 1 - \epsilon\bar{B} > 0, \qquad a_0 = 1 - \epsilon\bar{c}^2 > 0.$$

Let I_C be the component containing zero of $\{\chi : A_C(\chi) > 0\}$, where

$$A_C(\chi) = a_0 + \bar{c}(1 - 2\epsilon\bar{B})\chi + b\bar{B}\chi^2, \tag{5.1}$$

and let

$$J_C = \begin{cases} (-\epsilon, \infty), & \bar{B} \geq 0, \\ (-\epsilon, -1/\bar{B}), & \bar{B} < 0. \end{cases}$$

The change of state coordinates

$$\chi = -\frac{X + \bar{c}Y}{1 + \bar{B}Y}, \qquad X = -\bar{c}Y - \chi(1 + \bar{B}Y) \tag{5.2}$$

is a diffeomorphism on $1 + \bar{B}Y > 0$. The inverse image Ω_C of $I_C \times J_C$ is filled by the origin and periodic ovals of the original field. More precisely,

$$\mathcal{E}_C(\chi, Y) = \int_0^\chi \frac{s}{A_C(s)}\,\mathrm{d}s + \int_0^Y \frac{y}{(y + \epsilon)(1 + \bar{B}y)}\,\mathrm{d}y \tag{5.3}$$

is a real analytic first integral on this domain. It has one critical point, a nondegenerate minimum at the origin, and compact sublevel sets. Every level $\mathcal{E}_C = h > 0$ is one smooth connected oval, and the vector field is nonzero on it. The construction includes $\bar{B} = 0$ and both signs of the quadratic discriminant. On the positive axis $Y = 0$, membership in this domain is precisely $-p \in I_C$.

Proof. The original scaled equations on this branch are exactly

$$X_\xi = -Y(1 + \bar{B}Y) + \epsilon(\bar{B}X^2 - \bar{c}X), \qquad Y_\xi = (Y + \epsilon)(X + \bar{c}Y).$$

The change (5.2) is regular since $1 + \bar{B}Y > 0$; its X derivative is $-(1 + \bar{B}Y)^{-1} \neq 0$. Differentiating its quotient and substituting $X = -\bar{c}Y - \chi(1 + \bar{B}Y)$ gives, without a time change,

$$\chi_\xi = YA_C(\chi), \qquad Y_\xi = -(Y + \epsilon)(1 + \bar{B}Y)\chi. \tag{5.4}$$

The derivatives of the two potentials in (5.3) are $\chi/A_C(\chi)$ and $Y/((Y + \epsilon)(1 + \bar{B}Y))$. Their Lie derivatives along (5.4) are χY and $-\chi Y$. Thus the first integral is exact.

Each potential decreases strictly towards zero from its left endpoint and increases strictly from zero towards its right endpoint; its minimum is zero. We verify divergence at all endpoints. For the Y potential the lower endpoint is $-\epsilon$ and $1 + \bar{B}Y \to b > 0$. Its derivative is asymptotic to $-\epsilon/[b(Y + \epsilon)]$, so the potential tends to infinity. If $\bar{B} < 0$, the upper endpoint $-1/\bar{B} > 0$ is a simple zero of the other denominator factor and gives logarithmic divergence. If $\bar{B} > 0$, the potential at infinity is $\bar{B}^{-1} \log Y + O(1)$; if $\bar{B} = 0$, it is $Y - \epsilon \log((Y + \epsilon)/\epsilon)$ and again diverges.

A finite endpoint of I_C is a simple or double zero of A_C and is not zero, because $A_C(0) = a_0 > 0$. On the selected side its integrand has the sign of that endpoint and absolute size bounded below by a positive constant times the inverse distance to it. Integration from zero, in the appropriate direction, therefore tends to positive infinity. At an infinite endpoint with $\bar{B} > 0$, the integrand is $1/(b\bar{B}\chi) + O(\chi^{-2})$, giving logarithmic divergence. With $\bar{B} = 0$ and $\bar{c} \neq 0$, its infinite-end asymptotic is $1/\bar{c}$, giving linear growth of the potential in the outward direction. With $\bar{B} = \bar{c} = 0$ the potential is $\chi^2/2$. Finally, if $\bar{B} < 0$, the concave quadratic A_C has two finite roots of opposite signs, already covered above. These cases exhaust the endpoints. Since both potentials are nonnegative and diverge at their endpoints, every sublevel is compact in $I_C \times J_C$.

The only critical point is $(\chi, Y) = (0, 0)$. In (X, Y) its Hessian is

$$D^2\mathcal{E}_C(0, 0) = \frac{1}{1 - \epsilon\bar{c}^2} \begin{pmatrix} 1 & \bar{c} \\ \bar{c} & 1/\epsilon \end{pmatrix} > 0. \tag{5.5}$$

Write the potentials as $U(\chi), V(Y)$. For $h > 0$, the two roots $\chi_- < 0 < \chi_+$ of $U = h$ bound two graphs $V(Y) = h - U(\chi)$, one on either side of $Y = 0$. They join at $(\chi_\pm, 0)$ with nonzero gradient, forming one smooth embedded oval. The field has its only equilibrium at the origin, so its nonzero tangent field on each compact oval has a finite positive period. At $Y = 0$, $\chi = -X = -p$, giving the axis criterion. □

Proposition 5.2 (The reversible well)**.** *On $\mathcal{C}_r$, assume $1 + \epsilon\bar{m} > 0$ and $|\epsilon\bar{B}| < 1/4$. Put $\theta = \epsilon\bar{B}$, $Z = Y + \epsilon > 0$ and*

$$\mathcal{E}_r(X, Y) = X^2 Z^{-2\theta} + V_r(Y), \qquad V_r(Y) = 2\int_0^Y u(1 - \bar{m}u)(u + \epsilon)^{-1-2\theta}\,\mathrm{d}u. \tag{5.6}$$

Let $Y_s = 1/\bar{m}$ if $\bar{m} > 0$, and $Y_s = \infty$ otherwise. On $J_r = (-\epsilon, Y_s)$ define

$$h_- = \lim_{Y \downarrow -\epsilon} V_r(Y), \qquad h_+ = \lim_{Y \uparrow Y_s} V_r(Y), \qquad h_* = \min(h_-, h_+).$$

Both limits are positive or infinite. The set $\{Y \in J_r,\ 0 < \mathcal{E}_r < h_*\}$ is a period annulus: every level in this range is one compact periodic oval. If $\bar{B} \geq 0$ and $\bar{m} \leq 0$, then $h_* = \infty$ and this annulus fills the half-plane $Y > -\epsilon$, apart from its center. If $\bar{B} = 0$ and $\bar{m} > 0$, its upper boundary is the homoclinic loop through $(0, 1/\bar{m})$.

Proof. Here the field is $X_\xi = Y(\bar{m}Y - 1) + \theta X^2$, $Y_\xi = (Y + \epsilon)X$. Differentiating $X^2 Z^{-2\theta}$ cancels its cubic terms and gives $2XY(\bar{m}Y - 1)Z^{-2\theta}$; the derivative of V_r is its negative. This proves the first integral. On J_r the factor $1 - \bar{m}Y$ is positive. Thus V_r decreases strictly from its left endpoint to its minimum zero at $Y = 0$ and increases strictly thereafter. Its second derivative at zero is $2\epsilon^{-1-2\theta} > 0$. At the lower boundary its derivative has a negative nonzero multiple of $(Y + \epsilon)^{-1-2\theta}$ as its leading term. Hence $h_- = \infty$ when $\theta \geq 0$; when $\theta < 0$ the finite limit is positive. If $\bar{m} > 0$, h_+ is finite and positive. If $\bar{m} \leq 0$, the restriction $|\theta| < 1/4$ makes the positive-end integral divergent, both for the linear and quadratic numerator.

For $0 < h < h_*$, the regular coordinate $u = XZ^{-\theta}$ changes the energy into $u^2 + V_r(Y)$. Its level is one smooth compact oval, by the same two-graph construction as above. The field vanishes only at the origin in J_r, so the oval is periodic. The restriction $h < h_*$ retains compactness when $\bar{B} < 0$ or $\bar{m} > 0$.

When $\bar{B} = 0$ and $\bar{m} > 0$, $h_- = \infty$ and the upper critical point of V_r at Y_s is a nondegenerate maximum. The energy $h_+ = V_r(Y_s)$ has a negative turning point and two arcs meeting at the saddle $(0, Y_s)$. The field has no other zero on those arcs. They form the bounding homoclinic loop of the central well. For $\bar{B} \geq 0$, $\bar{m} \leq 0$, both limiting energies are infinite, proving the full half-plane assertion. □

5.2 The three-saddle triangle and the signed weak boundary

The separated integral covers either sign of $\bar{B}$. We now extract two boundary descriptions needed later: exact upper power coordinates on the three-saddle triangle, and a common finite strong and upper weak intersection through $\bar{B} = 0$.

Lemma 5.3 (The triangle and its upper saddle)**.** *On $\mathcal{C}_c$ write $\bar{B} = -\beta < 0$, $\bar{m} = \beta$, $\bar{c} = -\bar{a}$, and assume $1 - \epsilon\bar{a}^2 > 0$. Set*

$$b = 1 + \epsilon\beta, \qquad \Delta = \sqrt{\bar{a}^2 + 4\beta b}, \qquad k_\pm = \frac{\bar{a} \pm \Delta}{2b}, \qquad p_+ = \frac{\bar{a} - k_-}{\beta}, \quad p_- = \frac{k_+ - \bar{a}}{\beta}.$$

Then $p_\pm > 0$ and the period domain Ω_C is the triangle

$$\mathcal{T} = \{L_0 = Y + \epsilon > 0, \quad L_+ = p_+ - X + k_- Y > 0, \quad L_- = p_- + X - k_+ Y > 0\}. \tag{5.7}$$

Its vertices are $((\bar a \pm \Delta)/(2\beta), -\epsilon)$ *and* $(\bar a/\beta, 1/\beta)$. *The positive function*

$$\mathcal{H} = L_0 L_+^{\lambda_+} L_-^{\lambda_-}, \qquad \lambda_\pm = \frac{\Delta \pm \bar a(1+2\epsilon\beta)}{2\epsilon\beta\Delta} \tag{5.8}$$

is a first integral on $\mathcal{T}$. *Near the upper vertex the analytic coordinates*

$$U = L_+ L_0^{\epsilon\beta}, \qquad V = L_- L_0^{\epsilon\beta}$$

have a regular inverse and give the exact orbital equation

$$\frac{\mathrm{d}V}{\mathrm{d}U} = -r\frac{V}{U}, \qquad r = \frac{\lambda_+}{\lambda_-} = \frac{p_+}{p_-} > 0. \tag{5.9}$$

On compact β *intervals bounded away from zero, these local coordinate changes have bounded finite jets for small scales and bounded* $\bar a$.

Proof. The identity

$$\Delta^2 - \bar a^2(1+2\epsilon\beta)^2 = 4\beta b(1-\epsilon\bar a^2) > 0$$

proves positivity of $p_\pm$ and $\lambda_\pm$. The lines in (5.7) meet at the stated vertices. In the separated coordinate the roots of A_C are $-p_+$ and p_-; its leading coefficient is negative. For $Y < 1/\beta$, the inequalities $-p_+ < \chi < p_-$ translate exactly into $L_+ > 0, L_- > 0$. Thus the triangle agrees with Ω_C.

Direct differentiation of its three affine factors gives their cofactors

$$X - \bar a Y, \qquad -\epsilon\beta X - bk_- Y, \qquad -\epsilon\beta X - bk_+ Y.$$

The identities $\lambda_+ + \lambda_- = 1/(\epsilon\beta)$ and $\lambda_+ k_- + \lambda_- k_+ = -\bar a/b$ make their weighted sum zero. This proves the first integral. Its logarithm is strictly concave and tends to negative infinity at the boundary. At the origin

$$\frac{\lambda_+}{p_+} = \frac{\lambda_-}{p_-} = \frac{b}{\epsilon\Delta},$$

so both components of its gradient vanish. The energy $-\log(\mathcal{H}/\mathcal{H}(0,0))$ is consequently proper and strictly convex on the triangle. Its positive levels give the same periodic ovals described by Proposition 5.1.

At the upper vertex $L_0 = \epsilon + 1/\beta > 0$ and the two other factors have independent gradients. Multiplication by the positive analytic unit $L_0^{\epsilon\beta}$ preserves their invertible Jacobian. Since $\epsilon\beta(\lambda_+ + \lambda_-) = 1$,

$$\mathcal{H} = U^{\lambda_+} V^{\lambda_-}.$$

Differentiation along the nonzero saddle orbit gives (5.9). The ratio equals p_+/p_- by the preceding gradient identity. Near this finite positive-height vertex, L_0 is bounded away from zero on each stated compact block; all finite derivatives of the units and their inverses are bounded. □

Lemma 5.4 (A common real weak boundary)**.** *Fix* $\bar c > c_* > 0$ *in a compact interval and let* $\bar B$ *vary in a neighborhood of zero on* $\mathcal{C}_c$. *For sufficiently small* $\epsilon > 0$ *define*

$$\Delta_b = \sqrt{\bar c^2 - 4b\bar B}, \qquad k_s = \frac{-\bar c - \Delta_b}{2b}, \qquad \widehat{p}_C = \frac{2(1-\epsilon\bar c^2)}{\bar c(1-2\epsilon\bar B) + \Delta_b}. \tag{5.10}$$

These are regular through $\bar B = 0$ *and* $\bar B\widehat{p}_C = \bar c + k_s$. *The invariant line* $L = \widehat{p}_C - X + k_s Y = 0$ *is the boundary of* Ω_C *met by its positive finite axis inputs. Thus inputs* $p = \widehat{p}_C - t$, $0 < t \ll 1$,

belong to the period domain. In the reversed upper chart $R = 1/Y$, $y = X/Y - k_s$, the same line is $y = \widehat{p}_C R$, and its restricted equation is

$$\dot{R} = \widehat{p}_C R(R + \bar{B})(1 + \epsilon R), \qquad \gamma_C = \bar{B}\widehat{p}_C. \tag{5.11}$$

In particular $\partial_{\bar{B}}\gamma_C > c_1 > 0$ on a common smaller block. The finite strong and upper weak intersections on this center are the same actual point $p = \widehat{p}_C$.

Proof. The discriminant of A_C is $\bar{c}^2 - 4b\bar{B}$. Rationalizing its root nearest zero on the negative side gives $-\widehat{p}_C$. The displayed denominator has a fixed positive margin. Substitution in $bk_s^2 + \bar{c}k_s + \bar{B} = 0$ proves $\bar{B}\widehat{p}_C = \bar{c} + k_s$, including its continuous value at $\bar{B} = 0$. The equation $\chi = -\widehat{p}_C$ from (5.2) is precisely $L = 0$. Its cofactor is $\epsilon\bar{B}X - bk_sY$, so the line is invariant under the original field.

The first positive zero on $Y = -\epsilon$ of the horizontal field is

$$p_\epsilon = \frac{2b}{\bar{c} + \Delta_b}.$$

The identity $p_\epsilon - \epsilon\bar{c} = b\widehat{p}_C$ shows that this zero lies on $L = 0$. Its branch up to $Y = 0$ is the same finite strong boundary just identified by the scalar center domain. In the reversed upper chart substitute $y = \widehat{p}_C R$ in the exact projective field and use $\bar{c} + k_s = \bar{B}\widehat{p}_C$. This gives (5.11) on the actual weak face. At $\bar{B} = 0$,

$$\partial_{\bar{B}}\gamma_C = \widehat{p}_C = \frac{1 - \epsilon\bar{c}^2}{\bar{c}} > 0.$$

Continuity on the compact block proves the uniform positive margin. The actual invariant line can be preserved in any auxiliary extension used to construct its weak graph, so the two constructions retain this same intersection. □

For any oval in either of the stated domains, the two paths from its axis intersections meet at the same original central height $z = 1 + Y/\epsilon > 0$. Their complete central maps therefore agree on every common section interval parametrizing these ovals. This identity holds throughout each stated real period domain. The constructions below provide the signed extensions and coefficient segments needed for division. The triangle gives an exact local saddle power, while the reversible well retains its own period annulus; the separated domain remains the one used across signed $\bar{B}$ faces.

5.3 The three-cycle variation

The reversible center also gives a direct route to the lower bound. Keep the original coefficients $B = 0$, $m = \epsilon > 0$ and start at $a = c = d = 0$. Its bounded period annulus has the upper saddle and the regular lower paths prepared in Section 4. Use the actual upper ratio $r = 1 + \kappa$ as a coefficient coordinate. Section 13 constructs a single coefficient curve

$$(\bar{a}, \kappa, \nu) = (h_\epsilon\vartheta, \vartheta, \beta_\epsilon\vartheta), \qquad h_\epsilon \longrightarrow \tfrac{3}{2},$$

with exponentially small β_ϵ and positive $\delta_\epsilon, S_\epsilon$. The ratio variation produces the logarithmic term in the closing difference, and the line-breaking coefficient fixes its constant term. For the actual central-height closing function the resulting profile is

$$\lim_{\epsilon\downarrow 0}\lim_{\vartheta\downarrow 0}\frac{F(\delta_\epsilon x;\vartheta)}{\vartheta S_\epsilon} = P(x) := \frac{x^2}{2} - x\log x - x - \frac{1}{20},$$

with C^1 convergence on compact subintervals of $(0, 10]$. The function P has signs $-, +, -, +$ at $10^{-3}, 10^{-1}, 1, 10$; its derivative $x - \log x - 2$ has exactly two simple zeros. It therefore has three simple positive roots.

For each fixed sufficiently small ϵ, one sufficiently small positive ϑ preserves all three roots on three disjoint compact bands of real center ovals. These are three cycles of the same original field. Proposition 13.2 proves the first-variation profile; Theorem 13.3 proves the fixed-scale perturbation limit, simultaneous persistence, positive complete returns and convergence of each cycle to the entire graphic. The forward return also determines their alternating stability in Corollary 13.4. This construction uses the reversible center and its prepared paths independently of the upper count.

5.4 Coordinates defined by the actual passages

Coefficient paths must control both the slow factors and their moving entries on the original sections. We define the gaps and splittings from the actual weak graphs and strong fibres, then obtain common inverse boxes at fixed normalization. To retain order N through the minimum, Morse coordinate and time unit, reserve $N+4$ reduced-phase orders before choosing the larger domains. Connections to $Y=0$ use Euler scale derivatives.

We use the following elementary uniform inverse construction repeatedly.

Lemma 5.5 (Common inverse boxes)**.** *Let F_ϵ be a family of coefficient maps on a fixed real neighborhood, with bounded derivatives through a prescribed order, uniformly for sufficiently small $\epsilon \geq 0$, and suppose $F_\epsilon \to F_0$ in the coefficient C^1 norm. Suppose its Jacobian at a base point and $\epsilon = 0$ is invertible. After reducing the neighborhood, there are nested fixed product boxes $Q \Subset Q^+$ in the new coordinates on which all F_ϵ have single-valued inverses with bounded finite jets. Any additional passive coefficients can be included in these boxes. The assertion remains valid with Euler scale jets when that is the regularity of F_ϵ.*

Proof. Write $A = DF_0$ at the base point. On a small convex source ball arrange $\|A^{-1}(DF_\epsilon - A)\| < 1/2$, uniformly in the passive parameters and scale. The equation $F_\epsilon(x) = y$ is then the contraction

$$x = x_* + A^{-1}\{y - F_\epsilon(x_*) - [F_\epsilon(x) - F_\epsilon(x_*) - A(x - x_*)]\}.$$

Choose a common target ball whose contractions stay in the source ball, and nested product boxes Q, Q^+ inside it. The inverse Jacobian is bounded by $2\|A^{-1}\|$; differentiating $F_\epsilon \circ F_\epsilon^{-1} = \mathrm{id}$ gives all higher jets inductively, including passive and Euler derivatives. □

Proposition 5.6 (Actual gap and splitting coordinates)**.** *On the joint repeated and weak blocks specified below, the actual passages admit the following coefficient coordinates on common nested boxes, with uniformly bounded selected finite inverse jets:*

$$(\bar{B}, \delta, \tau, \zeta, \nu), \qquad (\bar{B}, \bar{c}, d_s, \varsigma, \nu),$$

respectively. The joint gaps are exactly $\delta - \zeta$ and $\delta + \zeta$; their center axis is $\tau = \zeta = \nu = 0$. The weak splitting is $d_s = P_L - \widehat{p}$, and its center axis is $d_s = \varsigma = \nu = 0$. The joint entries can be calibrated on the original finite section so that both equal $\Xi - \bar{B}p$ on the center and have bounded coefficient Lipschitz constants. On the weak block the regular horizontal input $\chi(u)$ has bounded coefficient jets, while the exact moving eigenvalue γ and the composed horizontal factor $F_I(\chi(u))$ have the positive center $\bar{B}$ derivatives stated below. All these constructions precede comparison of coefficient paths or zeros of a displacement.

Proof.

The joint repeated block. Fix $g_* > 0$ and the limiting coefficients

$$\bar{B} = g_*^2, \qquad \bar{m} = -g_*^2, \qquad \bar{a} = -2g_*, \qquad \bar{c} = 2g_*, \qquad \nu = \epsilon = 0.$$

Write $b = 1 - \epsilon\bar{B}$ and keep $\bar{B}$ and b bounded away from zero. The upper gap is the exact quantity already prepared in (4.40):

$$\mu_U = -\frac{(\bar{c}+\nu)^2 + 4b\bar{m}}{4b^2}. \tag{5.12}$$

For the lower passage we select the phase within the freedom of the construction in Proposition 4.1. The following calibration both specifies its gap and identifies the center input.

On the second center $\bar{a} = -\bar{c}$, $\bar{m} = -\bar{B}$, $\nu = 0$, define

$$\Xi = \frac{\bar{c}(1 - 2\epsilon\bar{B})}{2b}, \qquad \delta_c = \frac{4\bar{B}b - \bar{c}^2}{4b^2}, \qquad V = \frac{\Xi/\bar{B} - X - \bar{c}Y/(2b)}{Y + 1/\bar{B}}. \tag{5.13}$$

Its denominator has a positive margin on each strip under consideration. In the notation of the real center construction, $V = \bar{B}\chi + \Xi$; alternatively direct differentiation in (2.16) gives

$$V_\xi = bY(V^2 + \delta_c), \qquad V(X, -\epsilon) = \frac{\bar{c}/2 - \bar{B}X}{b}, \qquad V(p, 0) = \Xi - \bar{B}p. \tag{5.14}$$

These identities hold for positive, zero and negative δ_c.

In the lower strip choose a fixed $0 < z_h < 1$ and reverse time. Division by the positive factor $b(1 - z)$ gives $\dot{V} = \epsilon(V^2 + \delta_c)$ on the center. The normal coordinate z contracts there: its linear rate on $z = 0$ is $(X - \epsilon\bar{c})/b$, with a positive margin near $X = 1/g_*$. In a smaller strip both this rate and the derivative of z times its rate remain positive. Extend the equations first in these center coordinates, retaining the exact scalar equation and normal contraction, and retaining $z = 0$ invariance whenever $\nu = 0$; extend only the off-center defect by cutoffs. This is an admissible choice in the graph and phase construction of Section 4. It keeps the exact quotient on the center and preserves the factor ϵ in the slow equation. The latter also follows directly: $V_X = O(1)$, $V_z = O(\epsilon)$, while the original X velocity contains ϵ.

The resulting actual lower weak equation is $\dot{w} = \epsilon f_L^0(w; \mathbf{b}, \epsilon)$, with uniformly bounded finite jets and $(f_L^0)_{ww} > 1$. Its minimum w_0 is therefore a finite-smooth coefficient function. Set

$$\mu_L = f_L^0(w_0), \qquad u = (w - w_0)\left[\int_0^1 (1 - s)(f_L^0)_{ww}(w_0 + s(w - w_0))\, ds\right]^{1/2}. \tag{5.15}$$

This is an increasing state coordinate, and $f_L^0 = \mu_L + u^2$. Dividing by its positive derivative gives exactly $\dot{u} = \epsilon(\mu_L + u^2)$ while preserving the normal margin. On the center, $w = V$, $w_0 = 0$ and $u = V$.

The normalization in (5.15) is fixed by the preceding time and phase choice. To calculate it, take $\nu = 0$. The original line $z = 0$ is invariant, and on it $w = (\bar{c}/2 - \bar{B}X)/b$. Restriction of the reversed field divided by $b(1 - z)$ gives, with $\tau = \bar{a} + \bar{c}$,

$$f_L^0(w) = w^2 - \frac{\tau}{b}w + \frac{\bar{B}(1 + \epsilon\bar{m}) + \bar{a}\bar{c}/2 + \bar{c}^2/4}{b^2}, \qquad \mu_L = \frac{\bar{B}(1 + \epsilon\bar{m}) - \bar{a}^2/4}{b^2}. \tag{5.16}$$

Thus μ_L is the finite-X minimum multiplied by the positive unit $\bar{B}/b^2$; the compact horizontal factor F_I has a different normalization. For $\nu \neq 0$, (5.15) continues to use the actual weak graph and the chosen phase.

Set

$$\delta = \frac{\mu_U + \mu_L}{2}, \qquad \zeta = \frac{\mu_U - \mu_L}{2}, \qquad \tau = \bar{a} + \bar{c}, \qquad \varsigma = \bar{B} + \bar{m}. \tag{5.17}$$

At $\nu = 0$ the exact identity

$$\zeta = -\frac{\bar{c}\tau}{4b^2} + \frac{\tau^2}{8b^2} - \frac{\varsigma}{2b^2} \tag{5.18}$$

follows by subtraction of the two gaps. At $\tau = \varsigma = 0$,

$$\det \frac{\partial(\delta, \tau, \zeta)}{\partial(\bar{c}, \tau, \varsigma)} = \frac{\bar{c}}{4b^4} > c_* > 0. \tag{5.19}$$

For example, on this axis $\delta_{\bar{c}} = -\bar{c}/(2b^2)$ and $\zeta_\varsigma = -1/(2b^2)$, while $\zeta_{\bar{c}} = 0$, which proves the determinant. Continuity of the actual phase and minimum derivatives retains a uniform nonzero margin for both signs of small ν and small positive scale.

Consequently $(\bar{B}, \delta, \tau, \zeta, \nu)$ are coefficient coordinates on common boxes as in Lemma 5.5. On these boxes

$$\tau = \zeta = \nu = 0 \quad \Longleftrightarrow \quad \bar{a} = -\bar{c}, \quad \bar{m} = -\bar{B}, \quad \nu = 0. \tag{5.20}$$

Indeed (5.18) first gives $\varsigma = 0$ when $\tau = \zeta = \nu = 0$; the converse follows from the gap formulas. The center with prescribed $(\bar{B}, \delta)$ has

$$\bar{c}_c = 2\sqrt{\bar{B}b - b^2\delta} > 0. \tag{5.21}$$

All these inverse jets have uniform finite bounds on the larger box. The two factors are exactly $u^2 + \delta - \zeta$ and $v^2 + \delta + \zeta$. Thus fixed full passages across both minima pass exactly when $\delta > |\zeta|$; moving entries use the whole-path tests of Section 4.

Entries on one initial section. The common boxes must also control the entry functions. At a finite joint contact fix p in a small interval about $P = 1/g_*$ on $Y = 0$. Continue the outgoing lower branch backwards to $z = z_h$ and the upper branch forwards to a fixed positive radial section. Both segments are chosen before their slow passages. On the lower segment use z as the independent variable; X has a positive margin and $z \geq z_h$, so its equation and its finite parameter variations have bounded coefficients. The upper segment first uses the low-height estimate following (4.38), then a compact regular arc. Even at the repeated limiting input $p = P$ it reaches the fixed radial section in finite affine time. The entry maps and their transverse inverses have bounded ordinary p, coefficient and Euler scale jets on a slightly larger common domain.

Choose the upper auxiliary phase to preserve (5.13) on the center as well. This agrees with its boundary coordinate at $R = 0$ and with its exact gap (5.12). On the center the positive upper time factor is bY, and the lower reversed time factor is $b(1 - z)$. Record their accumulated scalar times on the initial segments, denoted T_U and T_L. These have bounded selected jets: the upper time integral has integrand $bY/(Y + \epsilon)$ times a bounded regular factor, and the lower one is taken on a fixed z-interval. The upper original transit time may contain $|\log \epsilon|$, but this scalar time does not.

Let u_e, v_e be the prepared entries at the fixed fast sections. Apply the inverse scalar flows for these bounded recorded times to define entries e_L, e_U on the original initial section. Off the center use the same smooth time integrals and the same complete scalar extensions. Finite-time scalar flow and its inverse have bounded jets on the larger box. Their roots and their through components are invariant, so this change of entry does not alter the whole-path passing tests in a smaller contact neighborhood. On the center, the exact flow identities in (5.14) give

$$e_L(p) = e_U(p) = U_c(p) := \Xi - \bar{B}p. \tag{5.22}$$

The bounded regular segments therefore calibrate the entries before the weak Dulac passage.

With $\mathbf{q} = (\bar{B}, \delta, \tau, \zeta, \nu)$ in the common larger box, the chain rule and the inverse bounds just proved give, for some fixed L,

$$|e_i(p; \mathbf{q}_1) - e_i(p; \mathbf{q}_2)| \leq L|\mathbf{q}_1 - \mathbf{q}_2|, \qquad i = L, U. \tag{5.23}$$

The bound also holds for each lower-order selected input jet when one more coefficient derivative is reserved. Since $\partial_p U_c = -\bar{B}$, both entry derivatives have a fixed negative margin

after reducing the transverse parameter box. For $A = |\tau| + |\zeta| + |\nu|$, comparison at fixed $(\bar{B}, \delta)$ yields

$$|e_i(p; \mathbf{q}) - U_c(p; \bar{B}, \delta)| \le CA. \tag{5.24}$$

The same construction on a separated finite-entry patch uses fixed slow sections. At its radial end the common actual input of Lemma 4.12 supplies the transition jets. The comparison paths below use these patchwise coordinate and entry bounds to preserve an entire passing interval.

The weak boundary with either sign of $\bar{B}$. Fix $\bar{c}_0 > 0$ and the limiting coefficients $\bar{B} = \bar{m} = 0$, $\bar{a} = -\bar{c} = -\bar{c}_0$. Keep the exact γ of (4.43); it is a finite-smooth function on an ordinary coefficient neighborhood open in both $\bar{B}$ and $\bar{m}$. Let P_L be the actual outgoing lower strong intersection with $Y = 0$, and let $\widehat{p}$ be the actual upper weak intersection used in (4.48). Define

$$d_s = P_L - \widehat{p}, \qquad \varsigma = \bar{B} + \bar{m}, \qquad \mathbf{q}_w = (\bar{B}, \bar{c}, d_s, \varsigma, \nu). \tag{5.25}$$

These are actual section positions, before either one is changed to a normalized power coordinate.

Choose the upper auxiliary graph to preserve the affine branch of Lemma 5.4. In the present notation put

$$\Delta_b = \sqrt{\bar{c}^2 - 4b\bar{B}}, \quad k_s = \frac{-\bar{c} - \Delta_b}{2b}, \quad p_c = \frac{2(1 - \epsilon\bar{c}^2)}{\bar{c}(1 - 2\epsilon\bar{B}) + \Delta_b}. \tag{5.26}$$

Thus $p_c = \widehat{p}_C$ from Lemma 5.4; its denominator has a positive margin through $\bar{B} = 0$. That lemma identifies the invariant line $X = p_c + k_s Y$ and both of its actual intersections with $Y = 0$. Subtract this known graph before making the auxiliary extension and extend only its off-center defect. The graph transform then retains this exact branch on the center for either sign of $\bar{B}$.

The lower simple-root preparation and the upper weak graph with its finite connector supply the coefficient jets. At the zero-scale limiting point the lower root solves $1 + \bar{a}P_L + \bar{B}P_L^2 = 0$, while the affine upper weak line has intersection $\widehat{p} = 1/\bar{c}$ independent of $\bar{a}$. Implicit differentiation therefore gives

$$\partial_{\bar{a}} d_s \longrightarrow 1/\bar{c}_0^2 > 0. \tag{5.27}$$

The zero-scale limit holds in the coefficient jets on a fixed small block: the only nonregular connector correction is the integrable low-layer carrier $\epsilon/(Y + \epsilon)$. Hence this derivative retains a fixed positive margin. Together with $\partial_{\bar{m}}\varsigma = 1$ and $\partial_{\bar{a}}\varsigma = 0$, this verifies the hypotheses of Lemma 5.5 for the weak coefficient map.

The second center is exactly

$$d_s = \varsigma = \nu = 0. \tag{5.28}$$

One direction follows from the invariant branch. Conversely at $\varsigma = \nu = 0$ the actual center $\bar{a} = -\bar{c}$ gives $d_s = 0$; the nonzero splitting derivative and uniqueness in the common inverse box show it is the only such $\bar{a}$. On this center, direct substitution into the actual weak eigenvalue in (4.43) gives

$$\gamma_c = \bar{B}p_c, \qquad \partial_{\bar{B}}\gamma_c|_{\bar{B}=0} = \frac{1 - \epsilon\bar{c}^2}{\bar{c}} > c_* > 0. \tag{5.29}$$

This is also the eigenvalue of the restricted equation (5.11). Continuity retains its positive derivative margin on a smaller common center block.

Finally we specify the regular connection used together with the weak inverse map. Select the outgoing fast section of the reversed weak block on the side $\varphi < \varphi_s$, a fixed small distance from the weak point. Following the original forward field from this section to the incoming horizontal section $\varphi = -1/v_h$ gives a compact angular connection; choose v_h small enough.

At the limiting field its angular velocity is $-\varphi(\varphi+\bar{c}_0)<0$ on this closed arc, with a fixed margin, and it retains that margin nearby. Both sections are transverse. The radial equation is divisible by R, so its transfer is R times a positive unit with bounded jets and positive derivative at zero. The weak input u is likewise R times a positive unit on the first section. At the horizontal end $T=R/(-\varphi)$ and the lower phase is a positive regular function of T. Their composition therefore gives

$$x=\chi(u;\mathbf{q}_w,\epsilon),\qquad \chi(0)=0,\quad \chi_u>c_*>0, \tag{5.30}$$

with bounded finite jets and inverse jets on a common signed auxiliary input interval. It gives the horizontal entry before the slow passage, so its regular inverse remains bounded even when $t(u)$ is exponentially small.

Let $H(\mathbf{q}_w,u)=F_I(\chi(u;\mathbf{q}_w))$. It and $\gamma(\mathbf{q}_w)$ have bounded coefficient derivatives on the whole larger box. On the center, $\nu=0$ and (4.4) gives exactly

$$F_{I,c}(x)=\bar{B}+\bar{c}x+(1-\epsilon\bar{B})x^2.$$

Because $\chi_c(0)=0$ for all passive parameters, Hadamard integration gives $\partial_{\bar{B}}\chi_c(u)=O(u)$, with the required input jets. Consequently

$$\partial_{\bar{B}}H_c=1+\bar{c}\partial_{\bar{B}}\chi_c-\epsilon\chi_c^2+2b\chi_c\partial_{\bar{B}}\chi_c=1+O(u+\epsilon)>1/2. \tag{5.31}$$

Equations (5.29) and (5.31), together with the common inverse box and coefficient Lipschitz bounds, will select one weak comparison center for the whole input interval, independently of closing zeros. □

5.5 Finite saddle splittings and the two center axes

The two lower contacts at a finite upper saddle require independent splitting coordinates, including near the intersection of the two center families. Fix $\bar{B}=-\beta$ with $0<\beta_*\le\beta\le\beta^*$, and work near $\bar{m}=\beta$, $\bar{c}=-\bar{a}$, $\nu=0$. The lower roots are simple on this compact block. Let $\widehat{p}_+$ and $-\widehat{p}_-$ be their strong intersections with $Y=0$, and let P_s and $-Q_s$ be the intersections of the upper saddle's two separatrices with that same section. Set

$$\delta_+=\widehat{p}_+-P_s,\qquad \delta_-=\widehat{p}_--Q_s.$$

Each intersection is defined by a single transverse connecting arc. We can therefore define both splittings before forming the complete upper passage; the following Jacobian calculation makes them coefficient coordinates.

Lemma 5.7 (Finite splitting coordinates)**.** *The variables $(\delta_+,\delta_-,r,\nu)$, with β fixed, are coefficient coordinates on a common neighborhood of each compact part of the three-saddle center, where r is the actual upper eigenvalue ratio. Near the reversible intersection, choose reflected lower power inputs and put*

$$u_0=\chi_+(\delta_+),\quad z_0=\chi_-(\delta_-),\qquad \xi_1=(u_0+z_0)/2,\quad \xi_2=(u_0-z_0)/2,\quad \kappa=r-1. \tag{5.32}$$

Then (ξ_1,ξ_2,κ,ν) are coordinates with bounded finite inverse jets on a common product box. The two center components in this box are exactly

$$\xi_1=\xi_2=\nu=0,\qquad \text{and}\qquad \xi_2=\kappa=\nu=0. \tag{5.33}$$

At fixed κ near their intersection, $\partial_{\bar{a}}\xi_2>c>0$ after the other independent coefficients are held fixed. These statements use ordinary coefficient and Euler scale derivatives of any finite order reserved in the preparations.

Proof. The lower strong intersections and positive maps $\chi_\pm$ are supplied by Proposition 4.8. The upper axes and their outer connections are supplied by Section 4.4. For the present coordinate assertion only its invariant-axis construction is needed: the denominators $n\lambda_u - \lambda_s$ and $\lambda_u - n\lambda_s$ have the same positive linear margins on every fixed compact set of positive saddle ratios. Thus the axes and regular legs apply on the stated compact center block without the later resonant shears. The compensator estimates remain restricted to the neighborhood of ratio one. Continuation to $Y = 0$ uses the low-layer carrier estimate, which gives convergence of the coefficient jets as $\epsilon \to 0$. Since $|X|$ has a positive lower bound on the final low layer, the connection remains uniformly transverse. We compute the coordinate Jacobian at zero scale and preserve its margin by one common reduction of the scale and coefficient domains.

Write $\Delta = \sqrt{\bar{a}^2 + 4\beta}$. The zero-scale lower roots and upper thresholds are

$$\begin{aligned} p_l &= \frac{\bar{a} + \Delta}{2\beta}, & q_l &= \frac{-\bar{a} + \Delta}{2\beta}, \\ P_s &= \frac{2}{\bar{c} + \sqrt{\bar{c}^2 + 4\bar{m}}}, & Q_s &= \frac{2}{-\bar{c} + \sqrt{\bar{c}^2 + 4\bar{m}}}. \end{aligned} \tag{5.34}$$

The upper formulas follow by intersecting the affine saddle's two eigenlines with $Y = 0$. On the center put $p = p_l = P_s$, $q = q_l = Q_s$. Implicit differentiation of $\bar{m}P_s^2 + \bar{c}P_s = 1$ and $\bar{m}Q_s^2 - \bar{c}Q_s = 1$ gives

$$\frac{\partial(\delta_+, \delta_-)}{\partial(\bar{m}, \bar{c})} = \frac{1}{\Delta}\begin{pmatrix} p^2 & p \\ q^2 & -q \end{pmatrix}, \qquad \det = -\frac{1}{\beta^2\Delta}.$$

On the zero-splitting axis, $\bar{m} = \beta$, $\bar{c} = -\bar{a}$, the ratio is $r = (\Delta + \bar{a})/(\Delta - \bar{a})$, with $\mathrm{d}r/\mathrm{d}\bar{a} = 2r/\Delta > 0$. The two splitting directions and this remaining tangent direction therefore give an invertible three-variable map. The exact triangle identity of Lemma 5.3 supplies the positive-scale zero-splitting axis.

At $\bar{a} = \bar{c} = 0$, choose the two reflected power inputs to have the same positive linear coefficient $k_l(\beta)$ at zero. The Jacobian in $(\bar{m}, \bar{a}, \bar{c})$ is then

$$\frac{\partial(\xi_1, \xi_2, \kappa)}{\partial(\bar{m}, \bar{a}, \bar{c})} = \begin{pmatrix} k_l/(2\beta^{3/2}) & 0 & 0 \\ 0 & k_l/(2\beta) & k_l/(2\beta) \\ 0 & 0 & -1/\sqrt{\beta} \end{pmatrix}, \qquad \det = -\frac{k_l^2}{4\beta^3}. \tag{5.35}$$

Coefficient-jet convergence preserves these margins, so Lemma 5.5, with β and scale passive, supplies the common inverse boxes and jets. The positive $\bar{a}$ derivative at fixed κ follows from the middle diagonal entry and continuity.

On the second center both separatrices are the triangle sides, so $u_0 = z_0 = 0$. Conversely the inverse coordinates and the nonzero tangent ratio derivative identify this entire nearby axis with that component. On the reversible component, reflection gives $u_0 = z_0$ and $r = 1$. Along it $\partial_{\bar{m}}\xi_1$ has the positive limit in (5.35); it parametrizes the whole second axis in (5.33) on a smaller box. Thus the two center components fill precisely the two axes in (5.33). □

Division of the upper orbital remainder requires one coordinate choice whose remainder vanishes on both center components. We now construct that choice together with its regular outer legs.

Lemma 5.8 (One upper coordinate choice on both centers)**.** *Near the intersection above, the analytic upper coordinates can be chosen so that their orbital equation is*

$$\frac{\mathrm{d}V}{\mathrm{d}U} = -r\frac{V}{U} - V^2 R(U, V), \qquad R = \epsilon\widetilde{R},$$

and $R = 0$ identically on each of the two center components. Their finite jets and inverse jets have the same uniformity as the finite saddle preparation. The two regular outer legs, expressed in the lower power inputs, have the form

$$G_+(t) = u_0 + tA_0(t), \qquad G_-(v) = z_0 + vB_0(v), \qquad G'_+, G'_- > c > 0,$$

on fixed signed connection intervals and on the common coefficient box.

Proof. On the triangle use the coordinates from Lemma 5.3; its first integral is $U^{\lambda_+}V^{\lambda_-}$, so its orbital equation is exactly $V_U = -rV/U$. On the reversible component the product coordinates constructed in Section 4.4 give $V_U = -V/U$. At their intersection $r = 1$. The transition between these two analytic coordinates preserves the foliation $UV =$ constant. To see the regularity at the axes, express either analytic first integral in the other product coordinates. Invariance under $U\partial_U - V\partial_V$ makes its power series a function of UV alone. Its derivative in UV is nonzero, by the saddle Hessian. Its analytic one-variable inverse shows that the transition and its inverse preserve the product foliation on a whole saddle box, including the axes.

Apply this intersection transition, with its passive β, ϵ dependence, to the entire reversible family. Its orbital equation stays exactly linear and its coordinates now agree with the triangle choice at the intersection. In coefficient variables $\tau = \bar{a} + \bar{c}$, $\eta = \bar{m} - \beta$, extend the two coordinate maps by taking their sum minus their common restriction: the triangle prescription is evaluated at $\eta = \tau = \nu = 0$, the reversible prescription at $\bar{a} = \bar{c} = \nu = 0$, and the subtracted intersection prescription at all four zero values. Use $\bar{a}$ as the remaining coordinate on the triangle and η on the reversible family, then extend independently in τ, ν. The maps agree on the intersection and retain an invertible state Jacobian on a smaller common neighborhood. Subtract the image of the actual saddle to center it; this subtraction vanishes on both prescribed families.

Straighten the neighboring invariant axes and normalize the orbital coefficient on them, choosing every change to be the identity when the axes and coefficient are already normalized. The two center equations are preserved. At zero scale both prescriptions and the field are affine-linear, so the normalized remainder vanishes there; division by ϵ gives its bounded analytic quotient. The shear preparation and reciprocal estimates of Proposition 4.10 apply with the same finite ratio restriction, now preserving both centers.

Each outer leg to $Y = 0$ lies in a compact regular strip except for its final low layer, whose uniform jets were proved in Section 4. Compose it with the regular connection into the lower power input. The orientation is inward from the corresponding strong barrier, so the resulting derivatives G'_+, G'_- are positive with fixed margins. Reflect the sections and coordinate prescriptions together to obtain the two legs. The signed intervals extend these individual regular connections; the upper passage retains its original saddle quadrant. Choose $|u_0|, |z_0|$ smaller than a fixed fraction of the signed connection-image radius. The common inverse-box argument then gives the inverse jets on a smaller image. □

5.6 Strong fibres and positive root factors

Adjacent phase preparations must give the same passing condition on their overlap. We identify their strongly contracting root fibres on one original transverse section, then divide along the common root surface. The resulting positive unit and its inverse have bounded jets because the section coordinates are regular.

Proposition 5.9 (Compatibility of prepared slow factors)**.** *Consider two of the finite-order phase preparations on a common smaller original tube and a common initial section. Assume their input derivatives have nonzero margins, with compatible orientations, and their positive*

time changes have bounded ratios and inverses. Use the normal domination chosen in Section 4 for the prescribed finite order. Then the root fibres of the two preparations agree. Near a double root, their pulled-back quadratic slow factors differ by a positive unit with bounded finite jets and inverse jets. The same conclusion holds after removing the common invariant-axis factor at a weak or horizontal cubic boundary. The unit estimates use one additional derivative for each division performed.

Proof. We first identify the root fibre at a positive scale. If $x = r$ is an equilibrium of the scalar phase equation, its normal fibre consists of orbits approaching that equilibrium at the normal contraction rate. Conversely, for an orbit with $x \neq r$, uniqueness for the scalar equation and its Lipschitz bound give

$$|x(t) - r| \geq |x(0) - r|e^{-C\varpi t}.$$

Here $\varpi = \epsilon$ for the lower phase and is the selected small angular width for the upper preparation. A regular phase projection is Lipschitz in the original state. Such an orbit cannot approach the equilibrium at the faster normal rate. Thus the strongly contracting orbit set is exactly the root fibre of either preparation. Bounded positive time changes preserve the strict separation after the widths are chosen to include their bounds. This proves equality of the two root fibres in the original tube, within the unextended field. The finite-smooth parameter limit gives their identity at zero scale.

Choose the first Morse input s and its gap μ so that its factor is $s^2 + \mu$. Pull the second preparation back to that same section:

$$F_2(s, \mu, \mathbf{c}) = h(s, \mu, \mathbf{c})^2 + \eta_2(\mu, \mathbf{c}).$$

Here $\mathbf{c}$ denotes the passive coefficients and η_2 is the second slow gap, independent of s and distinct from ν. At the common double root $h = \eta_2 = 0$ and $|h_s| > c > 0$. Equality of root fibres gives the exact identity

$$F_2(s, -s^2, \mathbf{c}) = 0. \tag{5.36}$$

Its second derivative at the double root is

$$(F_2)_{ss} - 2(F_2)_\mu = 0, \qquad (F_2)_{ss} = 2h_s^2.$$

Consequently $(F_2)_\mu = h_s^2 > 0$ there, with a uniform positive margin on a smaller coefficient block. Choose a product box $|s| < r$, $|\mu| < 2r^2$ inside that block. The segment between $-s^2$ and μ stays in the larger box, and

$$F_2 = (s^2 + \mu)U, \qquad U = \int_0^1 (F_2)_\mu(s, -s^2 + \theta(\mu + s^2), \mathbf{c})\, \mathrm{d}\theta > u_* > 0. \tag{5.37}$$

Differentiation of this integral gives the finite jets of U; the positive margin gives those of U^{-1}. This also proves the comparison on the positive-gap side, where neither factor has a root.

For a weak factor $u(\gamma + u)$, the common invariant axis and the regular orientation give a second input $v = uh(u, \gamma, \mathbf{c})$, where $h > h_* > 0$. Its full pulled-back factor is $uh(\gamma_2 + uh)$, up to a positive time unit. Remove the factor u first. The other root gives $\gamma_2(-u, \mathbf{c}) + uh(u, -u, \mathbf{c}) = 0$. At $u = \gamma = 0$, differentiation shows $\partial_\gamma \gamma_2 = h(0, 0, \mathbf{c})$; hence the derivative in γ of the divided factor is $h(0, 0, \mathbf{c})^2 > 0$. Integration from $\gamma = -u$ now gives a positive unit times $\gamma + u$, as in (5.37). A bounded positive time unit preserves its sign and finite jets.

At the horizontal cubic boundary, first remove the invariant factor x. The first remaining factor is the exact F_I of (4.4), with $\partial_{\bar{B}} F_I = 1 + O(\epsilon) > 1/2$ near the origin. It has a unique root surface $\bar{B} = \mathcal{B}(x, \bar{a}, \mathbf{c}, \epsilon)$. At the cubic point the two divided factors have positive quadratic coefficients, since the state and time orientations are positive. At zero scale and $\bar{a} = 0$, the first surface is $\bar{B} = -x^2$. Twice differentiating the second divided factor on this surface gives

its positive transverse $\bar{B}$ derivative, exactly as above. Continuity preserves that margin for small scale and coefficients. Divide each factor in $\bar{B}$ from the same surface $\mathcal{B}$; the ratio of their two positive integral units proves the assertion. All these segments are chosen in one larger coefficient box before restricting to actual passing points. A simple root uses the same one-variable division in its nondegenerate splitting coordinate. □

Use the finite-root comparison where the time-change ratio is bounded; use the horizontal weak/cubic coordinates when a root escapes to infinity. These comparisons identify the slow passing conditions and their regular factors. To form a logarithmic derivative ratio, both paths must also end in the same original central-height coordinate.

5.7 One comparison center on the original input interval

We seek a coefficient path ending at one real center valid for every input of the original passing interval. On separated strips the mean gap can stay fixed. At a contact we increase it so that the root retreats faster than the entry moves; at a weak boundary we increase $\bar{B}$. The inverse boxes of Proposition 5.6 bound the corresponding variations of the original coefficients.

Fix the normalization, inverse charts, phases and regular connections on larger domains $\mathcal{B}^+$, including the signed central and relative and absolute extensions of Section 4. Take $\mathcal{B} \Subset \mathcal{B}^+$ so that all the coefficient paths below stay there for every original input. Physical passages also impose the scalar whole-path tests, regular arc conditions and positive original central heights. We verify that each center endpoint lies in the real period domain of Proposition 5.1.

Proposition 5.10 (Fixed mean gap on the separated side)**.** *Use the joint coordinates* $\mathbf{b} = (\bar{B}, \delta, \tau, \zeta, \nu)$ *from Proposition 5.6, with exact normalized gaps*

$$\mu_L = \delta - \zeta, \qquad \mu_U = \delta + \zeta.$$

Choose both slow blocks between fixed sections on opposite sides of their respective minima. Let the section inputs range in a smaller prepared separated patch whose regular connections reach these sections. This includes finite entries separated from the joint threshold and the fixed horizontal section used at the infinite end. If $\delta > |\zeta|$*, the single coefficient path*

$$\mathbf{b}_\alpha = (\bar{B}, \delta, (1-\alpha)\tau, (1-\alpha)\zeta, (1-\alpha)\nu), \qquad 0 \le \alpha \le 1, \tag{5.38}$$

preserves both slow passages for every original input. It stays in the common construction domain. Its endpoint $\mathbf{b}_c = (\bar{B}, \delta, 0, 0, 0)$ *is one center, independent of the input, and its two complete central deviations agree on the whole represented input interval.*

Proof. Choose fixed sections on both sides of each minimum, with the outer sections between the threshold and the compact separated entry patch. The limiting regular arcs reach them transversally in finite time. The lower-layer and upper low-height estimates of Section 4 preserve these margins on larger domains, before the slow gaps vanish. At infinity use the fixed horizontal crosscut of Lemma 4.2 and its regular positive overlap. A smaller input patch gives the required product domains.

At every point of (5.38),

$$\mu_L(\alpha), \mu_U(\alpha) \ge \delta - (1-\alpha)|\zeta| \ge \delta - |\zeta| > 0.$$

Both full scalar blocks pass for arbitrary transit times and ratios of the two positive gaps. The regular connections stay in their fixed larger domains. Thus every point of the coefficient path retains the original prepared input interval.

The common gaps and sections give the same center on overlapping finite and horizontal patches. Hold the chosen common input fixed along each coefficient path; its original section

point is recovered by the bounded regular inverse. Identities at the radial endpoint refer to continuous extensions; positive inputs represent finite orbits.

At the endpoint $\tau = \zeta = \nu = 0$ the coefficient inverse gives $\bar{a} = -\bar{c}$, $\bar{m} = -\bar{B}$, and $4\bar{B}(1-\epsilon\bar{B}) - \bar{c}^2 = 4(1-\epsilon\bar{B})^2\delta > 0$. The real center polynomial of Proposition 5.1 is then positive on the full real axis. Every finite section input represented above belongs to its period annulus. Its lower arcs meet at the same positive original central height. The fixed references are zero at $\nu = 0$, so the complete central deviations agree. Continuity gives the same identity at an auxiliary infinite input. □

The endpoint analysis will join these separated patches through their common input coordinates. Near a contact, however, a gap may be nonpositive, so we must also control the motion of its root.

Proposition 5.11 (A lifted contact path)**.** *Near the joint threshold, use the same coefficient coordinates and the center-calibrated entry functions $e_L(p;\mathbf{b})$ and $e_U(p;\mathbf{b})$ from Proposition 5.6. Their scalar equations have factors $s^2+\mu_i$, $i = L, U$, and fixed positive exits, while their entries lie in a small neighborhood of zero. Let $\mathcal{I}(\mathbf{b})$ be their common original passing interval in the unchanged finite section coordinate p. Set*

$$A = |\tau| + |\zeta| + |\nu|.$$

After fixed restrictions of the coefficient and entry neighborhoods, the single straight path

$$\mathbf{b}_\alpha = (\bar{B}, \delta + 3\alpha A, (1-\alpha)\tau, (1-\alpha)\zeta, (1-\alpha)\nu), \qquad 0 \le \alpha \le 1, \tag{5.39}$$

stays in the larger construction domain and preserves both whole-path passing conditions at every $p \in \mathcal{I}(\mathbf{b})$. The endpoint $\mathbf{b}_c = (\bar{B}, \delta + 3A, 0, 0, 0)$ is a center whose actual period annulus contains every one of these section inputs. The complete central deviations agree there, for either sign or zero value of $\delta + 3A$.

Proof. The entry derivatives $\partial_p e_L, \partial_p e_U$ have the same negative sign and a nonzero margin on the common finite section neighborhood. For a nonpositive gap each passing set is therefore the interval $e_i(p) > \sqrt{-\mu_i}$; for a positive gap it is the whole entry neighborhood. Their intersection is the interval $\mathcal{I}(\mathbf{b})$ used in the statement.

Fix the gap-independent entry Lipschitz constant L on the larger inverse box, including the path-length factor, and choose r with

$$Lr < \tfrac{1}{2},$$

and less than half either positive exit. Reduce the original domains so that $|e_i| < r$, the paths and entries stay in their larger domains, and the gap magnitudes are less than a quarter of the squared exits. This fixes the exit margins before selecting a fibre.

Along its path A is a fixed number. The two gaps satisfy

$$\mu_i(\alpha) - \mu_i(0) \ge 2\alpha A, \qquad |e_i(p;\mathbf{b}_\alpha) - e_i(p;\mathbf{b})| \le L\alpha A. \tag{5.40}$$

If $\mu_i(\alpha) > 0$, its entire scalar path passes. Otherwise $\mu_i(0) \le \mu_i(\alpha) \le 0$. The original strict passing condition is $e_i(p;\mathbf{b}) > \sqrt{-\mu_i(0)}$. Both nonnegative roots are smaller than r, and therefore

$$\sqrt{-\mu_i(0)} - \sqrt{-\mu_i(\alpha)} = \frac{\mu_i(\alpha) - \mu_i(0)}{\sqrt{-\mu_i(0)} + \sqrt{-\mu_i(\alpha)}} \ge \frac{\alpha A}{r}.$$

For $\alpha A > 0$, combining this with (5.40) gives

$$e_i(p;\mathbf{b}_\alpha) > \sqrt{-\mu_i(\alpha)} + \alpha A(1/r - L) > \sqrt{-\mu_i(\alpha)}.$$

The case $\alpha = 0$ is the original strict passage; if $A = 0$, the fibre already is the required center. This proves the assertion for both slow blocks, for every α and for all original inputs with the same constants. The regular connecting pieces stay in their preselected larger domains, and the central connections use their fixed signed references.

To identify the center endpoint with an actual period annulus, put $b = 1 - \epsilon\bar{B}$ and $\Xi = \bar{c}(1 - 2\epsilon\bar{B})/(2b)$ there. The calibrated entry of either block on that center is

$$U_c(p) = \Xi - \bar{B}p.$$

If $\delta_c := \delta + 3A > 0$, its center polynomial is everywhere positive. If $\delta_c \le 0$, the just proved passing condition gives $U_c(p) > \sqrt{-\delta_c}$. More explicitly, write $\chi_C = -p$ for the value of the real center coordinate of Proposition 5.1 on this section. Its polynomial factors as

$$A_C(\chi_C) = \frac{b}{\bar{B}}[(\Xi + \bar{B}\chi_C)^2 + \delta_c].$$

Since $\Xi > \sqrt{-\delta_c}$ at $\chi_C = 0$, the condition places $\chi_C = -p$ in its positive component containing zero. Proposition 5.1 therefore supplies one actual periodic oval through each of these inputs and its positive central height. Its two central deviations agree, since both references vanish on this center. The same reasoning includes $\delta_c = 0$ at every strictly passing input; no limiting orbit on the root is added. □

The contact lift works because the root retreat $\alpha A/r$ exceeds the entry motion $L\alpha A$. At the weak boundary, positive $\bar{B}$ derivatives of the two slow factors will play the same role.

Proposition 5.12 (One center across the weak boundary)**.** *Use the weak coefficient coordinates*

$$\mathbf{b} = (\bar{B}, \bar{c}, d_s, \varsigma, \nu), \qquad d_s = P_L - \widehat{p}, \qquad \varsigma = \bar{B} + \bar{m},$$

from Proposition 5.6, in a neighborhood of $\bar{B} = \bar{m} = 0$, $\bar{c} = c_0 > 0$, $\bar{a} = -c_0$. Let u be the positive weak input, $t(u) > 0$ the upper displacement on its passing side and $\chi(u)$ the incoming horizontal phase. In this neighborhood the original prepared passing interval is

$$\mathcal{I}(\mathbf{b}) = \{0 < u < u_0 : \gamma(\mathbf{b}) + u > 0, \quad F_I(\chi(u; \mathbf{b}); \mathbf{b}) > 0, \quad d_s + t(u; \mathbf{b}) > 0\}. \tag{5.41}$$

There is a constant $K > 0$, fixed before the original fibre, such that, with $A = |d_s| + |\varsigma| + |\nu|$, the single center

$$\mathbf{b}_c = (\bar{B} + KA, \bar{c}, 0, 0, 0) \tag{5.42}$$

belongs to the larger coefficient domain and satisfies all conditions in (5.41) for every original passing u. Its complete central deviations agree on that whole interval. The straight coefficient segment from $\mathbf{b}$ to $\mathbf{b}_c$ lies in the common auxiliary domain used for finite-order division. No assertion that all its intermediate points are physical returns is required.

Proof. The upper factor is $u(\gamma + u)$ with fixed positive exit. Since $\bar{a} < 0$, $F_I' > c$ on the retained horizontal interval, so its entry positivity is the whole-path test. The simple lower deficit is $d_s + t(u) = P_L - p$. Increasing $t, \chi, F_I \circ \chi$ makes the intersection in (5.41) one interval.

Set $H(\mathbf{b}, u) = F_I(\chi(u; \mathbf{b}); \mathbf{b})$ and write the subscript c for restriction to $d_s = \varsigma = \nu = 0$. The coefficient and inverse-chart estimates preceding this subsection give constants L_γ, L_H on the larger domain such that

$$|\gamma(\mathbf{b}) - \gamma_c(\bar{B}, \bar{c})| \le L_\gamma A, \qquad |H(\mathbf{b}, u) - H_c(\bar{B}, \bar{c}, u)| \le L_H A.$$

The regular horizontal entry χ makes these constants uniform down to $u = 0$, including where t is flat. The margins (5.29) and (5.31) give on a common smaller box

$$\partial_{\bar{B}}\gamma_c \ge g_* > 0, \qquad \partial_{\bar{B}}H_c \ge h_* > 0. \tag{5.43}$$

Choose K so that $Kg_* > L_\gamma$ and $Kh_* > L_H$. If $A > 0$, integration of (5.43) gives

$$\gamma(\mathbf{b}_c) + u > \gamma(\mathbf{b}) + u > 0, \qquad H(\mathbf{b}_c, u) > H(\mathbf{b}, u) > 0$$

at every original passing input. If $A = 0$, these inequalities are the original strict ones. The upper center passage therefore exists and has $t_c(u) > 0$. Its lower deficit is $0 + t_c(u)$, which verifies the last condition as well. The original $u > 0$ is unchanged.

Choose the original box so that the displacement KA and transverse contraction stay in $\mathcal{B}^+$. Convexity keeps the segment there. The absolute zero and relative extensions of Section 4, with fixed signed central references, give its common Taylor domain even at nonpassing intermediate coefficients.

It remains to check that the comparison center represents the desired real center component. Its invariant affine branch has intersection $\widehat{p}_c = P_{L,c} > 0$ with $Y = 0$, and the point used above is $p = \widehat{p}_c - t_c(u)$. The input radius ensures $p > 0$. The center polynomial A_C has its adjacent negative root at $\chi_C = -\widehat{p}_c$, with a nonzero derivative, for either sign of small $\bar{B}$. Thus $-p \in (-\widehat{p}_c, 0)$ is in the positive component containing zero. Proposition 5.1 gives the real periodic oval through this point. Its central height is positive and the two complete central deviations agree. □

At each strictly passing center input, passage persists under small passive center variations. A vanishing comparison expression therefore has zero passive δ or $\bar{B}$ derivative there. Together with the larger-domain jet bounds, this local identity controls Taylor remainders even when the gaps have no uniform positive lower bound on the interval.

5.8 Division on the common domains

Vanishing on both center components gives a representation in the generators $\bar{a} + \bar{c}$, ν and $\bar{c}(\bar{B} + \bar{m})$. We prove a finite-order integral form of this statement, compare these generators with the passage coordinates, and apply it to transitions on the common input domains.

Proposition 5.13 (The center ideal and its local coordinates)**.** *Put $\tau = \bar{a} + \bar{c}$, $\varsigma = \bar{B} + \bar{m}$ and $\sigma = \bar{c}\varsigma$. Let f be C^{N+2} on a common product coefficient box, with any section variables and $\bar{B}$ treated as passive variables. Suppose*

$$f(0, 0, \varsigma, 0) = 0, \qquad f(0, \bar{c}, 0, 0) = 0,$$

where the four arguments are $(\tau, \bar{c}, \varsigma, \nu)$. Then

$$f = \tau A + \sigma B + \nu C, \tag{5.44}$$

where the coefficients have bounded total order N on a smaller common box. Their bounds follow from the indicated derivatives of f on the larger box. A common bound for those derivatives by a weight depending only on passive variables gives the same weight for the divided coefficients.

For the finite three-saddle coordinates of Lemma 5.7, there are constants $0 < c < C$ on a common smaller box such that

$$c(|\tau_1| + |\sigma_1| + |\nu|) \le |\xi_2| + |\kappa\xi_1| + |\nu| \le C(|\tau_1| + |\sigma_1| + |\nu|). \tag{5.45}$$

Near a single component, the joint and weak coordinate blocks instead give, respectively,

$$\begin{aligned} |\tau_1| + |\sigma_1| + |\nu| &\asymp |\tau| + |\zeta| + |\nu|, \\ |\tau_1| + |\sigma_1| + |\nu| &\asymp |d_s| + |\varsigma| + |\nu|. \end{aligned} \tag{5.46}$$

These comparisons concern coefficient functions on their inverse boxes. Dividing transition expressions additionally uses the common input domains constructed above.

Proof. First subtract $f(0,\bar{c},\varsigma,0)$ by integrating in (τ,ν) along its straight segment:

$$f - f(0,\bar{c},\varsigma,0) = \tau \int_0^1 f_\tau(s\tau,\bar{c},\varsigma,s\nu)\,\mathrm{d}s + \nu \int_0^1 f_\nu(s\tau,\bar{c},\varsigma,s\nu)\,\mathrm{d}s.$$

The remaining function vanishes on both axes of its $(\bar{c},\varsigma)$ plane. Two applications of the fundamental theorem give

$$f(0,\bar{c},\varsigma,0) = \bar{c}\varsigma \int_0^1 \int_0^1 f_{\bar{c}\varsigma}(0,s\bar{c},t\varsigma,0)\,\mathrm{d}s\,\mathrm{d}t.$$

The segments lie in the prescribed box. Total-order differentiation spends at most two coefficient derivatives and commutes with Euler scale differentiation, proving the formula and its weighted version.

By (5.33), apply this formula to ξ_2 and $\kappa\xi_1$ in the original coefficient box for the upper bound in (5.45). Conversely the same formula in the inverse product box (ξ_1,ξ_2,κ,ν) gives, for $f = \tau_1, \sigma_1$,

$$f = \xi_2 A_f + \kappa\xi_1 B_f + \nu C_f,$$

with bounded coefficients, proving the reverse bound.

In the joint block $\bar{c}$ has a positive lower bound, so $|\sigma_1| \asymp |\varsigma|$. The exact inverse coordinate map of Proposition 5.6 sends $\tau = \zeta = \nu = 0$ to $\tau_1 = \varsigma = \nu = 0$. Integrating in these independent transverse variables in each direction gives the first comparison in (5.46). In the weak block the same argument uses (d_s,ς,ν) and its inverse; $\bar{c}$ again is a positive unit. These are single-component divisions, each spending only one derivative. □

For transition division, retain larger signed intervals for the regular outer legs and lower power inputs. Reduce the coefficient box so all $G_\pm$ images stay there. Proposition 4.8 gives the absolute zero extension at nonpositive deficits; extend the relative coefficients by the same finite linear rule on both reflected sides. Lower-deficit jets are ordinary. After singular t^r composition, the upper-input jets are Euler jets $t\partial_t$.

The following extension preserves the mixed total order. Fix distinct positive numbers $a_0,\dots,a_N$ and solve the Vandermonde system $\sum_{j=0}^N c_j(-a_j)^k = 1$ for $0 \le k \le N$. For $x < 0$ near zero set

$$(\mathcal{E}f)(x;\mathbf{b}) = \sum_{j=0}^N c_j f(-a_j x;\mathbf{b}).$$

Choose the negative interval so all $-a_j x$ lie in the positive interval, with a cutoff farther from zero. The moments match input jets through order N; the formula commutes with coefficient and Euler derivatives and bounds mixed total order N by that same order. Constants depend only on the fixed order and interval.

The triangle identity holds on $u_0 = z_0 = \nu = 0$, where every small positive upper input has positive lower deficits. Reflection and the common extension preserve the reversible identity at signed deficits. Lemma 5.8 preserves both in its orbital remainder. Hence the integral paths

$$(s\xi_1, 0, t\kappa, 0), \qquad 0 \le s,t \le 1,$$

and those removing ξ_2,ν lie in the auxiliary domains, with no restriction to the physical max-norm face.

For the contact or weak comparison center, if $[\mathbf{b}_c,\mathbf{b}]$ lies in the larger construction domain and $f(p;\mathbf{b}_c) = 0$ for every original input, then

$$f(p;\mathbf{b}) = \int_0^1 D_{\mathbf{b}} f(p;\mathbf{b}_c + s(\mathbf{b}-\mathbf{b}_c))(\mathbf{b}-\mathbf{b}_c)\,\mathrm{d}s. \tag{5.47}$$

Propositions 5.11 and 5.12 bound the path length by the transverse size on the whole original interval. Second-order remainders use second derivatives on these same segments. The center choice is held fixed for the fibre.

Lemma 5.14 (Relative forcing from a closing zero)**.** *On one fixed coefficient fibre let the central closing function be*

$$F = Z - \rho\nu H, \qquad \rho = \sqrt{\epsilon}, \qquad H \geq h_* > 0,$$

where the reference gap H is independent of the section input. Suppose division on one of the common domains above gives

$$|Z| \leq C(I + |\nu|)\epsilon^{-M} e^{-S_*/\epsilon}, \qquad S_* > 0, \tag{5.48}$$

with a uniform positive action lower bound and an input-independent nonnegative size I of the relevant center generators. At any closing zero one has

$$|\nu| \leq C' I \epsilon^{-M-1/2} e^{-S_*/\epsilon}. \tag{5.49}$$

This is a constraint on the coefficients and hence holds on the whole input fibre. If $I = 0$, it forces $\nu = 0$; where the maps have the center identity there is no isolated closing zero.

Proof. At a zero of F, the left side of (5.48) is $\rho|\nu|H \geq \sqrt{\epsilon}h_*|\nu|$. Divide by this lower factor. Since $S_* > 0$ is fixed, the coefficient of $|\nu|$ on the right tends uniformly to zero. For a small enough common scale it is less than $1/2$ and can be absorbed. This proves (5.49), including $I = 0$. □

In particular, the two-center division preserves an absolute weight $C\epsilon^{-M}e^{-S_*/\epsilon}$ with $I = |\tau_1| + |\sigma_1|$. Section 4 supplies it when the retained lower segment has a uniform positive action margin; that margin is checked in each application.

The ideal comparisons transfer forcing to local generators: $|\nu| \leq \eta(I_{\text{global}} + |\nu|)$ and $I_{\text{global}} \leq C(I_{\text{local}} + |\nu|)$ imply $|\nu| \leq 2C\eta I_{\text{local}}$ for small η. The relation requires a closing zero.

On a block where $|\bar{B} + \bar{m}|$ has a positive lower bound, $\bar{c} = \sigma_1/(\bar{B} + \bar{m})$ and $\bar{a} = \tau_1 - \bar{c}$ also give a bounded comparison with $(\bar{a}, \bar{c})$. Near a finite upper saddle with regular lower legs, $\bar{a}$ and κ may replace these variables because $\partial_{\bar{c}}\kappa = -1/\sqrt{\bar{m}} + O(\epsilon)$ there. At a weak or a second-scale boundary the relevant generator retains its small scale as a weight. We specify those weighted coordinates when constructing the corresponding passages.

6 One return on the two initial energy regions

The first two energy classes must be counted on one return interval. We construct the large-energy return from independent core, tail and upper maps, using the direction $\ell \geq 0$ for each fixed field. Its exact logarithmic forcing gives a strict curvature sign. On a fixed finite-energy overlap, its closing function is a positive multiple of the compact displacement; matching their divided coefficients then gives one count for both classes.

6.1 Independent core, tail and upper maps

For each fixed field choose the reflection convention of (2.8) so that $\ell \geq 0$, and retain this direction throughout the count. The analytic constructions use signed independent forcing parameters before this fixed-field restriction. For a central height s put

$$h = s - 1 - \log s, \quad L = -\log s, \quad \vartheta = h^{-1/2}, \quad \widehat{\ell} = \ell/s, \qquad \beta = (Bh, mh, a\sqrt{h}, c\sqrt{h}). \tag{6.1}$$

For the construction and coefficient division, first regard $\vartheta, s, \beta, \widehat{\ell}$ as independent, with

$$0 < \vartheta < \vartheta_0, \qquad 0 < s \le e^{-1/\vartheta^2}, \qquad |\widehat{\ell}| \le 3, \qquad |\beta_i| \le b_0. \tag{6.2}$$

Here $b_0 > 0$ is fixed sufficiently small and $\vartheta_0 > 0$ is fixed afterwards. A selected jet is a word in $E_\vartheta = \vartheta\partial_\vartheta$, $E_s = s\partial_s$, ∂_{β_i}, $\partial_{\widehat{\ell}}$, and ordinary independent section derivatives. Every estimate refers to a prescribed finite number of such derivatives. The physical relations in (6.1) are imposed after these independent estimates. Since $\widehat{\ell} = \ell/s$, their weighted forcing jets give no bound for unweighted ∂_ℓ.

On the outgoing core use $X = \vartheta x$, $0 \le X \le 1$, and set

$$\begin{aligned} F(X) &= 1 + \beta_B X^2 + \beta_a X + \vartheta^2\beta_m, \\ D(X,z) &= F(X) - (1 + 2\vartheta^2\beta_m)z + \vartheta^2\beta_m z^2, \\ J(X) &= \exp\int_0^X \frac{t/\vartheta^2 - \beta_c}{F(t)}\,dt, \qquad I(X) = \int_0^X \frac{dt}{\vartheta F(t)J(t)}. \end{aligned} \tag{6.3}$$

The exact original graph equation is

$$z_X = \frac{s\widehat{\ell} + (X/\vartheta - \vartheta\beta_c - 2s\widehat{\ell})z + (\vartheta\beta_c + s\widehat{\ell})z^2}{\vartheta D(X,z)}, \qquad z(0) = s. \tag{6.4}$$

The incoming core is the same construction after replacing $(\beta_a, \beta_c, \widehat{\ell})$ by $(-\beta_a, -\beta_c, -\widehat{\ell})$. Denote their endpoint quantities by subscripts $+$ and $-$.

Proposition 6.1 (Source transition maps)**.** *After the preceding fixed choices the following statements hold on common neighborhoods of their real domains.*

(i) The signed normalized solution $U = z/(sJ)$ extends through $z = 0$ and satisfies

$$\begin{gathered} U(X) = 1 + \widehat{\ell}I(X) + R(X), \qquad |R|_N \le C_N e^{-1/(3\vartheta^2)}, \\ |U| < 10, \quad |z| < 1/8, \quad D > 27/32. \end{gathered} \tag{6.5}$$

The selected jets of I are bounded, $4/9 < I(1) < 2$, and $U_{\widehat{\ell}}(1) > 1/4$. There is a unique simple threshold $\widehat{\ell}_c \in (-3, 0)$; the outgoing core stays positive through $X = 1$ exactly when $\widehat{\ell} > \widehat{\ell}_c$. At equality its first zero is at $X = 1$.

(ii) For an independent entry $v \in [7/16, 9/16]$, the substitution $z = \exp[-(1-r)v/\vartheta^2]$ gives the tail equation

$$X_r = v\frac{1 - z + \beta_B X^2 + \beta_a X + \vartheta^2\beta_m(1-z)^2}{X + \vartheta^2\beta_c(z-1) + \vartheta s\widehat{\ell}(1-z)^2/z}, \qquad X(0) = 1. \tag{6.6}$$

It reaches $r = 1$, hence $z = 1$, in finite positive original time, with $3/4 < X < 2$ and denominator $> 3/4$. Its action map $T(v) = X(1)^2/2$ has bounded selected jets, $63/64 < T_v < 65/64$, and is uniformly close to $1/2 + v$. Its image contains the fixed interval $[31/32, 33/32]$. The inverse on that interval has the same jet bounds. Moreover

$$T_{\widehat{\ell}} - T_0 = \widehat{\ell}H_T, \qquad |H_T|_N \le C_N e^{-1/(3\vartheta^2)}, \tag{6.7}$$

and the corresponding inverse difference has the same divided bound.

(iii) With $Y = \vartheta^2 y$, $\lambda = \vartheta^2$ and $\alpha = s\widehat{\ell}/\vartheta$, the upper field is

$$X_\xi = Y(\beta_m Y - 1) + \lambda(\beta_B X^2 + \beta_a X), \qquad Y_\xi = (Y + \lambda)(X + \beta_c Y) + \alpha Y^2. \tag{6.8}$$

For sufficiently small $|\alpha| + \sum|\beta_i|$, each action $e = X^2/2 \in [3/4, 5/4]$ at $Y = 0$, $X > 0$, has a first upper passage to $Y = 0$, $X < 0$. It remains in $|X| < 2$, $0 \le Y < 3/2$. Its action map $\mathcal{U}_\alpha(e)$ satisfies $|\mathcal{U}_\alpha(e) - e| < 1/128$, with bounded finite jets in $e, \alpha, \beta, E_\lambda$.

Proof. Appendix D.3 proves the independent-entry jets and real passage conditions using the reciprocal Gaussian J^{-1} in the core, the exponentially small tail forcing s/z, and the integrable upper carrier $\lambda/(Y+\lambda)$. All three estimates precede substitution of logarithmic core values into the tail entries. □

Now impose the physical relation (6.1). Fix $\delta_0 = 1/8192$ and retain the positive-core set

$$U_+ \geq s^{2\delta_0}, \qquad U_- \geq s^{2\delta_0}. \tag{6.9}$$

The actual tail entries are

$$v_\pm = -h^{-1}\log(sJ_\pm(1)U_\pm).$$

Since $h^{-1}\log J_\pm(1) = 1/2 + O(b_0)$, $L/h = 1 + O(1/h)$, and $|U_\pm| < 10$, they lie in a fixed small neighborhood of $1/2$, strictly inside $[7/16, 9/16]$. Choose the coefficient box smaller once more and then h large so that their actions $e_\pm = T_{\pm,\widehat{\ell}}(v_\pm)$ satisfy $|e_\pm - 1| < 1/1024$. Both e_- and $\mathcal{U}_\alpha(e_+)$ then belong to the common tail-inverse interval. On this domain the exact shooting function is

$$F_{\widehat{\ell}} = \mathcal{U}_\alpha(e_+) - e_-. \tag{6.10}$$

Its zero joins the two independent lower arcs by the actual upper arc. Action is injective on the negative $Y = 0$ section, so this is exactly the closing condition for these five pieces. Near a closing orbit the backward incoming section map is an analytic diffeomorphism; its nonzero derivative shows that (6.10) has the same zero multiplicity as the Poincare displacement. This argument also applies at retained finite endpoints of (6.9), because all tube inequalities there are strict. Existence of a positive forward central return throughout this set is proved below in the fixed direction $\ell \geq 0$.

6.2 Center division and the physical derivative

At $\widehat{\ell} = 0$ write $L_\pm = T_{\pm,0}$, $v_{\pm,0} = v_\pm|_{\widehat{\ell}=0}$, and define, on a fixed entry interval about $1/2$,

$$\mathcal{M} = L_-^{-1} \circ \mathcal{U}_0 \circ L_+, \qquad G_0 = \mathcal{M}(v_{+,0}) - v_{-,0}. \tag{6.11}$$

The common tail-inverse domain makes this composition well defined. We now divide G_0 by the two center generators. The divided coefficients will give a positive amplitude and an increasing energy coordinate for the forced closing equation.

Lemma 6.2 (Divided unforced source function)**.** *On a fixed smaller coefficient box,*

$$G_0 = \tau_0\sqrt{h}\,A + \sigma_0 h^{3/2}C, \qquad \tau_0 = a + c, \quad \sigma_0 = c(B+m). \tag{6.12}$$

After $s = s(\vartheta)$ is imposed, A, C have bounded selected Euler and scaled-coefficient jets of every prescribed finite order. Their limiting functions as $\vartheta \to 0$ are analytic in β, and their errors in these jets are $O(r_s)$, where $r_s = \vartheta^2(1 + |\log\vartheta|)$. At $\beta = 0$ their limits are

$$A_* = 4\sqrt{2}/3, \qquad C_* = 8\sqrt{2}/15, \qquad \mathcal{M}_{\mathrm{pr}}(v, 0) = v. \tag{6.13}$$

Proof. Use the scaled coefficient coordinates $T = \beta_a + \beta_c$, $u = \beta_a$, $t = \beta_B + \beta_m$ and β_B. The cube $(T, u, t) \mapsto (q_1T, q_2u, q_3t)$, $0 \leq q_i \leq 1$, remains inside the larger independent coefficient box after a fixed initial shrinking. Indeed $\beta_m' = q_3\beta_m - (1-q_3)\beta_B$ and $\beta_c' = q_1\beta_c + (q_1 - q_2)\beta_a$. At $\widehat{\ell} = 0$ both normalized cores are $1 + R > 1/2$, so the entire cube has all the strict real passage margins.

The unforced shooting function vanishes on $T = u = 0$ by reversibility: reflection interchanges the lower arcs and the two halves of the upper arc. It also vanishes on $T = t = 0$. This is the second real center of Proposition 5.1, applied to the scaled coefficients $(\beta_B, \beta_m, \beta_a, \beta_c)$

and scale $\lambda = \vartheta^2$. Its polynomial A_C is positive on $[-2,2]$ in the small source box; hence the actual outgoing $Y = 0$ point lies in its period annulus. Uniqueness identifies its periodic arc with the constructed passages. Thus both center identities hold on the whole contraction cube used for division.

If $\Psi(T,u,t,\beta_B) = F_0$, successive fundamental-theorem integrals give

$$\begin{aligned} \Psi &= TM + utN, \quad M = \int_0^1 \Psi_T(qT,u,t,\beta_B)\,dq, \\ N &= \int_0^1\int_0^1 \Psi_{ut}(0,qu,rt,\beta_B)\,dq\,dr. \end{aligned} \tag{6.14}$$

Since $ut = Tt - \beta_c t$, the coefficients of T and $\beta_c t$ are $A_0 = M + tN$ and $C_0 = -N$. The mean-value formula for L_-^{-1} gives $G_0 = R_0 F_0$, with

$$R_0 = \int_0^1 (L_-^{-1})'(e_{-,0} + qF_0)\,dq > 0.$$

Thus $A = R_0 A_0$ and $C = R_0 C_0$ give (6.12). These specified integrals also fix the divided coefficients on a compact overlap used later.

To justify physical substitution, note that $E_\vartheta s = 2hs/(1-s)$. Each fixed number of derivatives adds only polynomial factors of h to an exponentially small core term. The exact unforced entries are

$$v_{\pm,0} = 1 + \vartheta^2(1-s) - \int_0^1 \frac{t}{F_\pm(t)}\,dt + \vartheta^2\beta_{c,\pm}\int_0^1 \frac{dt}{F_\pm(t)} - \vartheta^2\log(1+R_\pm).$$

They consequently have the stated bounded jets. Two extra coefficient orders before the Hadamard integrations supply any desired final order.

The limiting lower endpoints $\xi_+ > 0$, $\xi_- < 0$ solve

$$S_\beta(\xi_\pm) = 1, \qquad S_\beta(X) = \int_0^X \frac{t}{1 + \beta_a t + \beta_B t^2}\,dt.$$

The limiting upper system is $X' = -1 + \beta_m Y$, $Y' = X + \beta_c Y$. Appendix D.4 compares the exact and limiting integral equations in the reserved jets, giving error $O(r_s)$ on the whole coefficient cube. The same Hadamard formulas and common tail inverse give that error for A, C.

For the constants take $\beta_c = 0$, when the upper first integral is $X^2/2 + Y - \beta_m Y^2/2$. At input action e, put $q_e = e - X^2/2$ and $a_e = \sqrt{2e}$. Along this arc $Y = q_e + \frac{1}{2}\beta_m q_e^2 + O(\beta_m^2)$, and variation in β_c gives $\partial_{\beta_c}\mathcal{U}_{\mathrm{pr}} = \int_{-a_e}^{a_e} Y\,dX$. Hence

$$\int_{-a_e}^{a_e} q_e\,dX = \frac{4\sqrt{2}}{3}e^{3/2}, \qquad \frac{1}{2}\int_{-a_e}^{a_e} q_e^2\,dX = \frac{8\sqrt{2}}{15}e^{5/2}.$$

The lower endpoints are independent of β_m, β_c in this limit. The same Hadamard rule, in particular $C_0 = -N$, gives the positive constants in (6.13). At $\beta = 0$ the tail map is $v \mapsto 1/2 + v$ and $R_0 = 1$ in the limit. □

On the unforced coefficients, after physical substitution, define

$$\mathfrak{E} = E_\vartheta - 2\beta_B\partial_{\beta_B} - 2\beta_m\partial_{\beta_m} - \beta_a\partial_{\beta_a} - \beta_c\partial_{\beta_c}. \tag{6.15}$$

Holding the original field fixed gives $d/dh = -\mathfrak{E}/(2h)$. Analyticity of the principal coefficients and their jet comparison imply $C/A = 2/5 + O(|\beta| + r_s)$, with its first two $\mathfrak{E}$ derivatives $O(|\beta| + r_s)$. Thus, after fixed restrictions,

$$A > A_*/2, \quad C > C_*/2, \qquad q = hC/A, \qquad q_h > 1/5, \qquad q_{hh} = O((|\beta| + r_s)/h). \tag{6.16}$$

For instance, if $R = C/A$ then $q_h = R - \mathfrak{E}R/2$ and $q_{hh} = (\mathfrak{E}^2 R - 2\mathfrak{E}R)/(4h)$.

6.3 The exact logarithmic forcing

Let $U_{\pm,0} = U_{\pm}|_{\widehat{\ell}=0}$. Hadamard division of the signed core equations gives the exact secants

$$\frac{U_+}{U_{+,0}} = 1 + \widehat{\ell}\widetilde{I}_+, \qquad \frac{U_-}{U_{-,0}} = 1 - \widehat{\ell}\widetilde{I}_-, \qquad 1/3 < \widetilde{I}_\pm < 3. \tag{6.17}$$

Here $\widetilde{I}_\pm = I_\pm/U_{\pm,0} + O(e^{-h/3})$ in the selected independent jets. The actual fixed-field derivative, before suppressing any independent variable, is

$$D_L := \frac{d}{dL} = \widehat{\ell}\partial_{\widehat{\ell}} - E_s - \frac{1-s}{2h}\left(E_\vartheta - 2\beta_B\partial_{\beta_B} - 2\beta_m\partial_{\beta_m} - \beta_a\partial_{\beta_a} - \beta_c\partial_{\beta_c}\right). \tag{6.18}$$

In particular $D_L\widehat{\ell} = \widehat{\ell}$, $D_L h = 1 - s$, $D_L^2 h = s$, and $D_L\widetilde{I}_\pm = O(h^{-1})$, $D_L^2\widetilde{I}_\pm = O(h^{-2})$. Derivatives of $U_{\pm,0} - 1$ remain exponentially small.

The normalized cores can approach zero, so the tail entries require the full logarithms

$$\zeta_+ = \log(1 + \widehat{\ell}\widetilde{I}_+), \qquad \zeta_- = -\log(1 - \widehat{\ell}\widetilde{I}_-).$$

They are defined on (6.9); for h large, $|\zeta_\pm|/h < 1/1024$. Write $a_\pm = \widehat{\ell}\widetilde{I}_\pm$, $u_\pm = D_L\widetilde{I}_\pm/\widetilde{I}_\pm$ and $w_\pm = D_L^2\widetilde{I}_\pm/\widetilde{I}_\pm$. Exact differentiation gives

$$\begin{aligned} D_L\zeta_+ &= \frac{a_+(1+u_+)}{1+a_+}, \quad D_L^2\zeta_+ = \frac{a_+[1+2u_+ + w_+ + a_+(w_+ - u_+^2)]}{(1+a_+)^2}, \\ D_L\zeta_- &= \frac{a_-(1+u_-)}{1-a_-}, \quad D_L^2\zeta_- = \frac{a_-[1+2u_- + w_- + a_-(u_-^2 - w_-)]}{(1-a_-)^2}. \end{aligned} \tag{6.19}$$

For $0 < \widehat{\ell} \le 3$, their values and first two derivatives are positive. The smallness of $u_\pm, w_\pm$, $|a_\pm| \le 9$ and elementary logarithm inequalities give fixed constants such that

$$|\zeta_\pm| \le C|D_L\zeta_\pm| \le C^2|D_L^2\zeta_\pm|, \quad |D_L\zeta_\pm|^2 \le C|D_L^2\zeta_\pm|, \quad D_L^j\zeta_\pm \ge c\widehat{\ell} \quad (j = 0,1,2). \tag{6.20}$$

The denominators in (6.19) are retained in these comparisons, even if a core value is small.

Apply the unforced incoming tail inverse to the two actions in (6.10), keeping its two entries independent. The divided tail estimate and the bounded α derivative of the upper map yield an exact identity

$$\begin{aligned} &L_-^{-1}\mathcal{U}_\alpha T_{+,\widehat{\ell}}(v_+) - L_-^{-1}T_{-,\widehat{\ell}}(v_-) \\ &\qquad = \mathcal{M}(v_+) - v_- + \widehat{\ell}\mathcal{E}(v_+, v_-, \vartheta, s, \beta, \widehat{\ell}), \qquad |\mathcal{E}|_N \le C_N e^{-1/(3\vartheta^2)}. \end{aligned} \tag{6.21}$$

The quotient is removable at $\widehat{\ell} = 0$ by analytic dependence. The actual entries satisfy $v_+ = v_{+,0} - \zeta_+/h$ and $v_- = v_{-,0} + \zeta_-/h$. After their substitution,

$$|D_L^j(h\widehat{\ell}\mathcal{E})| \le C\widehat{\ell}e^{-h/4}, \qquad j = 0,1,2. \tag{6.22}$$

At most two inverse core factors occur: their cost is bounded by $s^{-4\delta_0}$, and $1/3 - 4\delta_0 > 1/4$ absorbs them and all fixed polynomial factors of h.

Multiply the left side of (6.21) by h and call the resulting exact function $\mathcal{D}$. Then

$$\begin{aligned} &\mathcal{D} = \tau_0 f + \sigma_0 q f - \Phi, \qquad f = h^{3/2}A > 0, \quad q = hC/A, \\ &\Phi = \zeta_+ S + \zeta_- - h\widehat{\ell}\mathcal{E}, \qquad S = \int_0^1 \mathcal{M}_v(v_{+,0} - t\zeta_+/h)\,dt. \end{aligned} \tag{6.23}$$

This exact section comparison retains the full dependence on $\widehat{\ell}$. Its zeros and their multiplicities agree with those of (6.10). On every fixed compact energy interval in its domain, Φ/ℓ has a

removable analytic extension at $\ell = 0$ with its ordinary energy derivatives: first divide the independent maps by $\widehat{\ell}$, then use $\widehat{\ell} = \ell/s$ and the positive lower bound of s on that fixed interval. The same statement holds for any selected coefficient derivatives after reserving their orders.

Proposition 6.3 (Source curvature on one interval)**.** *There are fixed $\eta > 0$ and $H_* < \infty$ such that for every fixed field in the selected direction $\ell \geq 0$, the set*

$$\mathcal{I}_s = \{h \geq H_* : |\beta_i| \leq \eta,\ 0 \leq \widehat{\ell} \leq 3,\ U_\pm \geq s^{2\delta_0}\}$$

is an interval, possibly empty. On it $q_h > 1/5$. If $\ell > 0$, the function $g = \Phi/f$ satisfies $g_{qq} > 0$. If $\ell = 0$, then $\Phi = 0$ identically and the closing equation divided by f is $\tau_0 + \sigma_0 q = 0$. All statements extend across retained finite endpoints as germs of analytic section maps.

Proof. At a fixed independent entry, the limiting-map comparison gives

$$|\mathcal{M}_v - 1| + |\mathcal{M}_{vv}| + |\mathcal{M}_{vvv}| + h|D_{L,\mathrm{ind}}\mathcal{M}_v| + h|D_{L,\mathrm{ind}}\mathcal{M}_{vv}| + h^2|D^2_{L,\mathrm{ind}}\mathcal{M}_v| \leq C(|\beta| + r_s).$$

The first two physical derivatives of $v_{+,0}$ have the corresponding h^{-1} and h^{-2} bounds. Appendix D.5 differentiates the integral for S, keeping also its $\mathcal{M}_{vvv}(D_L v)^2$ term. Using (6.20) gives

$$|D_L^j\{\zeta_+(S-1)\}| \leq C\varepsilon_1|D_L^j\zeta_+|, \qquad j = 0, 1, 2,$$

where ε_1 is made arbitrarily small by first shrinking η and then increasing H_*. Combining this with (6.22) proves

$$\Phi, D_L\Phi, D_L^2\Phi > 0, \qquad \Phi \leq CD_L\Phi \leq C^2D_L^2\Phi, \qquad D_L\Phi \geq c\widehat{\ell}. \tag{6.24}$$

In addition (6.16) gives $D_Lq \geq c_q > 0$, $D_L^2q = O(h^{-1})$, $D_Lf/f = O(h^{-1})$ and $D_L^2f/f = O(h^{-2})$. In the exact quotient

$$g_{qq} = \frac{(D_L^2g)(D_Lq) - (D_Lg)(D_L^2q)}{(D_Lq)^3},$$

the principal numerator is $(D_Lq)D_L^2\Phi/f > 0$. Every other term is bounded by $C(|D_L\Phi|/h + |\Phi|/h^2)/f$, less than half of that principal term when H_* is large. This proves strict curvature.

Before the core restrictions, $h, |\beta_i|, \widehat{\ell} = \ell e^L$ are nondecreasing functions of L, so the permitted energies form an interval. For $\widehat{\ell} \geq 0$ the outgoing core inequality is automatic. At any zero of $U_- - s^{2\delta_0}$, the secant identity gives exactly

$$D_L(U_- - s^{2\delta_0}) = -U_{-,0}(1 + u_-) + s^{2\delta_0}\,(1 + u_- + 2\delta_0 + D_LU_{-,0}/U_{-,0}) < -1/2.$$

Thus that boundary can only be crossed downwards, and its retained side is one interval. At $\ell = 0$ both restrictions are automatic. The strict tube margins prove the endpoint assertion. □

6.4 Positive return and localization in the chosen direction

To use the curvature bound for counted cycles, we need a return on the full interval and a core margin for every such cycle. We first complete the forward stable core, then use the closing condition to place each cycle inside the retained positive-core domain.

Proposition 6.4 (The actual source return)**.** *Choose the fixed constants and coefficient neighborhood smaller if necessary. In the direction $\ell \geq 0$, the shooting construction has a positive central first return throughout $\mathcal{I}_s$, before its closing equation is imposed. Every counted cycle with $h \geq H_*$ and $\mathscr{M}(h) \leq \eta$ satisfies the stronger bounds*

$$U_+, U_- \geq e^{-\delta_0 h} > s^{2\delta_0}, \tag{6.25}$$

and is one of its shooting zeros. The positive-return assertion is for this chosen direction; for an originally negative ℓ it represents the reflected forward, hence original backward, return.

Proof. For $0 \le \widehat{\ell} \le 3$, $U_+ \ge U_{+,0} > 1/2$. The outgoing entry differs from its unforced value by $-h^{-1}\log(1+\widehat{\ell}\widetilde{I}_+)$, of size at most $3\widehat{\ell}/h$. Therefore its positive core, tail and upper arc exist without any incoming-core assumption. Their negative-axis action is within $9/1024$ of one. The actual incoming tail inverse, whose domain was constructed with independent input, gives a unique $v_{\mathrm{out}} \in [7/16, 9/16]$ and a forward endpoint $x = -\vartheta^{-1}$, $z = e^{-hv_{\mathrm{out}}} > 0$.

On the remaining stable core, $-\vartheta^{-1} \le x \le 0$, $0 \le z \le 1/8$, the original graph denominator exceeds $27/32$. The field itself gives

$$z_x \le 2s\widehat{\ell} + 2\vartheta b_0 z.$$

At $z = 0$ the normal forcing is nonnegative. For $\ell > 0$ its sign prevents a positive trajectory from crossing downwards; for $\ell = 0$ uniqueness preserves positivity. Gronwall on this x-interval gives

$$0 < z(x) \le Ce^{-7h/16} + C\sqrt{h}\,e^{-h} < 1/8.$$

There is no exit through its upper boundary; the positive horizontal velocity carries the graph to $x = 0$ in finite time. Its height is below the fixed central-section cutoff after increasing H_*. This is the first return: both lower cores have increasing x, the tails keep their respective signs of x, and the upper intersection with $x = 0$ has $y > 0$. The core heights tend uniformly to zero, the tails have $|x| \ge 3/(4\vartheta)$ and $-1 < y \le 0$, and the upper action has a positive lower bound. Their compactifications therefore lie in the prescribed collar for sufficiently small scale.

Now consider a counted cycle. Lemma 2.3 gives $|\ell| < 3s$, so all its independent source parameters belong to the boxes above. The outgoing construction follows its actual orbit by uniqueness. At a cycle the returned central height is s; backward uniqueness on the incoming core first proves $U_- > 0$, without a quantitative core margin. It also gives

$$v_{\mathrm{out}} - v_{-,0} = -h^{-1}\log(U_-/U_{-,0}).$$

Center division, the bounded upper and tail derivatives, and the divided forcing estimates imply

$$|v_{\mathrm{out}} - v_{-,0}| \le C\big(|\beta_a + \beta_c| + |\beta_c(\beta_B + \beta_m)| + \widehat{\ell}/h\big).$$

The independently constructed tail inverse makes this estimate available before a quantitative incoming-core margin. Consequently

$$-h^{-1}\log U_- \le C/h + C(2\eta + 2\eta^2 + 3/h).$$

Choose η first to make its contribution less than $\delta_0/4$, and then H_* large enough for the remaining terms. The resulting bound is strictly smaller than δ_0. The outgoing bound follows from $U_+ \ge U_{+,0} > 1/2$. This proves (6.25) and the claimed identification with shooting zeros. All constants were selected in reflection-symmetric boxes; reflection exchanges the two independent cores and preserves this localization for the original negative sign. Only the local inverse-return identity near a periodic orbit is needed later to transfer its multiplicity. □

6.5 One count for the two initial energy classes

The compact and source functions describe the same return on their overlap. We identify their positive multiplier and show that their divided coefficient triples have the same limit there. This permits us to join the coefficients while preserving the Wronskian signs and count the two energy classes together.

Proposition 6.5 (The initial joint bound)**.** *Keep the collar and the base threshold of Section 2. There are constants $\eta > 0$ and $H_* > h(z_L) + 1$, followed by a reflection-invariant original coefficient neighborhood, such that the two initial classes in (2.11) together contain at most*

two isolated counted cycles, including multiplicity, for each fixed field. If $\ell = 0$, their joint bound is one, and that cycle is simple. If also $\tau_0 = \sigma_0 = 0$, the corresponding return is the identity and contributes no isolated cycle.

We use the same lower-section energy as in Appendix A:

$$h(s) = s - 1 - \log s, \qquad \Delta(h, \alpha) = h(P(h, \alpha)) - h.$$

Here P is the actual positive central return, analytic on the compact energy tube used in (A.3). The source construction uses the fixed-field direction $\ell \geq 0$; reflection transfers the count at periodic points afterwards.

Lemma 6.6 (Compact coefficient division)**.** *Fix $[h_1, h_2] \Subset (H_{\mathrm{ECT}}, \infty)$ and a slightly larger compact interval in that tail. On a sufficiently small coefficient neighborhood the analytic positive return on this larger interval has the exact representation*

$$\Delta = \tau_0 \mathcal{A} + \sigma_0 \mathcal{C} + \ell \mathcal{B}. \tag{6.26}$$

The coefficient triple $(\mathcal{A}, \mathcal{C}, \mathcal{B})$ converges in $C^2([h_1, h_2])$ as the coefficients tend to zero to $(\mathfrak{a}, \mathfrak{c}, \mathfrak{b})$ from Appendix A. Consequently its three ordered Wronskians have signs $+, +, -$ on this fixed interval.

Proof. Use independent coefficient coordinates (τ_0, ℓ, a, t, B), where $t = B + m$ and $c = \tau_0 - a$. First divide in the line-breaking coefficient:

$$\Delta = \Delta_0 + \ell \mathcal{B}, \qquad \Delta_0 = \Delta|_{\ell=0}, \qquad \mathcal{B} = \int_0^1 \partial_\ell \Delta(h, \tau_0, v\ell, a, t, B)\, \mathrm{d}v.$$

All coefficient segments stay in the common analytic return tube. On $\tau_0 = a = \ell = 0$ the field is reversible; on $\tau_0 = t = \ell = 0$ it is on the second center component. After a coefficient shrink, the real period domains of Section 5 contain the fixed compact family of base section points on both faces. The positive local return fixes these periodic inputs, since an increasing interval map has no nontrivial finite periodic orbit.

Apply the same specified integral rule (6.14), now to $\Psi(T, u, t, B) = \Delta_0(h, T, u, t, B)$ with h passive. It gives $\Delta_0 = \tau_0 M + atN$. Since $at = \tau_0 t - \sigma_0$, take $\mathcal{A} = M + tN$ and $\mathcal{C} = -N$. At zero parameter these are respectively $\partial_{\tau_0}\Delta$, $-\partial_a \partial_t \Delta$ and $\partial_\ell \Delta$, with precisely the normalization in (A.3). Analyticity on a neighborhood of the fixed compact product gives their C^2 convergence. Corollary A.2 preserves the strict Wronskian signs after a further coefficient shrink. □

Lemma 6.7 (Identification with the source displacement)**.** *On the positive-core source interval in the direction $\ell \geq 0$, let $\mathcal{D}$ be the exact source function in (6.23); thus*

$$\mathcal{D} = \tau_0 f + \sigma_0 q f - \Phi, \qquad f = h^{3/2} A > 0, \quad q = hC/A. \tag{6.27}$$

Then

$$\mathcal{D} = W(h, \alpha)\Delta(h, \alpha), \qquad W > 0. \tag{6.28}$$

The factor is locally analytic on the actual positive-return domain. On each fixed compact overlap with the analytic return tube, the source coefficient triple

$$(f, qf, -\Phi/\ell)$$

has a removable third member at $\ell = 0$ and converges in C^2 to

$$W(h, 0)(\mathfrak{a}, \mathfrak{c}, \mathfrak{b}). \tag{6.29}$$

The comparison is made after dividing that member by ℓ.

Proof. Fix the source label h and its actual field. Regard the core section $|x| = \sqrt{h}$ as fixed while changing a central comparison height. Let $E_h(u)$ be the incoming action at $y = 0$ obtained by following the incoming comparison path backwards from central height u. The outgoing core, tail and upper arc give an action E_{out}. For the actual positive return P,

$$E_{\mathrm{out}} = E_h(P), \qquad E_{\mathrm{in}} = E_h(s).$$

Scalar order keeps the core solutions, tail inputs and actions between their positive endpoint paths in the common inverse intervals of Proposition 6.1. The divided differences below therefore use actual paths. Scalar variation gives $z'_-(u) > 0$ for the incoming core endpoint. Its logarithmic entry $-h^{-1}\log z_-(u)$ decreases, and the tail action increases with that entry, so $E'_h(u) < 0$. Also $h'(u) < 0$ on this positive central interval.

Let L_- be the unforced incoming tail action map used to form the inverse-tail source difference. It is increasing on its common action-input interval. The source construction gives exactly

$$\mathcal{D} = h\{L_-^{-1}(E_h(P)) - L_-^{-1}(E_h(s))\}.$$

Put $d = P - s$ and $e = E_h(P) - E_h(s)$. The integral mean-value formula now gives (6.28) with

$$W = h\left[\int_0^1 (L_-^{-1})'(E_h(s) + ve)\,\mathrm{d}v\right] \frac{\displaystyle\int_0^1 E'_h(s + vd)\,\mathrm{d}v}{\displaystyle\int_0^1 h'(s + vd)\,\mathrm{d}v} > 0. \tag{6.30}$$

The nonzero denominators and finite-time analytic transitions give a positive analytic unit, including $P = s$, on the actual overlap.

At zero parameter, $E_h(u) = h(u)/h$ and the unforced tail is exactly

$$T_0(v) = \frac{1}{2} + v + \frac{e^{-hv} - 1}{h}, \qquad T'_0(v) = 1 - e^{-hv} > 0.$$

Writing $z_*(h)$ for the base height at $x = \sqrt{h}$, we obtain

$$W(h, 0) = \frac{1}{1 - z_*(h)} > 0, \qquad z_*(h) = s(h/2). \tag{6.31}$$

On a fixed overlap, shrink the original coefficient neighborhood so that all signed line-breaking segments through zero lie inside the analytic compact return tube and the independent source domains. Hadamard division in ℓ then defines Φ/ℓ with its two required energy derivatives. The unforced source uses the same (τ_0, a, t, B) division as the preceding lemma. Its first and mixed columns at zero are therefore the corresponding first and mixed derivatives of $W\Delta$. The only possible additional mixed terms are derivatives of W times Δ, $\partial_a\Delta$ or $\partial_t\Delta$ at zero, all of which vanish by the center identities and Appendix A. Thus the mixed column is $W(h, 0)\mathfrak{c}$; the other two columns are $W(h, 0)\mathfrak{a}$ and $W(h, 0)\mathfrak{b}$. Analytic division gives the stated C^2 convergence. □

Proof of Proposition 6.5. The constants are chosen in a fixed order. Appendix A first fixes H_{ECT}, and the collar gives $h_L := h(z_L) > H_{\mathrm{ECT}} + 2$. Choose the small source coefficient box and then increase its threshold to a fixed $H_* > h_L + 1$, retaining a little more source range to the left of H_*. The compact interval $[h_L, H_* + 1]$ and a fixed enlargement are now fixed. Only then shrink the original coefficient neighborhood. Make this shrink reflection invariant and small enough that the fixed overlap $[H_*, H_* + 1]$ belongs to all the retained domains: its scaled coefficients and $|\ell|/s(h)$ are uniformly small, and both core values are close to one there.

Fix a field and first use the direction $\ell \geq 0$. The source curvature and localization results give one positive-core source interval J_s starting at H_* and containing that overlap. Its upper

endpoint is determined by the scaled-coefficient and core conditions, not by any closing zero. It may be infinite. The compact return on $[h_L, H_* + 1]$ and the source return agree on their overlap, so together they give one actual positive return on the interval

$$J = [h_L, H_*] \cup J_s.$$

Every counted cycle of the first two classes is represented on J. For the source class, Lemma 2.3 gives $|\widehat{\ell}| < 3$, and Proposition 6.4 puts it strictly inside the required core inequalities. Hence no counted zero of those classes is discarded by forming J.

Extend W from the source part to a positive smooth function $\widetilde{W}$ on J, agreeing with W for $h \geq H_*$. For example, use its already retained slightly larger source overlap, multiply $\log W$ by a fixed cutoff which vanishes to the left of that overlap and is one by H_*, and exponentiate. Define one function

$$F = \widetilde{W}\Delta.$$

Choose a fixed smooth cutoff $\chi_0(h)$ equal to one for $h \leq H_*$ and zero for $h \geq H_* + 1$. On the fixed overlap both coefficient representations are defined. Extend their cutoff products by their respective formulas outside it, and put

$$\begin{aligned} f_1 &= \chi_0 \widetilde{W}\mathcal{A} + (1 - \chi_0) f, \\ f_2 &= \chi_0 \widetilde{W}\mathcal{C} + (1 - \chi_0) q f, \\ f_3 &= \chi_0 \ell \widetilde{W}\mathcal{B} - (1 - \chi_0)\Phi. \end{aligned} \tag{6.32}$$

The exact identities (6.26) and (6.28) give, throughout J,

$$F = \tau_0 f_1 + \sigma_0 f_2 + f_3. \tag{6.33}$$

Thus the cutoff has changed a representation of the same function.

We next verify, on the whole interval,

$$f_1 > 0, \qquad W[f_1, f_2] > 0, \qquad \operatorname{sign} W[f_1, f_2, f_3] = -\operatorname{sign} \ell \quad (\ell \neq 0). \tag{6.34}$$

On the compact left portion this follows from Lemma 6.6 and the identities $W[wa, wc] = w^2 W[a, c]$ and $W[wa, wc, wb] = w^3 W[a, c, b]$ for a common positive multiplier. On the source right portion, Proposition 6.3 gives $q_h > 0$ and, with $g = \Phi/f$, strict positive g_{qq} when $\ell > 0$. Direct differentiation gives

$$W[f, qf] = f^2 q_h > 0, \qquad W[f, qf, -\Phi] = -f^3 q_h^3 g_{qq} < 0.$$

These derivatives are taken at fixed original coefficients, as in the physical derivative specified in the source construction.

On $[H_*, H_* + 1]$, divide the third column in (6.32) by ℓ before taking a limit. By Lemma 6.7, the two triples being combined have the identical C^2 limit (6.29). Since the cutoff is fixed, their combination has that limit as well. Its ordered Wronskians have strict margins on this fixed compact interval, by (A.5) and $W(h, 0) > 0$. One common coefficient shrink therefore proves (6.34) also on the overlap. The same argument for the first two columns applies at $\ell = 0$.

Finally, set $r = f_2/f_1$. For $\ell > 0$, equation (6.33) and the Wronskian identities imply

$$r' > 0, \qquad \left(\frac{(F/f_1)'}{r'} \right)' = \frac{W[f_1, f_2, f_3]}{f_1^3 (r')^2} < 0.$$

The quotient and Rolle argument used in Proposition A.1 now gives at most two zeros of F, including finite endpoints. Although the cutoff multiplier is only smooth, it is positive: zero

orders up to two agree with those of Δ, and an order at least three would contradict the strict displayed derivative. On an unbounded J use each finite selection of zeros, as in Lemma 3.1.

When $\ell = 0$, $f_3 = 0$ and $F/f_1 = \tau_0 + \sigma_0 r$. If $\sigma_0 \neq 0$, its one possible zero is simple; if $\sigma_0 = 0 \neq \tau_0$, it has no zero. When both vanish, the exact center identities give $F = 0$ and no isolated return. Positive multiplication and the analytic energy coordinate preserve the Poincaré multiplicities.

For an original field with $\ell < 0$, apply the proved statement to its single reflected field. Reflection fixes the central height s, the energy h and the sizes of the four scaled coefficients, so it preserves the two initial classes. Near each periodic point its forward return is the inverse of the original forward return, as in (2.10); that identity preserves the order of every isolated displacement zero. These local inverse germs transfer the joint bound to the original field, while the interval of positive returns remains in the selected reflected direction. □

7 Finite endpoint configurations

A finite endpoint can arise when a lower scalar factor loses positivity, when the upper arc approaches a saddle, or when these events occur together. We first determine which combinations can be limits of actual cycles. The exact affine upper return then gives two action columns on the regular interval; the remaining configurations require comparisons in the moving lower deficits and the actual upper saddle ratio.

All comparisons retain the original central height. We fix the sections and coefficient boxes before the scale cutoff. An actual return must traverse each whole slow path and regular connection and have positive complete central outputs. The larger signed domains are used for coefficient division. The local counts proved here are inputs to the interval argument in Section 12: the three-saddle chart initially allows four closing zeros, and its derivative estimate must be combined with an entrance sign to obtain the sharp bound.

Table 1 gives the inputs and passing conditions for the endpoint comparisons in Sections 7–10. In each row K compares derivatives of the original central height. The scalar tests hold along the whole entry–exit path; regular connections and positive complete central outputs are also required.

Table 1: Inputs and passing tests for the endpoint comparisons.

Configuration	Input outward	Passing test	Role in the final count
Finite regular	Actual outgoing label $p\uparrow$	Both lower scalar factors positive; compact regular upper arc	At most one simple crossing of K; its departure sign enters the terminal piece.
Two lower roots	Strong deficit $t\downarrow$	Both lower deficits positive	At most one zero of the cleared derivative; ordinary or outward Rolle in Section 12.
Finite upper saddle	Saddle deficit $t\downarrow$	Regular lower legs, or each retained lower-loss passage	Actual ratio r; a regular-lower entrance uses Proposition 12.4.
Three finite saddles	$t\downarrow$, with the two transported lower deficits	All three passages positive	The preliminary four-cycle bound is sharpened by the seam signs and endpoint parity in Proposition 12.5.

Configuration	Input outward	Passing test	Role in the final count (continued)
One or two infinite ends	Finite deficit, weak $u\downarrow$, or a reciprocal input, as specified in Section 8	Strict, repeated or weak upper test and the inherited lower tests, including cubic horizontal factors	A strict derivative, a constant-sign connection, or the weak entrance of Proposition 12.6.
Joint repeated center	Contact $p\uparrow$; separated $s\downarrow$ with $s\to 1/p$	Contact signed-gap passages; both gaps positive on the separated part	One critical-point estimate on the contact–separated union in Proposition 9.5.
Secondary end	Actual tag $P\uparrow$; saddle $t\downarrow$ or weak $u\downarrow$	Actual upper first return and inherited lower passage on the normalized (C,M) face	The regular chain feeds the saddle or weak entrances in Section 12.4.

The regular finite generators are (τ_1,σ_1), whereas a regular-lower saddle uses $(\bar a, r-1)$. The three-saddle center near ratio one uses $(\xi_2,(r-1)\xi_1)$; away from one it uses (u_0,z_0). The joint block uses (τ,ζ); the secondary regular comparison uses $(\bar a,C)$ with $C=\bar d/\lambda$. Each pair belongs to its own coefficient chart. The local critical numerator M is a positive multiple of an input derivative of K. The secondary coefficient M in the normalized face (C,M) has a different meaning.

7.1 The upper return and the necessary lower geometry

The first task is to locate the possible endpoints of returning cycle data. The zero-scale upper return determines the regular input range. Whole-path positivity of the two lower factors and an action inequality then exclude some combinations of losses. Use the fixed-field direction $\ell\ge 0$ chosen in Section 2, and put

$$F_\pm(X)=1\pm\bar aX+\bar BX^2,\qquad A_\pm(P)=\int_0^P\frac{X}{F_\pm(X)}\,\mathrm{d}X$$

on intervals where the denominator is positive throughout. Approaching a root is allowed only from this passing side; the action is never continued through it.

At $\epsilon=\nu=0$ the upper field, divided by its positive factor Y, is

$$X'=\bar mY-1,\qquad Y'=X+\bar cY. \tag{7.1}$$

Let $S''-\bar cS'-\bar mS=0$, $S(0)=0$, $S'(0)=1$, and $I(T)=\int_0^T S(t)\,\mathrm{d}t$. An arc beginning at $(p,0)$ has $Y(t)=pS(t)-I(t)$ and $X=Y'-\bar cY$. For $\mathcal{D}=\bar c^2+4\bar m\ge 0$ write $\gamma_\pm=(\bar c\pm\sqrt{\mathcal{D}})/2$, and define

$$p_c=\begin{cases}1/\gamma_+, & \mathcal{D}\ge 0,\ \gamma_+>0,\\ \infty, & \text{otherwise,}\end{cases}\qquad q_c=p_c(\bar m,-\bar c).$$

Lemma 7.1 (The regular upper arc)**.** *The first return of (7.1) through $Y>0$ exists exactly for $0<p<p_c$. Its exit $(-q,0)$ defines an increasing analytic bijection $q=\mathcal{U}_0(p)$ from $(0,p_c)$ onto $(0,q_c)$. If T is the flight time, then*

$$\mathcal{U}_0'(p)=\frac{p}{q}e^{\bar cT},\qquad \mathcal{U}_0^{-1}(q;\bar m,\bar c)=\mathcal{U}_0(q;\bar m,-\bar c). \tag{7.2}$$

On every compact regular coefficient and input set the positive-scale return exists on a larger common tube and satisfies

$$\mathcal{U}_{\epsilon,\nu}(p)=\mathcal{U}_0(p)+O_N(\epsilon(1+|\log\epsilon|))+\nu O_N(1). \tag{7.3}$$

Here and below O_N includes selected finite ordinary input and coefficient derivatives and Euler scale derivatives, with a finite total order chosen before the domains. Common smaller image intervals have bounded inverse jets in the same class.

Proof. On the first positive interval of S put $W = S^2 - IS'$. Since $W' = \bar{c}W + S$,

$$W = e^{\bar{c}T} \int_0^T e^{-\bar{c}t} S(t)\,\mathrm{d}t > 0, \qquad (I/S)' = W/S^2 > 0.$$

Thus the flight-time equation $p = I(T)/S(T)$ has at most one solution. At that solution $Y'(T) = -W/S < 0$, so the exit is $(-q, 0)$ with $q = W/S > 0$. For $\mathcal{D} < 0$, $S = e^{\bar{c}T/2}\sin(\sqrt{-\mathcal{D}}T/2)/(\sqrt{-\mathcal{D}}/2)$; the quotient I/S tends to infinity at the first zero of S. For $\mathcal{D} > 0$ use $S = (e^{\gamma_+ T} - e^{\gamma_- T})/(\gamma_+ - \gamma_-)$. Its quotient tends to $1/\gamma_+$ if $\gamma_+ > 0$ and to infinity otherwise. At $\mathcal{D} = 0$, $S = Te^{\bar{c}T/2}$ gives the same rule, including $\bar{c} = 0$. These are exactly the stated ranges. The determinant formula for transverse sections has normal factors p, q and divergence $\bar{c}$, proving the derivative identity. Reflection gives its inverse and the range of q.

To estimate the positive-scale return on a common tube, split each compact reference arc into two layers $0 \le Y \le d_0$ and a middle arc with $Y \ge d_0$. On a layer $G = X + \bar{c}Y$ has a fixed nonzero margin. Set $g = \bar{m}Y - 1$, $F = \bar{B}X^2 + \bar{a}X$, $Q = \epsilon/(Y + \epsilon)$ and $r_0 = Y^2/(Y + \epsilon)$. The exact equation there is

$$X_Y = \frac{(1 - Q)g + QF}{G + \nu r_0}.$$

The perturbation at $\nu = 0$ is $Q(F - g)/G$, whose integral is $O(\epsilon \log(1 + d_0/\epsilon))$. In the middle, use the positive time change by $Y + \epsilon$; its field is $(g + Q(F - g), G + \nu r_0)$. Ordinary flow dependence and the transverse exit equation give the value estimate on larger compact tubes. For the selected jets use the bounded-layer variation argument (D.9), with $E_\epsilon = \epsilon\partial_\epsilon$ and the integrable carriers $E_\epsilon Q = Q(1-Q)$, $E_\epsilon r_0 = -Qr_0$; further Euler rows retain these carriers times bounded polynomials in Q. Ordinary ν integration gives the last term of (7.3). The positive compact margin in (7.2) then gives the inverse jets on fixed smaller common images. □

The lower maps must be expressed in the original central height to compare the two complete paths. Their absolute size is exponentially small, so we estimate a logarithmic amplitude after removing the action. The common parameter domains below also contain the center contractions of Section 5.

Lemma 7.2 (Regular lower amplitudes)**.** *Fix a compact set of positive finite inputs with $F_+ > 0$ on $[0, p]$. For p bounded away from zero, the signed outgoing backward map from $(p, 1)$ to $X = 0$ is*

$$L_+(p) = Z_+(p) - \ell H_+(p, \ell), \qquad Z_+(p) = a_+(p)e^{-A_+(p)/\epsilon}, \tag{7.4}$$

on a common enlarged input and coefficient domain. The logarithm of $a_+ > 0$ has bounded selected jets, while $H_+ = \sqrt{\pi/2} + O_N(\sqrt{\epsilon} + |\ell|)$ and its required ordinary ℓ derivatives are bounded. Reflection gives $L_-(q) = Z_-(q) + \ell H_-(q, \ell)$ with the same properties. Positive endpoints of either signed map imply a passage with $z > 0$.

With the fixed weak references of Section 4 one can instead write

$$L_+(p) = s_+ + U(p), \quad L_-(q) = s_- + V_0(q), \qquad U = \alpha_U e^{-A_+(p)/\epsilon}, \quad V_0 = \alpha_V e^{-A_-(q)/\epsilon}. \tag{7.5}$$

The positive amplitudes have bounded logarithmic jets on a small ordinary ν box, and $s_- - s_+ = \rho\nu H$, $H > h_* > 0$, is independent of input. In particular $U', V_0' < 0$ and their logarithmic derivative rates have the usual action limits on every smaller compact regular input interval.

Proof. Choose a small fixed $d_0 > 0$. On the regular layer $d_0 \le z \le 1$, use z as independent variable. Its denominator has a nonzero margin, so $P = p+\epsilon R$ has bounded selected jets. On the core use (4.21), with $\nu = \ell/\rho$ and $F_\epsilon = F_+ + \epsilon\bar{m}$. Its linear part is $(X/(\epsilon F_\epsilon) - \bar{c}/F_\epsilon)z + \ell/(\rho F_\epsilon)$. The backward kernel $K(X,u)$ on $0 \le X \le u \le P$ satisfies

$$|K(X,u)| \le Ce^{-c(u^2-X^2)/\epsilon}, \quad \int_X^P |K|\frac{u}{\epsilon}\,\mathrm{d}u \le C, \quad \int_X^P |K|\frac{\mathrm{d}u}{\rho} \le C.$$

The remainder has size $C((X/\epsilon+1)z^2 + |\ell z|/\rho)$. Choose d_0 and then a fixed ordinary ℓ disk so that the backward integral is a contraction. This constructs the signed map as well. On $z = 0$ the graph derivative has the sign of ℓ, so a trajectory with positive endpoints cannot leave and reenter $z > 0$. At $\ell = 0$ the line itself is invariant.

At zero forcing, $z(X) \le Cd_0e^{-c(P^2-X^2)/\epsilon}$, so $\int_0^P (X/\epsilon+1)z(X)\,\mathrm{d}X \le Cd_0$. Removing the action leaves a bounded logarithm: the endpoint shift $(A_+(P) - A_+(p))/\epsilon$ is bounded since $P - p = \epsilon R$, the linear correction uses $F_\epsilon - F_+ = O(\epsilon)$, and the nonlinear correction is the preceding integral. The normalized Volterra argument of Proposition 4.7 supplies its selected jets after $X = Py$: P stays in a compact positive interval, its regular endpoint jets are bounded, and the kernel satisfies the same weighted bound (D.6).

Differentiate the core integral with respect to ℓ at zero forcing. Up to an exponentially small layer endpoint term, its negative outgoing value is

$$\int_0^P \exp\left[-\int_0^X \mathfrak{f}_z(u, z_0(u))\,\mathrm{d}u\right] \frac{(1-z_0(X))^2}{\rho D(X, z_0(X))}\,\mathrm{d}X.$$

Near zero put $X = \rho v$. The integrand tends to $e^{-v^2/2}$; its error is $O(\rho)$ times a fixed polynomial and a Gaussian. Away from a fixed small central interval the positive linear rate makes the error exponentially small. The same kernels bound higher ordinary ℓ variations. Hadamard integration therefore proves the formula for H_+ and its jets. Reflection proves the incoming formula.

For (7.5), restrict the one chosen weak reference z_b at the variable endpoint P. Its continuation to $X = 0$ is the same reference for all inputs. On the weighted Gaussian core of Proposition 4.7, the difference $w = z - z_b$ satisfies $w_X = a(X)w + b(X,w)w^2$ with

$$a - X/(\epsilon F_+) = O(1), \quad \partial_\nu a = O(1), \quad |b| \le C(X/\epsilon + 1).$$

The weighted reference estimate gives $(X/\epsilon)|\partial_\nu z_b| = O(1)$, controlling the potentially large coefficient term. It is this weighted bound that is needed here, rather than bounds on the unweighted high X derivatives. The positive bounded endpoint deviation and the same normalized Volterra estimate give α_U; reflection gives α_V. The Gaussian variation above, now with $\ell = \rho\nu$, gives opposite reference signs and gap $\sqrt{\epsilon}\,\nu(\sqrt{2\pi} + O(\sqrt{\epsilon}))$. Finally $-\epsilon U'/U = A_+' - \epsilon(\log\alpha_U)'$ and its reflected identity give the positive derivative margins and logarithmic rates, using relative rather than absolute errors. □

Lemma 7.3 (Entry bounds and action ordering)**.** *There is a fixed $\chi > 0$ such that every parameter-dominant cycle in the chosen direction has $p > \chi/2$ and $q > \chi$ at small scale. If its incoming data stay in a compact set where $F_- > 0$ on $[0,q]$, then its outgoing data stay in a compact regular lower set and*

$$A_+(p) \le A_-(q) + C\epsilon. \tag{7.6}$$

Consequently outgoing loss with a regular incoming lower limit is impossible.

Proof. Take $\chi = \eta/4$, with the fixed energy threshold of Section 2, and decrease $\eta \le 1/16$ if necessary. The crossing argument in that section gives one intersection with each half of

$y = 0$. For $0 \le X \le \chi$, $0 \le z \le 1/8$, the lower denominator is greater than $3/4$. The cycle has $0 \le \ell < 3s$ and $s < e^{-\eta^2/\epsilon}$, so variation of constants gives

$$z_X \le (4X/(3\epsilon) + 4/3)z + 4s/\rho, \qquad z(X) \le 6se^{2X^2/(3\epsilon)+4X/3} < e^{-\eta^2/(2\epsilon)}.$$

Thus the orbit reaches $X = \chi$. A later decrease to $\chi/2$ cannot occur while $z \le 1/2$, where its horizontal velocity is positive. In $1/2 < z < 1$, $z_\xi \ge \chi z/4$ as long as $X \ge \chi/2$, and the possible decrease in X is at most $4\epsilon \log 2/\chi < \chi/2$. Hence $p > \chi/2$. For the incoming side, the reference arc with $q = \chi$ has $F_- > 3/4$ and $A_-(\chi) \le \eta^2/24$. Lemma 7.2 gives $s_{\mathrm{ref}} \ge ce^{-\eta^2/(24\epsilon)} > s$. This arc, the intervening central segment and the segment on $z = 1$ bound a flow region: the top is an entry, the central edge an exit, and uniqueness prevents crossing the reference arc. Every $q \le \chi$ would consequently give height at least s_{ref}, a contradiction.

Now fix incoming compact regular data, so $s \ge ce^{-A_{\max}/\epsilon}$ for a fixed $A_{\max}$. On a fixed outgoing strip $0 < X \le R$, a turn $D = 0$ is a strict maximum of X. Indeed, with $u = 1 - z$ its equation gives $u(1+\epsilon\bar{m}u) = -X(\bar{a}+\bar{B}X)$ and $u \le 2(1+R)X$. Thus $z_\xi > 0$ and $D_\xi = D_z z_\xi < 0$ at the turn, for small scale. Before a first rightward crossing of $X = R$ one has $D > 0$. Also $F_\epsilon - D > 0$, and the forcing term is nonnegative. On $[0, \chi]$ use $z_X/z \ge X/(\epsilon F_\epsilon) - 4/3$, and beyond χ use $z_X/z \ge (1 - \epsilon/\chi)X/(\epsilon F_\epsilon)$. Since $z(R) \le 1$, integration gives

$$(1 - \epsilon/\chi) \int_0^R \frac{X}{F_\epsilon(X)}\, \mathrm{d}X \le -\epsilon \log s + 4\chi\epsilon/3. \tag{7.7}$$

The incoming regular arc bounds the right side by $A_{\max} + o(1)$. Since $F_\epsilon(X) \le 2 + X + X^2$, the integral on the left is at least $\frac{1}{2}\log(R/2)$ for $R \ge 2$. A fixed sufficiently large R therefore bounds p. If F_+ then approached zero at or before a limiting endpoint, apply the same inequality for every R before its first zero and let R approach that zero. The integral diverges for both simple and double zeros, contradicting the bound. This argument also excludes nearby positive polynomials with minima tending to zero. Thus both lower data sets are compact and regular. At a closing zero, $L_+(p) = L_-(q)$ and $\ell \ge 0$ give $Z_+(p) \ge Z_-(q)$. Taking logarithms with their bounded amplitudes proves (7.6). □

Lemma 7.4 (Lower cones and simultaneous finite loss)**.** *At a limit* $(\mathbf{b}_*, p_*, q_*)$ *of cycle data,* F_+ *and* F_- *are nonnegative up to their respective endpoints. In particular*

$$p_* = \infty \Longrightarrow \bar{B} \ge 0,\ \bar{a} \ge -2\sqrt{\bar{B}}, \qquad q_* = \infty \Longrightarrow \bar{B} \ge 0,\ \bar{a} \le 2\sqrt{\bar{B}}.$$

If both finite lower limits lose positivity, then $\bar{B} = -\beta < 0$ *and their endpoints are the simple roots*

$$p_* = (\bar{a} + \Delta)/(2\beta), \qquad q_* = (-\bar{a} + \Delta)/(2\beta), \qquad \Delta = \sqrt{\bar{a}^2 + 4\beta}, \qquad p_* q_* = 1/\beta.$$

If the upper limit is regular, these roots satisfy $\mathcal{U}_0(p_*) = q_*$.

Proof. At an outgoing line $X = R > 0$, if $F_+(R) + \epsilon\bar{m} \le 0$, the horizontal velocity is negative for $0 < z < 1$. It prevents the first rightward crossing needed to reach $p > R$. Reflection gives the corresponding incoming obstruction. Passing to the limit proves nonnegativity and the two quadratic cone conditions. For $\bar{B} \ge 0$, a positive root of F_+ requires $\bar{a} < 0$, whereas one of F_- requires $\bar{a} > 0$. Hence both losses force $\bar{B} < 0$ and the displayed roots are unique and simple. The same barrier for the nearby exact quadratic roots prevents a cycle endpoint from lying beyond either root. Since each limiting root was assumed at or before its endpoint, they coincide. The upper identity follows from the regular return convergence. □

A strict real upper threshold also excludes returns beyond it, uniformly for unbounded inputs. We will use this fact at infinity. If $\mathcal{D} > 0$, $\gamma_+ > 0$ on a compact coefficient block, then for every fixed $\zeta > 0$ no input $p \geq p_c + \zeta$ returns, at a sufficiently small common scale. To prove this without bounding p, choose $r > 0$ between the two eigenvalues and close to γ_+, and choose $b_0 > 1/r$ with $b_0 < p_c + 2\zeta/3$. Then $A = \bar{m} + r\bar{c} - r^2 > 0$ and $rb_0 - 1 > 0$ have fixed margins. The limiting reference arc from $p_c + \zeta$ reaches a small $Y = d_0$ in $X + rY - b_0 > 0$. On the sloping boundary of the unbounded region $Y \geq d_0$, $X + rY - b_0 \geq 0$, its exact outward derivative in the time divided by $Y + \epsilon$ is

$$AY + rb_0 - 1 + Q\{F(b_0 - rY) - g(Y)\} + r\nu Y^2/(Y + \epsilon) > 0.$$

The error is bounded by $C_{d_0}\epsilon(1 + Y) + C|\nu|Y$; the bottom boundary points inward as well. This is a forward invariant region. Larger inputs either reach it, by uniqueness of the layer graphs, or escape affinely; neither returns. Reflection gives the exit exclusion. This proof requires a strict discriminant: it gives no exclusion at a repeated root.

Thus two infinite endpoints require $\bar{c}^2 + 4\bar{m} \leq 0$, because otherwise at least one reflected threshold is finite. The remaining upper limits have $\bar{m} < 0$, or $\bar{c} = \bar{m} = 0$. In the latter case the lower cones and primary normalization imply $\bar{B} > 0$. If only p_* is infinite and q_* finite, the incoming lower polynomial must lose positivity, by Lemma 7.3. These restrictions classify limits of actual cycle data. The passage constructions still use adjacent signed coefficient domains for their common comparisons.

7.2 Two columns on the regular interval

On a compact regular interval, the two center generators give two positive action columns with a strictly monotone quotient. This controls both the number of critical zeros and their crossing direction. We first state the elementary perturbation argument used to pass from the action to the actual return.

Lemma 7.5 (Two columns and their crossing direction)**.** *On one interval let A, B have uniformly bounded C^1 norms, with $|B| > b_* > 0$ and $(A/B)'$ of constant sign and magnitude greater than $r_* > 0$. Put $Q = aA + bB + e$, where a, b are constant, $I = |a| + |b| > 0$ and $\|e\|_{C^1} \leq \eta I$. For a sufficiently small common η, every zero is simple, $|a| \geq cI$ there, and its derivative has sign $aB(A/B)'$. Thus Q has at most one zero, including multiplicity. If moreover A, B, A', B' and $(A/B)'$ have positive margins, then Q' has sign a wherever $Q = 0$ or Q has sign a. When $a = 0$, both Q and Q' have sign b.*

Proof. At a zero, $a(A/B) + b = -e/B$, so the bounded ratio and nonvanishing B give $|a| \geq cI$. Differentiating at that zero gives

$$Q' = aB(A/B)' + e' - B'e/B.$$

The last two terms are $O(\eta I)$, proving the strict direction. Two crossings with this same direction are impossible. For the additional assertion, if a, b have the same sign, use the positive column derivatives. If their signs are opposite, at a point where $Q = 0$ or Q has sign a, $\operatorname{sign}(a)[a(A/B) + b] \geq -|e|/B$, again forcing $|a| \geq cI$. In $Q' = B'[a(A/B) + b] + aB(A/B)' + e'$ the second term has sign a and dominates the possible $O(\eta I)$ contribution of the opposite sign. The case $a = 0$ follows from the margins of B, B'. □

We keep the original central height as output. Write

$$\tau = \bar{a} + \bar{c}, \qquad \varsigma = \bar{B} + \bar{m}, \qquad \sigma = \bar{c}\varsigma.$$

Intersect the upper return range $0 < p < p_c$ with $F_+ > 0$ on $[0, p]$ and $F_- > 0$ on $[0, \mathcal{U}_0(p)]$. Since $\mathcal{U}_0$ increases, this regular limiting domain is one interval with left endpoint zero. On it

set $\mathcal{A} = A_-(\mathcal{U}_0(p)) - A_+(p)$. A polynomial quasi-invariant of the affine upper field expresses $\mathcal{A}$ in the two center generators. Its columns are positive integrals, and two weighted-average arguments prove that their quotient increases. The same calculation will determine the direction in which the actual critical function crosses zero.

Lemma 7.6 (The regular action columns)**.** *For the upper orbit of flight time T in Lemma 7.1, define*

$$J_1(T) = \int_0^T e^{-\bar{c}t}(1 - \bar{m}Y(t))\,\mathrm{d}t, \qquad J_2(T) = \int_0^T e^{-\bar{c}t}Y(t)\,\mathrm{d}t.$$

Then $J_1 = p + e^{-\bar{c}T}q > 0$, $J_2 > 0$, and

$$F_-(q) - e^{\bar{c}T}F_+(p) = -e^{\bar{c}T}(\tau J_1 + \sigma J_2). \tag{7.8}$$

On the regular interval,

$$\mathcal{A} = \tau A + \sigma C, \qquad A = \int_0^T \omega J_1\,\mathrm{d}t, \quad C = \int_0^T \omega J_2\,\mathrm{d}t, \quad \omega = \frac{pp_T e^{\bar{c}T}}{F_+(p)F_-(q)} > 0. \tag{7.9}$$

Both columns and their p derivatives are positive, and $(C/A)' > 0$. Every nonzero constant combination of A, C has at most one zero, which is simple. At the small endpoint, uniformly on bounded normalized coefficient sets,

$$A = \tfrac{2}{3}p^3(1 + O(p)), \qquad C = \tfrac{2}{15}p^5(1 + O(p)), \qquad C/A = \tfrac{1}{5}p^2(1 + O(p)). \tag{7.10}$$

Proof. Introduce the polynomial

$$V = 1 + \bar{a}X + \bar{B}X^2 + (2\bar{B} + \bar{c}\bar{a})Y + \bar{c}\bar{B}XY - \bar{B}\bar{m}Y^2.$$

Substitution of (7.1), grouping its constant and Y terms, gives $V' - \bar{c}V = -\tau(1 - \bar{m}Y) - \sigma Y$. Its endpoint values are $F_+(p)$ and $F_-(q)$, so integration proves (7.8). The upper arc has $Y > 0$ in its interior, giving $J_2 > 0$. Using $1 - \bar{m}Y = -X'$ and $e^{-\bar{c}t}X = (e^{-\bar{c}t}Y)'$, integration by parts gives $J_1 = p + e^{-\bar{c}T}q$.

For monotonicity, put $u_1(T) = \int_0^T e^{-\bar{c}t}\,\mathrm{d}t$. Then $J_1 = u_1 - \bar{m}J_2$. From $Y = p(T)S - I$ and the formulas for W, p_T, q above,

$$J_2' = p_T \int_0^T e^{-\bar{c}t}S(t)\,\mathrm{d}t = e^{-\bar{c}T}W^2/S^2 = e^{-\bar{c}T}q^2.$$

Thus $J_2'/u_1' = q^2$ is strictly increasing. Since both integrals vanish at zero, J_2/u_1 is its weighted average and is strictly smaller than its terminal value. Hence

$$(J_2/u_1)' = \frac{u_1'}{u_1}(q^2 - J_2/u_1) > 0, \qquad (J_2/J_1)' = \frac{(J_2/u_1)'}{(1 - \bar{m}J_2/u_1)^2} > 0.$$

Here $1 - \bar{m}J_2/u_1 = J_1/u_1 > 0$.

Differentiate $A_-(q) - A_+(p)$ and use $q_T = (p/q)e^{\bar{c}T}p_T$. The result is

$$\mathcal{A}_T = \frac{pp_T}{F_+(p)}\left(\frac{e^{\bar{c}T}F_+(p)}{F_-(q)} - 1\right) = \omega(\tau J_1 + \sigma J_2).$$

Both actions vanish at zero, so this proves (7.9). The ratio C/A is the weighted average of the strictly increasing J_2/J_1 with positive weight ωJ_1. The same averaging argument gives $(C/A)_T > 0$, hence also $(C/A)_p > 0$. It proves the zero assertion directly; positivity of the derivatives follows from the positive integrands. Finally scale $X = p\xi$, $Y = p^2\eta$, $t = p\theta$ in the affine field. It tends uniformly to $\xi' = -1$, $\eta' = \xi$, whose arc is $\eta = (1 - \xi^2)/2$. Thus $T = 2p(1 + O(p))$, $J_1 = 2p(1 + O(p))$, $J_2 = \frac{2}{3}p^3(1 + O(p))$ and $\omega = p/2(1 + O(p))$. Integration with $\mathrm{d}T = 2(1 + O(p))\mathrm{d}p$ gives (7.10). □

Proposition 7.7 (The regular critical function)**.** *Fix compact regular upper and lower data, with the input bounded away from zero. On common slightly larger tubes the closing function is exactly*

$$F(p) = U(p) - V(p) - \rho\nu H, \qquad V = V_0 \circ \mathcal{U}_{\epsilon,\nu}, \qquad K = \epsilon \log(V'/U').$$

Here $U', V' < 0$, and $H > h_* > 0$ is input independent. There are uniform choices of tubes, $\theta_0 > 0$ and a scale cutoff such that, on any one interval contained in the tubes, if

$$|\nu| \le \theta_0(|\tau| + |\sigma|), \tag{7.11}$$

then either the field is a center or K has at most one zero, which is simple. Consequently F has at most two isolated zeros, counting multiplicity, on the positive central passing interval. The coefficient condition holds with a factor tending to zero in place of θ_0 whenever the fibre contains a closing zero in one of these compact regular tubes.

Proof. Use the references of Lemma 7.2, fixed before the input is varied. Put $q = \mathcal{U}_{\epsilon,\nu}(p)$ and

$$r_U = A'_+ - \epsilon(\log \alpha_U)', \qquad r_V = A'_- - \epsilon(\log \alpha_V)'.$$

They have positive margins. Direct differentiation gives the exact identity

$$-K = A_-(q) - A_+(p) + \epsilon \log \frac{\alpha_U(p) r_U(p)}{\alpha_V(q) r_V(q) \mathcal{U}'_{\epsilon,\nu}(p)}. \tag{7.12}$$

The argument of the logarithm is positive and its logarithm has bounded selected finite jets. Thus the right side converges, with those jets, to $\mathcal{A}$ at $\nu = 0$, with error $O(\epsilon(1 + |\log \epsilon|))$; changing ν adds $O(\nu)$.

The real period domains of Section 5 contain each compact regular center tube under consideration. On the second component, the separated coordinate at $Y = 0$ is $\chi = -p$ and $A_C(-p) = F_+(p) + O(\epsilon)$ throughout the bounded interval. Its positive margin places the entire arc in the actual center domain. On the reversible component the compact upper arc lies strictly inside the central potential well. Hence in either case the complete central maps agree. Since the reference values vanish at $\nu = 0$, $K = 0$ there. The zero-scale action has the same two identities by (7.9).

Choose a common larger regular coefficient domain near each center tube and apply Proposition 5.13 to the difference between (7.12) and $\mathcal{A}$. Two additional coefficient derivatives suffice. Away from the center set the generator norm has a positive margin, so the error can instead be divided by that norm. On the resulting compact cover we have

$$-K = \tau A_\epsilon + \sigma C_\epsilon + \nu E_\epsilon, \qquad (A_\epsilon, C_\epsilon) \longrightarrow (A, C) \quad \text{in } C^2_p, \quad \|E_\epsilon\|_{C^2_p} \le C. \tag{7.13}$$

The common center domain is essential here: it allows division of the relative logarithmic error by the exact generators, with the stated input-derivative bounds.

On the larger tubes both actions have a common lower bound $S_* > 0$. The amplitude and upper-map estimates bound the required absolute coefficient derivatives by $C\epsilon^{-M} e^{-S_*/\epsilon}$, also on the small ν box after reducing S_*. The same center division preserves this weight. Lemma 5.14, with $I = |\tau| + |\sigma|$, therefore gives $|\nu| = o(1)I$ at a closing zero; for $I = 0$ it gives the center identity and no isolated zero.

Choose θ_0 and then the scale so that (7.13) satisfies Lemma 7.5, with columns (C, A) and coefficients (σ, τ). The strict quotient margin of Lemma 7.6 gives at most one simple zero of K. Lemma 3.2 gives the stated count for F. Since $L_+(p)$ and $L_-(q(p))$ decrease, their positive sets are initial subintervals. Intersecting them with the regular tube gives one positive-central passing interval, to which the count applies. □

Lemma 7.8 (Direction on the regular part)**.** *Under the preceding choices, if $\sigma \neq 0$, then at every point where $K = 0$ or K has sign $-\sigma$, its derivative in increasing p has sign $-\sigma$. After its possible zero it therefore keeps this sign and increases in absolute value on the rest of the regular passing interval. If $\sigma = 0$, $\tau \neq 0$, both K and K' have sign $-\tau$.*

Proof. The positive margins of A, A', C, C' and $(C/A)'$ on each compact regular tube allow the last clause of Lemma 7.5 to be applied to $-K$ in (7.13). At a zero, $-K$ crosses into the side of sign σ, and its derivative keeps that sign there. Thus it stays on that side. A regular input change preserves this direction for the same original-central function K. □

7.3 An incoming loss and two matched lower roots

An incoming lower loss can have unbounded transit time. To compare its central derivative with that of a regular outgoing arc, we use negative divergence on the outer arc and contraction through a fixed Gaussian core. This argument also allows an incoming endpoint at infinity. When both lower roots are lost, neither derivative provides a regular reference; we instead compare their logarithmic rates in the two moving deficits.

Lemma 7.9 (Incoming contraction)**.** *Suppose the outgoing lower data are compact and regular, with $A_+(p) \le A_{\mathrm{out}}$. Choose a fixed incoming crosscut $R > 0$ with $F_- > f_* > 0$ on $[0, R]$ and entries $q > R + 2\zeta$, where $\zeta > 0$ is fixed. On the actual common positive-central passing set, for either sign of ℓ, the incoming map $\mathcal{I}(q)$ satisfies*

$$0 < -\mathcal{I}'(q) \le \frac{C}{\epsilon} e^{-A_-(R)/\epsilon} \left(\frac{1+R^2}{1+q^2} \right)^{1/(24\epsilon)}. \tag{7.14}$$

If $A_-(R) > A_{\mathrm{out}} + 3$ and the upper derivative is bounded, then $K < 0$ on that set. The closing function is strictly monotone on each of its intervals. Every closing zero there is simple and has return multiplier less than one.

Proof. In the incoming coordinates $W = -X$, the forward divergence on $W \ge R$, $0 < z < 1$ is

$$-(1 + 2\epsilon\bar{B})W + \epsilon\{\bar{a} + \bar{c}(2z-1)\} - 2\rho\ell(1-z) \le -W/2.$$

The normal component satisfies $z' \le -(R/2)z + \rho \max(\ell, 0)$. On the fixed strip $R \le W \le R + \zeta$ the horizontal speed has magnitude at most $M\epsilon$. Any passing arc therefore reaches $W = R$ with

$$z_R \le e^{-R\zeta/(2M\epsilon)} + 2\rho \max(\ell, 0)/R. \tag{7.15}$$

An arc that has already reached $z = 0$ is outside the stated domain.

The fixed core from this crosscut to the center has denominator

$$D_- = F_- + \epsilon\bar{m} - (1 + 2\epsilon\bar{m})z + \epsilon\bar{m}z^2 > f_*/2$$

on a small height strip, and its graph equation is

$$z_W = \frac{Wz + \epsilon\bar{c}z(1-z) - \rho\ell(1-z)^2}{\epsilon D_-} =: f(W, z, \ell).$$

Let its signed output be $\mathcal{C}_R(z_R, \ell)$. The Gaussian estimate gives

$$0 \le \mathcal{C}_R(z_R, 0) \le C z_R e^{-A_-(R)/\epsilon}.$$

For this bound, integrate $\partial_W \log z = W/(\epsilon F_-) + O(1 + (W/\epsilon + 1)z)$; the nonlinear integral is bounded by Cz_R. For $\ell < 0$ with positive central output, the solutions for $\theta\ell$, $0 \le \theta \le 1$,

lie between this positive solution and the unforced one by scalar comparison. At fixed z_R, variation of constants gives

$$\partial_\ell \mathcal{C}_R(z_R, \theta\ell) = \int_0^R e^{-\int_0^u f_z(v,z(v),\theta\ell)\,\mathrm{d}v} \frac{(1-z(u))^2}{\rho D_-(u,z(u))}\,\mathrm{d}u \geq c > 0.$$

To obtain the positive lower bound, restrict the integral to $0 \leq u \leq \rho$. Its prefactor is at least c/ρ, and $|f_z| \leq C(W/\epsilon + 1 + |\ell|/\rho)$ bounds the exponent there. Positivity of the final output consequently implies $|\ell| \leq C z_R e^{-A_-(R)/\epsilon}$. For $\ell \geq 0$, positivity of the outgoing regular map instead gives $\ell \leq C Z_+(p)$. Thus ℓ is exponentially small on either part of the common passing set. Equation (7.15) now makes z_R exponentially small as well.

Set $\Phi(W) = \log(1+W^2)/(12\epsilon)$. Since $W' = -\epsilon D_-$ and $D_- \leq 2 + W + W^2 \leq 3(1+W^2)$, its time derivative is at least $-W/2$, hence at least the divergence. The determinant formula from $(q,1)$ to $W = R$ therefore bounds the transverse derivative by

$$\frac{Cq}{\epsilon}\left(\frac{1+R^2}{1+q^2}\right)^{1/(12\epsilon)}.$$

Half this power absorbs q uniformly even for unbounded entries. On the core, its normal variational exponent is $A_-(R)/\epsilon + O(1)$, because

$$f_z = \frac{W}{\epsilon(F_- + \epsilon\bar{m})} + \frac{\bar{c}}{F_- + \epsilon\bar{m}} + O((W/\epsilon+1)|z| + |\ell|/\rho),$$

and the last two quantities have the bounds just proved. Composition gives (7.14). The outgoing relative amplitude has $-L'_+(p) \geq (c/\epsilon)e^{-A_{\mathrm{out}}/\epsilon}$ on the same compact regular tube, including the small forcing just bounded. Thus the ratio of incoming and outgoing central derivatives is less than $1/2$ when $A_-(R) > A_{\mathrm{out}} + 3$ and the upper derivative is bounded. This proves $K < 0$ and the remaining claims. The estimate uses divergence integrated along the actual passing arc and remains valid for arbitrarily long residence near the incoming root. □

At a first finite simple or double incoming root, $A_-(R) \to \infty$ as R approaches the root from below. If the outgoing arc and upper return remain regular, choose R so that the preceding lemma applies. The resulting terminal part is monotone even though its incoming transit time diverges.

For simultaneous finite lower losses, Lemma 7.4 gives $\bar{B} = -\beta < 0$ and simple roots p_l, q_l with $p_l q_l = 1/\beta$. The next lemma determines when a regular upper return joins these roots and supplies the transverse parameter needed to measure their splitting.

Lemma 7.10 (The regular two-root match)**.** *For fixed $\beta > 0$, $\bar{a}$ and $\bar{m}$, the equation $\mathcal{U}_0(p_l; \bar{m}, \bar{c}) = q_l$ has a unique regular solution in $\bar{c}$ exactly when $\bar{m} < \beta$. At that solution the derivative with respect to $\bar{c}$ is strictly positive. If $\bar{a} \neq 0$, then*

$$\bar{a}\bar{c} < 0, \qquad \operatorname{sign}(\bar{a}+\bar{c}) = \operatorname{sign}\bar{c}, \qquad |\bar{c}| > |\bar{a}|. \tag{7.16}$$

For $\bar{a} = 0$ the solution is $\bar{c} = 0$.

Proof. On a returning upper arc $\bar{m}Y - 1 < 0$. Indeed the first maximum of Y must have $Y'' = \bar{m}Y - 1 < 0$; the region below that nullcline contains the complete arc. Its graph equation is $Y_X = (X + \bar{c}Y)/(\bar{m}Y - 1)$. Differentiation in $\bar{c}$, with integration from $X = p_l$ in the decreasing direction, has positive forcing after reversing the integration direction, because $Y > 0$ and $\bar{m}Y - 1 < 0$. It gives $Y_{\bar{c}} > 0$ at the exit and hence $\partial_{\bar{c}}\mathcal{U}_0 > 0$. For $\bar{m} > 0$, the formulas for S, I show that the range at fixed p_l is $(0, 1/(\bar{m}p_l))$: as the input approaches its saddle threshold, the exit approaches the reflected threshold and their product is $1/\bar{m}$. For

$\bar m \le 0$ that range is $(0,\infty)$, by the same exponential, repeated and trigonometric formulas, or their limits as $\bar c$ varies. Thus $q_l = 1/(\beta p_l)$ belongs to the range exactly when $\bar m < \beta$.

To determine the coefficient signs at the match, put $M = \beta - \bar m > 0$ and $\tau = \bar a + \bar c$. The endpoint identities from the solution $pS - I$ give

$$p_l + q_l = S(T)p_l(Mp_l - \tau) = e^{-\bar c T} S(T) q_l (Mq_l + \tau),$$

and hence

$$\tau = M \frac{e^{\bar c T} p_l^2 - q_l^2}{e^{\bar c T} p_l + q_l}. \tag{7.17}$$

The mechanical equation $Y'' - \bar c Y' - \bar m Y = -1$ has energy $\frac{1}{2}(Y')^2 + Y - \frac{1}{2}\bar m Y^2$ with derivative $\bar c (Y')^2$. Thus $q_l - p_l$ has the sign of $\bar c$. For the sign of (7.17), write $S(t) = e^{\bar c t/2} S_0(t)$. The positive fundamental solution S_0 is hyperbolic, linear or trigonometric. For $T/2 < u < T$ one has $S_0(u) > S_0(T-u)$: in the trigonometric case use $T < \pi/\sqrt{-\bar m - \bar c^2/4}$ and the sine difference formula. Pairing u and $T-u$ then shows that

$$\int_0^T e^{\bar c u/2} S_0(u)\,\mathrm{d}u - e^{\bar c T/2} \int_0^T e^{-\bar c u/2} S_0(u)\,\mathrm{d}u$$

has sign $\bar c$. Since $q_l/p_l = e^{\bar c T} I_{-\bar c}(T)/I_{\bar c}(T)$, this is precisely the sign of $e^{\bar c T} p_l^2 - q_l^2$. It proves the sign of τ. Finally $p_l - q_l = \bar a/\beta$ gives (7.16) and its zero case. □

Use the actual strong intersections of the two lower roots, their positive regular power inputs, and the regular upper map. Common finite-jet convergence preserves the preceding transverse derivative, so their splitting is a coefficient coordinate δ. The connector then has the form $g(t) = (t - \delta)k(t)$, with $k, g' > c > 0$ and bounded selected jets on larger signed intervals. The deficit comparison interval is $t > 0$, $g(t) > 0$; an actual return also requires positive complete central outputs. As in Section 5, extend each absolute map by zero on its own nonpassing side.

Proposition 7.11 (Two-root local count)**.** *Near every regular two-root match there are common coefficient and section neighborhoods and a scale cutoff such that the closing function has at most three isolated zeros, including multiplicity, on its positive central passing interval. The corresponding logarithmic ratio has at most two zeros and its derivative at most one. These assertions include either sign of splitting and the reversible match.*

Proof. The exact function is

$$F = D_+(t) - D_-(g(t)) - \rho\nu H, \qquad D_\pm(x) = K_\pm x^{h_\pm/\epsilon} e^{j_\pm(x)},$$

where $h_+ = p_l/\Delta + O(\epsilon)$, $h_- = q_l/\Delta + O(\epsilon)$ and $\Delta = \sqrt{\bar a^2 + 4\beta}$. All reference terms are input independent. Put

$$R_\pm = j_\pm' + \epsilon \frac{j_\pm' + x j_\pm''}{h_\pm + \epsilon x j_\pm'}.$$

The denominator has a positive margin. Thus $D_\pm''/D_\pm' = (h_\pm/\epsilon - 1)/x + R_\pm$ and the exact cleared numerator is

$$M = tgK' = (h_- - \epsilon)tg' - (h_+ - \epsilon)g + \epsilon tg\{g' R_-(g) - R_+(t) + g''/g'\}. \tag{7.18}$$

The multiplier tg is positive on the whole passing interval. Away from $\bar a = 0$, $h_- - h_+ = -\bar a/(\beta\Delta) + O(\epsilon)$ has a nonzero margin. Differentiating (7.18) gives

$$M' = (h_- - h_+)g' + (h_- - \epsilon)tg'' + O(\epsilon(t+g)).$$

First choosing the section radius small, then the coefficient neighborhood and scale small, makes M' have a fixed nonzero sign. Thus M has at most one simple zero there.

Near the reversible match use the independent transverse variables $(\bar{a}, \delta, \nu)$. The inverse coefficient chart follows from the strict splitting derivative just proved; its finite jets are bounded on common boxes. At their zero, the reflected power coordinates may be chosen equal, giving $D_+ = D_-$ and $g = t$. The relative coefficients extend by the common linear rule of Section 5, so the required coefficient segments are legitimate. At zero scale the columns of the numerator are

$$M_{\bar{a}} = -\frac{t}{2\beta^{3/2}} + O(t^2), \qquad M_\delta = \frac{1}{2\beta} + O(t^2).$$

Indeed $p_l = q_l = 1/\sqrt{\beta}$, $\Delta = 2\sqrt{\beta}$ and $(h_- - h_+)_{\bar{a}} = -1/(2\beta^{3/2})$; differentiating $g = (t - \delta)k$ with the common reflected unit $k(0) = 1$ gives the constant column. The same formulas hold after a common positive normalization if a different reflected unit was used. Consequently the second column has a positive margin and the derivative of their quotient is strictly negative on a short interval including zero. One coefficient Hadamard integral now gives

$$M = \bar{a}A_\epsilon + \delta B_\epsilon + \nu C_\epsilon, \qquad B_\epsilon > c, \quad (A_\epsilon/B_\epsilon)' < -c.$$

The finite-order preparations reserve two ordinary input derivatives for each logarithmic rate and one coefficient derivative for this division, which gives the displayed derivative bounds.

On the common small lower input intervals, the two large powers and their required absolute coefficient derivatives are bounded by $C\epsilon^{-N}e^{-b/\epsilon}$ with a fixed $b > 0$. Their exact reversible equality and the signed extensions give, at any closing zero, $|\nu| = o(1)(|\bar{a}| + |\delta|)$ by Lemma 5.14. If the latter two parameters vanish, the fibre is a center and has no isolated zero. Otherwise the strict quotient margin and relative C^1 forcing bound give at most one simple zero of M by Lemma 7.5. Two Rolle steps give the claimed bounds for K and F. Both full central outputs increase with t, so their positivity truncates the deficit interval to one interval. Excluded endpoints contribute no formal closing zero. □

With a finite transverse upper limit, the lower possibilities are now exhausted: both arcs are regular, the incoming arc alone loses positivity, or both arcs lose positivity. Lemma 7.3 excludes an outgoing loss alone. The remaining finite upper limit is a saddle and is treated next.

7.4 A finite upper saddle above regular lower arcs

Suppose $\bar{m} > 0$ and put

$$p_c = \frac{2}{\bar{c} + \sqrt{\bar{c}^2 + 4\bar{m}}}, \qquad q_c = \frac{2}{-\bar{c} + \sqrt{\bar{c}^2 + 4\bar{m}}}.$$

We first assume $F_+ > 0$ on $[0, p_c]$ and $F_- > 0$ on $[0, q_c]$. The perturbed separatrices meet $Y = 0$ at P_s, Q_s, and the returning quadrant of Section 4.4 has regular outer coordinates

$$p = P_s - \psi_+(t), \qquad q = Q_s - \psi_-(v), \qquad \psi_\pm(0) = 0, \quad \psi'_\pm > c > 0.$$

Retain the actual exponent r of the saddle passage $v = v(t)$. With weak references fixed independently of input, Lemma 7.2 gives complete central maps $s_+ + U(t)$, $s_- + V(v)$ and closing function

$$F(t) = U(t) - V(v(t)) - \rho\nu H, \qquad H > h_* > 0, \qquad U', V' > 0. \tag{7.19}$$

The gap H is independent of t. On a larger common domain the regular rates

$$\mathsf{A}_+(t) = \epsilon U''/U', \qquad \mathsf{A}_-(v) = \epsilon V''/V'$$

have positive margins and the finite mixed jets used below. A physical input must traverse the chosen quadrant and both regular outer legs, and both central outputs must be positive. These conditions restrict a single interval; positivity of each central output is a one-sided restriction because both maps are increasing.

The same scalar comparison will be used at secondary scale.

Lemma 7.12 (Saddle logarithms and their zero count)**.** *Let $\epsilon, r > 0$, $D = t\partial_t$, $\kappa = r - 1$, and let K be real analytic on $0 < t < t_0$. Fix positive constants c, C, c_1. K_c, R_*, T, W are independent of t. With common bounds and implied constants, each preparation gives, for sufficiently small ϵ, uniform section cutoffs and the stated bound with multiplicity.*

(i) *Suppose*

$$K = K_c + \epsilon\kappa \log t + \mathcal{E}(t), \qquad |K_c| \geq cR_*,$$
$$|\mathcal{E}| + |D\mathcal{E}| \leq CR_*\sqrt{t}, \quad R_* \geq c_1|\kappa|, \quad R_* > 0.$$

Then $Z(K) \leq 1$. If $\kappa = 0$ there is no zero; otherwise every zero is simple and K' has sign κ there.

(ii) *Suppose $\kappa \neq 0$ and, with the compensator of (4.34),*

$$DK = \epsilon\kappa + Tt - Wt\omega(t;\kappa) + R(t),$$
$$W/\kappa \geq c, \quad |T| \leq C|\kappa|, \quad |R'| \leq C|\kappa|\sqrt{t}.$$

First choosing t_0 small and then $|\kappa|$ small gives $(DK)'/\kappa < 0$, hence $Z(K) \leq 2$.

(iii) *Suppose r belongs to a compact positive interval with $|r - 1| \geq \delta > 0$, A, B, k_0 have common positive upper and lower bounds, and*

$$(DK)' = r^2 k_0 A t^{r-1} - B + O(t + t^{r-1}(t + t^r)) + O(\epsilon[1 + t^{r-1}]).$$

Then $(DK)'$ is positive for $r < 1$ and negative for $r > 1$, and $Z(K) \leq 2$.

The bounds persist on subintervals, without using a value at $t = 0$.

Proof. In (i), choose t_0 so $C\sqrt{t_0} < c/2$. For $\kappa = 0$ the constant fixes the sign. Otherwise, a zero requires

$$t \leq e^{-cR_*/(2\epsilon|\kappa|)}, \qquad \frac{|D\mathcal{E}|}{\epsilon|\kappa|} \leq \frac{CR_*}{\epsilon|\kappa|} e^{-cR_*/(4\epsilon|\kappa|)} < 1/2.$$

The last inequality is uniform since $R_*/(\epsilon|\kappa|) \geq c_1/\epsilon$. Thus every zero is simple and crosses with sign κ, giving at most one.

For (ii), put $\phi = (t\omega)'$. Then $\phi' = -rt^{\kappa-1} < 0$ and $\phi(t_0, \kappa) \to -\log t_0 - 1$ as $\kappa \to 0$. Choose t_0 so the latter exceeds $4C/c + 1$, and then choose the ratio neighborhood so this margin persists. Throughout the interval,

$$(DK)'/\kappa = T/\kappa - (W/\kappa)\phi + R'/\kappa < 0.$$

Thus DK has at most one zero. In (iii), t^{r-1} dominates for $r < 1$ and tends uniformly to zero for $r > 1$; choose the section and then the scale to absorb the errors. Rolle's inequality and $Z(K') = Z(DK)$ give both two-zero bounds. □

Proposition 7.13 (The regular-lower upper-saddle chart)**.** *On common coefficient and section neighborhoods of these data, for all sufficiently small positive scales, the actual closing function has at most three isolated zeros, including multiplicity. This bound is uniform through the actual ratio $r = 1$. Its critical function*

$$K = \epsilon \log \frac{V'(v)v'}{U'} \tag{7.20}$$

has at most two zeros on any nonidentity fibre containing a closing zero. More generally the critical-function bound holds whenever (7.25) is supplied on the fixed fibre. It holds on the whole common comparison interval $0 < t < t_0$, including inputs where a complete central output is nonpositive; central positivity only restricts this interval when actual returns are counted. The reversible identity fibre contributes no isolated zero.

Proof. By 3.2 it suffices to bound $Z(K)$ by two. Put $D = t\partial_t$ and $E = tv''/v' - (r-1)$. The exact differentiated critical function is

$$M = DK = \epsilon(r-1) + \mathsf{A}_-(v)tv' - t\mathsf{A}_+(t) + \epsilon E. \tag{7.21}$$

The outer coefficients have bounded jets through $t = v = 0$.

First keep r a fixed distance from one. Axis normalization as in (4.33), without its subsequent resonant shears, gives $v = k_0 t^r e^w$ with a bounded positive unit k_0. The reciprocal integral there implies $|D^j w| \le C\epsilon(t+t^r)$ for the finitely many input orders needed here. Indeed $t^r \int_t^1 s^{-r}\,ds = (t-t^r)/(r-1)$; the same estimate for its Euler derivatives follows from the integral equation and bounded state derivatives on the rectangle. The ratio separation bounds the denominator. Thus Lemma 7.12(iii) applies with $A = \mathsf{A}_-(0)$ and $B = \mathsf{A}_+(0)$, proving the count on this ratio range.

Near ratio one, use the compensator preparation (4.34), with $\kappa = r - 1$ and $|\kappa| \le 1/64$. Reserve eight total orders there, so the curvature formula retains four Euler and two coefficient orders. The actual ratio is a legal coefficient coordinate: at the reversible face $\partial_{\bar c}\kappa = -1/\sqrt{\bar m} + O(\epsilon)$. Integrating the regular rates gives the exact identity

$$\begin{aligned} K &= K_c + \epsilon\kappa\log t + \mathcal{E}(t), \qquad K_c = \epsilon\log\frac{rV'(0)}{U'(0)}, \\ \mathcal{E}(t) &= \int_0^{v(t)} \mathsf{A}_-(z)\,dz - \int_0^t \mathsf{A}_+(z)\,dz + \epsilon\left[w + \log\frac{r+Dw}{r}\right]. \end{aligned} \tag{7.22}$$

The passage estimates imply $|\mathcal{E}| + |D\mathcal{E}| \le Ct^{1/2}$. No restriction is placed on $|\kappa\log t|$.

To compare the constant with the logarithmic term, write $U = \alpha_U(p)e^{-A_+(p)/\epsilon}$ and $V = \alpha_V(q)e^{-A_-(q)/\epsilon}$ here with the action notation of Section 7.1, and put $r_U = A_+' - \epsilon(\log\alpha_U)'$, and similarly r_V. These factors are positive. At the separatrix inputs one obtains

$$K_c = A_+(P_s) - A_-(Q_s) + \epsilon\log\frac{r\alpha_V(Q_s)r_V(Q_s)\psi_-'(0)}{\alpha_U(P_s)r_U(P_s)\psi_+'(0)}. \tag{7.23}$$

The logarithmic factors have positive margins and bounded required jets. On $\nu = 0$ the regular connections give $K_c = A_+(p_c) - A_-(q_c) + O_N(\epsilon(1 + |\log\epsilon|))$. At a limiting ratio-one point with $\bar a \ne 0$ this limit is

$$-2\bar a\int_0^{1/\sqrt{\bar m}} \frac{X^2}{F_+(X)F_-(X)}\,dX \ne 0.$$

It stays bounded away from zero in a smaller neighborhood.

The coordinates of Section 4.4 satisfy $v = t$ and $U = V$ on $\bar a = \kappa = \nu = 0$. Choose the coefficient contractions within the same quadrant and enlarged regular lower tubes, so these exact identities can be used for division. With $n = |\bar a| + |\kappa| + |\nu|$, division along these contractions gives

$$|\mathcal{E}| + |D\mathcal{E}| \le Cnt^{1/2}, \qquad K_c = \bar a A_c + \kappa B_c + \nu C_c, \quad A_c < -c, \quad |B_c| + |C_c| \le C. \tag{7.24}$$

To see the strict sign, differentiate (7.23) at fixed actual κ; there $\partial_{\bar a}\bar c = O(\epsilon)$ and

$$A_c \longrightarrow -2\int_0^{1/\sqrt{\bar m}} \frac{X^2}{(1+\bar B X^2)^2}\,dX < 0.$$

Coefficient derivatives of t^r only add fixed powers of $|\log t|$, absorbed by the gap between $t^{1-1/64}$ and $t^{1/2}$.

On these regular lower tubes the actions have a common positive lower bound $S_* > 0$. The absolute quantities U, V and their finite coefficient derivatives are bounded by $C\epsilon^{-M}e^{-S_*/\epsilon}$; composition with t^r and its coefficient derivatives preserves this bound uniformly on the section. The center identity and coefficient division therefore give the hypothesis of Lemma 5.14 with $I = |\bar{a}| + |\kappa|$. Every fibre containing a closing zero satisfies

$$|\nu| \le \eta(\epsilon)(|\bar{a}| + |\kappa|), \qquad \eta(\epsilon) \longrightarrow 0. \tag{7.25}$$

If both generators vanish, this forces the exact identity fibre.

For a nonzero limiting $\bar{a}$, the constant and remainder estimates satisfy Lemma 7.12(i) with $R_* = 1$. Near the reversible face fix a large K_0. In the cone $|\bar{a}| > K_0|\kappa|$, the divided constant and relative forcing give the same hypotheses with $R_* = |\bar{a}|$.

In the remaining cone $|\bar{a}| \le K_0|\kappa|$, the compensator controls the derivative. A nonidentity fibre has $\kappa \neq 0$ and $n \le C_{K_0}|\kappa|$. Set $A = \mathsf{A}_-(0)$ and $B_0 = \mathsf{A}_+(0)$. Using $t^r = t - \kappa t\omega(t;\kappa)$ in (7.21) and the curvature formula gives

$$\begin{aligned} M &= \epsilon\kappa + Tt - Wt\omega + R, \\ T &= rA - B_0 + \epsilon\sigma(2 + 1/r), \qquad W = r(A\kappa + 2\epsilon\sigma). \end{aligned} \tag{7.26}$$

The prepared coefficients vanish at zero scale and on the reversible face, so $|\sigma| \le C\epsilon n$. All remaining terms vanish on that face as well. The reserved mixed jets of the passage and the regular rates, followed by one coefficient division, give $|D^j R| \le Cnt^{3/2}$, $j \le 2$. Explicitly, the undivided products have powers at least $t^{2-2/64}$ times fixed logarithmic powers; their coefficient variations retain a power strictly larger than $3/2$. Also $|T| \le Cn$, and the positive margin of A gives $W/\kappa \ge c > 0$ for small scale. Hence $|R'| \le C_{K_0}|\kappa|\sqrt{t}$. Lemma 7.12(ii) now gives $Z(K) \le 2$ in the remaining cone. Together with the separated-ratio case this proves the critical-function bound; 3.2 gives $Z(F) \le 3$ on the physical subinterval. □

Lemma 7.14 (Coefficient and endpoint signs)**.** *On a regular-lower saddle fibre with* (7.25)*, suppose an incoming regular overlap supplies*

$$\tau\sigma_1 < 0, \quad |\tau| \le C|\sigma_1|, \qquad \operatorname{sign} K = -\operatorname{sign}\sigma_1, \qquad \tau = \bar{a} + \bar{c}, \quad \sigma_1 = \bar{c}(\bar{B} + \bar{m}).$$

Then $\bar{a}\bar{c} < 0$, $|\bar{a}| \le C_1|\bar{c}|$, $\operatorname{sign}\kappa = -\operatorname{sign}\bar{c}$, and $K(0+)$ has sign $\bar{c}$, opposite to the supplied overlap sign. The approach toward this saddle is increasing p, hence decreasing t. The endpoint limit refers to the same relative critical function on its common extension $0 < t < t_0$; if a central-output condition cuts off that interval, it does not assert a positive central return at the excluded endpoint. These are entrance data for the interval argument of Section 12.

Proof. Regular lower positivity gives the identity and inequality

$$0 < F_+(p_c)/p_c + F_-(q_c)/q_c = (\bar{B} + \bar{m})(p_c + q_c).$$

Thus σ_1 has the sign of $\bar{c}$. The asserted coefficient cone follows from the two hypotheses on τ. For the actual saddle, $\kappa = -\operatorname{tr}/\lambda_u$. The zero-scale divided field has trace $\bar{c} + \nu$, so near the reversible face the expansion is

$$\kappa = -\bar{c}/\sqrt{\bar{m}} + O(\bar{c}^2 + \epsilon|\bar{c}| + \epsilon|\bar{a}| + |\nu|).$$

The coefficient cone and the relative forcing absorb the error, first giving $|\kappa| \le C|\bar{c}|$ and then its opposite sign. Away from $\bar{c} = 0$ the limiting formula $\kappa_0 = -2\bar{c}/(\sqrt{\bar{c}^2 + 4\bar{m}} + \bar{c})$ has a fixed nonzero margin, preserved at small scale and forcing. Finally (7.22) has bounded remainder and $\epsilon\kappa\log t$ has sign $-\kappa$ as $t \downarrow 0$. The coefficient cone and the supplied overlap sign therefore give the required endpoint sign on the comparison continuation. □

7.5 An upper saddle with one incoming lower loss

Keep the outgoing lower arc regular and let the incoming arc lose positivity. A simple first incoming root must coincide with q_c; a double root can also occur before it. In both cases use the whole-interval nonnegativity required by Section 7.1, rather than only the value at q_c.

Proposition 7.15 (The incoming-loss upper-saddle chart)**.** *On common neighborhoods of either such configuration, the actual closing function has at most three isolated zeros, with multiplicity. For the actual ratio $r \geq 1$, its critical function is strictly negative and the closing function is strictly increasing, so its bound is one. For $r < 1$ the multiplied critical numerator below is strictly increasing, giving at most two critical zeros and three closing zeros. These assertions hold on the entire actual passing interval, even when its boundary moves away from the upper separatrix input.*

Proof. For a simple incoming root, Proposition 4.8 gives $V(v) = D_-(g(v))$, where $g' > c > 0$ and passage requires $g(v) > 0$. Write $\lambda = h_-/\epsilon$, $L = \epsilon\lambda$ and

$$\frac{D_-''(g)}{D_-'(g)} = \frac{\lambda - 1}{g} + R_-(g), \qquad C(v) = R_-(g(v))g'(v) + g''(v)/g'(v).$$

The remainder and its needed ordinary jets are bounded, by the power preparation and the positive large-exponent factor in D_-'. The critical function remains $K = \epsilon \log(V'(v)v'/U')$. Its exact numerator, multiplied by the positive factor tg, is

$$M_s = (L - \epsilon)tg'v' + \epsilon g(\kappa + E) + \epsilon tgv'C - \mathsf{A}_+(t)tg. \tag{7.27}$$

For a double root, use the increasing Morse slow entry $g(v)$, $g' > c > 0$, and write $f(v) = \mu + g(v)^2$. If the root precedes q_c, choose one phase interval containing both the minimum and the compact upper-entry range. The bound $F_I'' > 1$ and the positive finite horizontal coordinate give one increasing Morse coordinate and one bounded positive time unit on that entire interval. Fixed regular pieces may be included in the connections. The moving-entry relative formula, including any regular normal input as part of its relative amplitude, gives

$$\frac{V''(v)}{V'(v)} = \frac{\mathsf{A}_-(g(v))g'(v)}{\epsilon f(v)} + \frac{g''(v)}{g'(v)}.$$

Here A_- has a positive margin and bounded jets by Lemma 4.5. Put $J = \mathsf{A}_-(g)g'$ and $C = g''/g'$. The exact numerator, now with positive factor tf, is

$$M_d = Jtv' + \epsilon f(\kappa + E) + \epsilon ftv'C - \mathsf{A}_+(t)tf. \tag{7.28}$$

Passage requires $\mu + s^2 > 0$ at every point between $g(v)$ and the fixed exit. Since g and v increase, the permitted inputs form an interval. The physical interval is its intersection with the upper quadrant, regular connection domains and positive central-output conditions. An excluded $t = 0$ supplies no physical endpoint value.

For $r \geq 1$ choose a fixed incoming section R before its limiting first root such that $A_-(R) > A_{\text{out}} + 3$. The incoming action diverges at either a simple or double root, whereas the outgoing action stays bounded. Choose a positive section gap and shrink the neighborhood so that every incoming endpoint lies beyond R by that gap. The upper derivative $v' = t^{r-1}e^w(r + Dw)$ is uniformly bounded for $r \geq 1$, including $r = 1$; the regular outer derivatives are bounded too. Lemma 7.9 therefore gives $V'(v)v' < U'(t)$ on the whole passing interval, hence $K < 0$. It applies to the actual incoming arc and its reference difference; no derivative estimate on its long transit time is required.

For $r < 1$ separated from one, the first term in the derivative of either numerator is at least ct^{r-1}. The other terms are $O(1) + o(t^{r-1}) + O(\epsilon t^{r-1})$ by the reciprocal upper estimate

used above. Choosing section and then scale proves $M' > 0$. As $r \uparrow 1$, this argument needs the sign of the resonant curvature term. We compute that sign in the original field. At a limiting ratio-one configuration, $\bar{c} = 0$ and $p_c = q_c$. Incoming loss with outgoing positivity implies $\bar{a} > 0$. For a simple loss this follows from $F_+(p_c) - F_-(p_c) = 2\bar{a}p_c > 0$; for a double loss its positive root has $\bar{a} = 2\sqrt{\bar{B}} > 0$. Appendix C.1 derives from the original field and its exact zero-trace equation

$$\sigma|_{\kappa=\nu=0} = -\chi_{\mathrm{loc}}^2 \epsilon \bar{a}[\bar{m}^{5/2} + O(\epsilon)].$$

Consequently $\sigma < -c\epsilon$ on fixed smaller ratio and coefficient neighborhoods of this loss configuration. The actual curvature formula is therefore $E = -\sigma Q + R_E$, where $Q, Q' > 0$ and $|D^j R_E| \le C\epsilon t^{7/4}$ for the orders used here.

For the simple numerator, group its differentiated terms as

$$M_s' = (rL - \epsilon)g'v' - \mathsf{A}_+(0)g(0) + Lg'v'E + \epsilon g E' + \mathcal{R}_s,$$

where $|\mathcal{R}_s| \le Ct^{r-1}(v + t + \epsilon)$ on the smaller near-resonant section. This bound follows directly by differentiating (7.27): the additional quadratic velocity terms contain $tv' = O(v)$, and $g(v) - g(0) = O(v)$. Terms with C carry ϵ; terms with its extra factor E use $|E| \le C\epsilon t^{1-1/64}(1 + |\log t|)$. The first term is at least ct^{r-1}, whereas $g(0)$ tends to zero at the limiting simple loss. The principal parts of the two displayed curvature terms are nonnegative because $g > 0$ on passage. Their remainders are bounded by $C\epsilon t^{3/4}(1 + v')$ and are absorbed by the positive margin. Thus $M_s' > 0$.

For the double numerator the same differentiation gives

$$M_d' = r^2 J(0)t^{r-1} - \mathsf{A}_+(0)f(0) + (J + \epsilon f_v)Ev' + \epsilon f E' + \mathcal{R}_d, \qquad |\mathcal{R}_d| \le Ct^{r-1}(v + t + \epsilon).$$

Here $f(0)$ need not tend to zero when the double root precedes the upper endpoint, so compare the two constant terms together. At zero scale and ratio one the incoming phase is $g = 1/q - \bar{a}/2$, $\mu = \bar{B} - \bar{a}^2/4$. Put $k_0 = g'(0) > 0$. The symmetric upper connections then have $p_t(0) = q_v(0) = -k_0 p_c^2$. Division of the lower slow time by $x = 1/q$ makes its limiting normal rate $1/x = q$. Consequently

$$J(0) = p_c k_0, \qquad \mathsf{A}_+(0) = \frac{k_0 p_c^3}{F_+(p_c)}, \qquad f(0) = \frac{F_-(p_c)}{p_c^2}.$$

Their exact limiting difference is

$$J(0) - \mathsf{A}_+(0)f(0) = \frac{2\bar{a}k_0 p_c^2}{F_+(p_c)} > 0. \tag{7.29}$$

For nearby $r < 1$, the factor $t^{r-1} \ge 1$ preserves a fixed positive margin after reducing the ratio neighborhood; if $f(0) < 0$ the estimate improves. Thus this argument includes a double root before the limiting upper endpoint. Since $J + \epsilon f_v > c$ and $f > 0$ on the actual interval, the principal curvature terms are again nonnegative. Their remainder and $\mathcal{R}_d$ are smaller than this margin after fixed section and scale restrictions. Hence $M_d' > 0$ on the entire passing interval.

In both cases M is a positive multiple of K' and is strictly increasing. Thus 3.2 gives at most two critical zeros and three closing zeros on the entire physical interval, including multiplicities and analytic endpoint germs. The entrance argument in Section 12 joins this interval to its preceding regular part. □

7.6 Two lower roots at a finite upper saddle

When both simple lower roots meet an upper saddle, all three passages can lose their exits together. Fix $\bar{B} = -\beta$, $0 < \beta_* \le \beta \le \beta^*$, near the triangle center $\bar{m} = \beta$, $\bar{c} = -\bar{a}$, $\nu = 0$. The actual upper ratio r stays in a fixed compact subset of $(0, \infty)$. Retaining both lower deficits as well as this ratio allows all three passages to degenerate independently. Use the actual splitting coordinates of Lemma 5.7, and the upper coordinates and regular legs of Lemma 5.8 near the reversible intersection. On the remaining compact parts, extend just the triangle coordinates $L_\pm L_0^{\epsilon\beta}$ from Lemma 5.3, and straighten and normalize the neighboring axes by the same construction. The axis graph denominators have positive margins on every compact positive-ratio interval; no resonant shear is needed for this extension. Its orbital remainder is ϵ times a bounded function and vanishes on the triangle component. The same regular outer legs therefore give

$$u = G_+(t) = u_0 + tA_0(t), \qquad z = G_-(v) = z_0 + vB_0(v), \qquad v = t^r e^{w(t)},$$

where $A_0, B_0, G_+', G_-' > c > 0$. The lower maps are those of Proposition 4.8:

$$D_\pm(s) = K_\pm s^{h_\pm/\epsilon} e^{j_\pm(s)}, \qquad K_\pm > 0, \quad h_\pm > h_* > 0, \quad j_\pm(0) = 0. \tag{7.30}$$

The upper passage and both lower deficit passages define the signed central comparison on

$$0 < t < t_0, \qquad u > 0, \qquad z > 0. \tag{7.31}$$

Since u, z increase, this is an interval. The closing function there is

$$F(t) = D_+(u) - D_-(z) - \sqrt{\epsilon}\,\nu H, \qquad H > h_* > 0.$$

The weak references, and hence H, are independent of t. The complete central outputs are $s_+ + D_+(u)$ and $s_- + D_-(z)$. Their positivity is an additional requirement for an actual return, beyond the deficit conditions in (7.31). The Gaussian reference calculation in Lemma 7.2 applies on the fixed inner cores: $F_\pm$ have positive margins there even when the outer roots are lost. It gives $H = \sqrt{2\pi} + O(\sqrt{\epsilon})$ and, more precisely,

$$s_+ = -\sqrt{\epsilon}\,\nu H_+, \qquad s_- = \sqrt{\epsilon}\,\nu H_-, \qquad H_+, H_- > c > 0, \qquad H = H_+ + H_-.$$

At a zero of F the two outputs agree. For $\nu \ge 0$ the incoming one is positive, and for $\nu \le 0$ the outgoing one is positive, since $D_\pm > 0$. Thus their common output is always positive at a closing zero. In each core the original height $\mathfrak{z} = 1 + y$ has derivative of fixed sign on $\mathfrak{z} = 0$, namely the sign of ν (with the reflected sign on the other core). An orbit with positive entrance and exit cannot cross that line and return. Together with the strict deficit passages and finite transverse connections, this proves that each closing zero gives an actual positive-height return. At nonclosing inputs we continue to use the signed comparison.

The numerator calculation below uses coefficient division as well as input differentiation. Reserve twelve total finite orders in the lower and upper preparations before shrinking their construction domains. This supplies six ordinary input and two coefficient derivatives of the regular connectors and lower relative coefficients, four Euler input and four coefficient derivatives of the upper relative logarithm, and the two coefficient derivatives spent in product division below. The lower graph can be constructed at order 23 by the $N + 11$ reserve in Proposition 4.8. Scale derivatives are Euler derivatives. The ordinary input derivatives of the lower connectors are not asserted for the singular composite t^r at zero.

Use the common signed relative extension and absolute zero extension after Proposition 5.13 when a coefficient segment changes a deficit's sign. For any fixed desired $S_* > 0$, first reduce

the input radius $d < 1$ so that a half of $d^{h_*/\epsilon}$ absorbs the upper bound for $\exp(\epsilon \log K_\pm/\epsilon)$, and then reduce ϵ so that the remaining half-power absorbs all reserved derivatives. This gives

$$|\partial_s^i \partial_{\mathbf{b}}^\alpha \widehat{D}_\pm| \le C\epsilon^{-M} e^{-S_*/\epsilon} \tag{7.32}$$

through the required finite orders. These extensions preserve both center identities by that construction. Keep the original t fixed on the coefficient segments and impose central positivity on actual returns.

Put

$$\mathcal{K} = \epsilon \log \frac{D'_-(z)G'_-(v)v'}{D'_+(u)G'_+(t)}, \qquad D = t\partial_t, \qquad \kappa = r - 1, \qquad E = \frac{tv''}{v'} - \kappa.$$

Ordinary primes on $G_\pm$ and $D_\pm$ mean their own input derivatives. For use in the chain rule define

$$R_\pm(s) = j'_\pm(s) + \frac{j'_\pm(s) + sj''_\pm(s)}{h_\pm/\epsilon + sj'_\pm(s)}.$$

Then $D''_\pm/D'_\pm = (h_\pm/\epsilon - 1)/s + R_\pm$ on $s > 0$. The displayed formula also defines bounded auxiliary coefficients on the signed interval; no logarithm of a zero extension is taken.

Proposition 7.16 (Complete three-saddle numerator)**.** *On the comparison interval the following function is exactly $tuz\,\mathcal{K}'$:*

$$\begin{aligned} M = {} & (h_- - \epsilon)uDz - (h_+ - \epsilon)zDu + \epsilon uz(\kappa + E) \\ & + \epsilon uz\left[R_-(z)Dz - R_+(u)Du + \frac{G''_-(v)}{G'_-(v)}Dv - \frac{G''_+(t)}{G'_+(t)}t\right]. \end{aligned} \tag{7.33}$$

This formula defines its auxiliary extension. Its endpoint value is

$$M(0+) = \epsilon\kappa u_0 z_0.$$

Near ratio one put $\xi_1 = (u_0 + z_0)/2$, $\xi_2 = (u_0 - z_0)/2$ and $I = |\xi_2| + |\kappa\xi_1|$. Then

$$\begin{aligned} M &= C_0 + Tt - Wt\omega(t;\kappa) + \mathcal{R}_3, \\ C_0 &= \epsilon\kappa u_0 z_0, \\ P &= u_0 z_1(rh_- - \epsilon) + \epsilon u_0 z_0 r J_-, \\ N &= z_0 u_1(h_+ - \epsilon r) + \epsilon u_0 z_0 J_+, \\ T &= P - N + \epsilon u_0 z_0 \sigma_s(2 + 1/r), \\ W &= \kappa P + 2\epsilon u_0 z_0 \sigma_s r, \end{aligned} \tag{7.34}$$

where $u_1 = G'_+(0)$, $z_1 = G'_-(0)$ and

$$J_+ = u_1 R_+(u_0) + G''_+(0)/u_1, \qquad J_- = z_1 R_-(z_0) + G''_-(0)/z_1.$$

Here σ_s denotes the resonant coefficient σ of (4.34) in the present center-preserving coordinates. For $|\kappa| \le 1/64$,

$$|D^j \mathcal{R}_3| \le C(I + |\nu|)t^{7/4}, \qquad 0 \le j \le 2. \tag{7.35}$$

If this fibre has one actual closing zero, then

$$|\nu| \le \delta_\epsilon I, \qquad \delta_\epsilon = C\epsilon^{-M-1/2} e^{-S_*/\epsilon} \longrightarrow 0. \tag{7.36}$$

Proof. Differentiate $\mathcal{K}$ before clearing its denominators. The two power rates give the first two terms in (7.33); the upper second derivative gives $\epsilon uz(\kappa + E)$. The ordinary curvature of each regular leg gives the last two terms in its bracket. Thus the numerator retains every coordinate contribution to the complete central derivative.

To verify the expansion, first regard t and t^r as independent small quantities. The endpoint Taylor formulas give

$$u = u_0 + u_1 t + O(t^2), \qquad z = z_0 + z_1 t^r + \text{terms of order greater than one.}$$

The terms linear in these two quantities are

$$C_0 + Pt^r - Nt - \epsilon u_0 z_0 \sigma_s Q(t), \qquad Q = 2rt\omega - (2 + 1/r)t.$$

In particular the $J_\pm$ contributions are present even when the connectors are nonlinear. The exact equality $t^r = t - \kappa t\omega$ gives (7.34).

We first estimate the remainder uniformly on the auxiliary box, before dividing by the center generators. Write $\mathcal{F}_a$ for a function whose selected $D^j \partial_{\mathbf{b}}^\alpha$ derivatives are $O(t^a)$, $j \le 2$, $|\alpha| \le 2$. With $a = 15/16$ the functions $t, t^r, t\omega$ belong to $\mathcal{F}_a$ on $|\kappa| \le 1/64$; every fixed parameter logarithm is absorbed by $63/64 - a > 0$. The upper preparation gives $v - t^r, Dv - rt^r \in \mathcal{F}_{2a}$ and $E + \sigma_s Q \in \mathcal{F}_{7/4}$. Integral Taylor remainders of $G_\pm$, $R_\pm$ and their regular curvatures then show that every omitted product has at least two $\mathcal{F}_a$ factors or the displayed curvature remainder. As $2a = 15/8 > 7/4$, it follows that $D^j \partial_{\mathbf{b}}^\alpha \mathcal{R}_3 = O(t^{7/4})$ on the entire common auxiliary box.

The triangle center gives vanishing of M and its retained coefficients on $\xi_1 = \xi_2 = \nu = 0$; the reflected extension identities give the same vanishing on $\xi_2 = \kappa = \nu = 0$. These two axes explain the generator pair $(\xi_2, \kappa\xi_1)$. Apply the product-ideal division of Proposition 5.13 to $\mathcal{R}_3$. Its explicit mixed integral in ξ_1, κ spends two coefficient derivatives and preserves $t^{7/4}$, proving (7.35).

Apply the identical division to $\widehat{D}_+(G_+(t)) - \widehat{D}_-(G_-(v(t)))$. Equation (7.32) bounds its divided coefficients by $C\epsilon^{-M} e^{-S_*/\epsilon}$. At a physical closing zero Lemma 5.14 now proves (7.36). This is a coefficient constraint on the whole fibre. If $I = 0$, it forces $\nu = 0$ and the exact center identity contributes no isolated zero. □

The scalar compensator calculation. The same scalar calculation controls the derivative at a single upper saddle and the numerator here. Keeping $t\omega$ intact makes its bounds uniform even when $|\kappa \log t|$ is unbounded.

Lemma 7.17. *For $|\kappa| \le 1/64$ set $r = 1 + \kappa$ and*

$$M = C + Tt - Wt\omega(t;\kappa) + R, \qquad L = |T| + |W| > 0.$$

Assume $|R'| \le C_* L t^{1/2}$ and $|R''| \le C_* L t^{-1/2}$. On a common sufficiently small $(0, t_0)$, M has at most two zeros, with multiplicity. If it has two, then

$$TW > 0, \qquad (C + R(0))W > 0. \tag{7.37}$$

There is also the following one-zero version. For fixed positive c_*, C_*, if

$$W/\kappa \ge c_*, \quad |T| \le C_*|\kappa|, \quad |R'| \le C_*|\kappa| t^{1/2}, \quad \kappa \ne 0,$$

then first choosing t_0 small and then $|\kappa|$ small makes M' have sign $-\operatorname{sign}\kappa$ throughout the interval. The constant C is unrestricted in both assertions.

Proof. Let $\phi = (t\omega)'$. Direct differentiation gives $\phi' = -rt^{\kappa-1}$. On $0 < t < e^{-6}$, $1 \le \phi \le 16t^{-1/8}$. At any zero of M',

$$T = W\phi - R', \qquad |W| \ge \frac{L}{2(1+\phi)}$$

once $C_* t_0^{1/2} < 1/2$. Thus $W \neq 0$ at such a zero and

$$\frac{|R''|}{|W| r t^{\kappa-1}} \le \frac{2C_*}{r}(1+\phi)t^{1/2-\kappa} \le CC_* t^{23/64} < 1/2$$

after a second common reduction. Every zero of M' is therefore simple and crosses with sign W. There can be at most one such crossing; Rolle's theorem proves the two-zero bound, including multiplicity. If $W = 0$, the estimate on R' excludes a zero of M'.

When $TW \le 0$, $|T - W\phi| \ge L > |R'|$, so two zeros are impossible. If $W > 0$, the unique possible critical point is a minimum. Two zeros force strict decrease to the first of them and hence $M(0+) > 0$. For $W < 0$ all signs reverse. The integrable bound on R' supplies $R(0)$ and proves (7.37). The last assertion is the ϕ estimate in the proof of Lemma 7.12(ii), applied to $M' = T - W\phi + R'$. □

The coefficient columns near ratio one. To apply the scalar lemma, we must control its coefficient norm $|T| + |W|$ by the center generators. Work on the same inverse box as above. Transversality of T to the reversible component and an exact factorization of W give the required determinant margin. At the reversible intersection, let $k_* = G'_+(0) = G'_-(0) > 0$. There $h_+ = h_- = 1/(2\beta)$ exactly. Put

$$A_c = k_*(1/(2\beta) - \epsilon) > c > 0.$$

The coefficient T has $\partial_{\xi_2} T = 2A_c$ there. Moreover the functions τ_1, σ_1 and σ_s/ϵ vanish on both center axes, so their divisions have bounded columns:

$$\begin{aligned} \tau_1 &= a_\tau \xi_2 + b_\tau \kappa \xi_1 + c_\tau \nu, \quad a_\tau > c > 0, \\ \sigma_s/\epsilon &= -a_s \xi_2 + b_s \kappa \xi_1 + c_s \nu, \quad a_s > c > 0, \\ |\tau_1| + |\sigma_1| &\le C(I + |\nu|). \end{aligned} \tag{7.38}$$

The positive a_τ is the inverse of the positive transverse column in (5.35), continued with its coefficient derivative margin. We verify the second direction from the original upper field.

The coordinates of Lemma 5.8 are affine at zero scale, so the first-order calculation in Appendix C.1 applies. Absorbing their positive linear section factors in χ_{loc}^2, it gives on $\kappa = \nu = 0$

$$\sigma_s = -\chi_{\mathrm{loc}}^2 \epsilon \bar{a}[\bar{m}^{5/2} + O(\epsilon)], \qquad \chi_{\mathrm{loc}} > c.$$

The factors ϵ and $\bar{a}$ are exact, by the affine zero-scale field and the reversible identity. The finite coordinate Jacobian gives $\partial_{\bar{a}} \xi_2 > c$ at fixed κ. Hence the ξ_2 column is strictly negative at the intersection. The integral division, with the uniform second coefficient jets, retains that sign on a smaller box and proves (7.38).

The W column has an exact positive factor on the finite-scale coefficient box. Define

$$Q_* = z_1(rh_- - \epsilon) + \epsilon z_0 r J_- > c > 0, \qquad Q^* = u_1(h_+ - \epsilon r) + \epsilon u_0 J_+ > c > 0.$$

Exactly, $P = u_0 Q_*$ and $N = z_0 Q^*$. On $\xi_2 = \nu = 0$ write $\sigma_s = \epsilon b_s \kappa \xi_1$. Then

$$W = \kappa \xi_1 [Q_* + 2\epsilon^2 r b_s \xi_1^2]. \tag{7.39}$$

The bracket has a uniform positive margin. Remove ξ_2, ν by single integral division. On $\nu = 0$ this gives

$$\begin{pmatrix} T \\ W \end{pmatrix} = \begin{pmatrix} a_T & b_T \\ a_W & b_W \end{pmatrix} \begin{pmatrix} \xi_2 \\ \kappa\xi_1 \end{pmatrix}, \qquad a_T > c, \quad b_W > c, \quad |b_T| \le C.$$

The exact formula for W also gives $|a_W| \le C|\kappa| + C\epsilon^2(|\xi_1| + |\xi_2|)^2$. Consequently the determinant has a positive margin after reducing the box; restoring ν and using (7.36) yields

$$cI \le |T| + |W| \le CI \tag{7.40}$$

on every noncenter fibre with a closing zero. The division uses the coordinates κ, ξ_1 themselves; their product $\kappa\xi_1$ is a generator.

Critical points on the deficit domain. The deficit inequalities now constrain a critical point of M. This step uses $u, z > 0$ and therefore applies on the comparison interval, rather than on the whole auxiliary box.

Lemma 7.18. *Near ratio one, under (7.36) on a noncenter fibre, every zero of M' in the comparison interval has $u_0 z_0 > 0$.*

Proof. By (7.34) and $1 - \kappa(t\omega)' = rt^\kappa$,

$$M' = rPt^\kappa - N - \epsilon u_0 z_0 \sigma_s Q' + \mathcal{R}_3'.$$

Suppose $u_0 z_0 \le 0$, and put $S(t) = |u_0|t^\kappa + |z_0|$. The principal terms have the same sign and modulus at least $cS(t)$; the noncenter assumption and forcing exclude $S = 0$. If $u_0 < 0$, the condition $u > 0$ gives $|u_0| \le Ct$. If $z_0 < 0$, $z > 0$ gives $|z_0| \le Cv \le Ct^r$. In either case

$$|u_0 z_0| \le CtS(t).$$

Since $|Q'| \le Ct^{-1/8}$, its curvature contribution is bounded by $C\epsilon|\sigma_s|t^{7/8}S(t)$. Also $I \le C(|u_0| + |z_0|)$ and $S(t) \ge t^{1/64}(|u_0| + |z_0|)$, whence

$$|\mathcal{R}_3'| \le CIt^{3/4} \le Ct^{47/64}S(t).$$

A common small t_0 makes both errors less than half the principal margin. Therefore M' cannot vanish. □

Ratios separated from one. Away from ratio one, a power comparison replaces the compensator. We must also recover the signs of the original coefficients: the curvature crossing alone does not determine them. After reflection it is enough to fix $r \ge 1 + \delta$, $\delta > 0$, on a compact ratio interval. Put $I_a = |u_0| + |z_0|$. The triangle-preserving coordinates give the unsheared equation

$$V_U = -rV/U - V^2 R(U, V), \qquad R = \epsilon(u_0 R_u + z_0 R_z + \nu R_\nu),$$

with bounded finite jets. Its exact reciprocal integral yields

$$v = t^r e^w, \quad |D^j w| \le C\epsilon(I_a + |\nu|)t, \quad v' \asymp t^{r-1}, \quad v'' \ge c\, v'/t.$$

Indeed $t^r \int_t^1 s^{-r}\, \mathrm{d}s \le Ct$, and its finitely many Euler derivatives have the same bound. The differentiated integral equations have small integral coefficients; iteration retains these bounds. Fixed coefficient logarithms are absorbed with a portion of the positive margin δ. No near-resonant shear is used here.

Single-center division of the absolute closing difference gives $|\nu| \le \delta_\epsilon I_a$ on every fibre with a closing zero. Direct integral division of (7.33) then gives, on the full positive t interval of the auxiliary extension,

$$M' = u_0 v' \mathcal{A} - z_0 \mathcal{B} + \nu v \mathcal{C}, \tag{7.41}$$

where

$$\mathcal{A}, \mathcal{B} > c > 0, \quad |\mathcal{A}'| \le C, \quad |\mathcal{B}'| \le C(1 + |u_0|/t), \quad |(v\mathcal{C})'| \le Cv/t.$$

These bounds follow by separating the two unmixed deficit terms from their mixed connector term. On $u_0 = z_0 = 0$, the numerator is $tv\nu$ times a bounded coefficient, by division from the exact triangle identity. On $z_0 = 0$, divide M by v and then by u_0; its endpoint coefficient is $(rh_- - \epsilon)z_1 > c$. On $u_0 = 0$, divide by t and then by z_0; its endpoint coefficient is $-(h_+ - \epsilon r)u_1 < -c$. The remaining mixed term has its constant $\epsilon\kappa u_0 z_0$ explicitly removed. Its first Euler derivatives are $O(t)$ and its second ordinary derivative is $O(1/t)$. The bounded connector derivatives and the preceding bounds on w prove these estimates term by term in (7.33); integral division in u_0, z_0 preserves them. Differentiating the two unmixed terms gives (7.41); the mixed second derivative accounts for $|u_0|/t$ in $\mathcal{B}'$. Ratios of v at points of a coefficient integration segment are positive bounded units, since r is held fixed and the corresponding difference of w is $O(\epsilon(I_a + |\nu|)t)$.

At a zero of M', its two principal terms must balance. The νv term is smaller than their sum since $v \le Ctv'$ and $v \le Ct$. Thus

$$u_0 z_0 > 0, \qquad |z_0| \asymp |u_0| v' \asymp |u_0| t^{r-1}. \tag{7.42}$$

At such a point, $M'' = u_0 v'' \mathcal{A}$ with relative error $O(t + |u_0| + \delta_\epsilon t)$. All zeros of M' are simple and cross with sign u_0, so there is at most one. This controls the auxiliary numerator. Actual returns remain subject to the strict deficit conditions.

Suppose M has two zeros in the comparison interval. The auxiliary interval contains both zeros and extends to $t = 0$. If u_0, z_0 were both negative, its only possible critical point would be a maximum, whereas $M(0+) = \epsilon\kappa u_0 z_0 > 0$. Such a function has at most one positive zero. Hence

$$u_0 > 0, \quad z_0 > 0, \quad z_0 \le C u_0 t_*^{r-1} \le C u_0 t_0^{\delta} \tag{7.43}$$

for a critical point t_* between the two zeros. In particular, after reducing t_0, $\xi_2 = (u_0 - z_0)/2 > 0$.

To recover τ_1 with a uniform sign, work first at $\epsilon = \nu = 0$ on $z_0 = 0$. Let $q = q_l = Q_s$ be the common incoming root. Its lower and upper equations are

$$\beta q^2 + \bar{a} q = 1, \qquad \bar{m} q^2 - \bar{c} q = 1.$$

The outgoing intersections are $p_l = 1/(\beta q)$ and $P_s = 1/(\bar{m} q)$. Therefore exactly

$$\tau_1 = (\bar{m} - \beta) q, \qquad \delta_+ = p_l - P_s = \frac{\bar{m} - \beta}{\beta \bar{m} q}. \tag{7.44}$$

Since $u_0 = \chi_+(\delta_+)$ with $\chi_+' > c$, these formulas give $\tau_1 = A_0 u_0$ on this surface, where $A_0 > c$ has bounded jets. In particular the derivative in $\bar{m} - \beta$ is a positive unit at the center, even when the actual ratio is used as the remaining coordinate.

In the common inverse coordinates define, at positive scale, $A_\epsilon = \int_0^1 \partial_{u_0} \tau_1(s u_0, 0, r, 0; \epsilon)\, \mathrm{d}s$. The exact triangle identity makes the omitted constant zero. Convergence of the lower strong intersections and the upper low-layer jets to their zero-scale values gives $|A_\epsilon - A_0| \le C\epsilon(1 + |\log \epsilon|)$. Single division in z_0, ν now gives

$$\tau_1 = A_\epsilon u_0 + B_\epsilon z_0 + C_\epsilon \nu, \qquad A_\epsilon > c, \quad |B_\epsilon| + |C_\epsilon| \le C.$$

Use (7.43) and $|\nu| \le \delta_\epsilon(u_0 + z_0)$. Choose t_0 with Ct_0^δ small first, then the scale with $C\delta_\epsilon$ small. The two errors are absorbed by $A_\epsilon u_0/2$, and hence $\tau_1 > 0$. The lower and upper intersection equations, together with the critical-point balance, therefore determine the coefficient direction. The same argument applied to $\bar m - \beta$ gives $\bar m - \beta > 0$: on the zero-scale surface in (7.44) its quotient by u_0 is also a positive bounded unit, and it vanishes on the exact positive-scale triangle axis. On the compact center part with $r \ge 1 + \delta$, one has $\bar c = -\bar a < -c_\delta < 0$; the actual ratio coordinates preserve this margin on a smaller neighborhood. Therefore the original primary generator has $\sigma_1 = \bar c(\bar m - \beta) < 0$ on this nonresonant two-zero branch. All ν terms in these divisions are $O(|\nu|)$ and are absorbed by the stated forcing; no extra ϵ factor is assigned to the ν dependence of the ratio.

The local calculation gives an upper bound of four closing zeros. It also identifies the only branch on which the numerator can have two zeros. The signs on that branch will be used at the already fixed regular seam in Section 12; they do not yet determine the value of $\mathcal{K}$ at that seam.

Proposition 7.19 (Local count and the two-zero terminal branch)**.** *On common small neighborhoods of every compact triangle-center stratum, the closing equation has at most four isolated zeros on* (7.31)*, with multiplicity. More precisely, on a noncenter fibre satisfying the local forcing bound supplied by one closing zero,*

$$Z(M) \le 2, \qquad Z(\mathcal{K}) \le 3, \qquad Z(F) \le 4.$$

If a terminal subinterval contains two zeros of M, counted with multiplicity, orient the whole field by reflection if necessary so that $r > 1$. Then

$$u_0 > 0, \qquad z_0 > 0, \qquad \xi_2 > 0, \qquad \tau_1 > 0, \qquad M(0+) > 0, \qquad \mathcal{K}(t) \longrightarrow -\infty \quad (t \downarrow 0). \tag{7.45}$$

Near ratio one the same branch also has $T > 0$, $W > 0$ and $\sigma_s < 0$; away from one it has $\bar m > \beta$ and $\sigma_1 < 0$. In particular $t = 0$ is the end of the deficit comparison interval in this branch. It need not be an end of the positive-central return domain. If a seam t_s in the comparison interval is fixed away from the zeros and its tail contains exactly these two zeros, then $M(t_s) > 0$, $\mathcal{K}_t'(t_s) > 0$, and $\mathcal{K}_p'(t_s) < 0$ in the outward coordinate $p = P_s - \psi_+(t)$, $\psi_+' > 0$. These conclusions do not assume or provide a sign for $\mathcal{K}(t_s)$.

Proof. Near one, (7.35) and (7.40) satisfy Lemma 7.17. Away from one, the single crossing established above gives $Z(M) \le 2$. On the comparison interval, $M = tuz\,\mathcal{K}'$ with $tuz > 0$, and $F' = D_+'(u)G_+'(t)(1 - e^{\mathcal{K}/\epsilon})$. Two applications of Rolle's theorem with multiplicity therefore give the displayed bounds. If there is no closing zero, the bound for F is immediate. If forcing gives the exact center, there is no isolated closing zero.

The away-from-one terminal signs were proved in (7.43) and (7.44). For the near-one signs, first restrict to a smaller ratio interval so that its reflected ratio remains in $|r - 1| \le 1/64$. Fix a large K_0 and separate the cone $|\xi_2| \le K_0|\kappa\xi_1|$. On this cone, the exact positive column (7.39), the small ξ_2 column, and relative forcing give

$$\frac{W}{\kappa\xi_1} > c, \qquad |T/\xi_1| \le C|\kappa|, \qquad |\mathcal{R}_3'/\xi_1| \le C|\kappa|t^{1/2}.$$

Thus the one-zero part of Lemma 7.17 applies to M/ξ_1. For $\kappa = 0$ this cone is the reversible center. Neither case permits the two-zero branch.

In the complementary cone choose K_0 large and then reduce the coefficient box so that T has the sign of ξ_2. Two zeros of M give a zero of M' in the deficit interval, so Lemma 7.18 implies $u_0 z_0 > 0$. The case $\kappa = 0$ would have $C_0 = 0$, contrary to (7.37). For $\kappa > 0$, $C_0 > 0$,

so that same lemma gives $W > 0$ and $T > 0$, hence $\xi_2 > 0$. Equation (7.38), this cone and forcing imply $\tau_1 > 0$ and $\sigma_s < 0$. The exact formula

$$W = \kappa u_0 Q_* + 2\epsilon u_0 z_0 \sigma_s r$$

then excludes $u_0, z_0 < 0$: both terms would be negative. Thus both deficits are positive.

Reflection exchanges the lower legs and replaces the actual upper passage by its inverse. The new ratio is $1/r$, the new critical function is $-\mathcal{K}$ composed with the inverse input, and its derivative numerator is a nonzero signed unit times the old one. It preserves zeros with multiplicity and the comparison interval. Apply the preceding argument in that complete reflected coordinate choice when $r < 1$; no inequality is transported while keeping the old entry coordinate fixed.

Positive endpoint deficits now place all sufficiently small $t > 0$ in the comparison interval. Both complete central outputs must still be positive for an actual return. At fixed coefficients, the lower derivatives at u_0, z_0 are finite and nonzero, while $v' \sim rt^{r-1}$. This proves the endpoint limit for $\mathcal{K}$ in (7.45). Finally $M(0+) > 0$ and exactly two zeros before the fixed seam, counted with multiplicity, imply $M(t_s) > 0$ by parity. The positive multiplier and $dp/dt < 0$ give the two derivative signs. The remaining step is to determine $\mathcal{K}(t_s)$ from the regular overlap. Proposition 12.5 combines that value with the derivative signs and endpoint parity to count zeros on the whole union. The local estimate alone leaves the four-zero closing bound. □

8 Nonparabolic infinite endpoint configurations

We now count returns for which at least one ray endpoint tends to infinity. The upper passage can approach a strict real pair, a repeated pair or a weak boundary. With one finite endpoint, its actual intersection determines the moving slow input. With two infinite endpoints, horizontal phases expose the exactly equal lower powers. Canceling these powers leaves the relative terms bounded while the constant contraction actions may diverge independently.

The resulting comparisons give local bounds on the whole passing domains. The weak lower-loss charts also retain the entrance signs needed to sharpen their four-zero bounds in Section 12. The joint repeated center requires the common domain of Section 9; the remaining parabolic corner is treated at the second scale in Section 10.

8.1 One finite endpoint and one infinite endpoint

Label the infinite endpoint by $q \to \infty$ and the finite outgoing input by $p \to p_* > 0$. If this labeling reverses the fixed-field direction, both signs of the resulting ℓ are retained on the same actual positive-central passing set; reflection is transferred back by the inverse return germ of Section 2. The lower restrictions are those of Lemma 7.4. The upper alternatives are a strict real pair with $\bar{m} < 0$, a repeated pair $\bar{m} = -\gamma^2$, $\bar{c} = 2\gamma > 0$, or the weak boundary $\bar{m} = 0$, $\bar{c} > 0$. In the repeated case $p_* \geq 1/\gamma$ is retained, since the strict barrier does not exclude these larger inputs.

At a regular finite lower endpoint, incoming contraction dominates any polynomial growth of the upper derivative in the large exit. The next lemma gives this polynomial bound through a repeated eigenvalue. Lower losses will then require a comparison of the two slow rates.

Lemma 8.1 (Upper expansion with negative $\bar{m}$)**.** *On a compact coefficient block with $\bar{m} < -m_0 < 0$ and $\bar{c} > c_0 > 0$, and a compact interval of positive finite inputs, every actual upper return with sufficiently large exit q satisfies $0 < q_p \leq C(1+q)^N$. The constants are uniform for small scale and either sign of ν, including at zero discriminant.*

Proof. Put $Z = Y + \epsilon$, $\theta = \epsilon\bar{B}$, $d = \bar{c} + \nu$ and

$$g(Z) = dZ - \epsilon(\bar{c} + 2\nu) + \nu\epsilon^2/Z, \quad U = \frac{Z^{1-\theta}}{1-\theta}, \quad W = Z^{-\theta}(X + g(Z)).$$

Divide time by Z. Differentiating these coordinates in the original upper field gives

$$U' = W, \qquad W' = -G(U) + f(Z)W,$$

where

$$f(Z) = (1-2\theta)d + \frac{\epsilon\bar{a} + 2\theta\epsilon(\bar{c} + 2\nu)}{Z} - \frac{(1+2\theta)\nu\epsilon^2}{Z^2},$$
$$G(U) = Z^{-\theta-1}Y[1 + \epsilon\bar{a}e_Y - (\bar{m} + \theta e_Y^2)Y], \qquad e_Y = \bar{c} + \nu Y/Z.$$

Since $|f| \le C$ and $G > 0$ for $Y > 0$, every zero of W is a downward crossing. A returning arc therefore has one height maximum. For large Z, separate the linear part:

$$\begin{pmatrix} U \\ W \end{pmatrix}' = \mathcal{L}\begin{pmatrix} U \\ W \end{pmatrix} + \begin{pmatrix} 0 \\ \mathcal{E} \end{pmatrix}, \qquad \mathcal{L} = \begin{pmatrix} 0 & 1 \\ -k & \tau \end{pmatrix}, \quad k = -(1-\theta)(\bar{m} + \theta d^2) > 0, \quad \tau = (1-2\theta)d > 0,$$

where $|\mathcal{E}| \le CZ^{-1}(|U| + |W|)$. The eigenvalues of $\mathcal{L}$ have uniformly positive real parts, since $k, \tau > 0$ on this compact block. The Lyapunov integral

$$\mathcal{S} = \int_0^\infty e^{-\mathcal{L}^{\mathsf{T}}t} e^{-\mathcal{L}t}\, dt$$

gives a positive matrix with uniform bounds for $\mathcal{S}$ and $\mathcal{S}^{-1}$, also at zero discriminant, and $\mathcal{L}^{\mathsf{T}}\mathcal{S} + \mathcal{S}\mathcal{L} = I$. Choose a fixed large Z_0. Then $\mathcal{V} = (U, W)^{\mathsf{T}}\mathcal{S}(U, W)$ satisfies $\mathcal{V}' \ge c\mathcal{V}$ on $Z \ge Z_0$.

On each monotone-height portion below Z_0,

$$\left| \frac{d\sqrt{1 + W^2}}{dU} \right| = \left| \frac{-G + fW}{\sqrt{1 + W^2}} \right| \le C.$$

The positive margin of the initial value $W = \epsilon^{-\theta}p$ bounds the outgoing time in a small initial strip. Away from its bottom, G has a positive margin: whenever positive W is small, $W' < -c$, so that such a visit has bounded duration and cannot be repeated before the height maximum. Otherwise W stays away from zero. On the incoming low strip the preceding inequality gives $|W| \ge cq - C$, so its time is $O(1/q)$. For large q the arc must reach Z_0, since the same inequality would bound q if it stayed below it. At its first crossing $\mathcal{V} \ge c$, and at its last crossing $\mathcal{V} \le C(1 + q^2)$. Exponential growth on the high part and the low-part bounds yield $T \le C(1 + \log(1 + q))$. The divergence of the (U, W) equation is f. Its section determinant formula gives

$$q_p = \frac{p}{q} \exp \int_0^T f\, dt \le C(1 + q)^N.$$

The endpoint factor $\epsilon^{-\theta}$ cancels exactly. The quadratic estimate uses no inverse eigenvalue difference, which gives the required uniformity at the repeated eigenvalue. □

Lemma 8.2 (A coarse count for attracting collar cycles)**.** *Suppose every periodic orbit in the collar whose ray labels lie between two retained label bounds has return multiplier less than one. Require the same property for every periodic orbit nested between two of them, without assuming isolation beforehand. Then there are at most four such cycles, all hyperbolic attracting.*

Proof. The family (2.3) has at most four affine equilibria. For $\ell \neq 0$, elimination using $x = -y(dy + c)/(1 + y)$ gives $yR(y) = 0$, where

$$R(y) = -(1 + y)^2 + By(dy + c)^2 + my(1 + y)^2 - a(dy + c)(1 + y).$$

This has degree at most three and $R(0) = -(1 + ac) \neq 0$; there is no equilibrium on $y = -1$, since $Q(x, -1) = \ell$. The elimination divides only by $1 + y$, so a zero of $dy + c$ elsewhere simply gives $x = 0$ and is retained. For $\ell = 0$, the branches $y = -1$ and $x = -cy$ give respectively $Bx^2 + ax + 1 + m = 0$ and $y[(Bc^2 + m)y - (1 + ac)] = 0$. Each has at most two roots, since $1 + m$ and $1 + ac$ have positive margins. The origin lies inside every counted cycle, leaving at most three equilibria between them.

Two nested collar cycles bound a compact affine annulus in the collar; the connected southern hemisphere lies outside it. If the annulus has no equilibrium, the boundary of the inner attracting cycle's basin is a nonempty compact invariant subset of its interior, separated from both attracting boundary collars. The Poincare–Bendixson theorem [5, Chapter 7] gives a periodic orbit on that boundary. It cannot attract a neighborhood, since that would make its basin boundary point interior to a disjoint basin. Its intermediate ray labels contradict the hypothesis. Hence the disjoint annuli between any finite ordered selection of cycles each contain an equilibrium. There are at most three such equilibria and therefore at most four cycles, even without an initial finiteness assumption. □

Proposition 8.3 (A regular finite arc with negative $\bar{m}$)**.** *If $\bar{m} < 0$, $\bar{c} > 0$ and $F_+ > 0$ on $[0, p_*]$, then near this one-infinite configuration all counted cycles are simple and attracting and there are at most four of them. On each common positive-central passing interval the ratio of incoming to outgoing central derivatives is less than $1/2$, so $K < 0$ and there is at most one closing zero on that interval. The coarse bound of four does not require connected saturation. The statement holds for either sign of ℓ.*

Proof. Choose a fixed incoming crosscut R before its first root with $A_-(R) > A_{\text{out}} + 3$. Such a choice is possible: the action diverges at a first root, and if no root occurs it diverges as $R \to \infty$, since $F_-(R) \leq C(1 + R + R^2)$. Lemma 7.9 applies on the actual common positive-central passing set. Its power $[(1 + R^2)/(1 + q^2)]^{1/(24\epsilon)}$ absorbs the fixed polynomial from Lemma 8.1. The ratio of the complete incoming and outgoing central derivatives is therefore less than $1/2$. At any periodic closing point this is the return multiplier, including one not initially assumed isolated. By the ray ordering of Lemma 2.8, an intermediate collar cycle has input in the same finite interval and exit above the same large q threshold. Thus every intermediate cycle satisfies the multiplier estimate, and Lemma 8.2 gives the bound of four across all passing components. On each common passing interval, $K < 0$ gives the stated one-zero bound directly. □

8.2 A strict real upper pair and a finite lower loss

When the finite lower arc loses its exit, its contraction can compete with the upper passage. We compare their logarithmic rates, beginning with $\bar{m} < 0$, $\bar{c} > 0$ and $\bar{c}^2 + 4\bar{m} > 0$. In the projective field (4.39) put

$$b = 1 - \epsilon\bar{B}, \quad d = \bar{c} + \nu, \quad \Delta = \sqrt{d^2 + 4b\bar{m}}, \quad \gamma_s = \frac{d(1 - 2\epsilon\bar{B}) - \Delta}{2b}, \quad r_s = \gamma_s/\Delta > 0.$$

At $R = 0$, $\varphi = (-d - \Delta)/(2b)$ the eigenvalues are $-\gamma_s, \Delta$. Its strong intersection is $P_s = p_c + O_N(\epsilon(1 + |\log\epsilon|)) + \nu O_N(1)$, and its returning side is $p < P_s$. The analytic invariant axes, constructed on a small projective rectangle with fixed hyperbolic margins and continued through the compact arc and low layer, separate the returning quadrant from the adjacent node. Axis normalization and the reciprocal integral give the complete radial output

$$s(t) = kt^{r_s}e^{w_s(t)}, \qquad k > c, \qquad |D^i\partial_{\mathbf{b}}^{\alpha} w_s| \leq C_N t^{\alpha_0}.$$

Here $D = t\partial_t$ and a fixed $\alpha_0 > 0$ is smaller than half the minimum of $1, r_s$ on the chosen block. Integrals of $t^{r_s}s^{-r_s}$ and their finite logarithmic derivatives prove this bound also when a higher saddle resonance is present; the strict exponent reduction absorbs those logarithms. Fix all section and coefficient derivative orders needed below before choosing the rectangle.

If $\bar{B} > 0$, the incoming half-line is regular whenever the finite outgoing lower arc loses positivity, since then $\bar{a} < 0$. The horizontal preparation is $D_I(s) = K_I s^{h_I/\epsilon} e^{P_I(s)/\epsilon}$, $h_I = b/\bar{B}$, $P_I(0) = 0$, with bounded selected jets of P_I. The bounded amplitude is P_I; its quotient by ϵ need not be bounded. Composing with the upper passage gives

$$J(t) = K_J t^{h_\infty/\epsilon} e^{\Psi(t)/\epsilon}, \quad h_\infty = r_s h_I > c, \qquad \epsilon t J''/J' = h_\infty - \epsilon + E(t), \tag{8.1}$$

where $\Psi = h_I w_s + P_I(s(t))$ after constants have been removed, and $E = D\Psi + \epsilon D^2\Psi/(h_\infty + D\Psi)$. Its prescribed Euler and coefficient jets are $O(t^{\alpha_0})$. The denominator has a positive margin on a fixed smaller section.

Proposition 8.4 (Strict real pair with lower loss)**.** *For $\bar{B} > 0$, a first simple lower root at p_c has local closing count three. A double lower root at p_c also has count three; a double lower root strictly before p_c has count two. The bounds include multiplicity and either sign of ℓ.*

Proof. At a simple root choose its regular power input as $u = \delta + t$, where $t > 0$ is the upper returning coordinate. The exact splitting is the difference of the two strong intersections in this coordinate. Write $\lambda_+ = h_+/\epsilon$ and $g = h_+ - h_\infty$. The physical interval has $t > 0$, $\delta + t > 0$, and positive central outputs. The logarithmic critical function has the full cleared numerator

$$M = \delta(h_\infty - \epsilon) - gt + (\delta + t)E(t) - \epsilon t(\delta + t)R_+(\delta + t), \qquad M = t(\delta + t)K'.$$

Here R_+ is the bounded ordinary correction to $D_+''/D_+' = (\lambda_+ - 1)/u + R_+(u)$.

On the zero-scale matching surface, with $p = p_c$, the exponent difference is

$$g_0 = \frac{p^2(\bar{B} + \bar{m})}{\bar{B}(1 - \bar{B}p^2)(1 + \bar{m}p^2)}.$$

Indeed the lower exponent is $p^2/(1 - \bar{B}p^2)$ and the incoming exponent is $-\bar{m}p^2/[\bar{B}(1 + \bar{m}p^2)]$. At $\bar{m} = -\bar{B}$, $\bar{a} = -\bar{c}$, the Jacobian of $(\delta_{\mathrm{raw}}, g)$ in $(\bar{a}, \bar{m})$ tends to $p^4/[\bar{B}(1 - \bar{B}p^2)^3] > 0$. This follows by multiplying $\partial_{\bar{a}}\delta_{\mathrm{raw}} = p^2/(1 - \bar{B}p^2)$ by the derivative of g_0 along the matching surface. The input change multiplies this determinant by a positive unit. Thus (δ, g, ν) give uniform inverse coefficient boxes. The second real center of Proposition 5.1 lies exactly on $\delta = g = \nu = 0$: its actual maps agree up to the common strong endpoint, and comparison of their positive leading powers forces $\lambda_+ = r_s h_I/\epsilon$ there. Conversely the nonzero Jacobian makes this the local center axis.

The center identity removes the constant transverse term in the critical numerator. Extend the relative coefficients to the larger box as in Section 5 and divide there to obtain

$$M = \delta A(t) - gtB(t) + \nu t Q(t), \quad A, B > c, \quad |A'| + |B'| \le Ct^{\alpha_0 - 1}, \quad |Q| + |tQ'| \le C.$$

For example first set $\delta = 0$ and divide M/t in g, ν, then integrate its δ derivative. The constant columns are $h_\infty - \epsilon$ and -1, and the differentiated remainders are the Euler-small terms of (8.1). Each absolute lower map has a fixed positive action before its central output. Its finite derivatives are exponentially small; center division and Lemma 5.14 therefore give $|\nu| = o(1)(|\delta| + |g|)$ on a fibre with a closing zero. At a zero of M these estimates force $\delta g > 0$ and $|\delta| \asymp |g|t$; differentiating the displayed division then gives sign $M' = -\operatorname{sign} g$. All zeros are therefore simple crossings in the same direction, so there is at most one. Away from $g_0 = 0$, the undivided numerator gives the same conclusion after a fixed section restriction. Since M is a positive multiple of K', two Rolle steps give three closing zeros, with multiplicity.

For a double root, retain the complete moving slow input $u(t)$, $u' > c > 0$, and $f_L = \mu_L + u^2$. Its rate after the regular input change has numerator $\widehat{A} = A_L(u)u' + \epsilon f_L u''/u' > c$. The whole-path condition on f_L is imposed. The exact critical numerator is

$$M = f_L(h_\infty - \epsilon + E) - t\widehat{A}.$$

At a double root at p_c, $u(0)$ tends to zero. At a zero, $f_L \asymp t$, and differentiation gives

$$M' = 2uu'(h_\infty - \epsilon + E) + f_L E' - \widehat{A} - t\widehat{A}' < 0.$$

Indeed $tE' = O(t^{\alpha_0})$, and all other positive-size errors are small compared with $\widehat{A}$. Hence M has at most one zero. If the double root is strictly before p_c, the limiting entry is negative and $f_L(0)$ has a positive margin, so $M > 0$ on a smaller section. Thus the critical numerator has at most one zero at contact and none in the separated case, giving the asserted counts. The scalar and central-output restrictions are monotone passing conditions, whose intersection is one interval. □

8.3 Repeated upper roots and signed horizontal parameters

At $\bar{c} = 2\gamma$, $\bar{m} = -\gamma^2$ put $P = 1/\gamma$. Section 4.5 provides the exact upper factor $f_U(v) = \mu_U + v^2$, its normally contracting radial map, and the finite entry $v_e(p)$ with $v_e' < -c$. The limiting entry is zero at $p = P$ and negative for $p > P$. Throughout this subsection retain the full test

$$\mathcal{P}(\mu, v): \quad \mu > 0 \quad \text{or} \quad [\mu \le 0 \text{ and } v > \sqrt{-\mu}], \qquad v < v_R. \tag{8.2}$$

This tests positivity on the entire interval to the fixed right exit: a root between two positive endpoint values still blocks passage. For $\bar{B} > 0$, the horizontal power and the regular angular connector give the complete incoming rate

$$J(v) = K_J \exp\left[-\frac{h_I}{\epsilon}\int_v^{v_R} \frac{\beta_U(s,0)}{f_U(s)}\,ds + \frac{\Psi(v)}{\epsilon}\right], \qquad \frac{J''}{J'} = \frac{A_U(v)}{\epsilon f_U(v)}. \tag{8.3}$$

Here $h_I = (1 - \epsilon\bar{B})/\bar{B}$ and

$$N = h_I\beta_U + f_U\Psi', \qquad A_U = N - \epsilon f_U' + \epsilon f_U N'/N > c.$$

The relative integral of Section 4, composed with independent normal inputs, bounds the prescribed mixed jets of Ψ, the regular connector terms and the rate coefficients. Any section-independent divergent contraction remains in K_J. The input-dependent normal output enters only through its bounded absolute jets.

Proposition 8.5 (Repeated upper root away from the joint center)**.** *Suppose $\bar{B} > 0$, the finite lower polynomial loses positivity at or before $p_* \ge P$, and $(\bar{B}, \bar{a}) \ne (\gamma^2, -2\gamma)$. The local closing count is at most three, and at most two unless a simple lower root lies at $p_* = P$.*

Proof. For a simple root at P, use its positive power input t. The upper input $v(t)$ has $v' > c$ and tends to zero at the limiting endpoint. Clearing the critical derivative gives

$$M = tC - f_U(v)H_o, \qquad C = A_U(v)v' + \epsilon f_U(v)v''/v' > c, \quad H_o = h_+ - \epsilon + \epsilon t R_+(t) > c.$$

Since t, v, f_U are small, the bounded jets and $H_o' = O(\epsilon)$ give $M' > c$ on a fixed smaller interval. It has at most one zero. If a simple or double lower root lies at $p_* > P$, its rate has vanishing denominator while the upper entry factor has a positive margin. After multiplying by the lower denominator, the critical numerator is strictly negative. If $p_* = P$ and a double lower

root lies strictly before P, the lower entry factor instead has a positive margin, and the upper rate dominates with a strictly positive numerator. These cases give two closing zeros.

It remains to compare a double lower root $r = 1/\sqrt{\bar{B}}$ and the repeated upper threshold P when both precede p_*. The entries then stay a fixed distance beyond their slow minima. Subtract the divergent actions, which are constant in p, before differentiating. The limiting logarithmic rates in this coordinate are

$$\epsilon\partial_p \log|D_{+,p}| \longrightarrow -\frac{p}{\bar{B}(p-r)^2}, \qquad \epsilon\partial_p \log|J_p| \longrightarrow -\frac{p}{\bar{B}(p-P)^2}.$$

For the first, the moving lower rate at its regular entry is $-A'_+(p)$. For the second take the exact affine upper field with $\bar{m} = -\gamma^2 - \omega^2$. At fixed $p > P$ its return has

$$T = \pi/\omega - 1/[\gamma^2(p-P)] + o(1), \qquad q = (p-P)e^{\gamma T}(1+o(1)).$$

Thus $\partial_p \log q \to p/(p-P)^2$, and the incoming horizontal rate multiplies it by $-1/\bar{B}$. To pass from the affine calculation to the original maps, use the positive margins of the reduced denominators on these entry intervals and the common integrable majorants for the differentiated relative normal integrals from Section 4. At the repeated limiting field its exact $1/V$ normal coordinate makes the upper relative logarithm zero, as established in Section 4.5. The connector's nonconstant terms are evaluated at a normal output tending to zero. These facts prove convergence in the finitely many ordinary input jets used here, without differentiating the divergent constant action. Subtracting the two displayed rates gives

$$\frac{p(r-P)(2p-r-P)}{\bar{B}(p-r)^2(p-P)^2},$$

whose sign is nonzero. Equality $r = P$ is exactly the excluded joint center. The strict rate difference gives two closing zeros. All the counts are on the intersection of the whole-path conditions and the positive central-output interval, and include multiplicity. □

When $r = P$, neither rate dominates. Both signed gaps must then be retained in the joint-center chart of Section 9. We next cover $\bar{B} = 0$, where replacing the horizontal factor by a power with exponent $1/\bar{B}$ would not be legitimate. Here a finite lower loss is simple and $\bar{a} < 0$. The exact phase of Proposition 4.1 has $F'_I > c$ near zero. Set $u = F_I(x) - \bar{B}$ and divide time by the positive unit $xF'_I(x)/u$. This produces

$$\dot{u} = \epsilon u(\bar{B} + u), \qquad \dot{Z} = -Z\widetilde{\beta}(u, Z), \qquad \widetilde{\beta},\ \widetilde{\beta} + Z\widetilde{\beta}_Z > c.$$

The unit extends through $u = 0$ and both signs of $\bar{B}$. The exact normal input supplies

$$\frac{D''_I}{D'_I} = \frac{A_I(u)}{\epsilon u(\bar{B}+u)}, \qquad A_I > c, \qquad u > \max(0, -\bar{B}).$$

The moving-entry formula bounds the ordinary and mixed jets of A_I. This exact quadratic factor remains valid through $\bar{B} = 0$ without dividing by $\bar{B}$.

For a strict upper pair write the input as $u = u(t)$ and put $\mathcal{R} = tu'/u = r_s + O_D(t^{\alpha_0})$, $tu''/u' = r_s - 1 + E_1$, $E_1 = O_D(t^{\alpha_0})$. With the simple lower input $\delta + t$, set

$$Q = A_I(u)\mathcal{R} + \epsilon(\bar{B}+u)(r_s - 1 + E_1) > c, \quad H_o = h_+ - \epsilon + \epsilon(\delta+t)R_+(\delta+t) > c.$$

The full numerator is

$$M = (\delta+t)Q - t(\bar{B}+u)H_o.$$

Its clearing factor is positive on $t > 0$, $\delta + t > 0$, $u > \max(0, -\bar{B})$. At a zero $(\delta+t)/t \asymp \bar{B} + u$ is small. Since $|tQ'| \le Ct^{\alpha_0}$, $tu' = u\mathcal{R}$ and $H'_o = O(\epsilon)$, differentiation gives $M' = Q + o(1) > c/2$

there. The same-direction crossing argument gives at most one numerator zero and three closing zeros, for both signs of nearby $\bar{B}$.

For a repeated upper pair let $u = u(v)$ include its regular angular connector. Write

$$f(v) = \mu_U + v^2, \quad S(v) = \bar{B} + u(v), \quad L(v) = fu_v/u > c.$$

The input-independent normal contraction is separated first. Its relative formula and the bounded absolute jets of u show that L and its needed ordinary jets are bounded. The exact incoming rate is

$$\mathcal{B} := \epsilon f S \frac{J_{vv}}{J_v} = A_I(u)L + \epsilon S(L - f' + fL'/L) > c.$$

In a positive finite lower power input t, the regular entry $v(t)$ has $v' > c$. Set $C = \mathcal{B}(v)v' + \epsilon f(v)S(v)v''/v'$ and $H_o = h_+ - \epsilon + \epsilon t R_+(t)$. The numerator and derivative are

$$M = tC - f(v)S(v)H_o, \qquad M' = C + tC' - [2vv'S + uLv']H_o - fSH_o'.$$

Use $fu_v = uL$ before estimating to cancel the apparent singularity in the composition derivative. Fix the finite upper entry neighborhood and its bounded jets first, then shrink t, $|\bar{B}|$ and the normal output u. The result is $M' > c$ on the entire passing interval, hence three closing zeros. Both (8.2) and $u > \max(0, -\bar{B})$ remain imposed. This also covers finite entries strictly beyond P.

8.4 The inverse upper map at the primary weak boundary

At the primary weak boundary, the upper return is best described by its inverse: the absolute output becomes flat, while the scalar factor still records the passing side. Fix $\bar{m} = 0$, $\bar{c} = c_0 > 0$ and $P = 1/c_0$. Use the actual weak parameter and normal form of (4.43) and Section 4.5; the actual weak face is $\bar{m} + \epsilon\bar{B}(\bar{c} + \nu)^2 = 0$. In reversed projective time the exact scalar factor is $f(u) = u(\gamma + u)$ and the fast direction contracts uniformly. The complete inverse upper displacement, including its regular finite connection, is

$$p = \widehat{p} - t(u), \qquad \frac{t'}{t} = \frac{N}{f}, \quad \frac{t''}{t'} = \frac{A_U}{f}, \qquad N, A_U > c, \quad u > \max(0, -\gamma). \tag{8.4}$$

Here $N = \beta_U + f\widetilde{j}'$, $A_U = N - f' + fN'/N$. Their required ordinary and mixed jets are bounded. The absolute output t and its selected derivatives tend uniformly to zero when the input and γ neighborhoods shrink. At a lost exit, extend the absolute output by zero and the relative coefficients by the complete auxiliary flow. Coefficient division uses these extensions rather than an inverse of the flat map.

The incoming horizontal entry $x = \chi(u)$ is the regular connector of Proposition 5.6. Thus

$$\chi(0) = 0, \quad \chi' > c, \quad L = u\chi'/\chi, \quad L(0) = 1, \quad R_\chi = \chi''/\chi'$$

have bounded required jets. Keep the exact raw factor of Proposition 4.1, with no division by $\bar{B}$ or $\bar{a}$. Its complete rate is

$$\frac{D_I''(x)}{D_I'(x)} = \frac{A_I(x)}{\epsilon x F_I(x)}, \qquad A_I = 1 + O_{C^N}(\epsilon), \quad A_I(0) = 1 - 2\epsilon\bar{B}. \tag{8.5}$$

The whole condition is $F_I(s) > 0$ for $\chi(u) \le s \le x_R$. Since $F_I'' > 1$ and $F_I(x_R) > c$, its retained side is the interval to the right of its last root, or the whole input interval if there is no root. This remains valid at the cubic boundary $\bar{B} = \bar{a} = 0$, where checking the entry value alone would not exclude a later root.

If an incoming finite double root is separated from the horizontal input, put fixed sections on both sides of it. Its whole-passage condition is a positive minimum, independent of u. The relative normal integral separates its possibly divergent contraction into an input-independent linear factor. Estimate the horizontal output first and then compose the bounded relative unit. This preserves (8.5) and its required ordinary input jets. The divergent constant is not differentiated in coefficients. If the whole-passage condition fails, the corresponding input set is empty.

Proposition 8.6 (A regular outgoing arc at the weak boundary)**.** *At $\bar{m} = 0$, $\bar{c} > 0$, $p_* = P$, $q_* = \infty$, a regular finite outgoing lower arc has local closing count three. An incoming finite double root, either sign of ℓ, and the reflected case are included.*

Proof. The regular outgoing deviation $D_o(p)$ has $R_o = -\epsilon D_o''/D_o' > c$ and bounded needed ordinary input jets, by Lemma 7.2. Put $I(u) = D_o(\widehat{p} - t(u))$ and $J(u) = D_I(\chi(u))$. Both have positive derivatives. For $K = \epsilon \log(J'/I')$, direct differentiation gives the exact numerator

$$\begin{aligned} M &= uF_I(\chi)(\gamma + u)K' \\ &= A_I(\chi)L(\gamma + u) - \epsilon F_I(\chi)A_U + \epsilon uF_I(\chi)(\gamma + u)R_\chi - F_I(\chi)R_o(\widehat{p} - t)Nt. \end{aligned} \tag{8.6}$$

Apply Lemma 4.11 with $h = F_I(\chi)$, $A = A_I L$, $R = F_I(\chi)R_\chi$ and $E = F_I(\chi)R_o N$. Their required C^1 bounds follow from the prepared rates and regular outgoing arc; $A = 1 + O(u + \epsilon)$ and $\|t\|_{C^1} \to 0$. It gives $M' > 1/2$ and three closing zeros by Lemma 3.2, on the intersection of the whole upper, horizontal and positive-central passing conditions. A separated double passage imposes its additional coefficient condition. All nearby counted cycles with $q \to \infty$ reach this horizontal crosscut with $u \to 0$ and follow the unique inverse upper transition. □

8.5 Finite lower loss at the weak boundary

The weak upper output may also meet a lower loss. First keep $\bar{B}_0 > 0$. The necessary outgoing alternatives are a first simple root at P, with $\bar{a} = -\bar{c} - \bar{B}/\bar{c}$ and $0 < \bar{B} < \bar{c}^2$; a double root at P, with $\bar{a} = -2\bar{c}$, $\bar{B} = \bar{c}^2$; or a double root strictly before P. Here $\bar{a} < 0$, so the incoming half-line is regular. For a root at P, choose a decreasing lower slow input $v_L(p)$, put $\delta_L = v_L(\widehat{p})$ and replace t by $v_L(\widehat{p} - t) - v_L(\widehat{p})$. This is a positive regular change of the absolute upper displacement; its rates still have the form (8.4) with bounded positive N, A_U. The lower rate is

$$\frac{D_L''(v)}{D_L'(v)} = \frac{A_L(v)}{\epsilon f_L(v)}, \qquad f_L(v) = v \quad \text{or} \quad f_L(v) = \mu_L + v^2, \qquad A_L > c.$$

For a double root impose positivity of f_L throughout its entire slow path. Both inputs now have positive derivatives: $I(u) = D_L(\delta_L + t(u))$ and $J(u) = D_I(\chi(u))$.

Their critical numerator is

$$\begin{aligned} M &= f_L(\delta_L + t)C - Dt, \\ C &= A_I(\chi)L(\gamma + u) - \epsilon F_I(\chi)A_U + \epsilon uF_I(\chi)(\gamma + u)R_\chi, \\ D &= F_I(\chi)A_L(\delta_L + t)N. \end{aligned} \tag{8.7}$$

Its clearing factor is $uF_I(\chi)(\gamma + u)f_L(\delta_L + t) > 0$. The upper inverse curvature contributes $-\epsilon F_I A_U$ to C; it remains in all subsequent rate comparisons.

Proposition 8.7 (Weak boundary with positive $\bar{B}$ and lower loss)**.** *At a root at P the unanchored local bounds are $Z(K') \le 2$, $Z(K) \le 3$ and $Z(F) \le 4$, including multiplicity. A double root strictly before P has closing count three.*

Proof. The coefficients in (8.7) have bounded C^2 input norms, $C' > 1/2$, and arbitrarily small $|C|$. Moreover $D > c$ because $\bar{B}_0 > 0$. Write $f_L(\delta_L + t) = A + rt$, with $(A, r) = (\delta_L, 1)$ or $(\mu_L + \delta_L^2, 2\delta_L + t)$, and put $E = D - rC > c$. Then $M = AC - tE$. At any zero the original expression $f_L C = Dt$ forces $C > 0$, hence also $A > 0$. For $A > 0$ define $R_0 = C/(tE)$ on the entire passing interval. Its derivative has sign opposite to

$$V = NCE + f(CE' - EC'), \qquad R_0' = -V/(ftE^2).$$

One differentiation gives

$$V' = NC'E + N'CE + NCE' + f'(CE' - EC') + f(CE'' - EC'') > 0.$$

The first term has a positive margin and the others are $O(|C| + |f'| + |f|)$, so $V' > c$ on a common neighborhood. The argument uses only small C, $C' > c$, $E > c$ and their bounded C^2 norms; this is the quotient estimate used again at the weak center. Every critical point of $C/(tE)$ is a strict maximum, since $R_0'' = -V'/(ftE^2) < 0$ there. Its strict superlevel sets are intervals and each level has at most two intersections with multiplicity. At $C \le 0$, $V = C(NE + fE') - fEC' < 0$, so every nonpositive-level crossing is strictly increasing. In particular the level $1/A$ proves the first count. If the double root precedes P, use fixed sections around it and separate its input-independent contraction. The outgoing entry rate is regular with positive R_o; the proof of Proposition 8.6 then applies unchanged. □

8.6 Center division through $\bar{B} = \bar{m} = 0$

At $\bar{B} = \bar{m} = 0$, the preceding positive lower bound for D is lost. Near the remaining threshold $\bar{a} = -\bar{c} < 0$, divide by the center generators in the weak coefficient box of Proposition 5.6:

$$(\bar{B}, \bar{c}, d_s, \varsigma, \nu), \qquad d_s = P_L - \widehat{p}, \quad \varsigma = \bar{B} + \bar{m}.$$

It includes both signs of $\bar{B}$ and $\bar{m}$ and has center axis $d_s = \varsigma = \nu = 0$. The lower coordinate is the unnormalized deficit $v = P_L - p$. Its simple-root map has the form

$$D_L(v) = K_L v^{\lambda} e^{\Psi_L(v)/\epsilon}, \qquad \frac{D_L''}{D_L'} = \frac{A_L(v)}{\epsilon v}, \quad A_L(0) = \epsilon(\lambda - 1), \quad \epsilon\lambda > c.$$

The scaled unit Ψ_L has bounded mixed jets, since a regular power coordinate $vk(v)$ adds $\epsilon\lambda \log k(v)$. The unscaled logarithm need not be bounded. Since $\bar{a} < 0$ has a fixed margin, the raw horizontal factor is increasing here.

The actual interval is precisely

$$u > 0, \qquad \gamma + u > 0, \qquad F_I(\chi(u)) > 0, \qquad d_s + t(u) > 0, \tag{8.8}$$

together with positive central outputs. All four displayed conditions are one-sided monotone restrictions. Put $v = d_s + t(u)$ and reuse C, D from (8.7), with $f_L(v) = v$. The exact numerator is

$$M = d_s C - tE, \qquad E = D - C, \qquad M = uF_I(\chi)(\gamma + u)(d_s + t)K'. \tag{8.9}$$

We have $C' = 1 + O(u + |\gamma| + \epsilon)$ and a small bounded C. The following center division replaces the missing positive lower bound for D across $\bar{B} = 0$.

Proposition 8.8 (The weak-center local bound)**.** *On a common neighborhood of this threshold, every noncenter fibre has at most four isolated closing zeros, counting multiplicity, on its actual passing interval. On a fibre with a closing zero,*

$$|\nu| \le \eta(\epsilon)(|d_s| + |\varsigma|), \qquad \eta(\epsilon) \to 0. \tag{8.10}$$

If the two generators vanish, the field is a center and contributes no isolated zero.

Proof. Reserve sufficient primitive orders for $C_u^2 C_{\mathfrak{b}}^2$ of the multiplied rates, including the two input derivatives spent in each logarithmic derivative. For example total order ten in the relative maps and connections suffices. The lower large power, on a fixed small deficit interval, and the horizontal normal contraction followed by the fixed central core give

$$\|D_L\|_{C_v^4 C_{\mathfrak{b}}^2} + \|D_I\|_{C_x^4 C_{\mathfrak{b}}^2} \le C e^{-c/\epsilon}$$

after reducing c to absorb the finite inverse-scale powers. For the first term choose the deficit radius so its logarithm dominates the bounded scaled amplitude. For the second keep a fixed positive-time final segment of the horizontal/core passage. Its normal contraction supplies a positive action independent of all weak gaps. Finite parameter variations are bounded by the normally dominated relative construction; the same exponential absorbs their polynomial losses. Both absolute maps have the zero extensions of Section 4. Their rate coefficients have bounded higher mixed extensions on the larger auxiliary coefficient domain.

For $A = |d_s| + |\varsigma| + |\nu|$, Proposition 5.12 supplies the single real center $(\bar{B} + K_* A, \bar{c}, 0, 0, 0)$ through every original passing input. Its complete deviations agree there, and therefore $E = 0$. Comparison of the absolute maps on that common domain yields $|I - J| \le CAe^{-c/\epsilon}$. With the fixed central gap $\rho\nu H$, $H > h_*$, Lemma 5.14 proves (8.10).

The same center supplies the rate identity needed for division. Each point of the original interval lies in a strictly passing open center neighborhood, so both E and its passive $\bar{B}$ derivative vanish there. Taylor expansion gives

$$E = d_s E_d^c + \varsigma E_\varsigma^c + \nu E_\nu^c + R, \qquad \|R\|_{C_u^2} \le CA^2.$$

For a fixed nonzero transverse vector $g = (d_s, \varsigma, \nu)$ set $e_i = E_i^c + g_i R/|g|^2$. These provide an exact division $E = d_s e_d + \varsigma e_\varsigma + \nu e_\nu$ with bounded C_u^2 norms and error $O_{C_u^2}(A)$ from the center columns. Only the fixed-fibre division is used; smooth dependence of this choice at $g = 0$ is unnecessary.

To determine the sign of the ς column, use the exact endpoint identity. At $u = t = d_s = 0$, the actual coefficients satisfy

$$N(0) = \Delta, \quad A_U(0) = \Delta - \gamma, \quad F_I(0) = \bar{B}, \quad A_I(0) = 1 - 2\epsilon\bar{B},$$

so

$$E(0) = \epsilon\bar{B}\lambda\Delta - (1 - \epsilon\bar{B})\gamma. \tag{8.11}$$

At $\bar{B} = 0$ this is $-\gamma$ for every d_s. Since $\bar{m} = \varsigma - \bar{B}$, at the zero center $\partial_\varsigma E(0) = 1/\bar{c} > 0$ and $\partial_{d_s} E(0) = 0$, at each small positive scale. The bounded mixed extension then gives $e_\varsigma > c > 0$ on a fixed smaller domain.

Absorb the d_s column into $\overline{C} = C - te_d$. It is small, bounded in C^2, and $\overline{C}' > 1/2$. The numerator becomes

$$M = d_s\overline{C} - t(\varsigma e_\varsigma + \nu e_\nu).$$

On a fibre with a closing zero put $I_0 = |d_s| + |\varsigma|$. If $|\varsigma| \ge \sqrt{\eta} I_0$, then $e = e_\varsigma + (\nu/\varsigma)e_\nu > c/2$. For $d_s = 0$ there is no numerator zero. Otherwise apply the quotient calculation of Proposition 8.7 with $(C, E) = (\overline{C}, e)$. It makes $R_0 = \overline{C}/(te)$ unimodal and gives at most two intersections with the level ς/d_s. Its hypotheses hold because $\overline{C}$ is small, $\overline{C}' > 1/2$, $e > c/2$ and both have the required bounded C^2 norms. In the complementary case $|d_s| \ge (1 - \sqrt{\eta})I_0 > 0$, and

$$M/d_s = \overline{C} - t[(\varsigma/d_s)e_\varsigma + (\nu/d_s)e_\nu]$$

is strictly increasing. Hence M has at most two zeros in either case. Its positive multiplier and two Rolle steps give four closing zeros. The auxiliary extensions enter only the coefficient estimates. The zeros counted by Rolle's theorem lie in (8.8). □

Lemma 8.9 (Weak-chart entrance data)**.** *The weak charts above supply the following data independently of any preceding zero count. Outward motion is increasing p and decreasing u, and* $\operatorname{sign} K_p = -\operatorname{sign} M$*. In the positive-$\bar{B}$ lower-loss chart, a numerator zero requires $A > 0$ and $C > 0$, and the strict superlevel sets of $C/(tE)$ are intervals. The limiting regular critical function is positive on its whole regular interval; this gives a positive actual margin on any retained compact regular overlap.*

In the weak-center chart, on each noncenter fixed fibre satisfying (8.10)*, on the same actual passing interval, the coefficient division in Proposition 8.8 gives*

$$M = d_s\overline{C} - t(\varsigma e_\varsigma + \nu e_\nu), \qquad \overline{C} = C - te_d, \qquad e_\varsigma > c > 0, \quad \overline{C}' > 1/2.$$

The divided coefficients have bounded C^2_u norms and $\overline{C}$ is small on the same common neighborhood; the division is chosen on each fixed nonzero transverse vector. Put $I_0 = |d_s| + |\varsigma|$. If $|\varsigma| < \sqrt{\eta} I_0$, then $d_s \neq 0$ and M/d_s is strictly increasing. Otherwise $\varsigma \neq 0$ and $e = e_\varsigma + (\nu/\varsigma)e_\nu > c/2$. For $d_s \neq 0$ in this alternative, $R_0 = \overline{C}/(te)$ increases and then decreases, either portion possibly absent. Its strict superlevel sets are intervals, and every nonpositive-level crossing is strictly increasing. Each level has at most two intersections, counted with multiplicity; in particular $Z(M) \leq 2$. The identity

$$M = d_s te(R_0 - \varsigma/d_s)$$

records which of those superlevel sets contains an entrance with a prescribed sign of M/d_s. The norms of (d_s, ς) and (τ, σ_1), $\sigma_1 = \bar{c}(\bar{B} + \bar{m})$, are comparable under the relative forcing condition. These facts will be used to anchor the terminal interval in Section 12.

Proof. The first statement follows from $p = \widehat{p} - t(u)$, $t' > 0$, and the positive multipliers defining M. The quotient calculation in Proposition 8.7 gives the superlevel, multiplicity and nonpositive-level assertions for both (C, E) and $(\overline{C}, e)$. Norm comparison is (5.46), with its $|\nu|$ term absorbed using (8.10).

For the remaining regular sign at $\bar{B} > 0$, the threshold coefficients have $\tau < 0$ and $\sigma_1 = \bar{c}\bar{B} > 0$. At $\bar{m} = 0$ the exact limiting upper return is parametrized by

$$p = P - \frac{T}{e^{\bar{c}T} - 1}, \qquad q = T - p.$$

For large T, $A_-(q) = O(\log T)$. At a simple outgoing root $A_+(p) \sim \bar{c}T/(\bar{c}^2 - \bar{B})$; at a double root $A_+(p) \sim e^{\bar{c}T}/(\bar{c}^3 T)$. These follow by integrating the respective simple or double pole of $p/F_+(p)$ and inserting $P - p \sim Te^{-\bar{c}T}$. Thus $K^0 = A_+(p) - A_-(q) > 0$ near P. It is positive near zero as well, by (7.10) and $\tau < 0$. Lemma 7.6 allows at most one simple zero, so there is none between these two ends. The finite-scale regular convergence proves the positive margin on each fixed compact overlap. This is the entry sign used in Section 12 to count the regular and weak parts together; their separate local bounds are not added. □

Proposition 8.10 (Local exhaustion for one infinite endpoint)**.** *Every necessary limiting configuration with exactly one finite and one infinite ray endpoint, except the joint center reserved for Section 9, has a neighborhood with a common small-scale bound of four isolated counted cycles, including multiplicity. The neighborhoods are open in the original normalized coefficients and include their signed slow gaps and both line-breaking signs.*

Proof. Lemma 7.1 and its strict barrier give the three upper alternatives stated above. For $\bar{m} < 0$, apply Proposition 8.3 to a regular finite lower arc. A lower loss has $\bar{a} < 0$, $\bar{B} \geq 0$ and is a first simple root at the endpoint or a double root at or before it. Propositions 8.4 and 8.5, together with the signed horizontal construction at $\bar{B} = 0$, cover these losses except for the reserved joint center.

At $\bar{m} = 0$, regular finite data are covered by Proposition 8.6. Positive-$\bar{B}$ losses are Proposition 8.7. At $\bar{B} = 0$ the two lower nonnegativity conditions give $-\bar{c} \le \bar{a} \le 0$: strict inequality at the left endpoint is regular, and equality is the weak-center chart of Proposition 8.8. The raw horizontal construction includes $\bar{a} = 0$ and its cubic factor. For an outgoing infinite endpoint reflect the field. The lower restrictions of Lemma 7.3 require a loss at or before its new finite outgoing endpoint; all those proofs allowed either sign of the reflected ℓ.

An empty whole-path domain contributes no cycle. On a nonempty physical interval the original maps are analytic transverse finite-time maps. The finite-smooth auxiliary coordinates and positive multipliers used in Rolle's theorem preserve their zero multiplicities. Reflection preserves the same return-germ multiplicity. The neighborhood and scale cutoff in each argument are fixed near one one-infinite limit. When the finite endpoint also tends to infinity, the next subsection supplies the required neighborhoods. □

8.7 Two infinite endpoints and one exact exponent

When both ray endpoints tend to infinity, the lower cones and upper barrier of Section 7.1 give the limiting conditions

$$\bar{B} \ge 0, \qquad |\bar{a}| \le 2\sqrt{\bar{B}}, \qquad \bar{c}^2 + 4\bar{m} \le 0.$$

The case $\bar{c} = \bar{m} = 0$ is reserved for the second scale in Section 10. In this subsection $\bar{m} < 0$. These inequalities describe the limiting point. The neighborhoods below are open in the coefficients and retain perturbations on both sides of the limiting boundaries.

For $\bar{B} > B_* > 0$, the two lower passages have exactly the same power exponent at every positive scale. By Proposition 4.8 and (4.50),

$$D_\pm(t) = K_\pm t^{h/\epsilon} e^{P_\pm(t)/\epsilon}, \qquad h = \frac{1 - \epsilon\bar{B}}{\bar{B}}, \quad P_\pm(0) = 0. \tag{8.12}$$

The selected jets of the scaled amplitudes $P_\pm$ are bounded; a regular radial coordinate change need not bound $P_\pm/\epsilon$. All input-independent constants, including slow actions, remain in $K_\pm > 0$. With the two weak references fixed, the complete central equation is

$$F = D_+(t) - J(t) - \rho\nu H_c, \qquad J = D_- \circ f, \quad H_c = H_+ + H_- > h_* > 0, \tag{8.13}$$

where $s_+ = -\rho\nu H_+$, $s_- = \rho\nu H_-$ and $H_\pm > 0$. The upper connector has $f' > 0$. Both deviation derivatives are positive, so $K = \epsilon \log(J'/D'_+)$ satisfies $F' = D'_+(1 - e^{K/\epsilon})$. Impose every slow-path and regular-connection condition on the comparison domain, and central positivity on actual returns. A zero of (8.13) has positive common height and positive core arcs by the argument of Section 7.6. With the collar and stated passage conditions it gives an actual return.

On a strict focus neighborhood the upper angular velocity is bounded away from zero between fixed horizontal sections. Its bounded-time radial equation gives $f = tk$, with positive bounded k, k^{-1}, f' and inverse derivative, in all selected jets. Apply Lemma 4.13, with the common exponent h, to obtain (4.51). Its coefficient-only constant is $C = \epsilon \log(K_-/K_+)$. All logarithm arguments have positive margins on a fixed small section. The cancellation uses equality of the exponents for the same field, before differentiation. On each endpoint block, Lemma 4.13 bounds the input-dependent terms through relative connector units. Input-independent slow actions stay in C; their possibly divergent coefficient derivatives are outside these finite-jet estimates.

8.8 Strict focus: values and slopes after center division

Proposition 8.11. *At a limiting point with $\bar B > 0$, $\bar a^2 < 4\bar B$, $\bar m < 0$ and $\bar c^2 + 4\bar m < 0$, one common neighborhood of the two infinite endpoints contains at most two isolated counted cycles, with multiplicity. The statement includes both center families and their intersection.*

Proof. Use the actual common input from Lemma 4.12, so that its limit is $t = 1/p$. On compact positive input intervals, the upper regular convergence and lower rate integrals give, on $\nu = 0$,

$$K(t) \longrightarrow -\mathcal{A}(1/t), \qquad \mathcal{A}(p) = A_-(\mathcal{U}_0(p)) - A_+(p), \qquad A_\pm(P) = \int_0^P \frac{x\,\mathrm{d}x}{1 \pm \bar a x + \bar B x^2}.$$

This convergence holds with the selected coefficient derivatives. It extends through $t = 0$: reserve one further input derivative in (4.51), compare on a short interval to a fixed positive input, and use the common Lipschitz bound before sending that input to zero. On this strict domain C and its selected coefficient jets are bounded; no slow gap is tending to zero in this assertion.

To count near the two center families, we need the value and slope of the limiting action after division by their generators. Put $M = -\bar m > 0$, $\Omega = \sqrt{4M - \bar c^2}$ and $\Upsilon = e^{\pi\bar c/\Omega}$. The upper linear solution in Section 7.1 has

$$S(T) = e^{\bar c T/2}\frac{\sin(\Omega T/2)}{\Omega/2}, \qquad I(T) = \int_0^T S(u)\,\mathrm{d}u, \qquad pS(T) = I(T).$$

At $T_0 = 2\pi/\Omega$, $S = 0$, $S' = -\Upsilon$, $S'' = -\bar c\Upsilon$, and $I = (1 + \Upsilon)/M$. Expanding $S(T) = I(T)/p$ by the implicit function theorem yields

$$\mathcal{U}_0(p) = \Upsilon p - \frac{\bar c(1 + \Upsilon)}{M} + \frac{\Upsilon - \Upsilon^{-1}}{2Mp} + O(p^{-2}). \tag{8.14}$$

Substitution in $q = (S^2 - IS')/S$ gives this expansion. The lower integrands give

$$A_\pm(p) = \frac{\log p}{\bar B} + C_\pm \pm \frac{\bar a}{\bar B^2 p} + \frac{\bar B - \bar a^2}{2\bar B^3 p^2} + O(p^{-3}), \quad C_- - C_+ = \frac{\pi\bar a}{\bar B\sqrt{4\bar B - \bar a^2}}.$$

The constant follows by separating the logarithmic derivative of $1 \pm \bar a x + \bar B x^2$ from its reciprocal. Consequently $\mathcal{A}(1/t) = J_0 + J_1 t + O(t^2)$, where

$$\begin{aligned} J_0 &= \frac{\pi}{\bar B}\left(\frac{\bar c}{\sqrt{4M - \bar c^2}} + \frac{\bar a}{\sqrt{4\bar B - \bar a^2}}\right), \\ J_1 &= \frac{1 + \Upsilon^{-1}}{\bar B^2}\left(-\bar a - \frac{\bar B\bar c}{M}\right). \end{aligned} \tag{8.15}$$

These vanish simultaneously only on $\bar a = \bar c = 0$ or $M = \bar B, \bar a = -\bar c$. Indeed $J_0 = 0$ fixes the opposite signs and gives $M\bar a^2 = \bar B\bar c^2$; inserting the resulting $\bar a = -\bar c\sqrt{\bar B/M}$ in J_1 proves the assertion.

The real period domains in Section 5 contain every represented input at either center. On a smaller product neighborhood, Proposition 5.13 therefore gives

$$K = \tau K_\tau + \sigma_1 K_\sigma + \nu K_\nu, \qquad \tau = \bar a + \bar c, \quad \sigma_1 = \bar c(\bar B - M).$$

The independent dividing coordinates are $(\tau, \bar c, \bar B - M, \nu)$; no inverse for σ_1 is used. The value and slope distinguish the two transverse action columns. Differentiating (8.15) by the successive differences of Proposition 5.13 gives the value–slope determinants

$$-\frac{2\pi(1 + \Upsilon^{-1})}{\bar B^3(4\bar B - \bar c^2)^{3/2}}, \qquad -\frac{\pi}{\bar B^3 M(\sqrt{\bar B} + \sqrt{M})}$$

on the second and reversible center, respectively. They agree at the intersection and have uniform nonzero margins on compact blocks. Reserve the two coefficient orders for this division, in addition to the three input orders and the one derivative spent in the cancellation formula. The primitive order can be fixed before the common sections by Section 4.

Here $\epsilon \log K_{\pm}$ are bounded. Choose t_0 so that half of $t^{h/\epsilon}$ absorbs these constants; the other half absorbs the finite differentiated powers and logarithms, exactly as in (7.32). Center division therefore gives the absolute weight required by Lemma 5.14, with $I = |\tau| + |\sigma_1|$. Every fibre with a closing zero satisfies $|\nu| \le \eta(\epsilon) I$, $\eta \to 0$. If $I = |\tau| + |\sigma_1| > 0$, the rank just computed yields

$$|K(0)| + |K'(0)| \ge cI, \qquad \|K\|_{C^2} \le CI.$$

On a common short interval these estimates force either a nonzero value throughout or a first derivative of one strict sign. Thus K has at most one zero. Off the centers the same alternative follows directly from (J_0, J_1). If $I = 0$, forcing gives the center identity. Finally the displayed formula for F' gives the claimed two zeros, with multiplicity. □

8.9 Fixed slow strips and their relative limits

As a lower double root or an upper repeated root is approached, transit times diverge and the strict-focus bounds on the action constants no longer apply. Fixed sections separate these contractions from the relative terms that will be differentiated.

Lemma 8.12. *Near an incoming lower double root $\bar{a} = 2\sqrt{\bar{B}}$, choose fixed lower phase sections on opposite sides of its minimum. The passage exists precisely when its exact scalar factor f_L is positive between them. Its map, about the selected weak reference, is $L_L u e^{r_L(u)}$, with bounded selected relative jets and*

$$\epsilon \log L_L = -A_L + O(\epsilon), \qquad A_L \longrightarrow +\infty.$$

Near an upper repeated root $\bar{c} = 2\gamma > 0$, $\bar{m} = -\gamma^2$, fix the upper slow sections $v = \pm v_R$. That passage exists precisely for $\mu_U > 0$ and has

$$G(u) = e^{-A_U} u e^{r_U(u)}, \qquad A_U = \frac{2\Xi}{b\sqrt{\mu_U}} \arctan \frac{v_R}{\sqrt{\mu_U}}, \tag{8.16}$$

where $b = 1 - \epsilon \bar{B}$, $\mu_U = -[(\bar{c} + \nu)^2 + 4b\bar{m}]/(4b^2)$ and $\Xi = (\bar{c} + \nu)(1 - 2\epsilon\bar{B})/(2b)$. The complete upper connector is $f(t) = \alpha t e^{\psi(t)}$, with $\alpha = e^{-A_U} k_0$, $k_0 > c$, and bounded selected jets of ψ and of the absolute factor α. These bounds do not bound coefficient derivatives of A_L or A_U.

Proof. Apply the exact phases and relative integral estimates of Section 4. On the lower strip, the normal rate has a positive margin and the scalar velocity is ϵf_L. At its limiting minimum r_0, positivity and coefficient closeness give $f_L(w) \le \delta + C(w - r_0)^2$. For fixed sections $R < r_0 < Q$,

$$A_L = \int_R^Q \frac{\beta_L(w, 0)}{f_L(w)} \, \mathrm{d}w \ge \frac{2\beta_*}{\sqrt{C\delta}} \arctan\left(c_1 \sqrt{C/\delta}\right) \longrightarrow \infty, \quad c_1 = \min(r_0 - R, Q - r_0).$$

The weak-branch reference may arrive at the outer section with nonzero normal value. Subtract its solution before applying the relative integral, so the difference still has the contracting normal factor. Regular central composition puts $-A_L + O(1)$ into $\epsilon \log K_-$ and leaves P_- bounded in the selected input-dependent jets. Reflection gives the corresponding assertion for K_+.

The upper exact equations are $v' = \mu_U + v^2$ and $R' = -R\beta_U(v, R)$, with $\beta_U(v, 0) = (\Xi - v)/b$. Integration between $\pm v_R$ gives (8.16). Both endpoints are fixed and separated, so the lost-exit

extension of the relative-core proposition applies. On adjoining sections $v_R \le |v| \le v_{\mathrm{out}}$, $\mu_U + v^2 \ge v_R^2/2$ after shrinking coefficients. The regular maps $\Phi_{\mathrm{in}}, \Phi_{\mathrm{out}}$ have bounded positive units. The absolute factor e^{-A_U} and the relative integral have the selected extensions from Proposition 4.3. Lemma B.3 gives the complete connector and its asserted jets. □

Lemma 8.13. *After subtraction of the input-independent contractions in the preceding lemma, the relative input derivatives converge to their repeated-root limits, uniformly even when the transit times diverge. On compact intervals $p > 1/\gamma$ the incoming upper logarithmic rate has limit $-p/(p-1/\gamma)^2$.*

Proof. Apply Proposition 4.3 to the fixed strips. The upper strip has $q = 1$, fixed transverse exits and quadratic time jets; the lower strip has $q = \epsilon$ and $T \ge c/\epsilon$. Their auxiliary flow jets converge on bounded time intervals, so Corollary 4.4 gives the relative differentiated limits even as the transit times diverge.

Its limiting rate is explicit. For the repeated affine field set $V = Y + \gamma^{-2}$, $Z = X + \gamma Y - \gamma^{-1}$ and $r = Z/V$. Then $V' = \gamma V + Z$, $Z' = \gamma Z$, $r' = -r^2$, and $\log(Vr) - \gamma/r$ is constant. At $(p, 0)$ it is $\log(p - P_0) - P_0/(p - P_0)$, where $P_0 = 1/\gamma$. The incoming relative logarithm has the negative of this entry value; its derivative is $-p/(p - P_0)^2$. Fixed regular connectors preserve this rate on the original entry coordinate. The preceding tail argument supplies its perturbed limit. Reflection uses its own forward contracting chart. □

8.10 Domination and competing slow actions

Proposition 8.14. *At an incoming lower double root with a strict upper focus, the local bound is one cycle, necessarily simple and attracting. Its reflected outgoing configuration is repelling and has no closing zero for $\ell \ge 0$. At a positive repeated upper root with a regular outgoing lower path, the bound is again one simple attracting cycle; it is absent for $\ell \le 0$. This includes an additional incoming lower double root. The reflected upper configuration reverses both stability and the excluded sign of ℓ.*

Proof. Every fixed slow strip must pass; outside its scalar passing condition there is no represented orbit. In the first case (4.51) is $K = -A_L + O(1)$, uniformly in input and transit time. In the second it is $K = -hA_U - A_L + O(1)$, where $A_L = 0$ if the incoming lower path is regular. Shrink the neighborhoods until the divergent negative terms absorb the bounded remainder. This uses no ratio between the actions and gives $K < -1$ and $F' > 0$ throughout the comparison interval. At a zero the first-return multiplier in central height is $J'/D'_+ < 1$, giving simplicity and attraction. The deviations both tend to zero as $t \downarrow 0$, so in the second case $D_+ > J$ excludes $\ell \le 0$ in (8.13). Reflection gives $D_+ < J$ in the outgoing-only lower case and excludes $\ell \ge 0$. These inequalities are valid on the extended comparison interval; they do not assert a positive central return at $t = 0$. □

Proposition 8.15. *Suppose the limiting data are*

$$\bar{B}_* > 0, \qquad \bar{a}_* = -2\sqrt{\bar{B}_*}, \qquad \bar{c}_* = 2\gamma > 0, \quad \bar{m}_* = -\gamma^2, \qquad \bar{B}_* \ne \gamma^2.$$

A common neighborhood of the two infinite endpoints contains at most two isolated counted cycles, with multiplicity. The same holds after reflection and for either sign of the line-breaking parameter.

Proof. Here the lower and upper contractions enter K with opposite signs. Both fixed slow strips must pass. The lower contraction belongs to K_+, so cancellation gives

$$K(0) = A_L - hA_U + O(1), \qquad \|K - K(0)\|_{C^2([0,t_0])} \le C.$$

The value $K(0)$ can be arbitrary because the two gaps vanish independently. Keep it undifferentiated in coefficients. To bound the nonconstant part, reserve at least three input derivatives of the primitive relative maps before the derivative spent in the critical logarithm, and use the same actual outgoing coordinate before either slow strip.

Put $r_0 = 1/\sqrt{\bar{B}_*}$ and $P_0 = 1/\gamma$. The regular outgoing entry integral has logarithmic rate $-p/[\bar{B}_*(p - r_0)^2]$. Combining this with the upper rate in Lemma 8.13, and with the limiting horizontal factor $h = 1/\bar{B}_*$, gives on compact subsets of $p > \max(r_0, P_0)$

$$K_p \longrightarrow \frac{p}{\bar{B}_*}[(p - r_0)^{-2} - (p - P_0)^{-2}].$$

In the reciprocal input this is

$$K_t \longrightarrow \frac{(P_0 - r_0)[2 - (P_0 + r_0)t]}{\bar{B}_*(1 - r_0 t)^2(1 - P_0 t)^2}. \tag{8.17}$$

Compare with a fixed positive input and use the uniform second derivative bound to extend this convergence to $t = 0$. Since $P_0 \neq r_0$, the right side has one strict sign on a common short interval. Thus K has at most one zero regardless of its constant value, and F at most two. The excluded equality is exactly the joint center treated in Section 9. □

8.11 The cubic horizontal boundary

Proposition 8.16. *At $\bar{B}_* = \bar{a}_* = 0$, $\bar{m}_* < 0$, $\bar{c}_*^2 + 4\bar{m}_* \leq 0$, the two-infinite-endpoint local bound is three isolated cycles, with multiplicity. The open neighborhoods include both signs of $\bar{B}$, both signs of ν, and a repeated upper boundary when present.*

Proof. At the cubic boundary the power representation is singular. Use instead the undivided phases of Proposition 4.8, written as

$$x' = \epsilon g_\pm(x), \quad g_\pm(x) = xF_\pm(x), \qquad Z' = -Z\beta_\pm(x, Z).$$

The exact coefficients satisfy

$$F_\pm(0) = \bar{B}, \quad F'_\pm(0) = \pm\bar{a}, \quad \beta_\pm(0, 0) = 1 - \epsilon\bar{B},$$

and on $\nu = 0$, $F_\pm(x) = \bar{B} \pm \bar{a}x + (1 + \epsilon\bar{m})x^2$. Fix a small exit $x_R > 0$ with $F_\pm(x_R) > x_R^2/2$ and a positive fixed fast input. An entry $x < x_R/2$ passes precisely when $F_\pm > 0$ on $[x, x_R]$. This is the positive scalar component adjoining x_R; it is one interval. Lost exits have infinite time, so the relative and absolute extensions of Section 4 apply here. No extension across a collision of entry and exit is needed.

After regular continuation to the central section set

$$N_\pm = \beta_\pm(x, 0) + \epsilon g_\pm \tilde{j}'_\pm, \qquad A_\pm = N_\pm - \epsilon g'_\pm + \epsilon g_\pm N'_\pm/N_\pm.$$

The exact logarithmic rate is

$$D'_\pm > 0, \qquad \frac{D''_\pm}{D'_\pm} = \frac{A_\pm(x)}{\epsilon x F_\pm(x)}, \quad A_\pm = 1 + O(\epsilon), \quad A_\pm(0) = 1 - 2\epsilon\bar{B}. \tag{8.18}$$

The last two assertions hold with the reserved jets on the auxiliary product box. Normal contraction absorbs the finite inverse scale powers from the central connection, giving the absolute bounds required for one coefficient division and the subsequent forcing estimate.

For a strict upper focus, $f = tk$ and its limiting map is $f_0(t) = 1/\mathcal{U}_0(1/t)$. At a positive upper repeated root, Lemma 8.12 gives instead $f = \alpha t e^{\psi}$, $\alpha \to 0$. Its proof only requires bounded $\bar{B}$, so remains valid here. In both cases

$$L = tf'/f, \qquad R_f = f''/f', \qquad L(0) = 1$$

have bounded input jets. The compositions depend on f and ψ, not on α^{-1}. The two scalar passing intervals at t and $f(t)$ intersect in one interval, since $f' > 0$; at the repeated root impose also $\mu_U > 0$. The signed central comparison may again be larger than the actual return domain.

The connector curvature enters when differentiating the complete critical function. Keeping this term gives the exact identity

$$\begin{aligned} \mathcal{P} &= tF_+(t)F_-(f(t))K' \\ &= F_+(t)A_-(f(t))L - F_-(f(t))A_+(t) + \epsilon t F_+(t)F_-(f(t))R_f. \end{aligned} \tag{8.19}$$

The exact endpoint values give $\mathcal{P}(0) = 0$, allowing one division by the radial input on the auxiliary box. Define $Q = \mathcal{P}/t = \int_0^1 \mathcal{P}'(ut)\,du$ on the whole auxiliary box. Reserve primitive total order eight: the rate and curvature formulas spend at most two orders, and this division one. The remaining five supply three input and two coefficient orders for Q. On the passing interval $Q = F_+F_-K'$, with positive multiplier.

If $\bar{c}_* \neq 0$, reflect to $\bar{c}_* > 0$. At zero lower coefficients and zero scale, $Q = t[L - (f/t)^2]$ and

$$Q'(0) = 1 - \alpha_0^2 > 0, \qquad \alpha_0 = e^{-\pi\bar{c}_*/\sqrt{-4\bar{m}_* - \bar{c}_*^2}}$$

in the strict focus case. In the repeated case the same exact numerator gives

$$|Q'(0) - (1 - \alpha^2)| \le C(|\bar{B}| + |\bar{a}| + \epsilon), \qquad \|Q''\| \le C.$$

Since $\alpha \to 0$, a common small interval has $Q' > c > 0$. It follows that Q has at most one zero, K at most two, and F at most three.

It remains to include $\bar{c}_* = 0$, in a strict focus neighborhood. Choose the reflected phase and graph extensions symmetrically. On $\bar{a} = \bar{c} = \nu = 0$ the connector is an increasing involution and hence the identity; the two relative maps and the extended numerator coincide. This is an identity on the common auxiliary product box, even if a contracted coefficient point loses a horizontal exit. Thus Hadamard division gives

$$Q = \bar{a}Q_a + \bar{c}Q_c + \nu Q_\nu.$$

The same division of the absolute deviations and Lemma 5.14 give, at a closing zero, $|\nu| \le \eta(\epsilon)(|\bar{a}| + |\bar{c}|)$, $\eta \to 0$.

Compute the transverse rank from the undivided limiting numerator, then transfer it to the actual divided columns. Put $M = -\bar{m}$ and expand the reciprocal of (8.14) as $f_0(t) = \alpha_0 t + f_2 t^2 + f_3 t^3 + O(t^4)$. Then

$$f_2 = \frac{\bar{c}\alpha_0(1 + \alpha_0)}{M}, \qquad f_3 = \frac{\bar{c}^2\alpha_0(1 + \alpha_0)^2}{M^2} - \frac{\alpha_0 - \alpha_0^3}{2M}.$$

At zero scale the value and slope of Q are

$$Q(0) = \bar{B}f_2/\alpha_0 + \bar{a}(1 + \alpha_0),$$

$$Q'(0) = \bar{B}(2f_3/\alpha_0 - f_2^2/\alpha_0^2) + \bar{a}f_2(1 + \alpha_0^{-1}) + 1 - \alpha_0^2.$$

Differentiating in $(\bar{a}, \bar{c})$ on the reversible face gives

$$\begin{pmatrix} 2 & 2\bar{B}/M \\ 0 & \pi(M-\bar{B})/M^{3/2} \end{pmatrix}.$$

Its determinant has a fixed nonzero margin near $\bar{B} = 0$. Mixed convergence of the undivided extended numerator transfers this rank to the actual columns. With $I = |\bar{a}| + |\bar{c}|$, forcing and the reserved bounds give $|Q(0)| + |Q'(0)| \geq cI$, $\|Q\|_{C^2} \leq CI$. The value-or-slope alternative on a fixed short interval gives at most one zero of Q, followed by the same two Rolle steps. If $I = 0$, forcing gives the reversible identity. Apply the finite-selection convention of Section 3. □

9 The joint center on a single domain

Fix the limiting coefficients

$$\bar{B} = -\bar{m} = \gamma^2, \qquad \bar{a} = -\bar{c} = -2\gamma, \qquad \epsilon = \nu = 0, \qquad P = 1/\gamma, \quad \gamma > 0.$$

We use throughout the actual coefficient coordinates of Proposition 5.6:

$$\mathbf{b} = (\bar{B}, \delta, \tau, \zeta, \nu), \qquad \mu_L = \delta - \zeta, \quad \mu_U = \delta + \zeta, \quad \tau = \bar{a} + \bar{c}.$$

The lower gap is (5.15). Near P, either gap may be nonpositive, so contact inputs must satisfy both whole-path passing tests. Beyond P, the separated description requires two positive gaps and continues to $1/p = 0$. The difficulty is to control one comparison through both descriptions as the gaps vanish at unrelated rates. We use the same original central outputs on their regular overlap and prove one critical-point direction on their union.

All comparisons below return to the original central height z. Use one fixed outgoing weak reference and one fixed incoming reference, continued from their fixed inner sections. Write their complete central maps as $s_+ + D_+$ and $s_- + J$, and put

$$F = D_+ - J - \sqrt{\epsilon}\,\nu H_c, \qquad K = \epsilon \log(J'/D'_+), \qquad s_- - s_+ = \sqrt{\epsilon}\,\nu H_c. \tag{9.1}$$

The prime refers to the same input in the two derivatives. Both derivatives are nonzero with the same sign on each open passing interval. A common change of input composes K with that change; it introduces no extra logarithm. Signed central continuations are used for coefficient comparison. Positive total central heights and the actual collar conditions are additionally imposed on counted orbits.

Proposition 9.1 (The joint block)**.** *There are a contact neighborhood of $p = P$, a separated interval continuing to the radial endpoint, a common coefficient neighborhood and a positive scale cutoff with the following properties.*

The two descriptions use the same gaps, phases, central outputs and coefficient inverse box. Their passing domains here impose the actual slow passages and fixed regular arcs, while permitting the signed central comparisons just described. With two positive gaps these domains have an open overlap and form one interval. With a nonpositive gap only the retained contact passing interval is used. No restriction on the ratio of two positive gaps is imposed.

Set $I = |\tau| + |\zeta|$. For one fixed sufficiently small $\vartheta > 0$, every noncenter fibre satisfying $|\nu| \leq \vartheta I$ has at most one zero of K' on this joint interval, including multiplicity. Each such critical point is simple and the sign of the second derivative in any regular input coordinate is $\operatorname{sign}\tau$*. Consequently K has at most two zeros there, with multiplicity.*

At an actual closing zero, $|\nu| \leq \eta(\epsilon) I$, where $\eta(\epsilon) \to 0$. Thus this joint block contains at most three isolated counted cycles, including multiplicity. If $I = 0$ and a closing zero exists, then $\nu = 0$ and the comparison is a center identity, contributing no isolated cycle.

Two estimates give the critical-point bound. On the separated interval, we remove the input-independent action constant and cancel the exact common powers before forming Hadamard columns. At contact, where division by a slow factor may be singular, we instead multiply by both factors. The resulting numerator has bounded jets through signed gaps. Its limiting value and slope give the same critical-point direction as the separated columns. A closing zero then supplies the relative coefficient bound needed to apply these estimates. The estimates include the radial endpoint; counted finite orbits have strictly positive radial input.

9.1 Separated expressions before coefficient division

Choose a small contact radius $r > 0$ and then $0 < d < r/4$. The separated finite entries satisfy $p \geq P + d$, and their limiting radial interval is

$$J = [0, (P + d)^{-1}].$$

Retain slightly larger neighborhoods at every fixed finite boundary. Place the fixed lower slow sections on opposite sides of the minimum, with the outer section between P and $P + d$. Choose the upper sections in the same way. In the upper block first choose their width for the desired finite normal domination; only afterwards choose the primary scale for the lower block. These are the $q = 1$ and $q = \epsilon$ choices of Proposition 4.3.

On a fixed positive annulus in $1/p$, apply Lemma 4.12 before either slow block. It gives a single coordinate s_ϵ, equal to the actual radial phase near infinity and to $1/p$ on the finite side. Its limit on J is $s = 1/p$, and its selected transition and inverse jets are bounded on the overlap. The actual inverse coefficient chart and these regular input charts are chosen on larger boxes before the coefficient neighborhood is reduced. Proposition 5.10 then supplies the whole contraction

$$(\bar{B}, \delta, \tau, \zeta, \nu) \longmapsto (\bar{B}, \delta, t\tau, t\zeta, t\nu), \qquad 0 \leq t \leq 1,$$

with this common input fixed. Every represented input passes through both slow blocks along the entire coefficient path. Thus the columns below compare functions on one input interval, without shrinking that interval as the coefficients move toward the center.

Lemma 9.2 (The nonconstant separated logarithm)**.** *On this common domain, write $K = C(\mathbf{b}, \epsilon) + R$ by subtracting its value at one fixed input. The function R has uniformly bounded mixed derivatives of total order four, in the ordinary common input, ordinary coefficients and Euler scale. The complete deviations have, after possibly decreasing $c > 0$,*

$$\|D_+\|_{C^1_{s_\epsilon} C^1_{\mathbf{b}}} + \|J\|_{C^1_{s_\epsilon} C^1_{\mathbf{b}}} \leq Ce^{-c/\epsilon}. \tag{9.2}$$

No coefficient derivative of the subtracted constant is included.

Proof. We treat the finite and radial inputs separately, removing only constants in the input. On a compact finite patch the first lower regular piece gives a small normal input $e^{-S(p)/\epsilon}$, where $S, S_p > c$ with bounded finite jets. Here the bounded logarithms of regular prefactors have been absorbed into S with their factor ϵ. The fixed lower slow map and central connection give

$$D_+(p) = C_+ L_L e^{-S(p)/\epsilon + r_+(p)}.$$

Here $C_+ L_L$ is input independent. Relative differentiation in Proposition 4.3, followed by the normalized central connection, bounds the required jets of r_+ by $Ce^{-c/\epsilon}$. The upper fixed-section map and its regular connectors give an incoming horizontal phase

$$\chi(p) = \alpha\phi(p), \qquad \alpha > 0, \quad r(p) = \log \phi(p), \quad -r_p > c,$$

with bounded jets of r and bounded absolute jets of α. The coefficient derivatives of $\log\alpha$ need not be bounded. The incoming horizontal central map is $D_I(\chi) = K_I\chi^{\mu}e^{P_I(\chi)/\epsilon}$, where

$$\mu = \frac{1-\epsilon\bar{B}}{\epsilon\bar{B}}, \qquad h_I = \epsilon\mu > c.$$

The reciprocal preparation in Section 4 supplies bounded selected jets of the action P_I. This form includes the regular changes of weak coordinate; it does not require bounded ordinary jets of P_I/ϵ. Normalize $P_I(0) = 0$ by changing K_I. The margin for $-r_p$ follows from its regular normalized upper entry and the small relative derivative on the chosen normal tube. Equivalently its zero-scale value is the strictly positive return rate (C.5) with the opposite sign; the bounded gap jets extend that margin uniformly to the repeated limit.

Differentiating the actual amplitudes, before taking logarithms, gives the exact finite-patch expression

$$\begin{aligned} K &= C_0 + S(p) + h_I r(p) + P_I(\chi) + \epsilon E(p), \\ E &= -r_+ + \log(-r_p/S_p) + \log(1+\chi P_I'(\chi)/h_I) - \log(1-\epsilon r_+'(p)/S_p). \end{aligned} \tag{9.3}$$

All logarithm arguments have positive margins. The possibly divergent actions $\epsilon\log L_L$ and $h_I\log\alpha$ belong to C_0. The nonconstant composition $P_I(\alpha\phi)$ uses absolute jets of $\alpha\phi$, so its bounds require no derivative of $\log\alpha$.

At the radial end the two horizontal exponents agree exactly for the same field, by (4.50). Write the complete outgoing map and the incoming map before their upper connector as

$$D_{\pm}(t) = K_{\pm}t^{h_I/\epsilon}e^{P_{\pm}(t)/\epsilon}.$$

All slow contractions independent of t belong to $K_{\pm}$. The upper connector, including its fixed repeated passage, has $f(t) = \alpha t e^{\psi(t)}$, where the selected jets of ψ and the absolute jets of f are bounded. Normalize $P_{\pm}(0) = \psi(0) = 0$ by transferring constants to $K_{\pm}, \alpha$. Apply (4.51) with $k = \alpha e^{\psi}$, grouping $h_I\log\alpha$ into $C_1 = \epsilon\log(K_-/K_+) + h_I\log\alpha$. The remaining logarithm arguments have positive margins on a fixed small radius. Its primitive jets follow from the horizontal preparation, relative slow integrals and central connections. Each block is first estimated with independent normal input, then composed using absolute jets of f.

Primitive total order five leaves total order four after the one derivative in (4.51) or (9.3), including $s^3\mathbf{b}^1$ and $s^2\mathbf{b}^2$ after the common input change. The radial and finite patches form a finite cover of J and describe the same central maps. Subtracting at a fixed input removes their constants.

Finally each finite lower path contains a fixed positive action before its slow strip, and the radial input has the common large power on a fixed interval smaller than one. The fixed inner central connection contributes a further positive action. The absolute variational estimates of Section 4 therefore bound the selected derivatives of both complete deviations by a finite inverse power of ϵ times $e^{-c_0/\epsilon}$. Decrease $c < c_0$ to absorb that power. This proves (9.2), including its continuous radial endpoint. □

9.2 Actual columns on the whole separated interval

Proposition 9.3. *For $\delta > |\zeta|$, on the common interval J,*

$$H := \partial_{s_\epsilon}K = \tau H_\tau + \zeta H_\zeta + \nu H_\nu. \tag{9.4}$$

The three columns are uniformly bounded in $C^2(J)$. As the scale and coefficient neighborhood tend to the joint limit,

$$\|H_\tau - U_0\|_{C^1(J)} + \|H_\zeta - V_0\|_{C^1(J)} \longrightarrow 0,$$

where U_0, V_0 are (C.9). This convergence permits every ratio of the two positive gaps.

Proof. At the center endpoint, Proposition 5.10 gives $K = H = 0$. Thus the Hadamard columns on this same domain are

$$H_i(s_\epsilon; \mathbf{b}) = \int_0^1 \partial_i H(s_\epsilon; \bar{B}, \delta, t\tau, t\zeta, t\nu)\, \mathrm{d}t, \qquad i = \tau, \zeta, \nu.$$

Overlapping descriptions give the same integrals. After removing the constant in K, Lemma 9.2 supplies their C^2 bounds from $s^3\mathbf{b}^1$ and their C^1 coefficient Lipschitz bounds from $s^2\mathbf{b}^2$, within total order four.

At a fixed positive center gap g, the scale limit on every compact positive subinterval of J is the derivative of $A_+(p) - A_-(q(p))$. To verify this directly, the lower logarithm integrates its reduced normal rate to A_+ or A_-; its relative logarithms and regular endpoint corrections are multiplied by ϵ and tend to zero after input differentiation. The upper limit is the finite-time return of (C.3). At fixed positive gap and finite p, its time and crossings are transverse, so the original upper equations and their coefficient variational equations converge there. This proves the limit also after the one coefficient and one input derivatives used for the columns.

Lemma C.2 gives that limiting function in a single expression on the closed interval, with the exponentially small reciprocal return retained. The finite-scale columns have a uniform C^2 bound. Comparing their values and first derivatives on $[0, s_2]$ to those at a fixed $s_2 > 0$, and then decreasing s_2, extends the fixed-gap convergence to $C^1(J)$, including zero. The limiting formula supplies the same Lipschitz control there.

The gap derivative bound makes the order of limits explicit. For a center column $\mathsf{H}_\epsilon(g)$ and $0 < g \le g_0$, the actual gap derivative bound gives

$$\|\mathsf{H}_\epsilon(g) - \mathsf{H}_*\|_{C^1} \le Cg_0 + \|\mathsf{H}_\epsilon(g_0) - \mathsf{H}_0(g_0)\|_{C^1} + \|\mathsf{H}_0(g_0) - \mathsf{H}_*\|_{C^1}.$$

Choose g_0 small first and then the scale. Formula (C.9) controls the last term, while the fixed-gap convergence controls the middle term. The same retained mixed derivatives compare an off-center Hadamard column to its center value with error $C(|\tau| + |\zeta| + |\nu|)$ in $C^1(J)$. All coefficient segments remain in the fixed-mean passing domain. Thus no estimate depends on $(\delta + \zeta)/(\delta - \zeta)$. □

9.3 The contact numerator with two signed gaps

Use the center-calibrated moving entries of Proposition 5.6, including their fixed regular initial legs. Denote the entries at the fixed fast sections by $u(p), v(p)$. They have bounded finite jets and $u_p, v_p < -c < 0$. Their exact slow factors are

$$f_L(u) = u^2 + \delta - \zeta, \qquad f_U(v) = v^2 + \delta + \zeta.$$

Each scalar path ends at a fixed positive exit. A positive gap permits the whole path; for a nonpositive gap the calibrated entry must exceed the positive root. Including the finite initial leg gives exactly the passing test of Proposition 5.11. Since both entries decrease with p, the two tests select one common interval.

The moving-entry estimates of Lemma 4.5, followed by the incoming horizontal and central maps, give

$$\frac{D_{uu}}{D_u} = \frac{A_L(u)}{\epsilon f_L(u)}, \qquad \frac{J_{vv}}{J_v} = \frac{A_U(v)}{\epsilon f_U(v)}. \tag{9.5}$$

The two numerators have positive margins and bounded finite mixed jets on the larger auxiliary product domain. More explicitly,

$$N_L = \beta_L(u, 0) + \epsilon f_L \widetilde{j}'_L, \qquad N_U = h_I \beta_U(v, 0) + f_U \Psi',$$
$$A_i = N_i - \epsilon f_i' + \epsilon f_i N_i'/N_i, \qquad i = L, U. \tag{9.6}$$

Here $\widetilde{j}_L$ is the complete lower relative logarithm. Writing the upper normal output as $\chi = CL_U e^{j_U}$, one may take $\Psi = h_I j_U + P_I(\chi)$, with the fixed regular units included. The absolute derivatives of χ are bounded even when its slow exit is lost, so the displayed expression has the required mixed extension. Choose the contact radius and gaps small enough that $f_U \Psi'$ does not remove the positive margin of $h_I \beta_U$. Formula (9.6) is the direct identity obtained from $\partial \log D = N/(\epsilon f)$. In particular the upper block retains its $q = 1$ normal dynamics; the ϵ in this logarithmic rate comes from the incoming horizontal exponent.

For the complete maps $D_+(p) = D(u(p))$, $J_+(p) = J(v(p))$, differentiate in the unchanged finite input:

$$\begin{aligned} \mathcal{M} &= f_L(u) f_U(v) K_p \\ &= f_L(u) A_U(v) v_p - f_U(v) A_L(u) u_p \\ &\quad + \epsilon f_L(u) f_U(v) \left(\frac{v_{pp}}{v_p} - \frac{u_{pp}}{u_p} \right). \end{aligned} \tag{9.7}$$

Both multiplying factors are positive on the actual passing interval, so $\mathcal{M}$ and K_p have the same zeros there, with the same multiplicities. The right side extends smoothly to the larger auxiliary domain even when a passage is absent. Reserve primitive total order eight here. The two derivatives spent on the logarithmic rates and connection curvature leave $C_p^3 C_{\mathbf{b}}^3$ bounds for $\mathcal{M}$; the ensuing Taylor argument uses only $C_p^2 C_{\mathbf{b}}^2$. This is a separate budget from the order-five separated construction.

Proposition 9.4 (Contact comparison and transverse jets)**.** *Let $A = |\tau| + |\zeta| + |\nu|$ and compare with the one center*

$$\mathbf{b}_c = (\bar{B}, \delta + 3A, 0, 0, 0).$$

On the whole original contact passing interval,

$$\mathcal{M} = \tau C_\tau + \zeta C_\zeta + \nu C_\nu + \mathcal{R}, \qquad \|\mathcal{R}\|_{C_p^2} \le C A^2, \tag{9.8}$$

where $C_i = \partial_i \mathcal{M}|_{\mathbf{b}_c}$ have bounded C^2 norms. At the joint limit the value and first input jets of (C_τ, C_ζ) at $p = P$ are those of

$$U_*(p) = p^2 - p/\gamma, \qquad V_*(p) = 4p^2/\gamma - 2p/\gamma^2 \tag{9.9}$$

multiplied by one smooth positive unit. Their Wronskian there is therefore a positive multiple of $-2/\gamma^4$.

Proof. Proposition 5.11 supplies the entire lifted coefficient segment and the inclusion of every original passing input in its slow domain. It also supplies the real center identity $D_+ = J_+$ at its endpoint, for either sign of $\delta + 3A$. Thus $\mathcal{M} = 0$ there. Varying δ along strictly passing centers preserves this identity, so its passive δ derivative also vanishes, as explained after that proposition. Taylor's formula on the same larger domain gives (9.8): the displacement in δ has length $3A$ but contributes only to the quadratic remainder. Here A is held fixed for the original fibre; the absolute values defining it are not differentiated.

We verify the limiting transverse jets before dividing by either slow factor. In this calculation abbreviate the barred coefficients by B, m, a, c, put $M = -m$, and use

$$F_L = 1 + ap + Bp^2, \qquad F_U = 1 - cp + Mp^2.$$

At zero scale and $\nu = 0$, the initial lower Morse coordinate is $-a/2 - Bp$, giving the factor BF_L. The upper linear field has the exact scalar coordinate

$$v_0 = \frac{c/2 - MX - cMY/2}{1 + MY}, \qquad v_0' = v_0^2 + M - c^2/4, \qquad v_0(p, 0) = c/2 - Mp.$$

Its initial factor is MF_U. These are the zero-scale limits of the calibrated entries, up to their finite initial scalar flows. They use the same strong-fibre normalization fixed in Section 5; they are not independent choices of gap.

The calibrated entries differ from these explicit entries by finite scalar flows whose times depend on p. Retaining this dependence, if $u_i = \phi^i_{T_i(p)}(e_i(p))$ and $\lambda_i = \partial_e \phi^i_{T_i}(e_i) > 0$, then exactly

$$f_i(u_i) = \lambda_i f_i(e_i), \qquad (u_i)_p = \lambda_i(e_{i,p} + f_i(e_i)T_{i,p}). \tag{9.10}$$

The first identity is the scalar variational equation. The times T_i and their selected derivatives are bounded on these finite regular legs; the lower scalar time includes its factor ϵ. After division of (9.7) by $\lambda_L\lambda_U$, its first two terms are

$$f_L(e_L)A_U(v)e_{U,p} - f_U(e_U)A_L(u)e_{L,p} + f_L(e_L)f_U(e_U)(A_U(v)T_{U,p} - A_L(u)T_{L,p}).$$

The curvature term also retains the product of the two initial factors. At the joint center each such factor is a positive multiple of $(1-\gamma p)^2$. A transverse derivative of their product has zero value and zero first p derivative at P. This explains why variable times and curvature are present in the full numerator but do not change the two limiting jets.

For a positive upper gap, the exact return rate (C.5) gives the zero-scale expression

$$F_L F_U K^0_p = pF_U - pF_L\,\frac{M + c/q + 1/q^2}{B - a/q + 1/q^2}.$$

The scalar identities (9.10) show that the zero-scale multiplier in (9.7) is this product times the positive smooth unit $BM\lambda_L\lambda_U$. Its long upper transit remains in q; it has not been included among the bounded initial times.

At the joint contact point the reciprocal $1/q$ and the finitely many mixed derivatives needed for these jets tend to zero. One way to verify this is the normally dominated moving upper passage of Section 4: its normal output is a positive regular multiple of $1/q$, its transit tends to infinity, and its differentiated absolute estimates tend to zero. In the linear field the same conclusion follows from

$$e^{cz/\Omega}\left(\cos z + \frac{2Mp - c}{\Omega}\sin z\right) = 1 :$$

at $p = P$, $z \to \pi/2$ as the gap decreases to zero, and the reciprocal output decays exponentially in $1/\Omega$. The mixed extension, rather than a pointwise asymptotic alone, permits the required transverse differentiation.

Consequently the limiting transverse value and first input jets are those of the master numerator

$$\mathscr{N} = F_L F_U\left(\frac{p}{F_L} - \frac{Mp}{BF_U}\right) = \frac{p}{B}\{\varsigma(1-cp) - M\tau p\}, \qquad \varsigma = B - M. \tag{9.11}$$

The columns in (τ, ς) are $-p^2$ and $p(1-cp)/B$, with Wronskian $p^2/B > 0$. At fixed mean gap the change of transverse coordinates at the joint center is $d\varsigma = -\gamma\,d\tau - 2d\zeta$. It gives precisely (9.9), whose Wronskian at P is $-2/\gamma^4$. Multiplication by a smooth positive unit multiplies the determinant by the square of its value; its derivative only adds a multiple of the value row to the derivative row.

The positive multiplier can also be identified from the initial factors. Their difference has transverse part $Bp\,\tau + (1 - cp + 2Bp^2)\varsigma$ on the center. It equals $(-B/p)\mathscr{N} + 2F_L\varsigma$ to first coefficient order. The last term has zero value and first input derivative at P. The common rate times the decreasing entry derivative is negative, so its product with $-B/p$ is the same positive unit. This also checks the sign without dividing by a vanishing factor.

Finally the uniform mixed bounds for the multiplied numerator transfer these jets to the finite-scale columns and to either sign of the nearby comparison gap. More explicitly, compare

first to a fixed positive center gap on the same larger domain, use finite scale convergence there, and then decrease that gap; the retained coefficient derivative bound controls the first comparison by the gap difference. This is the same fixed-gap order of choices used above and does not differentiate a divergent action constant. □

9.4 One direction across the actual overlap

Lemma 9.5. *On the union of the retained contact and separated domains, a noncenter fibre with $|\nu| \le \vartheta I$, for a common sufficiently small ϑ, has at most one zero of K', with multiplicity. At each critical point $|\tau| \ge cI$ and the second derivative has sign τ in every regular input coordinate.*

Proof. On the separated interval write $H = \tau U + \zeta V + \nu W$ using Proposition 9.3. Its limiting columns satisfy

$$V_0 < 0, \qquad (U_0/V_0)' = -\frac{\gamma^2}{2(2\gamma - s)^2} < 0$$

with fixed margins on J. The columns have bounded C^1 norms and retain these signs after the restrictions already made. Lemma 7.5, with error νW, gives $|\tau| \ge cI$ and a simple crossing of sign τ at every zero.

In contact divide the limiting columns by the positive factor p^2 and their common positive unit. They become

$$U = 1 - \frac{1}{\gamma p}, \qquad V = \frac{4}{\gamma} - \frac{2}{\gamma^2 p}.$$

At P, $V = 2/\gamma > 0$ and $(U/V)_p = \gamma^2/2 > 0$. The retained higher jets in Proposition 9.4 preserve these margins on one fixed smaller contact neighborhood. For the exact Taylor expression (9.8), its remainder divided by I has C^1 norm $O(I)$ when $|\nu| \le \vartheta I$. The same lemma applies with C^1 error $O(|\nu| + I^2)$, giving $|\tau| \ge cI$ and $\operatorname{sign} \mathcal{M}_p = \operatorname{sign} \tau$ at each zero. Since $f_L f_U > 0$ on the passing interval, (9.7) gives $\operatorname{sign} K_{pp} = \operatorname{sign} \tau$ there.

These statements concern the same K. If $p = p(s_\epsilon)$, then at a critical point

$$K_{s_\epsilon s_\epsilon} = K_{pp} (p_{s_\epsilon})^2.$$

The positive square, not the sign of the input orientation, transports its second-derivative sign. Near infinity use the increasing fibre coordinate $-s_\epsilon$; the same identity applies to this regular radial coordinate.

There is a genuine open overlap before this direction is used. Keep the contact radius r already fixed and $0 < d < r/4$. For two positive gaps every entry in a fixed interval, for example $(P + d, P + 2d)$, passes both complete slow paths. It belongs to the contact neighborhood and to the separated finite patch, with slightly enlarged intervals at both ends. Their finite initial and final regular legs are the same actual trajectories by uniqueness, and their phases and references were fixed together in Section 5. Hence their D_+, J, K agree on this open interval after the common regular input change. If one gap is nonpositive, the complete separated strip does not pass and is not adjoined; the contact whole-path conditions still select one interval.

The positive-gap union is therefore one interval on which all critical points are simple and have the same second-derivative sign. Between two such crossings, K' would have to cross in the opposite direction or have a nonsimple zero. Both are excluded. Thus the common direction gives at most one critical point on the whole union. The argument applies on every compact subinterval, hence on the whole half-open or compactified interval and at retained finite endpoint germs. □

Proof of Proposition 9.1. The preparations, domains and critical-point conclusions were proved above. We supply the forcing and the closing count. The fixed lower inner exit is on the central

side of the repeated minimum. Its input-independent weak reference therefore continues to the core independently of that slow passage. The Gaussian calculation of Lemma 7.2, on these fixed cores, gives $s_\pm = \mp\sqrt{\epsilon}\,\nu H_\pm$ and $H_c = H_+ + H_- = \sqrt{2\pi} + o(1) > h_*$ uniformly on the joint box.

On the separated contraction, (9.2) and its center identity give $|D_+ - J| \le C(I + |\nu|)e^{-c/\epsilon}$. The contact absolute maps have the same bound after one coefficient derivative: their separated entry and fixed exit give a normal flight of length at least c/ϵ in the lower block, and the incoming path has the fixed positive central action. The absolute estimates and their lost-exit extensions in Section 4 apply. Integrating along the entire lifted contact path of Proposition 5.11 therefore gives that same relative bound on every original contact input. These comparisons concern signed central extensions; they do not assert positive total heights at all intermediate coefficients.

Lemma 5.14 consequently gives $|\nu| \le C\epsilon^{-1/2}e^{-c/\epsilon}I = \eta(\epsilon)I$ on every fixed fibre containing a closing zero. At a zero the opposite signed references and positive deviations give positive common height, as in Section 7.6.

For a noncenter fibre the forcing bound meets the fixed threshold of Lemma 9.5. Its one-critical-point bound and Lemma 3.2 give at most three closing zeros on the common interval, including multiplicity. Restriction to actual counted orbits cannot increase this number. For $I = 0$ the comparison is the real center identity. Reflection uses its new outgoing label and inverse upper passage and preserves Poincare multiplicity by Section 2. □

10 Exact long matching and the secondary scale

This section treats the remaining corner

$$\bar{B} > 0, \qquad \bar{c} = \bar{m} = 0, \qquad |\bar{a}| \le 2\sqrt{\bar{B}}, \qquad p = q = \infty.$$

The primary affine upper field is parabolic; its returning arcs may have arbitrarily long passages on a scale determined by $\bar{d}$ and $\bar{m}$, independently of the primary scale. Follow each arc to its first exit, at either a fixed-height wedge section or a secondary-scale section. The former gives two long legs joined by a regular vertical passage; the latter gives a normalized affine upper problem. Attach the original lower passages in both cases and compare their outputs at the same central section. Throughout, $\bar{B}$ belongs to a fixed compact positive interval. Write

$$Z = Y + \epsilon, \quad \bar{d} = \bar{c} + \nu, \quad \theta = \epsilon\bar{B}, \quad b = 1 - \theta, \quad k = 1 - 2\theta, \quad \beta_\epsilon = k/b, \quad h_I = b/\bar{B}, \quad \mu = b/\theta.$$

In particular $\beta_\epsilon\mu = k/\theta$ exactly. The secondary scale is introduced at the second exit. Every counting interval is restricted by the complete lower passing conditions and positive original central heights. Coefficient division uses the signed central continuations with reference values fixed independently of input.

10.1 The weighted field and its exact long passage

The original upper equation, with its forcing retained, is

$$Z_\xi = Z[X + \bar{d}(Z - \epsilon)] - \epsilon\nu(Z - \epsilon). \tag{10.1}$$

This identity is the reason for using $\bar{d}$, rather than $\bar{c}$, in the new scale. For $r = Z^{-1/2}$, $v = Xr$ and $\mathrm{d}\tau/\mathrm{d}\xi = 1/r$, put

$$D_v = \bar{d}/r, \quad M_v = \bar{m}/r^2, \quad A_v = \bar{a}r, \quad N_v = \nu r, \quad E_v = \epsilon r^2.$$

The exact polynomial equations are

$$\begin{aligned} H_v &= v + D_v(1-E_v) - \epsilon N_v(1-E_v),\\ P_v &= (1-E_v)[M_v(1-E_v)-1] + \theta v^2 + \epsilon A_v v - vH_v/2,\\ r_\tau &= -rH_v/2, \quad v_\tau = P_v, \quad (D_v)_\tau = D_vH_v/2, \quad (M_v)_\tau = M_vH_v,\\ (A_v)_\tau &= -A_vH_v/2, \quad (N_v)_\tau = -N_vH_v/2, \quad (E_v)_\tau = -E_vH_v. \end{aligned} \tag{10.2}$$

Conserved products recover the original coefficients, for example $D_v r = \bar{d}$ and $N_v/r = \nu$, so retain these variables in the vertical passage.

For any fixed $V > 0$, small extended entries at $v = V$ pass uniquely to $v = -V$, with finite time and bounded finite jets. Indeed, at zero extended variables $P_v = -1 - kv^2/2$. On a slightly larger compact tube keep $P_v < -1/2$, bound the logarithmic rates of the five extended variables, and reduce the entry widths by their growth factor for time $4V$. First exit then gives passage in time at most $4V$. Finite variational equations and the transverse hitting equation give the stated jets, including the inverse. The radial component is $r_- = r_+ L_v$ with positive bounded unit $L_v = \exp[-\frac{1}{2}\int H_v \,\mathrm{d}\tau]$. Reflection makes $L_v = 1$ on the reversible leaves. These are jets in extended entries, not bounded original $\bar{d}$ derivatives at $r_+ = 0$.

On the outgoing adjoining chart set

$$\varphi = X/Z, \quad R = 1/Z = \varphi^2 w, \quad g = \bar{d}/\varphi, \quad j = \bar{m}/\varphi^2, \quad e = \epsilon\varphi^2 w.$$

Divide the field by $X > 0$. Direct substitution gives

$$\begin{aligned} G &= 1 + g(1-e) - \epsilon\nu\varphi w(1-e),\\ W &= (1-e)[j(1-e)-w] - b - g(1-e) + \epsilon\varphi w[\bar{a} + \nu(1-e)],\\ \varphi' &= \varphi W, \qquad w' = -w(G+2W), \qquad g' = -gW, \qquad j' = -2jW. \end{aligned} \tag{10.3}$$

Thus $g\varphi = \bar{d}$, $j\varphi^2 = \bar{m}$ remain constant. Its origin has contracting and expanding rates $-b, k$.

Proposition 10.1 (The exact long map)**.** *Fix an order N, a section $\varphi_0 > 0$, and a finite terminal height range $0 \le w_e \le w_*$. On sufficiently small common ordinary coefficient neighborhoods, the backward map from $(\varphi, w, g, j) = (H, w_e, \gamma, J)$ to $\varphi = \varphi_0$ is*

$$w_{\mathrm{in}} = w_e(H/\varphi_0)^{\beta_\epsilon} e^{\Psi}. \tag{10.4}$$

For one $q > 0$ depending only on the fixed order,

$$\begin{aligned} \Psi &= \Psi_0(\gamma, J, w_e; \theta) + O_N^{\mathrm{aux}}(H^q), \qquad H\partial_H\Psi = O_{N-1}^{\mathrm{aux}}(H^q),\\ \partial_x\Psi &= \epsilon S_x, \qquad S_x = O_{N-1}^{\mathrm{aux}}(H^q), \qquad x \in \{\bar{a}, \nu\}. \end{aligned} \tag{10.5}$$

In the auxiliary norms θ and ϵ are independent. The derivatives are ordinary terminal-height derivatives through zero, ordinary $\gamma, J, \bar{a}, \nu, \theta, \epsilon$ derivatives and $H\partial_H$, with the indicated total-order budgets. Ordinary ϵ differentiation of ϵS_x may remove its explicit factor. After $\theta = \epsilon\bar{B}$ is imposed, scale derivatives are taken with $E_\epsilon = \epsilon\partial_\epsilon + \theta\partial_\theta$; they preserve that factor. No bound on $\epsilon|\log H|$ is required.

Proof. Put $t = \log(\varphi/H)$ and $L = \log(\varphi_0/H)$. Exactly,

$$\varphi = He^t, \quad g = \gamma e^{-t}, \quad j = Je^{-2t}, \qquad w_t = -wQ, \quad Q = 2 + G/W, \quad 0 \le t \le L. \tag{10.6}$$

Choose the larger rectangle with $W < -c$, $Q > 1/2$; then $w \le w_e e^{-t/2}$ and no earlier exit is possible. Subtracting the exact saddle ratio gives

$$Q - \beta_\epsilon = (G-1)/W + (W+b)/(bW)$$

with a right side integrable uniformly in L. Integration gives $\Psi = -\int_0^L (Q - \beta_\epsilon)\,\mathrm{d}t$ and the exact power in (10.4). Appendix D.6 differentiates its renormalized integral equation, including the moving endpoint and the height $w_e = 0$, and proves (10.5). The limiting unit retains its terminal-height dependence: $\Psi_0(0, 0, w_e; \theta) = -(2b)^{-1} \log(1 + 2w_e/k)$. □

The exact power can also be checked on the reversible field. Its first integral is $Z^{-2\theta}(X^2 + \bar{B}^{-1} + 2Z/k)$. At fixed $Z/|X|$ its comparison with the maximum-height section gives the power b/k, inverse to β_ϵ. For positive scale $1 < b/k < 2$, so that single power need not have an ordinary second derivative at zero. This is why the long estimate uses Euler input derivatives. Keeping β_ϵ exact is also essential for unrestricted separation of scales: replacing it by its zero-scale value would leave an error proportional to $\epsilon \log H$, which need not be small.

10.2 Two actual exits and compact section labels

Locate the first exit of each returning upper arc to identify the middle passage and define tags at its actual section points.

Fix a small $w_0 > 0$. At the wedge exit $w = w_0$, write $\varphi = H$, $\gamma = \bar{d}/H$, $J = \bar{m}/H^2$. The vertical passage (10.2), followed by the reflected long map, has data

$$\widetilde{H} = HL_v, \qquad \widetilde{\gamma} = -\gamma/L_v, \qquad \widetilde{J} = J/L_v^2.$$

Consequently the two original horizontal power inputs are

$$u_\pm = H^{\beta_\epsilon} L_\pm, \qquad L_\pm, L_\pm^{-1} > c > 0. \tag{10.7}$$

Their logarithms have the finite jets above. On a fixed original coefficient fibre the derivative is

$$\mathcal{D} = H\partial_H - \gamma\partial_\gamma - 2J\partial_J, \qquad \mathcal{D}\log u_\pm = \beta_\epsilon + O(\delta + H^q) > c > 0$$

when $|\gamma|, |J| < \delta$. Reflection gives $u_+ = u_-$ on $\bar{a} = \bar{d} = \nu = 0$.

For $\lambda > 0$ define

$$\lambda = \max(|\bar{d}|, \sqrt{|\bar{m}|}), \quad C = \bar{d}/\lambda, \quad M = \bar{m}/\lambda^2, \quad \max(|C|, \sqrt{|M|}) = 1, \quad \Lambda = \lambda^{\beta_\epsilon}. \tag{10.8}$$

Coefficient differentiation is performed on ordinary open neighborhoods of these signed faces, keeping λ fixed rather than redifferentiating the maximum. Put $\hat{x} = \lambda X$, $\hat{z} = \lambda^2 Z$, $\delta_s = \epsilon\lambda^2$ and $\mathrm{d}t_s/\mathrm{d}\xi = 1/\lambda$. The exact field is

$$\begin{aligned} \dot{\hat{x}} &= (\hat{z} - \delta_s)[M(\hat{z} - \delta_s) - 1] + \theta\hat{x}^2 + \epsilon\bar{a}\lambda\hat{x}, \\ \dot{\hat{z}} &= \hat{z}[\hat{x} + C(\hat{z} - \delta_s)] - \epsilon\nu\lambda(\hat{z} - \delta_s). \end{aligned} \tag{10.9}$$

There is no coefficient ν/λ. At zero scale division by $\hat{z} > 0$ gives the affine field

$$\hat{x}' = M\hat{z} - 1, \qquad \hat{z}' = \hat{x} + C\hat{z}. \tag{10.10}$$

Proposition 10.2 (First exits of returning upper arcs)**.** *Consider actual upper arcs returning from $(p, 0)$ to $(-q, 0)$ with primary data tending to the stated corner, $\nu \to 0$ and $p \to \infty$. Closure at the central section is not required. After fixed choices of sections and one common coefficient restriction, each arc has either a first wedge exit*

$$w = w_0, \quad 0 < H < H_0, \quad |\bar{d}| < \delta H, \quad |\bar{m}| < \delta H^2,$$

or a passage between $\hat{\varphi} = \hat{x}/\hat{z} = L$ and $-L$ in one fixed compact projective rectangle $0 < R = 1/\hat{z} < R_{\max}$. For $\lambda = 0$ only the first alternative occurs. No uniform flight-time bound is asserted.

Proof. The arrival estimate in Appendix D.7 sends $p \to \infty$ to the fixed section $\varphi = \varphi_0$ with $w \to 0$, without any bound on $\epsilon \log p$. Also $q \to \infty$: otherwise a bounded positive subsequence of q and the reflected regular upper transition bound p; if $q \to 0$, the small-entry equation gives $p/q \to 1$. Both contradict $p \to \infty$. These arguments involve returning arcs, not their central closing values.

Choose the wedge constants first, then $L \geq 10$ so large that $1/L < \delta/2$, $1/L^2 < \delta/2$. Keep the adjoining signs $W < -c$, $G + 2W < -c$ on $\lambda L \leq \varphi \leq \varphi_0$, $0 < w \leq w_0$. Here φ decreases and w increases until a first terminal side is reached. At $w = w_0$ the required ratios are at most $1/L, 1/L^2$. Its value H tends to zero, since on every fixed $\varphi \geq \varphi_1 > 0$ strip the time and multiplicative height growth are bounded while the entry height tends to zero. The other side is $\varphi = \lambda L$; assign a corner hit to either alternative. If $\lambda = 0$, exponential height growth reaches $w = w_0$.

At the second side $0 < R \leq L^2 w_0$. The fixed first-exit barrier of Appendix D.7 confines the arc until $\hat{\varphi} = -L$. This crossing must occur for a returning arc, whose incoming endpoint has $\hat{\varphi} = -q/(\epsilon\lambda) < -L$. It then has $w < R_{\max}/L^2 < w_0 + 12$; reflected adjoining monotonicity continues the same arc to the original horizontal section. Uniqueness excludes another middle excursion. The spatial barrier is not a claim that arbitrary incident trajectories return. □

Lemma 10.3 (Actual secondary tags)**.** *On the fixed outgoing secondary section, its actual point (L, R) has a positive affine-flow tag $P \in [p_0, \infty]$. The reciprocal tag $t = 1/P$ is a regular increasing coordinate in R, including $R = 0$, with bounded finite jets and inverse jets. The incoming section has the reflected tag Q with the same properties. These tags are not defined as $\lambda p, \lambda q$.*

Proof. Follow (10.10) backward from the actual section point. Multiplying by its fixed initial radial value R gives

$$(\bar{x}, \bar{z})(0) = (L, 1), \qquad \dot{\bar{x}} = R - M\bar{z}, \quad \dot{\bar{z}} = -\bar{x} - C\bar{z}.$$

Before $\bar{z} = 0$, $0 \leq \bar{z} \leq 1$ and $\bar{x}(s) \geq L - s$. A first-exit comparison gives a transverse zero at $0 < T < 2/L$, with $\bar{x}(T) > L/2$. This solution is smooth on the entire compact set $0 \leq R \leq R_{\max}$, $|C|, |M| \leq 1$. Thus $P = \bar{x}(T)/R \geq L/(2R_{\max}) =: p_0$ and $t = R/\bar{x}(T)$ vanishes simply at zero. Away from zero the flow determinant gives

$$\det(f, a_R) = \frac{R - M + L(L + C)}{R^3} > 0, \qquad \det(f, (1, 0)) = -P < 0$$

at the two sections, respectively; hence $P_R < 0$ and $t_R > 0$. Compactness supplies a common inverse margin. Reflection proves the incoming statement. □

The actual section points and the first-exit choice give the compact data

$$(\bar{B}, \bar{a}, \epsilon, \lambda, C, M, t_P, t_Q), \quad \max(|C|, \sqrt{|M|}) = 1, \quad 0 \leq t_P, t_Q \leq p_0^{-1},$$

together with the wedge or secondary exit choice and its fixed section. At $\lambda = 0$ the secondary entries are limiting directions, not values of a divided original coefficient. The same construction applies to nonperiodic returning arcs and will be used for joint limits of an outer orbit and intermediate arcs.

10.3 Exact cancellation and regular comparisons

The original lower maps have one common horizontal exponent. Its absolute power supplies the bound on ν at a closing zero; cancellation in the derivative quotient supplies the relative expression used to count zeros on the fixed fibre.

The original horizontal lower preparation is

$$D_{\pm}(u) = K_{\pm} u^{\mu} e^{P_{\pm}(u)/\epsilon}, \quad P_{\pm}(0) = 0, \quad \|P_{\pm}\|_{C^N} \le C, \quad H_{\pm}(u) := h_I + uP'_{\pm}(u) > c. \tag{10.11}$$

This is (4.25), in the two reflected coordinates. A finite lower double root is crossed between fixed slow sections before the final fixed core. Its whole-path condition must hold; otherwise the passage is absent. On its passing side the additional action belongs to $\epsilon \log K_{\pm}$ alone. The relative normal estimate supplies the same ordinary input bounds for $P_{\pm}$. No bounded coefficient derivatives of that divergent action are inferred.

At $\hat{\varphi} = \pm L$ apply the variable-height long map, where $H = \lambda L$, $\gamma = C/L$, $J = M/L^2$. After regular outer connectors it gives

$$u_{\pm} = \Lambda v_{\pm}. \tag{10.12}$$

At a finite regular tag $v_{\pm}, v'_{\pm} > c$ after choosing a common oriented input. At strict-focus infinity, using $t = 1/P$, $v_{\pm} = tV_{\pm}(t)$ with positive bounded units, derivatives and inverse margins. The angular field there has a negative margin on the compact passage, so its time and its ordinary radial jets through $R = 0$ are bounded. At singular tags the two outer maps are constructed separately.

The long unit converges as $\lambda \to 0$ uniformly in θ; then $\theta \to 0$ identifies its limit with the affine tag flow:

$$v_-/v_+ \longrightarrow P/Q(P; M, C) \tag{10.13}$$

with the selected section and coefficient jets, including strict-focus reciprocal zero. With s denoting the long variable, the affine backward leg has $\hat{\varphi} = Le^s$, $\hat{x} = [Le^s w(s)]^{-1}$ and $w(s) \sim w_e e^{-s} e^{\Psi_0}$. Its axis endpoint therefore satisfies

$$P^{-1} = L w_e e^{\Psi_0(C/L, M/L^2, w_e; 0)}.$$

The original horizontal phase is $\varphi_0 w_{\mathrm{in}}$ times a unit tending to one, so its quotient by Λ also tends to P^{-1}. Reflection gives Q^{-1} on the other leg. The ordinary height estimates and the radial passage give the stated jets after removing the common simple factor at reciprocal zero.

Let $I_{\pm} = D_{\pm}(u_{\pm})$ and $F = I_+ - I_- - \sqrt{\epsilon}\,\nu H_c$, where $H_c > c$ is independent of input. For a common increasing input let $l_{\pm} = D \log u_{\pm}$; use $D = \mathcal{D}$ in the wedge and $D = t\partial_t$ on a secondary reciprocal interval. The complete derivative logarithm is

$$\begin{aligned} \mathcal{K} = \epsilon \log(I'_-/I'_+) &= \epsilon \log(K_-/K_+) + h_I \log(u_-/u_+) \\ &\quad + P_-(u_-) - P_+(u_+) + \epsilon \log(l_-/l_+) \\ &\quad + \epsilon \log \frac{H_-(u_-)}{H_+(u_+)}. \end{aligned} \tag{10.14}$$

In this quotient, cancel the identical power Λ, or H^{β_ϵ}, before any coefficient derivative; at reciprocal zero cancel the common simple factor t as well. The exact cancellation removes $(\partial\beta_\epsilon)\log\lambda$ from the relative terms. Single-sided compositions still contain factors $\Lambda|\log\lambda|^j$, which a reserved positive power absorbs. Thus the estimates allow unbounded $\epsilon|\log\lambda|$.

Lemma 10.4 (Absolute scale budget)**.** *On each regular-lower center comparison box, for arbitrary fixed $A, N > 0$, sufficiently small radial and scale cutoffs give*

$$|I_+| + |I_-| \le Ce^{-A/\epsilon}\lambda^N \quad \text{or} \quad Ce^{-A/\epsilon}H^N$$

with the finite coefficient derivatives used for center division. The same absolute upper bound holds on a passing lower-double overlay, whose extra action only decreases the amplitudes. The derivatives of that extra action are excluded.

Proof. Before any scale factor is discarded, the exact combined power is $\Lambda^{\mu} = \lambda^{k/(\epsilon\bar{B})}$. Bound the remaining amplitudes by $e^{C_0/\epsilon}$. Choose λ_0 so that $\frac{1}{2}\inf(k/\bar{B})\log(1/\lambda_0) > C_0 + A$, then choose ϵ so that $\epsilon N < \frac{1}{2}\inf(k/\bar{B})$. This proves the estimate. Further reserved powers absorb every fixed logarithm, inverse scale loss and coefficient derivative. The proof for H is identical; at focus infinity reserve part of the common input power as well. A fixed lower core has an additional positive contraction budget, uniformly on compact coefficient blocks, by Proposition 4.3. A fixed lower bottleneck contributes only further positive action. □

On a common regular-lower box reflection gives the exact identity $I_+ = I_-$ when $\bar{a} = \bar{d} = \nu = 0$ in the wedge, or $\bar{a} = C = \nu = 0$ at secondary scale. On that common box, divide the absolute difference by the transverse coefficients using this identity. Only then impose the closing equation. The absolute budget and Lemma 5.14 give, with $\eta_\epsilon \to 0$,

$$|\nu| \le \eta_\epsilon(|\bar{a}| + |\bar{d}|) \quad \text{in the wedge,} \qquad |\nu| \le \eta_\epsilon \lambda^N(|\bar{a}| + |C|) \quad \text{at secondary scale.} \tag{10.15}$$

The exponent N is any preselected finite exponent after reserving the required extra losses. In the first formula choose at least two powers of H before replacing $H^N|\bar{d}|/H$ by $|\bar{d}|$. These scale powers come from the absolute deviations. Once a closing zero has supplied the coefficient inequality, the critical-function estimates use only the weaker bounds

$$|\nu| \le \eta_\epsilon(|\bar{a}| + |\bar{d}|) \quad \text{in the wedge,} \qquad |\nu| \le \eta_\epsilon(|\bar{a}| + |C|) \quad \text{on regular secondary intervals,} \tag{10.16}$$

imposed on a fixed coefficient fibre, with $\eta_\epsilon \to 0$.

For regular lower coefficients the endpoint constant is

$$L(\bar{a}, \bar{B}) = -\frac{\pi\bar{a}}{\bar{B}\sqrt{4\bar{B} - \bar{a}^2}}. \tag{10.17}$$

Indeed its limiting difference is $-2\bar{a}\int_0^\infty X^2/(F_+F_-)\,\mathrm{d}X$. With $s = \sqrt{\bar{B}}X$, reciprocal symmetry of the quartic reduces the integral to $\pi/(2\bar{B}\sqrt{4\bar{B} - \bar{a}^2})$. The subtracted tail is integrable with coefficient derivatives on every compact subset of $|\bar{a}| < 2\sqrt{\bar{B}}$. The exact horizontal power subtraction and the finite-order lower phase estimates transfer this limit with the selected jets. At a lower double root, instead, retain $\mathcal{K} = \mp A_l + O(1)$ on its whole-path passing set; the sign corresponds to the contracting side. It is a strict constant-sign comparison, not a bound on $\mathcal{K}'$.

Proposition 10.5 (Wedge and regular secondary rows)**.** *The wedge has local bound $Z(F) \le 2$, and so does every finite regular secondary tag or strict-focus infinite tag. Each assertion holds on an open coefficient and section neighborhood, includes both signs of ν, and retains all lower passing conditions. Near a regular center the derivative logarithm has at most one zero under the respective bound in (10.16). For the closing-zero claim that bound follows from any one closing zero; if none exists, there is nothing to count. A lower-double action-dominated neighborhood has a strict sign of $\mathcal{K}$ and hence at most one closing zero.*

Proof. In the wedge, divide (10.14) on its common center box:

$$\mathcal{K} = \bar{a}A + (\bar{d}/H)C_w + \nu E, \qquad A \to -\frac{\pi}{2\bar{B}^{3/2}}, \quad C_w \to -\frac{2c_w}{3\bar{B}}, \quad c_w = \frac{\sqrt{1 + 2w_0}}{w_0}. \tag{10.18}$$

The $\mathcal{D}$ derivatives of these columns tend to zero. To compute the second column, the limiting center parabola is $Z = (p^2 - X^2)/2$ and $p = c_w/H$ at $w = w_0$. Variation of $Z'' - \bar{d}Z' = -1$ gives $\partial_{\bar{d}}(q - p) = 2p^2/3$. Its endpoint action contribution is $-2Hp^3/[3(1 + \bar{B}p^2)]$, with the displayed limit. The first column is the derivative of (10.17) at zero. The fixed-section endpoint variation cancels between the two center maps. The long estimates transfer these calculations with their $\mathcal{D}$ jets on small $|\gamma|, |J|, H, \epsilon$.

Absorb the forcing column by adding $\nu\bar{a}E/(\bar{a}^2+\bar{d}^2)$ to A and $H\nu\bar{d}E/(\bar{a}^2+\bar{d}^2)$ to C_w. The fixed-fibre operator has $\mathcal{D}\bar{a}=\mathcal{D}\bar{d}=\mathcal{D}\nu=0$. Thus (10.16) makes the corrections $O(\eta_\epsilon)$ and $O(H\eta_\epsilon)$ in the required $\mathcal{D}$ jets. Both columns retain negative margins and $\mathcal{D}\log[C_w/(HA)]<-1/2$. Thus their ratio is strictly monotone and $\mathcal{K}$ has at most one zero. Zero transverse coefficients give the exact center. Strict nonzero $\bar{a}$ and the lower-double boundary are instead covered by the value or action sign in (10.17). Compactness of the closed lower cone supplies a finite choice of such neighborhoods; one neighborhood counts the entire smaller wedge of a fixed fibre.

At regular secondary tags the limit is

$$\mathcal{K}_0 = L(\bar{a},\bar{B}) - \bar{B}^{-1}\log(Q/P). \tag{10.19}$$

We record the affine-flow facts that determine its columns. For return time T, set $\zeta(s)=e^{-Cs/2}\hat{z}(s)$ along the orbit. If $S_\chi''=\chi^2 S_\chi$, $S_\chi(0)=0$, $S_\chi'(0)=1$, $\chi^2=M+C^2/4$, the Dirichlet formula gives

$$P=\frac{\int_0^T S_\chi(T-s)e^{-Cs/2}\,\mathrm{d}s}{S_\chi(T)},\quad Qe^{-CT/2}=\frac{\int_0^T S_\chi(s)e^{-Cs/2}\,\mathrm{d}s}{S_\chi(T)}.$$

The first-return time precedes the first positive zero of S_χ. Pair s and $T-s$ for $0<s<T/2$: the kernel difference $S_\chi(T-s)-S_\chi(s)$ is positive by the hyperbolic sine formula or its linear limit, and for $\chi=i\omega$ it equals $2\cos(\omega T/2)\sin(\omega(T/2-s))/\omega>0$ because the first-return condition gives $T<\pi/\omega$. On this half-interval, $e^{-Cs/2}-e^{-C(T-s)/2}$ has sign C, so the paired integral gives $\operatorname{sign}(P-Qe^{-CT/2})=\operatorname{sign}C$. Together with $Q_P=(P/Q)e^{CT}$ this proves $\operatorname{sign}\partial_P\log(Q/P)=\operatorname{sign}C$. On $C=0$ the divided column is

$$G(P,M)=\partial_C\log(Q/P)|_{C=0}=\frac{2}{P^2}\int_0^{z_{\max}}\sqrt{P^2-2z+Mz^2}\,\mathrm{d}z,$$
$$G_P=2\int_0^1\frac{v^2}{1-MP^2v^2}\,\mathrm{d}v>0.$$

The vanishing endpoint integrand justifies its differentiation. For $M=-r^2$, direct integration gives $G(1/t,M)=\pi/(2r)-2t/r^2+O(t^2)$. More generally at strict-focus infinity the affine solution gives

$$\log(Q/P)|_{t=0}=\frac{\pi C}{\sqrt{-4M-C^2}},\quad \partial_t\log(Q/P)|_{t=0}=-\frac{C(1+e^{-\pi C/\sqrt{-4M-C^2}})}{-M}.$$

These follow by imposing the first zero in

$$\hat{z}(s)=\frac{1}{M}\left[1-e^{Cs/2}\left(\cos\omega s-\frac{MP+C/2}{\omega}\sin\omega s\right)\right],\qquad \omega=\sqrt{-M-C^2/4};$$

they have finite coefficient jets on strict-focus compact sets.

For $C\neq0$ these derivative margins transfer to $\mathcal{K}$. For $C=0,\bar{a}\neq0$ use its nonzero value. Near $\bar{a}=C=0$, divide into $\mathcal{K}=\bar{a}A+CD+\nu E$; the limiting columns are $A_0=-\pi/(2\bar{B}^{3/2})$ and $D_0=-G(1/t,M)/\bar{B}$. Their ratio has a strict derivative margin, also at $t=0$. The mixed jets of Proposition 10.1 and the horizontal preparation give $\|E\|_{C^1}\le C$ uniformly on this regular-lower box. To see this, cancel Λ in (10.14) before taking one ν and one input derivative. At focus infinity write $v_\pm=tV_\pm$ and cancel t as well. The remaining relative units have positive margins; one input derivative of their logarithmic rates and one coefficient derivative spend two input orders and one coefficient order, with one further input order reserved for the simple-factor division. At fixed λ,C, the original lower coefficient is $\bar{c}=\lambda C-\nu$, so the ν derivative is $\partial_\nu-\partial_{\bar{c}}$ in the original lower box. The single-sided terms retain a positive

scale power. For fixed horizontal and central sections, shrink the regular center box so that $|\bar{a}| \le \sqrt{B_*}/2$ and $\epsilon|\bar{m}| \le 1/4$. The exact formula (4.4), with a smaller scale cutoff, gives $F_I^{\pm}(s) \ge B_*/2 + s^2/2$ on $[0, x_R]$. Write $\beta_I^{\pm}/(sF_I^{\pm}) = h_I/s + q_{\pm}$ and use the fixed split x_h of Proposition 4.8. The horizontal proof and the curved and central connections give

$$\begin{aligned}\epsilon \log K_{\pm} = & -S_c^{\pm} - h_I \log x_h - \int_0^{x_h} q_{\pm}(s)\,\mathrm{d}s - \int_{x_h}^{x_R} \frac{\beta_I^{\pm}(s,0)}{sF_I^{\pm}(s)}\,\mathrm{d}s \\ & + \epsilon \log U_{\pm} - \epsilon \log \epsilon, \qquad U_{\pm} = Z_e^{\pm}(0) e^{j_{\pm}(0)} u_c^{\pm}(0) a_c^{\pm}.\end{aligned}$$

The units $U_{\pm}$ are positive with bounded logarithmic coefficient jets by Propositions 4.6 and 4.7. For $\mathscr{D}_\nu = \partial_\nu - \partial_{\bar{c}}$, fixed R gives $\mathscr{D}_\nu h_I = \mathscr{D}_\nu S_c^{\pm} = 0$. Differentiating the fixed integrals using the displayed positive denominator and Proposition 4.1 therefore bounds $\mathscr{D}_\nu[\epsilon \log(K_-/K_+)]$ uniformly on this regular center box. Define the forcing column by integrating this ν derivative from zero. The corrections

$$\frac{\nu \bar{a} E}{\bar{a}^2 + C^2}, \qquad \frac{\nu C E}{\bar{a}^2 + C^2}$$

are therefore $O(\eta_\epsilon)$ in C^1 under (10.16). Adding them to A, D preserves their margins and the strict derivative of their quotient. This gives $Z(\mathcal{K}) \le 1$ and $Z(F) \le 2$. The lower-double case uses its independent constant action sign. □

Lemma 10.6 (Outward direction and regular departure)**.** *The wedge's outward derivative is a positive multiple of $-\mathcal{D}$, and the secondary tag P increases outward. Under the corresponding relative forcing bounds, if $\bar{a}\bar{d} < 0$ and $\mathcal{K} = 0$ or $\operatorname{sign}\mathcal{K} = -\operatorname{sign}\bar{d}$, its outward derivative has sign $-\operatorname{sign}\bar{d}$ on the wedge and on a fixed compact regular secondary set.*

Proof. For the actual input $a(p) = (p, \epsilon)$ and wedge output $b(H) = (1/(w_0 H), 1/(w_0 H^2))$, the flow determinants are $-\epsilon p < 0$ and $\dot{w}/(w_0^3 H^4) > 0$. The flow Jacobian has positive determinant, so $H'(p) < 0$ and $\mathrm{d}/\mathrm{d}p = (H'(p)/H)\mathcal{D}$ exactly. The tag orientation follows from Lemma 10.3 and the ordering of the transverse outgoing intersections.

Put $s = \operatorname{sign}\bar{d}$. The hypothesis $s\mathcal{K} \le 0$ in the wedge first excludes the strict nonzero lower-cone values and then, by its two negative columns, bounds $|\bar{a}| \le C|\bar{d}|/H$. Inserting this bound into $-s\mathcal{D}\mathcal{K}$ leaves the strict negative main term $(|\bar{d}|/H)C_w$; the column derivatives are arbitrarily smaller. At secondary scale the same hypothesis and (10.19) give $|\bar{a}| \le C|C|$. The outward derivative of $\log(Q/P)$ has sign C and a margin relative to $|C|$, using $-t$ as the regular outward coordinate at focus zero. The divided relative errors are small compared with that margin. Thus the outward derivative of $\mathcal{K}$ has sign $-s$. This is a local direction statement on actual open regular overlaps; it does not add their zero counts. □

10.4 Singular secondary endpoints

For use at every singular row keep the exact projective equation. With $R = 1/\hat{z}$, $\hat{\varphi} = \hat{x}/\hat{z}$, $A_s = \epsilon\bar{a}\lambda$, $\nu_s = \epsilon\nu\lambda$, it is

$$\begin{aligned}R' = & -R(\hat{\varphi} + C) + (C\delta_s + \nu_s)R^2 - \nu_s\delta_s R^3, \\ \hat{\varphi}' = & M - b\hat{\varphi}^2 - C\hat{\varphi} + R[-1 - 2M\delta_s + (A_s + C\delta_s + \nu_s)\hat{\varphi}] \\ & + R^2[\delta_s + M\delta_s^2 - \nu_s\delta_s\hat{\varphi}].\end{aligned} \tag{10.20}$$

Its coefficients converge on fixed compact rectangles. The lower coefficient $\bar{B}$, however, is still measured at the original scale; replacing it by $\bar{B}/\lambda^2$ would lose the compact coefficient range required by the lower estimates. We therefore attach each new upper map to the original lower maps (10.11).

Proposition 10.7 (A finite secondary saddle)**.** *For $M > 0$, the limiting tags $P = P_c(M, C)$, $Q = P_c(M, -C)$ have local bound $Z(F) \le 3$. Let $\kappa = r - 1$ for the actual saddle ratio r. Under $|\nu| \le \eta_\epsilon(|\bar a| + |\kappa|)$, with $\eta_\epsilon \to 0$, their derivative logarithm also satisfies $Z(\mathcal{K}) \le 2$. Here*

$$P_c(M, C) = \frac{2}{C + \sqrt{C^2 + 4M}}.$$

The actual domain is the positive returning saddle quadrant, intersected with every lower whole-path condition and the positive central outputs. A lower double root may shorten this domain.

Proof. The limiting saddle is $(-C/M, 1/M)$, with ratio $r_0 = (\sqrt{C^2 + 4M} - C)/(\sqrt{C^2 + 4M} + C)$. Near its positive height, division by $\hat z$ turns (10.9) into an analytic $O(\epsilon)$ perturbation of the affine field, uniformly through $\lambda = 0$. The axis construction of Section 4.4 therefore applies on a common neighborhood. Its two outer maps are $u_+(t) = \Lambda v_+(t)$, $u_-(y) = \Lambda v_-(y)$ with $v_\pm, v_\pm' > c$; cancellation of Λ gives $v_-(0)/v_+(0) \to r_0$.

Near $C = 0$ preserve the exact reversible identity. On $\bar a = C = \nu = 0$ put $z_s = \delta_s + 1/M$ and

$$W_c(z) = (z/z_s)^{-2\theta}, \qquad I_c(z) = \int_{z_s}^{z} \frac{2(s - \delta_s)[M(s - \delta_s) - 1]}{s} W_c(s)\, \mathrm{d}s.$$

The first integral is $W_c\hat x^2 - I_c$. Since $I_c = (z - z_s)^2 J_c$ with $J_c(z_s) = 1/z_s > 0$, its analytic square root supplies reflected product coordinates. Extend them and normalize the neighboring axes, using the identity when they are already normalized. The reciprocal preparation then gives the actual ratio $\kappa = r - 1$ and

$$y = t^r e^w, \qquad w = -\sigma_s t\omega(t; \kappa) + O_E(\epsilon n t^{7/4}), \qquad |\sigma_s| \le C\epsilon n, \quad n = |\bar a| + |\kappa| + |\nu|.$$

The identity is $y = t$ on the center. Hadamard division on the common coefficient box supplies the factor n in the remainder. Finite mixed jets are obtained from the same equation and coefficient division on its fixed positive quadrant.

Put $U = D_+(u_+)$, $V = D_-(u_-)$ and $\mathsf{A}_\pm = \epsilon U''/U', \epsilon V''/V'$. For $u = \Lambda v$, set $Z_\pm = H_\pm(u)v'/v$. Then $\mathsf{A}_\pm = Z_\pm + \epsilon Z_\pm'/Z_\pm$. These rates have positive margins and bounded ordinary jets. A lower-double action is a constant multiplier and cancels here. It is not differentiated in coefficients.

Integrating these rates gives (7.22) with $(v, K) = (y, \mathcal{K})$:

$$\mathcal{K} = K_c + \epsilon\kappa \log t + \mathcal{E}(t), \qquad K_c = \epsilon \log[rV'(0)/U'(0)]. \tag{10.21}$$

The remainder and its Euler derivative are $O(t^{1/2})$, or $O(nt^{1/2})$ near the center. Cancellation of Λ before differentiating shows

$$K_c \longrightarrow L(\bar a, \bar B) + \bar B^{-1} \log r_0.$$

On a regular center box this yields $K_c = \bar a A_c + \kappa B_c + \nu C_c$, where $A_c \to -\pi/(2\bar B^{3/2})$ and the other columns are bounded. The exact center and the absolute budget give $|\nu| \le \eta(|\bar a| + |\kappa|)$ at a closing zero. The actual ratio coordinate is legitimate because $\partial_C \kappa \to -1/\sqrt{M}$.

We apply Lemma 7.12 to this preparation. For a nonzero limiting $\bar a$, clause (i) uses $R_* = 1$; for a dominating lower action it uses that action's magnitude. Near the center fix a large K_0. The divided constant and relative forcing give the same hypotheses with $R_* = |\bar a|$ in the cone $|\bar a| > K_0|\kappa|$.

In the remaining cone the constant may cancel, so we count the Euler derivative instead. For $|\bar a| \le K_0|\kappa|$, the coefficient bound gives $n \le (1 + K_0)(1 + \eta_\epsilon)|\kappa|$. If $\kappa = 0$, it first gives $\bar a = \nu = 0$, the exact center. On the noncenter part, differentiation gives (7.26) with $K = \mathcal{K}$, $A = \mathsf{A}_-(0)$,

$B_0 = \mathsf{A}_+(0)$ and $\sigma = \sigma_s$; the prepared remainder satisfies $|D^j R| \le C n t^{3/2}$, $j \le 2$. Since $\sigma_s/\kappa = O(\epsilon)$, the positive rate margin gives $W/\kappa > c$, $|T| \le C|\kappa|$ and $|R'| = |DR|/t \le C|\kappa| t^{1/2}$. Clause (ii) gives $Z(\mathcal{K}) \le 2$. Away from ratio one, the unsheared reciprocal equation gives $y = k_0 t^r e^w$ with a bounded positive k_0 and the derivative expansion of clause (iii), with $A = \mathsf{A}_-(0)$ and $B = \mathsf{A}_+(0)$. Thus the same bound holds there. By 3.2, $Z(F) \le 3$ on the lower-passing, positive-central subinterval; no endpoint value at a truncated passage is used. □

We also retain the actual coefficient identity needed by a later entrance argument. On $\bar{a} = \nu = 0$ let $q = M + \theta C^2$. The saddle Jacobian of the divided field is exactly

$$J_s^{\mathrm{ref}} = \begin{pmatrix} -2\theta C/(1+\delta_s q) & (M - \theta C^2)/(1+\delta_s q) \\ 1 & C \end{pmatrix}.$$

The remaining coefficients change it by $O(\epsilon\lambda(|\bar{a}| + |\nu|))$. Thus on a compact positive-M block

$$\kappa = -\alpha_s C + O(C^2 + \epsilon\lambda|\bar{a}| + \epsilon\lambda|\nu|), \qquad \alpha_s = \frac{1 + M\delta_s - 2\theta}{\sqrt{M(1+M\delta_s)}} > c > 0. \tag{10.22}$$

The reference ratio is exactly one at $C = 0$. This is an identity for the secondary parameters $C = \bar{d}/\lambda$; its small ν term does not justify dropping the order-one ν term in the primary ratio when $\bar{c}$ is held fixed.

Proposition 10.8 (Real, repeated and weak secondary rows)**.** *For $C > 0$ the following limiting rows have the stated local bounds; reflection supplies $C < 0$:*

row	*limiting tags*	$Z(F)$
$M < 0,\ C^2 + 4M > 0$	$P = P_c,\ Q = \infty$	≤ 2
$C = 2g,\ M = -g^2$	$P_c = 1/g \le P < \infty,\ Q = \infty$	≤ 2
$C = 2g,\ M = -g^2$	$P = Q = \infty$	≤ 2
$M = 0$	$P = 1/C,\ Q = \infty$	≤ 3

Every assertion holds in an open coefficient and section neighborhood. It includes neighboring signs of the unfolding parameters and either lower double root on its whole-path passing set. Positive central outputs are imposed in addition to the upper conditions below.

Proof. In every row the lower maps remain (10.11), with divergent lower actions confined to input-independent constants. The upper singularities require different estimates: a hyperbolic power for the strict row, a quadratic angular passage for the repeated rows, and the exact moving radial factor for the weak row. We keep the passing domains throughout these preparations and their adjacent unfoldings.

For the strict row define

$$\Delta = \sqrt{C^2 + 4bM}, \quad \varphi_s = (-C - \Delta)/(2b), \quad \gamma = \varphi_s + C = \frac{Ck - \Delta}{2b}, \quad r = \gamma/\Delta.$$

Both Δ, γ have positive margins. The projective saddle has eigenvalues $-\gamma, \Delta$. Straighten its axes and integrate its orbital reciprocal equation in the returning quadrant. The map is $k_0 t^r e^{j(t)}$, with three selected Euler derivatives of j tending to zero as the section shrinks. Compact positive-ratio margins absorb the fixed logarithmic losses, including internal resonances. The other quadrant enters the attracting node. After the two long legs,

$$u_+ = \Lambda v(t), \quad v, v' > c, \qquad u_- = \Lambda k_1 t^r e^{j_1(t)}.$$

In (10.14), $t\partial_t \log l_+ = 1 + O(t)$ and $t\partial_t \log l_- = o(1)$. Consequently $t\mathcal{K}' = h_I r - \epsilon + o(1) > 0$. The terms from $P_\pm$ are $O(\Lambda t) + O(\Lambda t^{r_*})$. This proves the two-zero closing bound.

Near a repeated root the boundary restriction of (10.20) gives exactly, after the positive division by b and a finite-order phase projection,

$$\begin{aligned} v' &= \mu_u + v^2 =: f(v), \qquad R' = -R\beta_u(v,R), \\ \beta_u(v,0) &= (\Xi - v)/b, \qquad \mu_u = -\frac{C^2+4bM}{4b^2}, \qquad \Xi = \frac{Ck}{2b}. \end{aligned} \tag{10.23}$$

The normal margin is positive. Fix the finite derivative order before taking the angular strip small enough that its slow variation growth is dominated by that margin, as in Section 4.5. The phase construction preserves this exact boundary equation. A moving entry v can reach a fixed positive exit v_R precisely when

$$f(s) > 0 \quad (v \le s \le v_R): \quad \mu_u > 0 \text{ or } [\mu_u \le 0,\ v > \sqrt{-\mu_u}]. \tag{10.24}$$

No scalar root can be crossed by using a different normal fibre.

For a finite repeated tag the entry coordinate decreases strictly with P. At the limiting field this follows from the coordinates V, Z_1 and $r' = -r^2$ in Appendix D.8; the fixed radial entry and transverse outer map preserve a negative derivative margin. The normal passage has

$$G(v) = k_0 \exp[-\int_v^{v_R} \beta_u(s,0)/f(s)\,\mathrm{d}s] e^{j(v)}, \qquad N = fG'/G = \beta_u(v,0) + fj' > c.$$

The selected ordinary jets of j are bounded by Proposition 4.3 and Lemma 4.5. Write $u_+ = \Lambda w(v)$, $w, w' > c$, $u_- = \Lambda k_1 G(v)$ and $\psi = w'/w$. The full numerator is

$$\begin{aligned} f\mathcal{K}' = {} & h_I N - h_I f\psi + u_- P_-' N - f u_+ P_+' \psi \\ & + \epsilon(fN'/N - f' - f\psi'/\psi) + \epsilon f \partial_v \log[H_-(u_-)/H_+(u_+)]. \end{aligned} \tag{10.25}$$

The last term is $O(\epsilon\Lambda)$; the other perturbations are $O(\Lambda + |f| + \epsilon|f'| + \epsilon|f|)$. First reduce the angular strip, then the scale, so $h_I N$ dominates. Thus $f\mathcal{K}' > 0$ on the whole actual branch.

For the two-infinite repeated row, fixed slow sections $-v_R, v_R$ require $\mu_u > 0$. Their contraction is

$$e^{-A_u}, \qquad A_u = \frac{2\Xi}{b\sqrt{\mu_u}} \arctan(v_R/\sqrt{\mu_u}).$$

The outer relative units have ordinary radial jets, giving $u_+ = \Lambda t V_+(t)$, $u_- = \Lambda e^{-A_u} t V_-(t)$ with $t = 1/P$ and positive units. Hence

$$\mathcal{K}(0) = \epsilon \log(K_-/K_+) - h_I A_u + O(1), \qquad \|\mathcal{K} - \mathcal{K}(0)\|_{C^2} \le C.$$

The explicit repeated first integral and the uniformly contracting tails of Appendix D.8 prove

$$\mathcal{K}_P \longrightarrow \bar{B}^{-1}[P^{-1} - P/(P-P_c)^2], \qquad \mathcal{K}_t \longrightarrow \frac{P_c(2-P_c t)}{\bar{B}(1-P_c t)^2} > 0. \tag{10.26}$$

The ordinary C^2 bound transfers this derivative limit to $t = 0$: compare first with a fixed small positive t and then let that input decrease to zero. The positive derivative gives the two-zero closing bound even when upper and lower divergent actions cancel in the constant.

For the weak row take $C > 0$; reflection is applied to the whole field before choosing the contracting chart when $C < 0$. Put $\Delta = \sqrt{C^2 + 4bM}$ and $\varphi_s = (-C - \Delta)/(2b)$. The exact radial eigenvalue is

$$\gamma = \varphi_s + C = -\frac{2(M + \epsilon\bar{B}C^2)}{Ck + \Delta}. \tag{10.27}$$

Thus the moving weak face is $M+\epsilon\bar{B}C^2=0$. In reversed projective time the normal eigenvalue is $-\Delta$. The weak graph gives a reduced equation $\zeta'=\zeta G(\zeta)$, where $G(0)=\gamma$ and

$$G'(0)=\frac{1+2M\delta_s-(A_s+C\delta_s+\nu_s)\varphi_s}{\Delta+\gamma}-C\delta_s-\nu_s\longrightarrow 1/C>0.$$

Choose the finite smooth complete extension and phase projection as in Section 4.5. Its inputs hold here: $R=0$ is invariant, the normal margin is Δ, and the displayed positive $G'(0)$ gives $u=G(\zeta)-\gamma=\zeta\int_0^1 G'(s\zeta)\,\mathrm{d}s$. Division by its positive time factor gives exactly $u'=f(u)=u(\gamma+u)$, spending one radial derivative. The derivative orders precede the weak width, so normal decay dominates every retained slow variation. The unique positive component reaching the fixed exit u_R is

$$\max(0,-\gamma)<u<u_0<u_R/2,\qquad |\gamma|<u_R/2. \tag{10.28}$$

Intersect this domain with the two original lower whole-path conditions and positive central outputs.

Let $t=t(u)$ be the inverse normal passage with its regular connector. Proposition 4.3 and Lemma 4.5 give

$$\frac{t'}{t}=\frac{N}{f},\qquad \frac{t''}{t'}=\frac{A_U}{f},\qquad A_U=N-f'+fN'/N,\qquad N,A_U>c,$$

with bounded ordinary C^2 rates. Moreover $\|t\|_{C^3}\to 0$ uniformly as $u_0,|\gamma|\to 0$, with u_R fixed: the transit time tends uniformly to infinity, and $f(u_R)/f(u)\le e^{\varpi T}$ when $|f'|\le\varpi$, so the reserved normal exponential absorbs every fixed inverse-speed loss. In the secondary upper construction retain joint order twelve before the positive coordinate/time change, leaving eleven; relative-core with $N=7$ uses nine primitive orders and gives seven orders for the relative logarithm, hence five for its rates, including two ordinary input and three coefficient orders. Independently retain original-lower field, reference and connection order ten for the Gaussian divided-ν bound in Appendix D.2. The lower-phase reserve gives raw graph order seventeen for this target, whereas the full lower-preparation reserve in Proposition 4.8 gives eighteen for $P_\pm$ through order seven. Choose the original lower graph at order eighteen. Retain order seven also in Φ,w for the rate and simple-factor divisions below. All orders count ordinary section inputs, coefficients and Euler scale derivatives together.

Write $u_+=\Lambda w(t)$, $w,w'>c$, and $u_-=\Lambda\Phi(u)$, $\Phi(0)=0$, $\Phi'>c$. Put $L_u=u\Phi'/\Phi$, $R_\Phi=\Phi''/\Phi'$, $\psi=w'/w$, and

$$\begin{aligned}A_I&=H_-(u_-)-\epsilon+\epsilon u_-H_-'(u_-)/H_-(u_-),\\ R_{\mathrm{out}}&=H_+(u_+)\psi+\epsilon[\Lambda w'H_+'(u_+)/H_+(u_+)+\psi'/\psi].\end{aligned}$$

The complete derivative quotient, with its common exact scale power already canceled, gives

$$\mathcal{M}=f\partial_u\mathcal{K}=A_IL_u(\gamma+u)+\epsilon fR_\Phi-R_{\mathrm{out}}(t)Nt-\epsilon A_U. \tag{10.29}$$

Here $A_I(0)=h_I-\epsilon=k/\bar{B}$, $A_I',A_I''=O(\Lambda)$ and $L_u=1+O(u)$. Lemma 4.11 applies with $h=1$, $A=A_IL_u$, $R=R_\Phi$ and $E=R_{\mathrm{out}}N$. Their bounded ordinary C^1 rates and $\|t\|_{C^2}\to 0$ give $\partial_u\mathcal{M}>c$ after the weak ranges and then the scales are reduced, retaining the term $-\epsilon A_U'$. Hence $Z(\mathcal{K}')\le 1$, $Z(\mathcal{K})\le 2$ and $Z(F)\le 3$ by 3.2. Lower-double constants do not enter these input rate derivatives.

Each numerator argument uses only input derivatives of lower bottleneck maps, so their divergent constants are harmless. Reflection exchanges both actual maps and the oriented inputs, takes $(\bar{a},C,\nu)$ to their negatives, and preserves return multiplicity. It is applied before choosing the contracting chart. □

The weak numerator supplies a useful direction without taking a formal endpoint value: in the outward orientation u decreases, so $\operatorname{sign} \mathcal{K}'_{\mathrm{out}} = -\operatorname{sign} \mathcal{M}$. This remains valid on a shortened lower-passing interval. For finite saddles (10.22) supplies the actual ratio direction. These directions give the entrance counts in Propositions 12.7 and 12.8.

10.5 Exhaustion of the normalized escape faces

It remains to show that these rows cover every returning-arc limit. We use the compact actual tags to include the repeated and weak boundaries, without assuming that the arc closes at the central section.

Proposition 10.9. *Every limit of nonwedge returning upper arcs in Proposition 10.2, with an inherited lower cone, belongs to one of the following rows or its reflection:*

limiting upper data	*tags*
regular	$0 < P < P_c, \quad Q = Q(P; M, C)$
$M > 0$	$P = P_c, \quad Q = Q_c$
$M < 0,\ C > 0,\ C^2 + 4M > 0$	$P = P_c, \quad Q = \infty$
$C > 0,\ C^2 + 4M = 0$	$P_c \le P < \infty, \quad Q = \infty$
$M = 0,\ C > 0$	$P = 1/C, \quad Q = \infty$
$M < 0,\ C^2 + 4M < 0$	$P = Q = \infty$
$M < 0,\ C^2 + 4M = 0$	$P = Q = \infty.$

Here $Q_c = P_c(M, -C)$ and $P_c = \infty$ when $C^2 + 4M < 0$ or $C + \sqrt{C^2 + 4M} \le 0$. The secondary signed normalization remains in every row. No third occurrence of the same zero corner is possible. The assertion applies to returning arcs without a closing condition.

Proof. Take a convergent subsequence in the compact normalized face, inherited lower cone and reciprocal tags. If $P_* < P_c$, the affine first return is regular and compact finite-time convergence gives $Q_* = Q(P_*; M, C)$. Reflection gives the converse when $Q_* < Q_c$.

Let $D = C^2 + 4M > 0$ and $P_c < \infty$. A supercritical $P_* > P_c$, including infinity, enters the boundary node

$$R = 0, \qquad \hat{\varphi}_n = (-C + \sqrt{D})/2,$$

whose eigenvalues are $-\sqrt{D}$ and $-(C + \sqrt{D})/2 < 0$. For a finite tag the affine solution has a positive growing coefficient; for an infinite tag follow the boundary angular equation from L. A small trapping neighborhood in $R \ge 0$ persists by its strict eigenvalue margins and invariant boundary. Continuous dependence on the preceding compact segment sends nearby arcs into it and excludes arrival at $-L$, contradicting their return. Reflection gives the opposite-end exclusion. This requires $D > 0$.

If both finite tags are nonregular, the affine eigenline classification therefore gives the two finite thresholds with $M > 0$. If just one tag is infinite, it gives the strict-real, finite-input repeated, or weak row. At a repeated root the full finite interval $P \ge P_c$ remains. A finite opposite tag would be a regular reflected return and would force $P < P_c$. If both tags are infinite, the node exclusions force $D \le 0$. Normalization then implies $M < 0$, since $M = 0$ would force $C = 0$ and violate $\max(|C|, \sqrt{|M|}) = 1$. These give precisely the two-infinite rows.

On the normalized repeated face $|C| = 1$, $M = -1/4$; on the limiting weak face $|C| = 1$, $M = 0$. Their adjacent signed unfoldings have the whole-path domains (10.24) and (10.28); the reciprocal repeated tag has its separate relative preparation. Together with the strict-real, finite-saddle and focus rows, these cover every boundary of the normalized face, with actual first-exit flags and tags retained. Normalization excludes $(C, M) = (0, 0)$, so another parabolic zero corner cannot occur.

An excluded returning sequence at arbitrarily small scales would contradict these exclusions. Each retained row has the open local neighborhood proved in its preparation, completing the returning-arc exhaustion. □

Theorem 10.10 (The local secondary atlas)**.** *The wedge and all rows of Proposition 10.9 have the preparations and unanchored closing-zero bounds proved above. The bounds are two for wedge, regular, strict-real and either repeated row, and three for finite-saddle and weak rows. They apply with multiplicity on one actual comparison interval with the lower whole-path and positive-central restrictions. The adjoining maps use the exact exponent β_ϵ; the two lower maps use the exact common exponent μ. All coefficient contractions precede restriction to the normalized faces, and no bound on $\epsilon|\log\lambda|$ is imposed.*

Proof. Combine Propositions 10.5, 10.7 and 10.8. The scalar through-domains are connected intervals, and a failed lower whole-path condition excludes the corresponding return. The common central maps increase in each chosen comparison input; positive-central restrictions therefore truncate an interval and do not add components. The exact center contributes no isolated zeros. The returning-arc routing and tag exhaustion supply the configuration list without using any of these counts. No conclusion here adds the bounds from distinct chart pieces. □

11 The physical fibre and its ordered cover

Fix an original field and follow its returning arcs outwards. An outer closing orbit bounds the line-breaking coefficient; three equilibrium-free strips then supply all intermediate returns, including the nonclosing ones. We order their local descriptions and transport this one coefficient inequality to the common interval used in Section 12. Figure 3 shows the regular and terminal parts. Their open overlap carries the same K and permits a seam beyond a preceding regular zero.

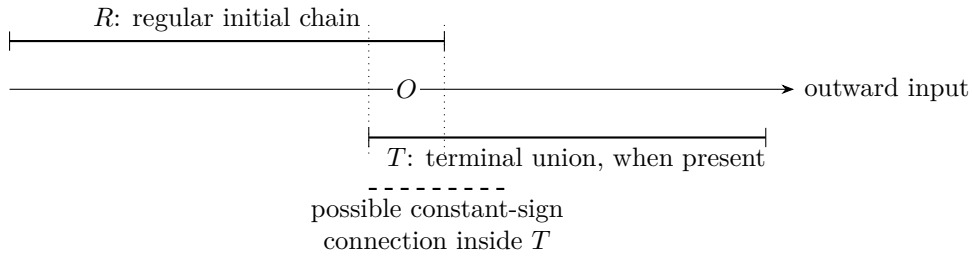


Figure 3: Schematic intervals on one actual physical fibre with an outer closing witness. O is an open overlap. Near the parabolic corner R includes the finite connection, wedge and regular secondary part. The depicted terminal lower-loss connection, when present, remains inside T. The lengths carry no metric information, and the arrow orders inputs rather than trajectories. Proposition 11.8 supplies these domains.

We use the original and normalized generators

$$\tau_0 = a + c, \quad \sigma_0 = c(B + m), \qquad \tau_1 = \bar{a} + \bar{c}, \quad \sigma_1 = \bar{c}(\bar{B} + \bar{m}),$$

so $\tau_0 = \rho\tau_1$, $\sigma_0 = \rho^3\sigma_1$, $\epsilon = \rho^2$ and $\ell = \rho\nu$. Except when explicitly discussing $\rho = 0$, all normalized coordinates have $\rho > 0$. Reflect the whole field once, if necessary, to arrange $\ell \geq 0$. The new outgoing endpoint is the old incoming endpoint; reflection does not preserve a previously selected numerical outgoing cutoff. It preserves the original generator norms and cycle multiplicities by Proposition 2.8.

11.1 Cycle candidates, retained scales, and outer forcing

Write

$$F_{\pm}(X) = 1 \pm \bar{a}X + \bar{B}X^2, \qquad \mathcal{P} = \{\bar{c} = \bar{m} = 0,\ \bar{B} > 0,\ |\bar{a}| \le 2\sqrt{\bar{B}}\}$$

on the normalized max-norm face. At $\mathcal{P}$ one has $\bar{B} \ge 1/4$. Indeed either $\bar{B} = 1$ or $|\bar{a}| = 1$, and the latter inequality gives $\bar{B} \ge 1/4$. The compactified interval $[0, \infty]$ is always given its ordinary topology through $p \mapsto p/(1+p)$. We first cover closing orbits, so that an outer orbit can supply a coefficient inequality. We will extend the cover to the intervening comparison arcs after proving that all of those arcs return.

Lemma 11.1 (Refined candidate space)**.** *The limiting data of parameter-dominant counted cycles have a compact refinement with the following properties. Away from $\mathcal{P}$ its coordinates are the normalized coefficients and the two compactified actual endpoints. Near $\mathcal{P}$ it additionally retains the secondary direction (C, M), the secondary scale and actual first-exit tags. Every point has one of the local descriptions of Sections 7–10, including their boundary faces. The same compactification can be applied simultaneously to an outer cycle and any returning upper arc between fixed positive initial endpoints and its endpoints. The assertion about such additional arcs is compactness only; their existence and local classification will be proved below.*

Proof. Let $\chi > 0$ be the fixed strong-entry constant in Lemma 7.3. For parameter-dominant cycles, $p \ge \chi/2$, $q \ge \chi$. Take the intersection, as $r \downarrow 0$, of the closures of their data with $0 < \rho < r$ in

$$\mathcal{S} \times [\chi/2, \infty] \times [\chi, \infty], \qquad \mathcal{S} = \{(\bar{B}, \bar{m}, \bar{a}, \bar{c}) : \max(|\bar{B}|, |\bar{m}|, |\bar{a}|, |\bar{c}|) = 1\}.$$

This is a compact set. Its points satisfy the nonnegative lower conditions of Lemma 7.4. The parameter ν tends to zero by Proposition 2.7.

Only the corner at $\mathcal{P}$ needs further coordinates. Put

$$\bar{d} = \bar{c} + \nu, \qquad \lambda = \max(|\bar{d}|, \sqrt{|\bar{m}|}), \qquad C = \bar{d}/\lambda, \quad M = \bar{m}/\lambda^2 \quad (\lambda > 0).$$

The direction belongs to the compact face $\mathcal{D} = \{\max(|C|, \sqrt{|M|}) = 1\}$. Use the fixed sections of Lemma 10.2. Retain a finite flag specifying a wedge exit, a secondary exit, or their overlap. For a wedge intersection retain $H \in [0, H_0]$ and, if $H + \lambda > 0$, the ratio $\chi_{\mathrm{exit}} = H/(H+\lambda)$. For a secondary intersection retain its actual radial coordinates $R_{\mathrm{in}}, R_{\mathrm{out}} \in [0, R_{\max}]$ on the sections $\phi = \pm\lambda L$, together with

$$t = 1/P, \qquad \widetilde{t} = 1/Q, \qquad 0 \le t, \widetilde{t} \le 1/p_*.$$

Here P, Q are the auxiliary linear-flow labels of Lemma 10.3, evaluated at those actual section points; in particular $t = R/\bar{x}(T)$ on the incoming section. They are not defined to be $\lambda p, \lambda q$. Use a fixed placeholder for an inactive coordinate. Allow all the indicated flags on an overlap, and take the closure of the resulting actual-data graph in the finite union of these compact products, also retaining $\lambda \in [0, \lambda_*]$. Taking the closure retains radial limits without inverting a map at zero radius. An orbit in an overlap may have two sets of tags; keeping both copies preserves compactness and suffices for a finite subcover.

If H/λ tends to infinity, routing is in the wedge and $|\bar{d}|/H + |\bar{m}|/H^2$ tends to zero. For a finite ratio the first-exit flag is retained: a first wedge exit has $H/\lambda \ge L$ and hence the fixed wedge inequalities, whereas a first secondary exit is in the secondary rectangle. The equality face has the open overlap proved below. Radial escape is $R = 0$, equivalently $1/P = 0$ or $1/Q = 0$, and is retained in the secondary infinity charts. The cases of a finite upper threshold and its repeated limit are retained with their actual deficits. Proposition 10.9 classifies the entire normalized face, including its repeated and weak boundaries, and excludes another zero

corner. For $\lambda = 0$ an actual field uses only the wedge; secondary directions there record limits with $\lambda > 0$. This classification applies to returning upper arcs.

For completeness the primary alternatives follow directly from the upper linear flow. A finite nonregular pair is (p_c, q_c) with $\bar{m} > 0$. One finite and one infinite endpoint is a strict real threshold, a repeated threshold with its whole interval $p \geq P$, or the weak boundary $\bar{m} = 0$; reflection gives the other orientation. At two infinite endpoints the strict barrier of Lemma 7.1 gives $\bar{c}^2 + 4\bar{m} \leq 0$. Strict focus, repeated upper root and the cubic horizontal boundary are covered in Section 8; the joint repeated center is Section 9. If $\bar{m} = 0$, then $\bar{c} = 0$ and the lower cones put the point in $\mathcal{P}$. At a finite lower loss the first root, its double limit, or the simultaneous pair gives the alternatives of Section 7. The two-root and one-infinite statements include roots that escape to infinity, so these faces are not omitted.

For an intermediate arc, take the product with a second set of actual tags, sharing the coefficients but retaining its own section points and first-exit flag. Use independent initial bounds $p \geq p_0 > 0$, $q \geq q_0 > 0$ for this copy; q_0 is uniform on the compact regular entrance. Impose $p \leq p^{\mathrm{out}}$, $q \leq q^{\mathrm{out}}$ before closing the actual-data graph. This remains a closed subset of a compact product. The independent tags are needed because an intermediate arc may take a wedge exit although the outer cycle takes a secondary exit. □

Lemma 11.2 (Uniform selection over compact fibres)**.** *In a compact metric product X with continuous outer projection $\pi : X \to B$ to a compact metric space, let X_r be the actual data with $0 < \rho < r$ and put $X_0 = \bigcap_{r>0} \overline{X_r}$. If a finite union U of prepared open pieces contains the limiting fibre $X_0 \cap \pi^{-1}(\{b\})$, then an outer neighborhood and a positive original scale cutoff can be chosen so that every actual datum over that neighborhood belongs to U. If this construction applies at every outer limit, finitely many such choices give common cutoffs and bounds.*

Proof. Choose V with $\overline{V}$ disjoint from the compact set $\pi(X_0 \setminus U)$. If no scale cutoff worked, excluded actual data over V at scales tending to zero would have a subsequential limit in $(X_0 \setminus U) \cap \pi^{-1}(\overline{V})$, a contradiction. A finite outer subcover gives the common minimum scale cutoff and maximum bounds. The same argument applies to the one-copy graph. □

Lemma 11.3 (An endpoint cutoff chosen after contraction)**.** *There is $p_\sharp > 0$, chosen before the primary scale cutoff, and finitely many local candidate neighborhoods that contain every counted cycle with $p \geq p_\sharp/16$ at sufficiently small original parameter. The regular member contains all sufficiently small endpoint data with $p_\sharp/16 \leq p \leq p_\sharp/2$. Each other member is either separated from its local center set on its closure or has a common center-comparison domain and a positive contraction exponent $b_j > p_\sharp^2$. Secondary center estimates retain any prescribed finite power of λ, or of the wedge height, before division.*

Proof. Start with the nonregular part of the compact refined limit set in Lemma 11.1. At a noncenter point shrink the closed coefficient neighborhood away from the center set. At a center point the classification gives a divided regular, two-root, finite-saddle, weak, focus or joint chart. There are no additional unprepared center losses: the exact quasi-invariant (7.8) implies that an incoming finite root on a center also has an outgoing root; a one-infinite center has $\bar{a} = -\bar{c}, \bar{m} = -\bar{B}$, so its finite lower factor equals the upper entry factor; and a lower double root in the secondary corner has $|\bar{a}| = 2\sqrt{\bar{B}} > 0$ and is off the secondary center. The repeated coincidence is precisely the joint chart.

On each center neighborhood, first retain a fixed lower subsegment with positive action. After reducing the coefficient neighborhood, normal contraction on this subsegment bounds the absolute coefficient derivatives of the complete central deviations by $C\epsilon^{-N} e^{-b_j/\epsilon}$, with $b_j > 0$. These are estimates of the absolute deviations; the logarithm of an input-independent slow contraction may diverge. The large-power preparation extends the same estimate by zero up to a lost exit. Section 5 supplies the real center and the whole input domain on which to

divide. Double division uses two ordinary coefficient derivatives of the absolute difference, and a single component uses one. The budgets retained in Sections 7–9 include these derivatives; this step takes no further input derivative. In the corner use the weighted absolute estimates of Lemma 10.4. For the present argument retain power two before the coefficient division; if a later local coordinate requires more, retain its specified finite power first. The exact exponent in Proposition 10.1 makes these choices uniform without bounding $\epsilon|\log\lambda|$. For the wedge choose its fixed widths $|\gamma|, |J| < \delta$ inside the regular-lower center box whenever $|\bar{a}|$ is small. The remaining wedge data have $|\bar{a}|$ bounded below and are primary off-center data. Thus an off-center secondary direction attached to a first wedge exit is not used to estimate that exit by a power of λ; its actual radial power is a power of H.

Take a finite cover of this refined nonregular set, and then choose $p_\sharp \le \chi/2$ so that $p_\sharp^2 < \min_j b_j$, and decrease it to meet the small-endpoint error bounds in the source and regular-column preparations. These choices precede the smallness choice for ϵ. The compact complement of the chosen open neighborhoods is strictly regular. Enlarge that compact regular set to contain $p_\sharp/16 \le p \le p_\sharp/2$ and the limits of cycles in the two initial energy classes with $p \ge p_\sharp/16$. The latter enlargement is regular: the source endpoints satisfy $p = \sqrt{2\epsilon h e_+} \le 2\sqrt{\eta}$ and $q/p = \sqrt{e_-/e_+} = 1 + O(\eta)$, while the lower and upper thresholds have a fixed positive lower bound. The compact-energy class has $p, q \to 0$ and causes no further endpoint limit. The regular upper return and the positive lower polynomials therefore give a common compact tube for this enlargement.

Apply Lemma 11.2 to this one-copy refined graph. Positive endpoint limits from the initial classes lie in the regular enlargement, with q bounded below when $p \ge p_\sharp/16$ by the same source estimates. Thus the selected finite family covers all the asserted cycle data. □

The candidate cover locates each outer closing orbit. To obtain one sharp count, we next extract its coefficient inequality; saturation then supplies all intervening comparison paths.

Proposition 11.4 (Forcing supplied by an outer closing zero)**.** *Choose the field orientation with $\ell \ge 0$. After a common original parameter shrink, a counted cycle with $p \ge p_\sharp$ either lies in a constant-sign local comparison whose full remaining physical passing interval is covered by that same comparison, or supplies the coefficient inequality*

$$|\ell| \le e^{-p_\sharp^2/(20\epsilon)}(|\tau_0| + |\sigma_0|). \tag{11.1}$$

Equivalently,

$$|\nu| \le e^{-p_\sharp^2/(20\epsilon)}(|\tau_1| + \epsilon|\sigma_1|). \tag{11.2}$$

An interior constant-sign piece is not sufficient for the first alternative. On a center fibre with a closing zero there are no isolated closing zeros.

Proof. Apply the candidate cover to the outer cycle. A constant-sign chart containing the entire remaining passing interval gives the first alternative. In every other chart we derive the coefficient inequality, including on constant-sign pieces separated from the center set.

The nonnegative lower cone and (7.7), applied at $p_\sharp$, give

$$s \le \exp[-p_\sharp^2/(5\epsilon)].$$

Indeed $A_+(p_\sharp) \ge p_\sharp^2/[2(1 + p_\sharp + p_\sharp^2)]$ and the error in the budget is absorbed after $p_\sharp$ is fixed. Since $|\ell| < 3s$, any closed primary neighborhood on which $|\tau_1| + |\sigma_1| \ge g_0 > 0$ satisfies (11.1): use $|\tau_0| + |\sigma_0| \ge \rho^3 g_0$ and absorb this fixed power of ρ in the strict gap between the two exponents.

At an off-center secondary first exit, the primary action estimate alone does not retain the vanishing secondary weight. We use the exact long match to recover it. Suppose first

$|C| \geq g_0/2$, where $|\bar{a}|+|C| \geq g_0$ on the chosen closed neighborhood. The exact long match gives an outgoing horizontal input $u_+ \leq C_1\lambda^{k/b}$, with $b = 1-\epsilon\bar{B}$, $k = 1-2\epsilon\bar{B}$. The lower common power has exponent $b/(\epsilon\bar{B})$. The bounded scaled action and scaled prefactor, including a possible additional contracting lower double passage, imply

$$s \leq I_{\mathrm{out}} \leq e^{C_0/\epsilon}\lambda^{k/(\epsilon\bar{B})}. \tag{11.3}$$

The first inequality uses the actual reference $s_+ = -\ell H_+ \leq 0$. Choose $\kappa_* > 0$ with $k/\bar{B} \geq \kappa_*$, then $\lambda_0 > 0$ so that

$$(\kappa_*/2)\log(1/\lambda_0) \geq C_0 + p_\sharp^2/5, \qquad \epsilon \leq \kappa_*/4.$$

For $\lambda \leq \lambda_0$, (11.3) yields $s \leq \lambda^2 e^{-p_\sharp^2/(5\epsilon)}$. For $\lambda \geq \lambda_0$ the primary budget gives the same bound with the fixed factor λ_0^{-2}. Consequently $|\nu| \leq \lambda|C|/2$ at small scale and

$$|\bar{c}| = |\lambda C - \nu| \geq \lambda g_0/4, \qquad |\sigma_0| \geq c\rho^3\lambda g_0.$$

The extra factor λ^2 proves (11.1) uniformly, even if $\epsilon|\log\lambda|$ is arbitrarily large. If $|C| < g_0/2$, then $|\bar{a}| \geq g_0/2$; the corner has $|\bar{c}| < g_0/4$, so this reduces to the primary off-center estimate.

On a retained center neighborhood, cancellation on the real center recovers the generator factor. Write the complete closing function as $Z - \rho\nu H$, with $H > h_*$ independent of the section input. The actual center division gives

$$|Z| \leq C\epsilon^{-N}e^{-b_j/\epsilon}(|\tau_1| + |\sigma_1| + |\nu|).$$

The local inverse charts and the two ideal comparisons in Proposition 5.13 justify the indicated generators. In secondary variables retain the weighted version before returning to primary coefficients: a term $\lambda^2|C|$ is bounded by $|\bar{d}| = \lambda|C|$, and the wedge version is identical with $H^2|\gamma| \leq |\bar{d}|$ and its retained height. The small-$|\bar{a}|$ wedge box was included in these center comparisons independently of its secondary direction. Since $|\bar{d}| \leq |\bar{c}|+|\nu|$ and $\bar{B}+\bar{m} > 1/8$ here, the resulting bound is again the displayed primary one. No inverse power of λ is introduced. At a closing zero Lemma 5.14 absorbs the ν term. Since $b_j > p_\sharp^2$, the fixed powers of ϵ needed to return to original generators are absorbed with room to spare.

Finally consider the compact regular member at $p \geq p_\sharp$. Retain a slightly larger actual tube with $p \geq p_\sharp/2$. The regular upper map has $q \geq p(1-p)$ at small endpoints; hence both retained actions have a lower bound greater than $p_\sharp^2/10$ after $p_\sharp$ is reduced. The regular absolute maps and double center division yield

$$|\ell| \leq C\epsilon^{-N}e^{-p_\sharp^2/(10\epsilon)}(|\tau_0| + |\sigma_0|),$$

which implies the required exponent $p_\sharp^2/20$. The finite candidate cover makes these choices uniform. If the generators vanish, absorption first gives $\nu = 0$, and the actual center identity excludes isolated zeros. □

11.2 Equilibria and the three returning strips

To continue the comparison paths between an initial input and an outer cycle, we must exclude an equilibrium that could trap an intervening orbit. We do this first, then fill the upper and two lower strips by trajectories.

Lemma 11.5 (Equilibria enclosed by counted cycles)**.** *On a common small original parameter neighborhood, the Jordan disk of every counted cycle contains the origin and no other equilibrium. The section points of any two counted cycles belong to one interval of the actual return domain.*

Proof. The origin has index one and remains the only equilibrium in a fixed inner disk. The fixed regular collar strips remain free of equilibria. Proposition 2.2 places every counted cycle essentially in the collar, surrounding that disk. The upper vertex calculation in that proposition leaves at most one additional equilibrium in the upper chart, and it is a saddle. It remains to exclude horizontal equilibria from the cycle's disk.

If $\rho = 0$, then $B = m = a = c = 0$ and $\dot{x} = -y$; the equilibrium equations give only the origin. Suppose $\rho > 0$. In a horizontal chart $x = \sigma/r$, $y = v/r$, $\sigma = \pm 1$, the equation $\dot{y} = 0$ is

$$\sigma(r + v) + dv^2 + crv = 0.$$

Its unique small solution is $v = rh(r)$ with $h = -1 + o(1)$. Thus $|z| = |1 + y| < \delta$ at any horizontal equilibrium. The other equilibrium equation reads

$$1 - z + \bar{B}X^2 + \bar{a}X + \epsilon\bar{m}(1 - z)^2 = 0.$$

Choose fixed small $R, \delta > 0$ and then small ϵ. The left side is positive for $|X| \le R$, $|z| \le \delta$, so the equilibrium lies in one of the two half-strips $|X| > R$.

In the incoming half-strip put $T = 1/(-X)$ and use the positive time change $\mathrm{d}\tau/\mathrm{d}\xi = -X$. With $\eta_\ell = \rho\ell$, the exact equations are

$$\begin{aligned} T_\tau &= \epsilon T[\bar{B} - \bar{a}T + \{1 - z + \epsilon\bar{m}(1 - z)^2\}T^2], \\ z_\tau &= -z + \epsilon\bar{c}Tz(z - 1) + \eta_\ell T(1 - z)^2. \end{aligned} \tag{11.4}$$

On $0 \le T \le 1/R$, $|z| \le \delta$, the tangential derivatives are $O(\epsilon)$ and the normal derivative is $-1 + O(\epsilon + |\eta_\ell|)$. Extend to a slightly larger rectangle, preserving $T = 0$ and these strict margins. The normally contracting graph construction used in Section 4 gives

$$z = g(T) = \eta_\ell T\, O_{C^2}(1), \qquad Z = z - g(T), \qquad Z_\tau = -\beta(T, Z)Z, \quad \beta > 1/2.$$

The two factors in g follow by solving on $\eta_\ell = 0$ and $T = 0$, respectively, and dividing these vanishing factors in the bounded graph family. Every equilibrium lies on $Z = 0$. The small original quantity is $\rho\ell$; no bound on $\nu = \ell/\rho$ is used.

A counted cycle has $z > 0$ by Proposition 2.3. At $T = 1/R$, $0 \le z \le \delta$, one has $T_\tau > 0$, while $|g| < \delta/2$. A cycle point with $Z < 0$ cannot leave the rectangle in backward time through $T = 1/R$; it cannot reach $z = \delta$ because $Z < 0$ is preserved and $g < \delta/2$. It cannot reach $z = 0$, and its compact affine orbit stays away from $T = 0$. It would therefore remain in the rectangle for all backward time, whereas $|Z|$ grows exponentially. This is impossible. A periodic orbit on $Z = 0$ would be a nonconstant periodic orbit of the scalar equation on that graph, also impossible. Every cycle point in this half-strip consequently lies strictly above g.

An equilibrium with $0 < z = g(T) < \delta$ is joined vertically to $z = 0$ without meeting the cycle: cycle points with that T are above g, including any points above the rectangle. The line $z = 0$ is outside the cycle's disk, since the whole cycle has $z > 0$. Thus the equilibrium is outside the disk. The same holds immediately if its height is nonpositive. Reflecting the field and reversing time gives the argument on the outgoing half-strip, for either sign of ℓ. The coordinate T includes finite roots and roots escaping to infinity, so no horizontal escape case remains.

The single upper saddle is now the only possible additional equilibrium inside the disk. Its index would change the enclosed total from one to zero, contradicting the index of a periodic orbit. Thus only the origin is enclosed. Two counted cycles are nested essential curves and bound an equilibrium-free annulus in the collar. Cut this annulus at the transverse central section. The compact-limit argument for the strips below shows that every intermediate section point returns, placing both cycles in one interval of the actual return domain. □

Proposition 11.6 (Saturation of the physical interval)**.** *Assume* (11.1)*, fix* $p_0 \in [p_\sharp/8, p_\sharp/4]$*, and let* γ *be a counted cycle with* $p_\gamma > p_0$*. Every input* $p \in [p_0, p_\gamma]$ *has an actual upper return* $q(p)$ *and both actual lower central intersections. They satisfy*

$$q'(p) > 0, \qquad L'_+(p) < 0, \qquad L'_-(q) < 0, \qquad L_+(p),\ L_-(q(p)) \geq s_\gamma > 0.$$

All three arcs, and the segment joining their central endpoints, lie in the fixed collar. This statement holds at nonclosing inputs as well as on periodic orbits.

Proof. At p_0 the fixed small-endpoint upper map and both regular lower maps exist. After reducing $p_\sharp$, their two actions are at most $p_\sharp^2/31$ uniformly in the stated interval for p_0. Their positive homogeneous amplitudes are bounded below by a fixed positive prefactor times $e^{-p_\sharp^2/(31\epsilon)}$. The actual signed reference contributions are $O(|\ell|)$. Since the original generator norm is bounded and the forcing exponent is $p_\sharp^2/20$, both complete central heights are positive for sufficiently small scale. Their arcs lie in the collar by the regular construction.

The axis endpoints of the three initial arcs lie inside the corresponding endpoints of γ. Since trajectories cannot cross γ, the arcs stay in its disk. Join their central endpoints by the central crosscut. Each lower path stays in its own half-plane until its first central intersection, so it meets the crosscut only there. The three arcs and the crosscut form a simple essential curve in the collar, inside γ, whether or not its two central values coincide.

The initial and outer upper arcs, joined by the two axis segments, bound a compact equilibrium-free strip by Lemma 11.5. Since $\dot{y} = x$ on $y = 0$, the positive axis segment is an entrance, the negative one an exit, and the orbit sides cannot be crossed. A nonexiting orbit would have an equilibrium-free compact ω-limit set, hence a periodic orbit by Poincaré–Bendixson. Its Jordan disk lies in $y > 0$ inside γ and contains no equilibrium, contradicting the periodic-orbit index. Thus every intermediate upper entry returns, with $q_0 \leq q(p) \leq q_\gamma$. The transverse flow determinant has a fixed sign and gives $q'(p) > 0$.

Construct the right and left lower strips from the corresponding initial and outer arcs, an axis segment, and a central crosscut. They are equilibrium-free as well. On the right use backward time, so the positive axis segment is an entrance; on the left use forward time and the negative axis segment is an entrance. The central crosscuts are exits because $\dot{x} > 3/4$ on the fixed central rectangle. The same compact-limit and index argument gives both central intersections at every intermediate endpoint. Transversality and endpoint order give $L'_+(p) < 0$, $L'_-(q) < 0$; their minimum heights occur on γ and equal $s_\gamma > 0$.

The three strips and the joining central segments lie in the annulus between the two essential curves, hence remain in the fixed collar. The nonexit contradiction uses compactness for each fixed field; flight times may diverge as a saddle or slow gap tends to zero. □

On this physical interval keep the actual central height as output. The fixed weak references give

$$F(p) = L_+(p) - L_-(q(p)) = U(p) - V(p) - \rho\nu H, \qquad H > 0,$$

where H is independent of p and $U', V' < 0$. Consequently

$$K = \epsilon \log(V'/U'), \qquad F' = U'(1 - e^{K/\epsilon}). \tag{11.5}$$

A common regular input change and a change of the fixed additive references preserve K on this orbit interval. We keep the original central height as output, since a nonlinear output change need not preserve it. The signed center-comparison domains used to estimate these functions are auxiliary; the physical interval of Proposition 11.6 has positive complete central heights.

11.3 The ordered cover of all intermediate arcs

Saturation supplies the intermediate paths. We must now order their local descriptions so that separate endpoint bounds become a bound on one interval. The whole-passage tests will give a regular prefix followed by at most one terminal union. At the parabolic corner this requires two connections: from the finite family to the wedge, and from the wedge to the secondary family.

Lemma 11.7 (The two parabolic connections)**.** *Near $\mathcal{P}$ the finite regular family connects by an actual open overlap to the wedge, possibly through a constant-sign lower loss interval. The wedge in turn has an actual open overlap with the regular secondary family. The construction applies to the nonperiodic returning arcs in Proposition 11.6.*

Proof. Consider first the incoming lower double boundary $\bar{a}_* = 2\sqrt{\bar{B}_*}$ and put $r = 1/\sqrt{\bar{B}_*}$. The limiting upper return has $q = p$. For $0 < p < r$,

$$A_-(p) - A_+(p) = 2\bar{a}_* \int_0^p \frac{X^2}{F_+(X)F_-(X)}\,\mathrm{d}X > 0.$$

Thus $K < 0$ on every fixed compact regular initial interval, uniformly in a smaller off-center neighborhood. Let $C_w = \sqrt{1 + 2w_0}/w_0$. The limiting upper parabola $Y = (p^2 - X^2)/2$ meets $w = w_0$ at $H = C_w/p$. Choose $L_1 > 2r$ so large that $C_w/L_1 < H_0/2$. The upper arcs for $L_1 \le p \le 2L_1$ form a compact regular family; their transverse intersections persist and belong to the wedge after the coefficients are reduced.

The incoming action tends to infinity as its first double root is approached from below. Choose $R < r$ with

$$A_-(R) > \max_{p \le 2L_1} A_+(p) + 3,$$

where the maximum is on the retained regular outgoing interval. Then choose a fixed $p_1 < r$ with $p_1 > R + 2\zeta$ for some $\zeta > 0$. Uniform convergence of the regular upper map to $q = p$ on $[p_1, 2L_1]$ gives $q(p) > R + \zeta$ there. Lemma 7.9 now gives $K < 0$ on its entire physical passing part, even beyond the lower root where the ordinary regular action formula no longer applies. This interval overlaps both the regular initial family and the wedge. The corresponding lower slow action in the wedge has the same strict sign and dominates its bounded remainder. Reflection gives the other lower double boundary and reverses the sign. Away from these boundary neighborhoods the lower factors have a common positive margin on every retained finite interval, so the regular family connects directly to the wedge.

For the wedge–secondary overlap, choose the fixed L in secondary routing large enough before decreasing any scale. Require $2/L < \delta$ and $4/L^2 < \delta$. At $w = w_0$, $H = k_0\lambda L$, with $1/2 < k_0 < 1$, the exact secondary coordinates are

$$\widehat{x} = \frac{1}{k_0 L w_0}, \qquad \widehat{z} = \frac{1}{k_0^2 L^2 w_0}.$$

The auxiliary linear flow defining the actual tags gives $P, Q = O(C_w/L)$ uniformly for bounded (C, M). At $C = M = 0$ the values are $C_w/(k_0 L)$; the perturbation on these short arcs is of higher order in $1/L$. Increase L so all these tags lie in one compact regular interval below the finite secondary thresholds. The wedge inequalities are strict, and the same actual arcs meet the secondary sections before reaching $w = w_0$. Transversality persists as k_0 varies on a smaller open subinterval of $(1/2, 1)$. Thus an open family of the same returning arcs has both descriptions, including the nonclosing comparisons. □

Proposition 11.8 (An ordered cover on each physical fibre)**.** *After a common original parameter shrink, every physical interval of Proposition 11.6 is covered by a connected regular*

initial part and an ordered union of the local passing charts of Sections 7–10. Their adjacent members overlap on actual orbits. At a joint repeated center one contact member and the full separated member form a single union. Near $\mathcal{P}$ the initial part includes the finite connection, the wedge and the regular secondary part. After grouping overlapping descriptions of the same end, there is at most one terminal union after this initial part. It may contain a constant-sign lower-loss connection followed by a singular upper end; that connection is retained as its own piece.

The coefficient neighborhoods and all cutoffs can be selected from a finite family, independently of the chosen outer cycle. The order includes signed split-root faces and the repeated one-infinite interval. Each local derivative estimate is applied on its stated whole-passage domain. This proposition makes no sum of local zero bounds.

Proof. Fix the finite derivative orders of Sections 7–10 and retain their larger construction domains before reducing sections or coefficient neighborhoods. Saturation supplies actual upper and lower returns with positive central heights on $[p_0, p_\gamma]$. Since $q' > 0$, their endpoints satisfy $p \le p_\gamma$, $q \le q_\gamma$ and inherit the outer lower cones.

The key ordering fact is that a lost passing component cannot reappear. The exact lower factor on the fixed phase interval is

$$F_I(x) = \bar{B} - \bar{a}x + (1 + \epsilon\bar{m})x^2 - \epsilon\nu(1 + 2\epsilon\bar{m})x^3 H(x) + \epsilon^3\nu^2\bar{m}x^4 H(x)^2.$$

By Proposition 4.1, it is strictly convex and positive near the fixed exit x_R. Lemma 4.2 requires $F_I(s) > 0$ at every point between the actual entry and x_R. Thus only the component adjoining x_R passes: if a root is present, the entry must lie to the right of its last root. The entry decreases as its actual endpoint increases, since $T = 1/p$ or $1/q$ and $t_T > 0$. Its allowed endpoints consequently form one outward initial interval. Entry positivity in the other component would not suffice. Also $F_I'(0) = -\bar{a}$ and its reflected derivative is $\bar{a}$; strict convexity permits an interior positive double minimum on at most one of the two sides.

The primary upper slow tests have the same one-sided form. For a repeated factor $v^2 + \mu_U$, passage to the positive fixed exit requires $\mu_U > 0$ or $v > \sqrt{-\mu_U}$; its actual finite entry decreases with p. This retains the negative entries beyond the repeated threshold whenever they pass. For the weak factor $u(\gamma + u)$, the whole domain is $\max(0, -\gamma) < u < u_0$, and $p = \hat{p} - t(u)$ with $t' > 0$. At joint contact both calibrated entries decrease, so the two whole-path tests are intersections of one-sided restrictions. General horizontal factors use the whole lower test above; a smaller monotone block alone permits an endpoint test. Signed auxiliary coefficient comparisons do not remove physical inputs, whose complete central heights are positive by saturation.

The increasing upper graph and its reflected inverse are those of Lemma 7.1. Its nonregular ends are classified in Lemma 11.1, with the full repeated interval $p \ge P$, $q = \infty$ retained. Following the graph outward gives these terminal templates:

- At a regular finite upper end, the first lower loss is a single root or the simultaneous simple-root pair. A simple root terminates passage; a double retains its signed slow alternatives.
- At a finite upper saddle, the lower paths are regular, in one or both contact neighborhoods, or have one incoming double-loss connection before the saddle. An outgoing double with regular incoming outer data is excluded by Lemma 7.3, applied to the outer cycle. The incoming connection is the whole constant-sign interval of Lemma 7.9.
- At a one-infinite end, a lower root can precede, coincide with, or follow the upper threshold. These give respectively the lower connection, the contact neighborhood, or the upper-tail piece followed by the lower loss. A root escaping to infinity retains its finite-root and cubic descriptions in the same family.

- At two infinite endpoints the lower cones exclude a prior simple root. Section 8 supplies the strict-focus, one lower-double, repeated, opposing-slow and cubic alternatives; Section 9 supplies the joint contact interval and its full separated tail.

A finite saddle has one returning quadrant, while the strict, repeated and weak preparations have the through components just described. Together with the monotone entry tests, this excludes a second independent upper end or a return after a split-root obstruction. The pieces of each listed end therefore form one terminal union, retaining a preceding constant-sign lower connection as a separate piece within it.

Near $\mathcal{P}$, use the actual first exits and tags of Proposition 10.9. Routing confines a returning arc to the fixed secondary rectangle and uniqueness excludes another middle excursion. The normalized face $\max(|C|, \sqrt{|M|}) = 1$ has the finite-saddle, strict real, repeated, weak and focus alternatives, including $1/P = 0$ and $1/Q = 0$. The repeated row retains its full label interval. At its weak end the actual tag is $P = P_s - \psi(t(u))$ with $\psi' > c$; again outward motion decreases u. The preceding lemma inserts the finite connection, wedge and regular secondary prefix. The same first-threshold reasoning then gives one secondary terminal family; a lower-double overlay can add its retained connection but cannot add an opposing double or another returning quadrant.

Retain actual open seams before reducing the scale: a fixed positive deficit crosscut between finite regular and singular pieces; the joint contact/separated overlap $P + d < p < P + 2d$ with both gaps positive; and a fixed reciprocal annulus before either slow block, using Lemma 4.12. These connectors are transverse with bounded inverses. The finite lower-double bridge has the strict action margin of the preceding lemma. That lemma also supplies the finite–wedge and wedge–secondary transverse overlaps. Within the secondary family retain its fixed deficit crosscuts and reciprocal annuli. Adjacent strict, repeated and weak faces use their own uniform preparations; no estimate at a fixed positive eigenvalue ratio is extended across zero or infinity. All seams compare the same original central function.

Apply Lemma 11.2 to the joint compactification of Lemma 11.1: the intermediate copy has independent bounds $p \geq p_0$, $q \geq q_0 > 0$, its own first-exit tags, and $p \leq p^{\mathrm{out}}$, $q \leq q^{\mathrm{out}}$. Each compact limiting fibre is covered by the ordered pieces and their whole regular prefix. The retained crosscuts, action margins and transverse seams make this an open cover, including reciprocal radial ends; the one-sided entry tests preserve order. Positive endpoint limits from the initial energy classes use the regular enlargement in Lemma 11.3. Thus the lemma gives common cutoffs after all sections, derivative budgets and overlap margins have been fixed. □

11.4 Transport in the original coefficient fibre

We transfer the outer orbit's inequality to the splitting, gap and saddle-ratio coordinates. The comparisons below may contain $|\nu|$; the exponential factor absorbs it on the fixed coefficient fibre.

Proposition 11.9 (Uniform coefficient and domain transport)**.** *On the ordered cover of Proposition 11.8, the fixed original inequality* (11.1) *supplies the relative coefficient bounds used in the local critical-function estimates, uniformly at all physical inputs, including nonclosing inputs. More precisely, for each local center-generator size I_{loc} used in Sections 7–10 there is a modulus $\eta_{\mathrm{loc}}(\epsilon) \to 0$ such that*

$$|\nu| \leq \eta_{\mathrm{loc}}(\epsilon) I_{\mathrm{loc}}.$$

The coefficient constants and moduli are common to the finite cover. The coefficient divisions take place on the larger comparison domains of Section 5 and the corresponding secondary domains. The counted inputs remain on the positive-central physical interval. If a local generator size vanishes, absorption first gives $\nu = 0$ and then the relevant exact center identity.

Proof. Set $\delta_\epsilon = e^{-p_\sharp^2/(20\epsilon)}$. The original inequality is

$$|\nu| \le \delta_\epsilon(|\tau_1| + \epsilon|\sigma_1|) \le \delta_\epsilon(|\tau_1| + |\sigma_1|). \tag{11.6}$$

We compare the original generators with the local transverse coefficients in each type of chart.

On regular and strict-focus charts take $I_{\mathrm{loc}} = |\tau_1| + |\sigma_1|$. At a two-lower-root match, the strict splitting derivative of Proposition 7.11 makes $(\bar a, \delta, \nu)$ a bounded inverse chart on a larger box. Here δ is the actual strong-intersection splitting. Single division from its exact reversible axis gives

$$|\tau_1| + |\sigma_1| \le C(|\bar a| + |\delta| + |\nu|).$$

The finite upper saddle uses $(\bar a, \kappa, \nu)$ with the actual $\kappa = r - 1 = -\operatorname{tr}/\lambda_+$; the derivative $\partial_{\bar c}\kappa = -1/\sqrt{\bar m} + O(\epsilon)$ has a nonzero margin. The same division, already described after Proposition 5.13, gives $|\tau_1| + |\sigma_1| \le C(|\bar a| + |\kappa| + |\nu|)$. Off these center axes the transverse size has a positive lower bound on each smaller closed chart.

At the finite three-saddle intersection, (5.45) gives

$$|\tau_1| + |\sigma_1| + |\nu| \asymp |\xi_2| + |\kappa\xi_1| + |\nu|.$$

Thus take $I_{\mathrm{loc}} = |\xi_2| + |\kappa\xi_1|$; the product retains both intersecting center branches. The double-division segments and derivative budget are those of Section 5.

At a strict one-infinite simple-root coincidence use the exact splitting δ and the actual scaled exponent difference g. On its strict compact row the determinant of this coordinate change tends to

$$\frac{p_c^4}{\bar B(1 - \bar B p_c^2)^3} > 0,$$

up to the positive unit from the regular power coordinate. The calculation and the equality of the actual leading powers on the second center are in Proposition 8.4. Their bounded inverse and single division give $|\tau_1| + |\sigma_1| \le C(|\delta| + |g| + |\nu|)$. At the repeated and weak faces use instead the two comparisons (5.46), with local sizes $|\tau| + |\zeta|$ and $|d_s| + |\varsigma|$, respectively. Their inverse charts and exact center axes were proved in Proposition 5.6; the repeated comparison uses (5.18), rather than the strict-row determinant. The fixed-mean and lifted-contact comparisons, and the single weak offset $\bar B + K_* A$ of Proposition 5.12, are defined on larger domains containing every original input and the whole coefficient segments. At a cubic two-infinite end $\bar B = 0, \bar m < 0$, $|\bar B + \bar m|$ has a positive margin. Hence $|\tau_1| + |\sigma_1| \asymp |\bar a| + |\bar c|$, the cubic generator size. On other constant-sign pieces either the same bounded comparison applies or the transverse size is bounded below; no division at a vanishing gap is needed.

At the parabolic corner we must retain the scale weights before comparing generator sizes. Since $\bar B + \bar m > 1/8$, bounded normalized coefficients give

$$|\tau_1| + |\sigma_1| \le C(|\bar a| + |\bar c|) \le C(|\bar a| + |\bar d| + |\nu|).$$

Inserting this in (11.6) and absorbing gives

$$|\nu| \le 2C\delta_\epsilon(|\bar a| + |\bar d|). \tag{11.7}$$

On the wedge, $\bar d = H\gamma$ with $H \le H_0 < 1$; on the secondary charts, $\bar d = \lambda C$ with $\lambda \le \lambda_* < 1$. It follows, uniformly at every actual input, that the forcing is small relative to the local sizes $|\bar a| + |\gamma|$ and $|\bar a| + |C|$. The stronger weighted inequality (11.7) is retained whenever those local expressions are returned to original coefficients. The wedge bound in (10.16) retains $|\bar a| + |\bar d|$; the secondary bound uses $|\bar a| + |C|$. The additional powers in (10.15) arise when an actual secondary closing witness supplies its absolute budget; they are stronger than the bounds used here.

Near a finite secondary upper saddle the exact ratio calculation (10.22) has a nonzero derivative in C and yields

$$|C| \le C_0(|\kappa| + \epsilon\lambda|\bar{a}| + \epsilon\lambda|\nu|).$$

Thus the preceding bound implies $|\nu| \le \widetilde{\eta}(\epsilon)(|\bar{a}| + |\kappa|)$ after one further absorption. The estimate therefore uses the actual ratio of the secondary field, including its coefficient error relative to the limiting linear field. On the weak and real repeated secondary rows $|C|$ has a positive margin on the normalized face, so their transverse size is bounded below. They need no inverse coordinate division at the vanishing weak eigenvalue. A strict focus center has its regular division in $(\bar{a}, C)$, and the finite saddle center was treated above. These neighborhoods cover the normalized face with the adjacent strict rows, so the constants are uniform on their common boundary faces.

In every unweighted comparison we have obtained $I_{\mathrm{global}} \le C(I_{\mathrm{loc}} + |\nu|)$. Choose the common scale with $C\delta_\epsilon < 1/2$; then (11.6) gives $|\nu| \le 2C\delta_\epsilon I_{\mathrm{loc}}$. The weighted comparisons have just given the same conclusion with their explicitly retained factors. If $I_{\mathrm{loc}} = 0$, this inequality gives $\nu = 0$ before any division by that size. The inverse coordinate identity then places the field on its real center axis, so its actual compared maps agree. This deals with the zero-size case as well.

These are coefficient inequalities, so they hold also at nonclosing inputs. All divisions use the larger domains of Section 5, while saturation supplies positive central heights for the original field. Taking the largest of the finite coefficient bounds makes the absorptions and moduli uniform on the whole interval. □

12 Entrance counts and the uniform upper bound

The physical interval of Section 11 carries one original-height derivative logarithm K. Its regular pieces have at most one crossing. The three-saddle and weak endpoint estimates alone allow four closing zeros, so they cannot give the sharp bound. The terminal arguments below use the sign left by the regular part to bound K by two zeros on the whole interval. If a small cycle supplies the stronger entrance sign, the regular part is zero-free and the bound improves to one. These are the two counts required by Section 3, in the fixed-field orientation of Section 2.

We first derive the entrance from a small cycle and carry it along the regular chain. The primary and secondary terminal arguments then give the whole-interval count and the uniform upper bound.

12.1 What a small cycle supplies at the common entrance

Use the fixed-field orientation $\ell \ge 0$ of Section 2, with the reflection-symmetric thresholds of Section 3. The coefficient divisions are those of the original signed auxiliary maps. The source representative counts the small cycles and determines the ratio of the two homogeneous generators. To obtain the entrance sign of K, we then evaluate the regular comparison in the original central height. This change of representation is necessary because a unit relating closing functions need not preserve their derivative zeros.

Proposition 12.1. *After a common neighborhood restriction, suppose $\rho > 0$ and (3.7) holds. There is at most one counted cycle with $p < p_\sharp/4$, and it is simple. A counted cycle with $p < p_\sharp/8$ implies*

$$\tau_1\sigma_1 < 0, \qquad 0 < -\tau_1/\sigma_1 \le p_\sharp^2/300. \tag{12.1}$$

The actual regular comparison at $p_0 = p_\sharp/4$ then satisfies $\operatorname{sign} K(p_0) = -\operatorname{sign}\sigma_1$. Thus it supplies the precise signed entrance of hypothesis (III) *in Section 3.*

Proof. The endpoint action bound of Section 7.1, with the fixed primary normalization, places every parameter-dominant cycle beyond $p > \eta/8 \geq p_\sharp$; the same bound is recorded in the physical cover of Section 11. Hence the small cycles here belong to the two initial energy classes, and are zeros of the one function constructed in Proposition 6.5. In the source class its outgoing action satisfies

$$p^2 = 2e_+\epsilon h, \qquad |e_+ - 1| < 1/1024.$$

For a cycle with $p < p_\sharp/4$ this gives $\epsilon h < p_\sharp^2/31$. On the compact energy class the same inequality follows from $h \leq H_*$ after the scale cutoff is reduced. Let J_0 be the initial-energy interval up to that upper bound; it contains all these cycles, with their multiplicities. Its lower energy bound is the fixed $h(z_L) > H_{\mathrm{ECT}} + 2$. We do not use any ECT assertion near energy zero.

Write the actual source representative as

$$F_{\mathrm{src}} = \tau_0 f_1 + \sigma_0 f_2 + f_3, \qquad Q = f_2/f_1, \quad g = f_3/f_1.$$

The compact overlap and the source derivative estimates in Section 6 give on this single initial-energy interval

$$Q \geq q_* > 0, \qquad Q' \geq q_0 > 0, \qquad Q \leq C_Q h.$$

On the compact interval these inequalities follow from the first two ordered Wronskians and fixed coefficient convergence; on the source tail they follow from $Q = hC/A$, $Q_h > 1/5$. The fixed overlap joins the same two columns, so their signs can be tested on the union.

On J_0 we need the line-breaking contribution and its ordinary energy derivative to be exponentially smaller than the homogeneous coefficient norm. The signed compact contribution is ℓ times a bounded coefficient. On the source part it is the exact logarithmic forcing divided by $h^{3/2}A$. Since $\widehat{\ell} = \ell/s$ and $s^{-1} = e^{h+1-s} \leq e^{h+1}$, (3.7) and the source logarithmic estimates yield

$$|g| + |g'| \leq C \exp\left(1 - \frac{11p_\sharp^2}{620\epsilon}\right)(|\tau_0| + |\sigma_0|) =: \vartheta_0(|\tau_0| + |\sigma_0|). \tag{12.2}$$

Here $1/20 - 1/31 = 11/620$. The same bound applies to the fixed cutoff terms on the compact overlap. The source logarithms are evaluated at $\widehat{\ell} = o(1)$ on this smaller interval, so their denominators have fixed margins. The derivative is the actual fixed-coefficient energy derivative, not an independent source derivative.

If both homogeneous generators vanish, forcing gives $\ell = 0$, and the center identity contributes no isolated zero. Otherwise

$$F_{\mathrm{src}}/f_1 = \tau_0 + \sigma_0 Q + g, \qquad (F_{\mathrm{src}}/f_1)' = \sigma_0 Q' + g'.$$

If $|\sigma_0|q_0 > \vartheta_0(|\tau_0| + |\sigma_0|)$, this derivative has a strict sign on all of J_0. Every zero is therefore simple and there is at most one. In the complementary case $|\sigma_0| \leq 2\vartheta_0|\tau_0|/q_0$ at small scale, and

$$|F_{\mathrm{src}}/f_1| \geq |\tau_0|\left(1 - \frac{2\vartheta_0 C_Q p_\sharp^2}{31 q_0 \epsilon} - 2\vartheta_0\right) > 0.$$

The exponential absorbs ϵ^{-1}, so there is no zero. This proves uniqueness and simplicity on the whole small-cycle set.

At the smaller cycle, with energy h_y, the same equation gives

$$-\frac{\tau_0}{\sigma_0} = Q(h_y) + g(h_y)/\sigma_0, \qquad |g/\sigma_0| \leq \vartheta_0(1 + |\tau_0/\sigma_0|).$$

For small ϑ_0, absorb the ratio on the right. The positive lower bound $Q \geq q_*$ proves the strict negative sign of τ_0/σ_0. Moreover $\epsilon h_y < p_\sharp^2/127$. The independent source constants give $C/A \to 2/5$. Choose a fixed $H_1 \geq H_* + 1$ so that its scale-tail error is below $1/100$, then choose $p_\sharp$ small so that the coefficient error on this energy range is below $1/100$. For $h \geq H_1$, $Q \leq 0.42h$; below it $Q \leq C_Q H_1$. Therefore

$$\epsilon Q(h_y) \leq \max(0.42p_\sharp^2/127, \epsilon C_Q H_1) \leq p_\sharp^2/302$$

after reducing the primary scale. The strict inequality $0.42/127 < 1/302$ and the exponential error leave the margin to $1/300$. Since $\tau_1/\sigma_1 = \epsilon\tau_0/\sigma_0$, this proves (12.1).

The generator ratio now determines the entrance sign of the original central comparison. The physical-fibre construction in Section 11 includes the regular point $p_0 = p_\sharp/4$. The two regular columns in Lemma 7.6 satisfy $C/A = p^2/5 + O(p^3)$, uniformly on the normalized compact coefficient set. Thus at this fixed point $C/A \geq p_\sharp^2/85$ after choosing $p_\sharp$ small. Let $w = (\tau_1, \sigma_1)/(|\tau_1| + |\sigma_1|)$. Equation (12.1) makes $w_1 + w_2 C/A$ have sign σ_1 and modulus at least $p_\sharp^2/200$. The exact regular representation, divided by this generator norm, is

$$K(p_0)/(|\tau_1| + |\sigma_1|) = -A(p_0)(w_1 + w_2 C(p_0)/A(p_0)) + e(p_0).$$

The positive $A(p_0)$ has a fixed margin, and regular coefficient convergence together with the fixed-field forcing makes $e(p_0)$ smaller than half the preceding margin. Consequently $K(p_0)$ has sign $-\sigma_1$, as required. All choices preceded the fixed field and its cycle. Reflection transfers this statement with its actual outgoing label and the identity (3.6). □

12.2 The complete regular chain

The regular primary, wedge and secondary comparisons form one chain. A common crossing direction on their actual overlaps gives a single zero bound for K. Near the parabolic corner, the primary coefficient cone shows that the outward direction is $-\operatorname{sign} C$, with $C = \bar{d}/\lambda$.

Proposition 12.2. *Assume the fixed-field inequality (3.7). The regular initial part of the ordered physical interval has at most one zero of K, and that zero is simple. Near $\mathcal{P}$ this assertion includes the finite connection, the wedge and the regular secondary tags, with their actual overlaps.*

If a primary regular zero has occurred, its subsequent sign and outward derivative have sign $-\sigma_1$ on the remaining regular chain. The same conclusion holds if its initial point has

$$\tau_1\sigma_1 < 0, \qquad \operatorname{sign} K = -\operatorname{sign}\sigma_1; \tag{12.3}$$

in that case the regular chain is zero-free. If the first zero instead occurs in the wedge or in the regular secondary part, its subsequent sign and outward derivative have sign $-\bar{d}$, and $\bar{a}\bar{d} < 0$. In particular a secondary terminal after that zero receives $\bar{a}C < 0$ and K of sign $-C$.

Proof. Saturation and coefficient transport give the positive-central interval with fixed-fibre forcing. The common-input identity (3.5) identifies K on its overlaps.

On the primary compact regular part Lemma 7.8 gives at most one simple zero and its outward direction $-\sigma_1$. At a zero, or under (12.3), the positive column quotient on a fixed regular overlap gives

$$|\tau_1| \leq C|\sigma_1|, \qquad |\nu| \leq Ce^{-p_\sharp^2/(20\epsilon)}|\sigma_1|. \tag{12.4}$$

The second inequality involves coefficients only, so it persists as the input moves along the chain. Near $\mathcal{P}$, $\bar{B} + \bar{m}$ has a fixed positive margin. Thus $\bar{c}$ has the sign of σ_1, $|\bar{a}| \leq C|\bar{c}|$, and $\bar{a}\bar{c} < 0$: the latter follows from $(\bar{a} + \bar{c})\bar{c} < 0$. Equation (12.4) makes $\nu = o(|\bar{c}|)$, so $\bar{d} = \bar{c} + \nu$

has that same sign and $\bar{a}\bar{d} < 0$. For $\lambda > 0$, $C = \bar{d}/\lambda$ has that sign too. No division of the error estimate by λ was used.

On the wedge the two columns in (10.18), after the fixed-fibre forcing column is absorbed, have negative margins and a strictly monotone quotient in the actual outward direction. Hence any zero is simple, requires $\bar{a}\bar{d} < 0$ and crosses in direction $-\bar{d}$. The regular secondary formula (10.19) has the same property: its lower constant has sign $-\bar{a}$, and $\log(Q/P)$ has sign C, with the derivative margin of Lemma 10.6. Near the center this follows from its divided columns; away from the center the nonzero value or derivative margin proves it directly. A zero with $C = 0$ would force the center identity, so does not give an isolated zero. The strict-focus reciprocal endpoint uses the stated value or derivative estimate in that same regular secondary preparation.

Apply Lemma 3.3 on the actual open overlaps. After a primary zero, the coefficient cone makes $\bar{d}$ and σ_1 have the same sign, so all later crossings have the same direction. Without a primary zero, all wedge and secondary crossings have direction $-\bar{d}$. This gives one zero on the union; the local direction lemmas keep K departing once it has sign $-\bar{d}$.

The finite lower-double connections in Lemma 11.7 require a separate observation. On the lower-cone boundary their regular prefix, finite bridge, wedge and every retained compact regular secondary interval have one strict action sign, opposite to $\bar{a}$. Their lower action dominates on every retained compact secondary part, so they contribute no zero; a singular end remains terminal. On the primary prefix τ_1 has the sign of $\bar{a}$. If $\tau_1\sigma_1 < 0$, the sign $-\sigma_1$ is therefore opposite to this actual value of K. Thus the strengthened entrance cannot occur in these boundary neighborhoods.

Under (12.3), the primary part departs from zero. Its cone (12.4) carries the same direction to the later pieces, excluding a first zero there. This proves the strengthened assertion. □

12.3 Primary terminal intervals with their regular entrances

Let R be the regular prefix of the physical interval and T its terminal union. On their actual open overlap, regular changes of input identify the same original-central K.

By Proposition 12.2, R has at most one simple zero. After a zero, or under the signed entrance below, it supplies the departure sign $-\sigma_1$ and the coefficient cone (12.4). More generally, Lemma 7.8 gives that derivative sign whenever K has sign $-\sigma_1$, without requiring the cone; if $\sigma_1 = 0$, K, K_p have sign $-\tau_1$ off the center. A regular zero requires $\tau_1\sigma_1 < 0$. These statements hold on the actual overlap, including nonclosing inputs.

If K is constant on a terminal interval, a nonzero constant contributes no zero; a zero constant extends into its regular overlap and, by the regular count under forcing, belongs to the exact center case, contributing no isolated closing zero. Otherwise choose a seam in the overlap away from zeros of K and the terminal derivative numerator, after a regular zero if one has occurred. The part of T before the seam is then assigned to R, with every multiplicity retained. We fix this seam before classifying the terminal critical zeros, and use the same convention for the secondary anchors and the final union.

Lemma 12.3 (Entrance counts)**.** *On a tail after this seam, a positive-multiplier numerator with at most j zeros gives $Z(K) \le j + 1$, or $Z(K) \le j$ when $KK'_{\mathrm{out}} > 0$ at the seam. On an explicitly constructed comparison continuation, a two-zero bound and opposite nonzero endpoint signs give at most one zero on every physical subinterval. If u decreases outward and a positive-multiplier numerator of K_u strictly increases in u, an entrance with $K < 0$ and $K'_{\mathrm{out}} < 0$ likewise leaves at most one zero, which is simple. All counts include multiplicity and apply to finite selections at excluded endpoints.*

Proof. Ordinary and away Rolle in Lemma 3.1 give the first assertion. Opposite endpoint signs force odd total multiplicity, so a two-zero continuation has one zero. In the last assertion the

seam numerator is positive and, moving outward, can become negative at most once. Thus K first decreases below zero and then strictly increases, allowing at most one simple crossing. □

Proposition 12.4 (Regular lower arcs at an upper saddle)**.** *In the chart of Proposition 7.13, the terminal part has at most two zeros of K, with multiplicity. If R has a zero, its remaining terminal part has at most one. The same improvement holds when R has no zero but the supplied entrance satisfies*

$$\tau_1\sigma_1 < 0, \qquad \operatorname{sign} K = -\operatorname{sign}\sigma_1. \tag{12.5}$$

Thus $Z(K; R \cup T) \le 2$ without a prescribed entrance and it is at most one under (12.5) if the preceding part has no zero.

Proof. Proposition 7.13 gives the unanchored bound. In either improved case, the regular overlap supplies (12.5) and the cone (12.4); coefficient transport supplies (7.25) on this fixed fibre. Lemma 7.14 then gives opposite signs at the seam and $t = 0$ for the same original-central K on its two-zero comparison continuation. Lemma 12.3 leaves one zero there and hence on its physical part. Add the possible single regular zero. □

At the three-saddle end the derivative numerator may have two zeros. Ordinary Rolle would then allow three zeros of K. We need the signs forced by the two-critical-zero case, together with the regular entrance, to remove this extra possibility.

Proposition 12.5 (The three-saddle union at a fixed seam)**.** *Suppose T is the three-saddle chart of Proposition 7.19, and impose its relative forcing on the fixed fibre. Then*

$$Z(K; R \cup T) \le 2.$$

After a zero in R, every seam satisfying the preceding conventions leaves at most one terminal zero. If there is no regular zero and (12.5) holds, the union has at most one zero. All counts include multiplicity and remain valid when central positivity or a moving lower boundary truncates the passing interval.

Proof. For the physical interval $J = [p_0, p_\gamma]$ of Proposition 11.6, the outer closing orbit supplies the local forcing in (7.36). Indeed, write the actual seam as $p_s = P_s - \psi_+(t_s)$, with $\psi_+' > 0$. The separatrix input P_s has no finite upper return. Saturation therefore excludes $p_\gamma \ge P_s$, and hence

$$p_s \le p_\gamma < P_s, \qquad 0 < t_\gamma \le t_s.$$

The fixed signed connecting maps constructed in Section 7.6 contain this smaller upper passage. Its actual lower returns remain on the passing sides of the two strong separatrices, by Proposition 4.8; thus $G_+(t_\gamma) > 0$ and $G_-(v(t_\gamma)) > 0$. Consequently γ is an actual closing zero in the same three-saddle comparison chart. Equation (7.36) holds on its fixed coefficient fibre, including the comparison continuation used below.

Write $u = G_+(t)$, $z = G_-(v(t))$, $v = t^r e^w$ as in Section 7.6. On the deficit comparison domain,

$$M = tuzK_t, \qquad t > 0, \quad u > 0, \quad z > 0.$$

The outward coordinate is $p = P_s - \psi_+(t)$ with $\psi_+' > 0$. Let T_s be the actual positive-central tail after the fixed seam t_s, and set

$$j_s = Z(M; T_s). \tag{12.6}$$

The local bound gives $j_s \le 2$ at this fixed seam. If $j_s \le 1$, Lemma 12.3 proves the assertions: regular departure gives $KK_p > 0$ after a regular zero or under the prescribed signed entry.

Suppose now that $j_s = 2$. Apply, if needed, the reflection identity to the whole union, its coefficients, and its already fixed seam so that $r > 1$. It sends K to its negative composed with the corresponding input change, and changes the signs of τ_1, σ_1 together. It preserves multiplicity, outward ordering and the signed-entry condition, and transports the already chosen seam and threshold to their corresponding actual points. The reflected outgoing label is the original incoming label; the numerical outgoing cutoff need not be unchanged.

With this local forcing fixed, apply Proposition 7.19 on the same deficit comparison interval, with the original central function $K = \mathcal{K}$ and the already fixed seam. Its two-zero classification gives

$$u_0, z_0 > 0, \qquad \tau_1 > 0, \qquad M(0+) = \epsilon\kappa u_0 z_0 > 0, \qquad K(0+) = -\infty. \tag{12.7}$$

The actual tail's two zeros exhaust the comparison bound. The same proposition gives $M(t_s) > 0$ and $K_p(t_s) < 0$ at its fixed seam.

The regular overlap determines the remaining sign, $K(t_s)$. For $\sigma_1 \geq 0$, $\tau_1 > 0$ and the positive regular columns give $K(t_s) < 0$, with relative forcing absorbed. For $\sigma_1 < 0$, a positive value of $K(t_s)$ would have sign $-\sigma_1$, and Lemma 7.8 would imply $K_p(t_s) > 0$, a contradiction. The seam avoids a zero, so again $K(t_s) < 0$. Together with (12.7), this gives negative signs at both ends of the continued tail. Ordinary Rolle bounds its total multiplicity by three, and parity reduces this to two. Restricting to the physical subinterval preserves that bound.

There cannot have been a regular zero: since $\tau_1 > 0$, such a zero would require $\sigma_1 < 0$ and would leave $K > 0$ at the seam, contrary to the preceding conclusion. Thus the union has at most two zeros. Finally the no-zero signed-entry alternative also excludes $j_s = 2$. If $\sigma_1 > 0$, it contradicts $\tau_1\sigma_1 < 0$; if $\sigma_1 < 0$, it would require a positive seam value of K. Only $j_s \leq 1$ remains, and the away estimate gives one. □

Proposition 12.6 (Primary weak thresholds with their regular part)**.** *Let T be either a positive-$\bar{B}$ weak threshold with a lower root at P, as in Proposition 8.7, or the weak-center chart of Proposition 8.8. Under the actual overlap and fixed-fibre forcing hypotheses above,*

$$Z(K; R \cup T) \leq 2.$$

After a zero in R, the remaining terminal part has at most one zero. With no regular zero and (12.5), the whole union has at most one. These statements concern the common physical passing interval, with all whole-path and central positivity conditions.

Proof. The upper input u decreases outward, so each positive-multiplier numerator of K_u has sign opposite to K_p. Throughout this proof we restrict the quotients to the physical interval where the lower whole-path test from $\chi(u)$, $u > \max(0, -\gamma)$, the finite lower deficit and central positivity all hold.

The weak center. Write $d = d_s$ and $s = \varsigma = \bar{B} + \bar{m}$ in this proof. Use the complete numerator (8.9), with the actual moving eigenvalue γ and the retained term $-\epsilon F_I(\chi)A_U$ in C. On this fixed fibre, Proposition 11.9 supplies $|\nu| \leq \eta(|d| + |s|)$, $\eta \to 0$. With $e_s = e_\varsigma$, Lemma 8.9 supplies the divided coefficients

$$M = d\overline{C} - t(se_s + \nu e_\nu), \qquad e_s > c, \quad \overline{C}' > 1/2, \tag{12.8}$$

on the same coefficient branch, with bounded C_u^2 coefficients and small $\overline{C}$. Under that forcing the coordinate norm is comparable to $|\tau_1| + |\sigma_1|$.

If $|s| < \sqrt{\eta}(|d| + |s|)$, the small-$s$ alternative of that lemma gives M/d strictly increasing, so $Z(M) \leq 1$. Lemma 12.3 gives all the assertions in this case.

Otherwise $s \neq 0$ and $e = e_s + (\nu/s)e_\nu > c/2$. If $d = 0$, $M = -tse$ has no zero and the assertions follow by the same Rolle inequalities. If $d \neq 0$, set

$$R_0 = \frac{\overline{C}}{te}, \qquad M = dte(R_0 - s/d).$$

Lemma 8.9 gives unimodality of R_0, interval strict superlevel sets, and strictly increasing crossings of nonpositive levels. In particular, if $ds < 0$, there is at most one zero of M, and the preceding ordinary and away Rolle arguments apply.

It remains that d and s have the same sign. Since $\bar{c} > 0$, this is the sign of σ_1. Suppose first that the seam has K of sign $-\sigma_1$. Its regular derivative has that same sign, so the seam numerator has sign d. Thus the seam lies in

$$\{u : R_0(u) > s/d\}.$$

Moving outward decreases u and can leave this interval at most once. Consequently the remaining terminal part has at most one numerator zero. Its away entrance gives at most one zero of K. This argument applies after the possible regular zero and under the prescribed signed entrance.

If instead the seam value of K has sign σ_1, no regular zero has occurred. If M there has sign d, the same superlevel argument gives at most one numerator zero and ordinary Rolle gives at most two zeros of K. If M has the opposite sign, then K_p has the sign of K at the seam; away Rolle and $Z(M) \le 2$ again give at most two. The seam avoids numerator zeros. If the original overlap point was such a zero, its initial segment was assigned to R before the seam was fixed, so the count still covers the entire union.

Positive $\bar{B}$. At a simple threshold $\bar{a} = -\bar{c} - \bar{B}/\bar{c}$ with $0 < \bar{B} < \bar{c}^2$, and at the double threshold $\bar{B} = \bar{c}^2$, $\bar{a} = -2\bar{c}$. Hence at either threshold $\tau_1 < 0$ and $\sigma_1 > 0$. Lemma 8.9 gives $K > 0$ on the retained compact regular overlap for this same original central function.

There was therefore no preceding regular zero, since its outward sign would be $-\sigma_1 < 0$. For the same reason the prescribed signed entry cannot occur on this union. It remains to bound the unanchored terminal continuation of this positive overlap. Use the numerator and quotient from Proposition 8.7,

$$M = AC - tE, \qquad E > c, \qquad R_0 = C/(tE).$$

Here $A = \delta_L$ for a simple lower loss, and $A = \mu_L + \delta_L^2$ for a double loss. Lemma 8.9 gives $A > 0$ and $C > 0$ at every numerator zero and interval strict superlevel sets of R_0. If $A \le 0$, M has no zero and the assertion follows. If $A > 0$, an entrance with $M > 0$ lies in the interval $\{R_0 > 1/A\}$ and encounters at most one numerator zero outward; ordinary Rolle gives at most two zeros of K. An entrance with $M < 0$ has $K_p > 0$ and is away from zero, so $Z(M) \le 2$ and away Rolle give the same bound.

Reflection transports the union, its signs and multiplicities. The improvement comes from this entrance; an arbitrary weak-chart subinterval still has the unanchored four-zero closing estimate of Section 8. □

12.4 Secondary terminal intervals with their entrances

Let R be the regular, wedge and regular-secondary prefix, T its finite-saddle or weak terminal interval, and O their actual open overlap. We use the common $\mathcal{K}$ of (10.14), the fixed-fibre forcing of Section 11 and the seam convention of Section 12.3.

By Proposition 12.2, the chain has at most one simple zero, after which

$$\bar{a}C < 0, \qquad \operatorname{sign}\mathcal{K} = -\operatorname{sign} C, \qquad \operatorname{sign}\partial_{\mathrm{out}}\mathcal{K} = -\operatorname{sign} C. \tag{12.9}$$

Here $C = \bar{d}/\lambda$ with $\lambda > 0$ fixed on the fibre. Lemma 10.6 gives the same departure from a supplied signed entrance; the positive column quotient on a compact regular-secondary overlap gives $|\bar{a}| \le K_1|C|$.

Proposition 12.7 (The finite secondary saddle with an entrance)**.** *Let T be the finite-saddle interval of Proposition 10.7, with its fixed-fibre relative forcing. Without a prescribed entrance it has $Z(\mathcal{K};T)\le 2$. If the preceding chain has a zero, then its remaining terminal part has at most one zero of $\mathcal{K}$. The same improvement holds without a preceding zero when O satisfies*

$$\bar{a}C<0,\qquad \operatorname{sign}\mathcal{K}|_O=-\operatorname{sign}C. \tag{12.10}$$

Thus $Z(\mathcal{K};R\cup T)\le 2$ without a specified entrance, and the terminal union after an entrance satisfying (12.10) has at most one zero if its preceding chain has no zero. All counts include multiplicities and use the actual passing and positive-central restrictions.

Proof. First separate the lower-double overlay. On a compact regular overlap before the saddle, the upper logarithms and the relative lower term are bounded, whereas (10.17) has arbitrarily large modulus and sign $-\operatorname{sign}\bar{a}$ near $\bar{a}=\pm 2\sqrt{\bar{B}}$. Under (12.10) this is $\operatorname{sign}C$, opposite to the entrance sign, so the anchored case is excluded. The value-domination argument of Proposition 10.7 already gives at most one zero in the overlay, all crossings having the same nonzero actual-ratio sign. This count survives restriction by every lower whole-path test.

On the remaining compact regular-lower block the signed comparison maps extend along the saddle quadrant to $t=0$, with

$$\mathcal{K}=K_c+\epsilon\kappa\log t+\mathcal{E}(t),\qquad \mathcal{E}(t)\longrightarrow 0. \tag{12.11}$$

Here $\kappa=r-1$ is the actual ratio difference. The two central outputs need not both be positive on this whole continuation.

To determine the endpoint sign of this continuation, we need the sign of the actual ratio difference κ. The overlap cone and the transported forcing give

$$|\bar{a}|\le K_1|C|,\qquad |\nu|\le\eta_\epsilon(|\bar{a}|+|C|)\le(K_1+1)\eta_\epsilon|C|.$$

Near $C=0$, identity (10.22) now reads

$$\kappa=-\alpha_sC+\mathcal{R}_s,\qquad |\mathcal{R}_s|\le C_0(|C|+\epsilon\lambda K_1+\epsilon\lambda(K_1+1)\eta_\epsilon)|C|,\qquad \alpha_s>c_0>0.$$

Choose the C neighborhood and then the scale restrictions so that the coefficient of $|C|$ in this error is less than $c_0/2$. Thus $\operatorname{sign}\kappa=-\operatorname{sign}C$ and $|\kappa|\ge c_0|C|/2$. Away from this neighborhood the same sign follows with a strict margin from

$$r_0-1=-\frac{2C}{\sqrt{C^2+4M}+C};$$

the exact divided field is a uniformly small perturbation on the compact positive-M block. The case $C=0$ is excluded by $\bar{a}C<0$. The retained coefficient error $\epsilon\lambda|\nu|$ is therefore small relative to $|C|$, which gives the ratio sign uniformly even as C tends to zero.

For $|\bar{a}|>K_0|\kappa|$ the same value-domination argument already gives at most one zero. In the complementary cone the unanchored estimate $Z(\mathcal{K})\le 2$ holds on the entire saddle comparison continuation. Equation (12.11) gives its endpoint sign $\operatorname{sign}C$, opposite to the sign at the fixed seam. The opposite endpoint signs and Lemma 12.3 give one zero on the continuation and at most one on its physical subinterval. The departure (12.9) keeps the overlap zero-free after a chain zero; adding that possible zero gives the union count. □

Proposition 12.8 (The weak secondary end with an entrance)**.** *Consider the weak row of Proposition 10.8, first with $C>0$. Without an entrance condition its terminal interval has $Z(\mathcal{K};T)\le 2$. If the preceding chain has a zero, its remaining terminal part has at most one zero. This improvement also holds when there is no chain zero but the open overlap satisfies*

$$\mathcal{K}|_O<0,\qquad \partial_{\mathrm{out}}\mathcal{K}|_O<0. \tag{12.12}$$

Consequently the whole chain and terminal union has at most two zeros of K; with the supplied entrance and no chain zero, it has at most one. Reflection supplies the $C < 0$ statement. These conclusions permit either lower double root to shorten the passing interval and permit positive-central truncation.

Proof. On the physical fibre Section 11 supplies

$$|\nu| \le C\delta_\epsilon(|\bar{a}| + |\bar{d}|), \qquad \bar{d} = \lambda C,$$

from the outer closing zero. After a chain zero, or under the entrance, $\mathcal{K} < 0$ and $\partial_{\text{out}}\mathcal{K} < 0$ there. The actual outgoing tag is $P = P_s - \psi(t(u))$ with $\psi' > c$ and $t' > 0$, so outward motion decreases u and $\operatorname{sign} \partial_{\text{out}}\mathcal{K} = -\operatorname{sign} \mathcal{M}$. Lemma 12.3, with the increasing numerator of Proposition 10.8, leaves at most one simple tail zero on the lower-passing, positive-central interval. Adding the possible chain zero gives the assertions; reflection supplies $C < 0$. □

12.5 Joining the one terminal union

We now join the regular departure to the terminal estimates. Each terminal union includes its constant-sign connections, so the result counts K on the whole physical interval.

Proposition 12.9 (The logarithmic-ratio count on the physical interval)**.** *Assume* (3.7)*, and let $J = [p_0, p_\gamma]$, with $p_0 \in \{p_\sharp/8, p_\sharp/4\}$, be the physical interval of Proposition 11.6. Then*

$$Z(K; J) \le 2. \tag{12.13}$$

If (12.3) *holds at p_0, then*

$$Z(K; J) \le 1. \tag{12.14}$$

All zeros are counted with their full analytic multiplicities. These assertions verify hypothesis (III) *of Section 3.*

Proof. Write $J = R \cup T$ by Proposition 11.8, with at most one terminal union and every internal constant-sign connection retained. Section 11 gives its domains and fixed-fibre forcing. Use the seam and center convention of Section 12.3. If there is no terminal part, Proposition 12.2 already gives both conclusions. For each direct derivative row below, Lemma 12.3 gives two tail zeros, or one after a regular zero or the supplied strong entrance: Proposition 12.2 gives $KK_p > 0$ at that seam. A direct bound $Z(K; T) \le 1$ gives the same union conclusions; a constant-sign piece contributes no zero.

The following table records the terminal estimates used in the proof. Every derivative numerator has a positive multiplier on its actual passing interval.

Terminal description	Estimate or entrance result used
Two finite lower roots	$Z(K') \le 1$; 7.11
One incoming loss at a finite saddle	$K < 0$ if $r \ge 1$, or $Z(K') \le 1$ if $r < 1$; 7.15
Strict or repeated one-infinite lower contact	$Z(K') \le 1$; 8.4, 8.5
Signed horizontal finite root or cubic end	$Z(K') \le 1$; signed-horizontal calculation in Section 8.3, or Proposition 8.16
Strict focus or nonjoint opposing tail	$Z(K) \le 1$; 8.11, 8.15
Joint contact and separated tail	$Z(K') \le 1$ on the whole union; 9.5
Finite saddle with regular lower legs	Anchored result 12.4
Primary three-saddle	Fixed-seam union result 12.5
Regular outgoing lower arc at a primary weak upper end	$Z(K') \le 1$; 8.6
Primary weak contact or weak center	Union result 12.6
Secondary finite saddle or weak end	12.7, 12.8
Other secondary end	$Z(K) \le 1$ or a strict derivative; 10.5, 10.8

The constant-sign lower or upper slow pieces are those of Lemma 7.9 and Propositions 8.3, 8.14, together with their reflections and secondary lower-double overlays. The joint estimate applies to its whole contact–separated union. At a cubic end the multiplier is $F_+(t)F_-(f(t)) > 0$ on passage. It remains to treat interior connections and the entrance-dependent anchors.

A lower loss before a finite saddle. Suppose an incoming double root precedes a finite upper saddle. Write its lower parameters as $\bar{B} = b_0^2, \bar{a} = 2b_0$ and the upper eigenvalues as $\alpha_0, -d_0$, with $\alpha_0, d_0 > 0$. The root inequality $1/b_0 < q_c = 1/d_0$ gives

$$\tau_1 = 2b_0 + \alpha_0 - d_0 > b_0 + \alpha_0 > 0.$$

The regular action difference is positive at small input and tends to $+\infty$ at the incoming lower root. The regular one-simple-zero result therefore makes $K < 0$ throughout the intervening limiting regular interval. Compact convergence gives its actual sign on the retained overlaps, and incoming contraction continues it through the finite connection. There was no regular zero. The remaining saddle interval has at most two zeros of K by Proposition 7.15; for $r \geq 1$ it has none. Hence the complete union has at most two. If the strong entrance held, $\tau_1 > 0$ would force $\sigma_1 < 0$ and a positive initial value of K, contradicting the negative regular sign. Thus this configuration cannot occur under that entrance. The coincident incoming loss needs no intervening connection and its derivative bound is applied directly. Reflection proves the opposite orientation with its corresponding interval and signs.

A lower root before or after a one-infinite threshold. A simple root before the threshold ends passage and cannot be followed by a second counted part. For a double outgoing root r strictly before a strict or repeated upper threshold P, the inverse incoming contraction estimate gives $K > 0$ through its finite connection. The outgoing action diverges while the incoming action there is regular; the small-input sign is positive as well, so the regular one-zero result excludes a preceding regular zero. For example at the repeated field $\bar{a} = -2/r, \bar{B} = r^{-2}, \bar{c} = 2/P, \bar{m} = -P^{-2}$, $r < P$ gives $\tau_1 < 0$ and $\sigma_1 > 0$. At a strict threshold write the positive roots of the upper characteristic polynomial as $\gamma_+ > \gamma_- > 0$. The inequality $r < 1/\gamma_+$ gives $\tau_1 = \gamma_+ + \gamma_- - 2/r < 0$ and $\sigma_1 > 0$ again. Thus the strong entrance would require a negative initial value and is excluded. The upper contact and separated tail after that connection have the same strictly negative K_p on their open overlap: this is the double-before-threshold numerator in Section 8 and, at a repeated tail, the sign in (8.17). Their union has at most one zero. Its preceding positive connection does not increase this bound.

At a repeated upper threshold with a first lower root $r > P$, the intervening returning tail is the constant-sign piece $K < 0$ of Proposition 8.3. The finite-root derivative and its infinite limiting description are alternative overlapping charts of the same end, as in Proposition 11.8. For a first simple root $\bar{a} = -1/r - \bar{B}r$ and $\bar{B}r^2 \leq 1$; hence

$$\sigma_1 = 2P^{-1}(\bar{B} - P^{-2}) < 0, \qquad \tau_1 = 2/P - 1/r - \bar{B}r > 0.$$

The same inequalities hold at a double root $r > P$. The small-input value and the last regular value are both negative, so again the regular prefix is zero-free. The strong entrance would have positive sign and is excluded. The final finite-root piece has a strict derivative, or the signed horizontal/cubic numerator has at most one zero. It consequently contributes at most two zeros, with no count from the preceding constant-sign connection. Root coincidence with P is one contact case already listed above.

The weak boundary with a lower double strictly before its threshold uses its constant-sign connection and the weak-regular numerator of Proposition 8.6. For an outgoing double, $r < P = 1/\bar{c}$ gives $\tau_1 = \bar{c} - 2/r < 0$, $\sigma_1 = \bar{c}/r^2 > 0$ and a positive initial/connection sign. For an incoming double, $\bar{a} = 2/r$ gives $\tau_1 = \bar{c} + 2/r > 0$ and $\sigma_1 = \bar{c}/r^2 > 0$; the incoming action gives the negative connection sign, and the strong coefficient condition is itself impossible. In either case the primary regular prefix has no zero, its value is incompatible with the strong

entrance, and the final numerator has at most one zero. This gives at most two zeros on the union. The weak contact exactly at the threshold is the anchored weak case below.

Primary anchors. Apply the three primary entrance propositions listed in the table. Coefficient transport supplies their fixed-fibre forcing, including the primary order-one ν term; the three-saddle proposition derives its stronger (7.36) from the same outer cycle. Their union counts already include the regular prefix and exclude the incompatible entrances in the two-critical three-saddle and positive-$\bar{B}$ weak branches.

Secondary anchors. The complete regular chain supplies (12.9), either after its zero or from (12.3). Together with coefficient transport, these are the saddle-anchor hypotheses. At the weak end, reflection to $C > 0$ gives $K, K_p < 0$, the weak-anchor entrance. The two propositions give the asserted union counts, retaining lower-double overlays and their incompatible signs. The other secondary rows use the table; their regular/focus pieces have already been counted in R.

Exhaustion in Section 11 completes the proof: a regular zero leaves at most one terminal zero; without one the union has at most two. Under (12.3) the chain is zero-free, each compatible terminal case leaves at most one zero, and the sign arguments above exclude the remaining connections.

Physical truncation restricts the counted interval. The endpoint signs used above belong to the stated comparison continuations. Under reflection the original central logarithm transforms as $K_{\mathrm{ref}}(q) = -K(p)$ and $K_{\mathrm{ref}}\partial_q K_{\mathrm{ref}} = KK_p/q'(p)$. Thus multiplicities and outward departure are preserved at the corresponding seam and threshold, while both generator signs reverse. The reflected outgoing label is the original incoming label. All noncompact counts use the finite-selection convention of Section 3. □

12.6 Verification of the four hypotheses

Theorem 12.10 (Uniform upper bound)**.** *There are a fixed open spherical neighborhood $\mathscr{U}$ of Γ and an open coefficient neighborhood $\mathscr{V}$ of X_0 in the twelve-dimensional space of quadratic vector fields such that every field in $\mathscr{V}$ has total Poincaré multiplicity at most three among its isolated affine periodic orbits contained in $\mathscr{U}$.*

Proof. First work in the five-parameter family (2.3). We check the four hypotheses with one common choice of constants. Choose the independent base-energy threshold and collar of Section 2. Fix the primitive finite derivative orders before choosing the slow sections and the larger comparison coefficient boxes of Sections 4–5. The endpoint preparations of Sections 7–10 and the refined compact cover of Section 11 then supply finitely many positive margins and section widths. Choose $p_\sharp \le \eta/8$ smaller than the finitely many bounds required there and in Proposition 12.1. Finally reduce the original coefficient neighborhood so that all scale cutoffs, relative errors, strict margins and absorptions hold. These choices precede both the field and any selection of its cycles. They are made on a reflection-symmetric neighborhood.

Hypothesis (I) is Proposition 6.5 together with the endpoint action bound of Section 7: the two initial classes have one joint bound two, and every parameter-dominant cycle has $p > \eta/8$. The branch $\rho = 0$ belongs entirely to the initial classes. No scale quotient is used on that branch.

For $\rho > 0$ choose the fixed-field orientation $\ell \ge 0$. Proposition 11.4 gives the original coefficient inequality (3.7) whenever its single monotone alternative does not already settle the remaining interval. Proposition 11.6 then supplies the entire physical segment from either prescribed entrance to any selected outer cycle. It includes every intermediate nonperiodic comparison point and extends to neighborhoods of its regular endpoints. The central section construction of Section 2 identifies closing-zero orders with Poincaré multiplicities. This verifies (II).

Propositions 11.8 and 11.9 put those same segments in the local comparison domains with the same fixed coefficient inequality. Proposition 12.9 bounds K by two zeros on the whole union, and by one when the specified coefficient signs and entrance sign hold. This verifies (III). The count on this segment already includes the initial-class cycles lying there. Proposition 12.1 proves the uniqueness, simplicity, coefficient ratio and regular-section sign required by (IV).

Proposition 3.4 now applies to every finite selection of counted cycles. Its proof adds actual outer and small witnesses when needed and does not assume that an outermost cycle among all cycles exists. Thus every finite selection has total multiplicity at most three. A larger or infinite isolated collection would contain a finite selection violating that bound. Identity returns contribute no isolated cycles.

The exact reflection identities in Section 3 transfer the result back to the original orientation, using $p_{\mathrm{ref}} = q$ and the inverse return germ. They preserve multiplicity and the actual entrance condition; an unchanged original p threshold is not required. Finally Proposition 2.1 supplies one smaller fixed spherical collar and one open neighborhood in the full coefficient space. Its near-identity affine conjugacies and positive time changes preserve isolated-cycle multiplicity. The resulting $\mathscr{U}, \mathscr{V}$ prove the theorem. □

13 Three simple cycles in every collar

We construct three cycles in one field on the regular block $\bar{B} = 0$, $\bar{m} = 1$. A variation at a reversible center gives a three-root closing profile. We first fix the scale and then choose a finite coefficient perturbation that preserves all three roots and their actual return paths. Finally, a sequence of scales makes each cycle converge to the entire hemicycle. This construction is independent of the upper bound.

13.1 The fixed-scale center and its actual saddle ratio

For $0 < \epsilon \ll 1$ take the original coefficients $B = 0$, $m = \epsilon$ and $a = c = d = 0$. The exact first integral is

$$\mathscr{H}_\epsilon(x, y) = \frac{x^2}{2} + V_\epsilon(y), \qquad V_\epsilon(y) = (1+\epsilon)y - \frac{\epsilon y^2}{2} - (1+\epsilon)\log(1+y), \quad y > -1. \tag{13.1}$$

Indeed $V_\epsilon' = y(1-\epsilon y)/(1+y)$, so its Lie derivative cancels $x\dot{x} = -xy + \epsilon x y^2$. The potential decreases from infinity to zero on $(-1, 0)$ and increases from zero to a nondegenerate maximum at $y = 1/\epsilon$. Consequently the bounded well consists of periodic ovals with $0 < \mathscr{H}_\epsilon < H_s$, where

$$H_s = \frac{1}{2\epsilon} + 1 - (1+\epsilon)\log(1 + 1/\epsilon). \tag{13.2}$$

Its boundary is the homoclinic loop through $(0, 1/\epsilon)$, as in Proposition 5.2. The two loop intersections with $y = 0$ have normalized coordinates $P_s = Q_s = \sqrt{2\epsilon H_s} \to 1$.

We keep $\bar{B} = 0$, $\bar{m} = 1$ and use $(\bar{a}, \kappa, \nu)$ as local coefficient variables, where $\kappa = r - 1$ and r is the actual upper saddle ratio. For the finite-order constructions of Section 4, reserve at least six total input, coefficient and Euler scale orders before choosing the common neighborhoods. The variation below uses ordinary coefficient derivatives at a fixed positive scale. The exact upper field before division by $Y + \epsilon$ is

$$X_\xi = Y(Y-1) + \epsilon\bar{a}X, \qquad Y_\xi = (Y+\epsilon)(X + \bar{c}Y) + \nu Y^2.$$

At the center the saddle is $(0, 1)$ and its Jacobian is $\left(\begin{smallmatrix} 0 & 1 \\ 1+\epsilon & 0 \end{smallmatrix}\right)$. Let S be the moving saddle and T the trace of its Jacobian. Differentiating both equilibrium equations gives

$$S_{\bar{a}} = (0, 0), \qquad S_{\bar{c}} = (-1, 0), \qquad S_\nu = (-1/(1+\epsilon), 0),$$

and, including this motion in the trace,

$$T_{\bar a} = \epsilon, \qquad T_{\bar c} = 1 + \epsilon, \qquad T_\nu = \frac{1 + 2\epsilon}{1 + \epsilon}.$$

Since $\kappa = -T/\lambda_u$ and $\lambda_u = \sqrt{1+\epsilon}$ at the center, the parameter inverse $\bar c = \bar c_\epsilon(\bar a, \kappa, \nu)$ satisfies exactly

$$\bar c_{\bar a} = -\frac{\epsilon}{1+\epsilon}, \qquad \bar c_\kappa = -\frac{1}{\sqrt{1+\epsilon}}, \qquad \bar c_\nu = -\frac{1+2\epsilon}{(1+\epsilon)^2}. \tag{13.3}$$

The equilibrium and parameter Jacobians have uniform nonzero margins on a fixed positive-height box. The implicit function theorem therefore supplies this inverse with the finite coefficient jets used below on a common neighborhood.

Choose the symmetric local saddle sections and coordinates of Section 4.4. Its passage is

$$v = t^{1+\kappa} e^{w(t)}, \qquad t > 0,$$

and at the reversible center it is exactly $v = t$. The two regular outer legs reach $Y = 0$ with positive labels $p(t)$ and $q(v)$ near one. Increasing t or v moves into the period annulus, giving $p_t, q_v < -c < 0$ on smaller fixed intervals.

Lemma 13.1 (The regular lower block)**.** *With one fixed weak reference on each side, the two complete central maps have the form $s_+ + U(t)$ and $s_- + V(v)$, where*

$$U(t) = e^{-S_+(t)/\epsilon + \ell_+(t)}, \qquad V(v) = e^{-S_-(v)/\epsilon + \ell_-(v)}. \tag{13.4}$$

On the indicated coefficient block the functions $S_\pm, \ell_\pm$ have bounded prescribed finite ordinary input, coefficient and Euler scale jets. Moreover $S_\pm > s_* > 0$ and $-S'_+, -S'_- > c > 0$. The reference values are independent of input, and

$$s_- - s_+ = \sqrt{\epsilon}\,\nu H, \qquad H > h_* > 0, \qquad H|_{\bar a = \kappa = \nu = 0} \longrightarrow \sqrt{2\pi}. \tag{13.5}$$

Thus $U', V' > 0$, and the rates

$$\mathsf{A}_+ = \epsilon U''/U', \qquad \mathsf{A}_- = \epsilon V''/V'$$

have positive margins and bounded finite mixed jets. For each of the finitely many coefficient derivatives used here,

$$|\partial^\alpha_{\mathbf{b}} U(0)| + |\partial^\alpha_{\mathbf{b}} V(0)| + u'_\epsilon(0) \le C\epsilon^{-M} e^{-c/\epsilon}, \qquad u_\epsilon(t) = U(t;0) = V(t;0). \tag{13.6}$$

At the center set $A_\epsilon = \epsilon u''_\epsilon(0)/u'_\epsilon(0)$; then $0 < c \le A_\epsilon \le C$.

Proof. Both axis labels lie in $(3/4, 5/4)$, and $1 \pm \bar a X$ have positive margins on a larger fixed interval. On the regular layer from $z = 1$ to a fixed small $z = d_0 > 0$, the normal speed has a nonzero margin. The original graph equation with independent variable z gives $P = p + \epsilon R(p, \mathbf{b}, \epsilon)$ with bounded finite jets, and likewise on the reflected side.

Choose one weak graph and keep it fixed as p varies. Proposition 4.1 and $w = \epsilon z T$ give its height as $\epsilon\nu$ times a function with bounded ordinary endpoint and coefficient jets on this compact positive X interval. Its restriction at each P has the same central reference. Apply Proposition 4.7 with P as an independent regular endpoint: $X = Py$ puts its weighted reference equation on a fixed interval, and P has a positive lower margin. Its relative amplitude and the positive input deviation $d_0 - z_b(P)$ therefore have bounded ordinary P jets. With

$$A_\pm(p) = \int_0^p \frac{X}{1 \pm \bar a X}\,\mathrm{d}X,$$

the outgoing deviation is $a_+(p)e^{-A_+(p)/\epsilon}$, with bounded mixed jets of $\log a_+$, since the layer correction

$$\frac{A_+(P)-A_+(p)}{\epsilon}=R(p,\mathbf{b},\epsilon)\int_0^1 A'_+(p+s\epsilon R)\,\mathrm{d}s$$

has those bounds. Reflection gives the incoming statement. Composing with the two regular upper legs gives (13.4); their jets and inverse margins are supplied by the low-layer estimate of Section 4.4, whose scale carrier has integral $\epsilon\log(1+\delta/\epsilon)$. These connections exclude the saddle transit. The signs $A'_\pm>c$ and $p_t,q_v<-c$ give the action derivative margins.

For the reference difference, both references vanish when $\nu=0$. Their Gaussian variational equations give a factor ν directly. At the center, the first ν variation η on the outgoing core satisfies

$$\eta_X=\frac{X}{\epsilon(1+\epsilon)}\eta+\frac{1}{1+\epsilon}.$$

A fixed positive endpoint contributes only an exponentially small terminal term at $X=0$. Thus

$$\eta(0)=-\frac{1}{1+\epsilon}\int_0^{R_0} e^{-X^2/(2\epsilon(1+\epsilon))}\,\mathrm{d}X+O(\epsilon^{-M}e^{-c/\epsilon}).$$

The reflected reference has the opposite derivative. The change $X=\sqrt{\epsilon}\,u$, with its Gaussian integrable majorant, proves the limit in (13.5). On the larger small coefficient box the same kernel argument for the exact difference quotient in ν retains this leading positive term, with an error $O(\sqrt{\epsilon})$ after division by $\sqrt{\epsilon}$. More explicitly, put $X=\sqrt{\epsilon}\,u$ and $z_b=\sqrt{\epsilon}\,\nu\zeta$ in (4.21), using the continuous quotient at $\nu=0$. On each bounded u interval its equation tends uniformly to $\zeta_u=u\zeta+1$; the weighted reference estimate and Gaussian kernel control the remaining tail. The outgoing value tends to $-\sqrt{\pi/2}$ and the reflected value to $\sqrt{\pi/2}$, uniformly in this coefficient box. This gives the uniform positive margin for H. These estimates use the weighted reference equation and its kernel, without requiring ordinary high-order X derivatives at the Gaussian center.

Finally put $Z_\pm=-S'_\pm+\epsilon\ell'_\pm>c$. Direct differentiation of (13.4) gives

$$\epsilon U''/U'=Z_++\epsilon Z'_+/Z_+,$$

and the corresponding identity for V. All required rate jets are therefore bounded and positive after reducing the scale. Each fixed coefficient derivative of the amplitudes costs only a finite inverse power of ϵ, multiplying $e^{-s_*/\epsilon}$. This proves (13.6) and the center rate bound. □

13.2 A first variation with three simple closing zeros

The actual closing difference in these coordinates is

$$F(t)=\mathscr{D}(t)-\sqrt{\epsilon}\,\nu H,\qquad \mathscr{D}(t)=U(t)-V(v(t)).\tag{13.7}$$

Define

$$K(t)=\epsilon\log\frac{V'(v(t))v'(t)}{U'(t)},\qquad D=t\partial_t.$$

Integration of the lower logarithmic rates gives the exact formula

$$\begin{aligned}K(t)&=K_c+\epsilon\kappa\log t+\mathcal{E}(t),\\ K_c&=\epsilon\log\frac{(1+\kappa)V'(0)}{U'(0)},\\ \mathcal{E}(t)&=\int_0^{v(t)}\mathsf{A}_-(s)\,\mathrm{d}s-\int_0^t\mathsf{A}_+(s)\,\mathrm{d}s+\epsilon\left[w+\log\frac{1+\kappa+Dw}{1+\kappa}\right].\end{aligned}\tag{13.8}$$

The constant term K_c has the selected bounded coefficient jets. Write, at the center and in the actual parameters,

$$a_c(\epsilon) = \partial_{\bar a} K_c, \qquad b_c(\epsilon) = \partial_\kappa K_c.$$

Their limits are

$$a_c \longrightarrow -2/3, \qquad b_c \longrightarrow 1. \tag{13.9}$$

Indeed Lemma 13.1 gives $K_c = A_+(p_c) - A_-(q_c) + O_{C^2_{\mathrm{b}}}(\epsilon(1 + |\log\epsilon|))$ on $\nu = 0$. At zero scale,

$$p_c = \frac{\sqrt{\bar c^2 + 4} - \bar c}{2}, \qquad q_c = \frac{\sqrt{\bar c^2 + 4} + \bar c}{2},$$

and

$$A_+(p) = p^2/2 - \bar a p^3/3 + O(\bar a^2), \qquad A_-(q) = q^2/2 + \bar a q^3/3 + O(\bar a^2).$$

Use (13.3) to differentiate these expressions at fixed κ or fixed $\bar a$, respectively. The first derivative is $-2\int_0^1 X^2\,\mathrm{d}X = -2/3$. For the second, $p_{c,\bar c} = -1/2$, $q_{c,\bar c} = 1/2$ and $\bar c_\kappa \to -1$, giving one.

We also record directly the upper first variation needed here. In the prepared upper rectangle the orbital equation is

$$V_U = -rV/U - \sigma V^2 - UV^3\mathcal{R}(U, V).$$

At the center $V = t/U$, $r = 1$ and $\sigma = \mathcal{R} = 0$. For any bounded coefficient direction, the derivatives $\dot\sigma$ and $\dot{\mathcal{R}}$ are $O(\epsilon)$ with their needed finite state jets, by the original-field construction in Section 4.4. Set $\dot V = (t/U)\eta(U)$. Linearization of this equation, with its fixed input, gives

$$\eta_U = -\dot r/U - \dot\sigma\, t/U - t^2\dot{\mathcal{R}}(U, t/U)/U, \qquad \eta(t) = 0.$$

At $U = 1$, after removing $\dot r \log t$, this is

$$\dot w(t) = \dot\sigma\, t\log t - t^2\int_t^1 \dot{\mathcal{R}}(u, t/u)\,\frac{\mathrm{d}u}{u}, \qquad |D^j\dot w| \le C\epsilon t(1 + |\log t|) \quad (0 \le j \le 3). \tag{13.10}$$

Euler differentiation keeps $u, t/u$ bounded and adds bounded endpoint terms, proving the displayed estimates for the true saddle variation.

The ratio variation supplies the logarithmic term in K. We choose the $\bar a$ direction to balance its constant term at a smaller input scale, and then use ν to shift the closing difference. These three choices produce the following profile.

Proposition 13.2 (The closing profile)**.** *For every sufficiently small $\epsilon > 0$, there is a bounded direction $h_\epsilon \to 3/2$, an exponentially small β_ϵ, and positive numbers $\delta_\epsilon, S_\epsilon$ such that along the original coefficient curve determined by*

$$(\bar a, \kappa, \nu) = (h_\epsilon\theta, \theta, \beta_\epsilon\theta)$$

the first variation $f_\epsilon = \partial_\theta F|_{\theta=0}$ obeys

$$\frac{f_\epsilon(\delta_\epsilon x)}{S_\epsilon} \longrightarrow P(x) = \frac{x^2}{2} - x\log x - x - \frac{1}{20}. \tag{13.11}$$

The convergence is uniform on $[0, 10]$ after continuous extension of the values at zero, and is C^1 on every compact subinterval of $(0, 10]$. The curve uses the actual ratio inverse (13.3). This proposition requires no bound on the number of cycles of other fields.

Proof. Put $L = \log(1/\epsilon)$ and choose

$$h_\epsilon = \frac{\epsilon(L + \log L + \log A_\epsilon + 2) - b_c}{a_c}, \qquad \delta_\epsilon = \frac{\epsilon}{A_\epsilon L}, \qquad S_\epsilon = u'_\epsilon(0)\delta_\epsilon > 0. \tag{13.12}$$

The limits (13.9) prove the assertion about h_ϵ. First vary $(\bar{a}, \kappa, \nu) = (h_\epsilon\theta, \theta, 0)$, and write $k_\epsilon = \dot{K}$ and $d_\epsilon = \dot{\mathscr{D}}$. At the center let A be the common lower rate; $\mathsf{A}(t) = A_\epsilon + O(t)$ with bounded finite derivatives. The variation of the endpoint is $\dot{v} = t(\log t + \dot{w})$. Differentiating every term in (13.8) gives

$$\dot{\mathcal{E}}(t) = \int_0^t (\dot{\mathsf{A}}_- - \dot{\mathsf{A}}_+)(s)\,\mathrm{d}s + \mathsf{A}(t)t(\log t + \dot{w}) + \epsilon(\dot{w} + D\dot{w}).$$

The explicit κ derivatives in the logarithm's numerator and denominator cancel, leaving the displayed moving-endpoint term. The mixed rate bounds and (13.10) imply

$$\begin{aligned} k_\epsilon(t) &= \epsilon(L + \log L + \log A_\epsilon + 2) + \epsilon\log t + A_\epsilon t\log t + R_\epsilon(t), \\ |R_\epsilon(t)| &\le C\{t + t^2|\log t| + \epsilon^2 t(1 + |\log t|)\}, \\ |R'_\epsilon(t)| &\le C\{1 + t(1 + |\log t|) + \epsilon^2(1 + |\log t|)\}. \end{aligned} \tag{13.13}$$

Here the rate-difference integral contributes $O(t)$, the error $\mathsf{A}(t) - A_\epsilon$ contributes $O(t^2|\log t|)$, and the upper relative terms give the last term in each bound.

The identity $\log\delta_\epsilon = -L - \log L - \log A_\epsilon$ now gives exactly

$$\frac{k_\epsilon(\delta_\epsilon x)}{\epsilon} - (2 + \log x - x) = \frac{x}{L}(\log x - \log L - \log A_\epsilon) + \frac{R_\epsilon(\delta_\epsilon x)}{\epsilon}.$$

It proves C^1 convergence to $G(x) = 2 + \log x - x$ on compact positive intervals. The value error also tends to zero in $L^1(0, 10)$: (13.13) supplies an integrable majorant $C(1 + |\log x| + x|\log x|)$ there, and quantitative integration gives an error $O((1 + \log L)/L)$.

The exact identity $\mathscr{D}' = U'(1 - e^{K/\epsilon})$, differentiated at the center, is

$$d'_\epsilon(t) = -u'_\epsilon(t)k_\epsilon(t)/\epsilon. \tag{13.14}$$

Furthermore

$$\log\frac{u'_\epsilon(\delta_\epsilon x)}{u'_\epsilon(0)} = \frac{1}{\epsilon}\int_0^{\delta_\epsilon x} \mathsf{A}(s)\,\mathrm{d}s = \frac{x}{L} + O(\epsilon/L^2), \qquad 0 \le x \le 10.$$

The value $d_\epsilon(0)$ exists: the regular lower maps at the separatrix inputs are smooth in coefficients, and $t\log t \to 0$. Integrating (13.14) from zero gives

$$\frac{d_\epsilon(\delta_\epsilon x) - d_\epsilon(0)}{S_\epsilon} \longrightarrow J(x) = \frac{x^2}{2} - x\log x - x. \tag{13.15}$$

The integrable majorant proves uniform convergence of values on $[0, 10]$; the same identity proves C^1 convergence on compact positive intervals, with error $O((1 + \log L)/L)$.

It remains to realize the constant shift by an actual line-breaking parameter. At fixed $(\bar{a}, \kappa)$ put $e_\epsilon(t) = \partial_\nu\mathscr{D}(t)|_0$ and $H_\epsilon = H|_0$, and define

$$\beta_\epsilon = \frac{d_\epsilon(0) + S_\epsilon/20}{\sqrt{\epsilon}\,H_\epsilon - e_\epsilon(0)}. \tag{13.16}$$

Lemma 13.1 implies

$$\sqrt{\epsilon}\,H_\epsilon - e_\epsilon(0) \ge (h_*/2)\sqrt{\epsilon}, \qquad |\beta_\epsilon| \le C\epsilon^{-M}e^{-c/\epsilon}.$$

On the curve in the statement, the first variation satisfies exactly

$$\frac{f_\epsilon(t)}{S_\epsilon} = \frac{d_\epsilon(t) - d_\epsilon(0)}{S_\epsilon} - \frac{1}{20} + \beta_\epsilon \frac{e_\epsilon(t) - e_\epsilon(0)}{S_\epsilon}.$$

An absolute bound for the last difference would lose the exponentially small normalization S_ϵ. Instead we use the differentiated closing identity. At fixed κ the ν derivative of (13.8) has no $\epsilon \log t$ term. Its constant, rate-integral and upper terms are bounded, so $|K_\nu(t)| \le C$ for $0 < t \le 10\delta_\epsilon$. The differentiated exact identity gives

$$e'_\epsilon = -u'_\epsilon K_\nu/\epsilon, \qquad \left\| \frac{e_\epsilon(\delta_\epsilon x) - e_\epsilon(0)}{S_\epsilon} \right\|_{C^1([a,b])} \le C_{a,b}/\epsilon \quad (0 < a < b \le 10).$$

The value bound also holds on $[0, 10]$. The derivative identity cancels the common factor $u'_\epsilon(0)$; the exponentially small β_ϵ then sends both normalized errors to zero. Since $\nu = 0$ at the center, differentiation of νH introduces only $\beta_\epsilon H_\epsilon$. Combining these facts with (13.15) proves (13.11); on each $[a, b]$ its error is at most

$$C_{a,b}\left(\frac{1 + \log L}{L} + \frac{|\beta_\epsilon|}{\epsilon}\right).$$

□

13.3 Finite perturbations and localization

Theorem 13.3. *For every open neighborhood $\mathscr{U}$ of the hemicycle Γ on the northern Poincare sphere, there is a sequence of quadratic fields tending to the base field, each with three distinct simple limit cycles whose northern images lie in $\mathscr{U}$. Each of the three sequences of cycles can be chosen to converge to the entire Γ in Hausdorff distance. Consequently the quadratic cyclicity of Γ is at least three.*

Proof. The profile has signs $-, +, -, +$ at $10^{-3}, 10^{-1}, 1, 10$, and

$$P'(x) = x - \log x - 2, \qquad P''(x) = 1 - 1/x.$$

Thus P' has exactly two simple zeros, one on each side of one. Rolle's theorem and the four signs give exactly three simple positive zeros. Choose disjoint compact root intervals with P' bounded away from zero and opposite endpoint signs.

First choose ϵ so that Proposition 13.2 preserves these margins. At this fixed scale, smooth coefficient dependence at positive section inputs gives

$$\frac{F(\delta_\epsilon x; \theta)}{\theta S_\epsilon} \longrightarrow \frac{f_\epsilon(\delta_\epsilon x)}{S_\epsilon} \quad \text{in } C^1$$

on the three intervals as $\theta \to 0$, preserving one simple closing zero in each for every sufficiently small positive θ.

At the fixed scale, the center ovals with $10^{-3}\delta_\epsilon \le t \le 10\delta_\epsilon$ form a compact family inside the bounded period annulus, with finite flights, transverse section crossings and a positive minimum central height. Continuous dependence preserves these properties for small θ, so all three closing zeros compare actual positive outputs for the same field. The forward first return on the incoming saddle section is

$$\mathcal{R}(t) = U^{-1}(s_- - s_+ + V(v(t))).$$

Transversality preserves the center's single-turn itinerary. At a closing zero,

$$\mathcal{R}'(t) = \frac{V'(v(t))v'(t)}{U'(t)} = e^{K(t)/\epsilon}, \qquad F'(t) = U'(t)(1 - \mathcal{R}'(t)). \tag{13.17}$$

Since $U' > 0$ and $F' \neq 0$, the three zeros give three distinct hyperbolic cycles, each of multiplicity one. The admissible θ may depend on ϵ.

We next locate the center ovals before making the final perturbation. At $\bar{B} = 0$, $\bar{m} = 1$, the saddle first integral used in the upper coordinates is

$$\mathcal{H} = X^2 - \int_1^Y \frac{2s(s-1)}{s+\epsilon}\,\mathrm{d}s = 2\epsilon(\mathscr{H}_\epsilon - H_s).$$

The symmetric product coordinates satisfy $\mathcal{H} = -4UV$. On the normalized local sections, $UV = \chi_{\mathrm{loc}}^2 t$, where the fixed section factor χ_{loc} and its reciprocal are bounded. Thus, uniformly for the selected ovals,

$$H_s - \mathscr{H}_\epsilon = \frac{2\chi_{\mathrm{loc}}^2 t}{\epsilon}, \qquad \epsilon\mathscr{H}_\epsilon \longrightarrow \frac{1}{2}. \tag{13.18}$$

At the central section $x = 0$, $z = s$, the identity

$$\mathscr{H}_\epsilon(0, s-1) = (1+\epsilon)(s - 1 - \log s) - \epsilon(s-1)^2/2$$

also shows $\epsilon(s - 1 - \log s) \to 1/2$.

Write $C_{\epsilon,t}$ for the center oval with section input t. We prove

$$\sup_{10^{-3}\delta_\epsilon \le t \le 10\delta_\epsilon} d_H\big(\iota(C_{\epsilon,t}), \Gamma\big) \longrightarrow 0, \tag{13.19}$$

where d_H is spherical Hausdorff distance. We prove both distance estimates, so that the limit is the entire hemicycle. First take any sequence of points on selected ovals with $\epsilon \to 0$. A bounded affine subsequence with a limit in $y > -1$ is impossible: (13.1) stays bounded on compact subsets there, whereas the energies diverge uniformly. A bounded affine limit must therefore lie on $y = -1$. An unbounded sequence has $y > -1$, so every spherical accumulation point lies on the upper equatorial semicircle. Both types lie in Γ. Compactness proves that the maximum distance from the selected ovals to Γ tends to zero.

Conversely, let E range over the selected energies, so that $\epsilon E \to 1/2$ uniformly. For every fixed M, the lower branch of $V_\epsilon(y) = E - x^2/2$, $|x| \le M$, exists in $-1 < y < 0$. The potential is uniformly bounded on compact subintervals of $(-1, 0]$, whereas $E - x^2/2 \to \infty$. Hence this branch tends uniformly to $y = -1$, covering every finite point of the lower arc. At $y = 0$ the two points $x = \pm\sqrt{2E}$ approach the horizontal equatorial endpoints.

Fix an upper equatorial direction $(\cos\phi, \sin\phi)$ with $0 < \phi < \pi$ and $\cos\phi \neq 0$. On its ray take $(x, y) = c\epsilon^{-1/2}(\cos\phi, \sin\phi)$. Uniformly for c in a fixed compact positive interval,

$$\epsilon\mathscr{H}_\epsilon(x, y) = \frac{c^2\cos^2\phi}{2} + o(1), \qquad 0 < y < 1/\epsilon.$$

Values of c on opposite sides of $1/|\cos\phi|$ bracket every selected E. The intermediate value theorem supplies an intersection on the bounded-well component, tending to that equatorial direction. For the vertical direction, its upper turning point y_+ satisfies $V_\epsilon(y_+) = E$ and $y_+ \ge E/(1+\epsilon) \to \infty$, since $V_\epsilon(y) \le (1+\epsilon)y$ for $y \ge 0$. Every point of Γ is therefore approached uniformly over the selected energy band. Applying these pointwise estimates to a finite net of the compact curve Γ gives the reverse distance estimate and proves (13.19).

Choose $\epsilon_n \downarrow 0$ so that the compact center families lie in smaller neighborhoods whose closures lie in $\mathscr{U}$. At each fixed ϵ_n, choose one $\theta_n > 0$ below the root-persistence and path-positivity cutoffs for all three intervals, small enough that the three perturbed cycles remain in $\mathscr{U}$, and also $\theta_n < 1/n$. Finite-time dependence over the compact transverse center family lets us require, at the same time, that each perturbed cycle have spherical Hausdorff distance

less than $1/n$ from its corresponding center oval. Equation (13.19) then gives convergence of each of the three cycles to the entire Γ. In original coefficients this sequence is

$$B = 0, \qquad m = \epsilon_n, \qquad a = \sqrt{\epsilon_n}\, h_{\epsilon_n}\theta_n,$$
$$c = \sqrt{\epsilon_n}\, \bar{c}_{\epsilon_n}(h_{\epsilon_n}\theta_n, \theta_n, \beta_{\epsilon_n}\theta_n), \qquad d = c + \sqrt{\epsilon_n}\, \beta_{\epsilon_n}\theta_n.$$

Reduce θ_n further so that $|\bar{a}|, |\bar{c}| < 1$. The primary scale remains exactly $\rho = \sqrt{\epsilon_n}$, and all original coefficients tend to zero. These fields lie in the original five-parameter family and hence prove the lower bound in the full quadratic family, independently of the upper bound and its forcing theorem. □

Corollary 13.4 (Stability of the three cycles)**.** *For the positive perturbation amplitudes in the construction of Theorem 13.3, the three selected cycles are, in their nesting order, attracting, repelling and attracting.*

Proof. At the three increasing simple roots of P, its derivative has signs $+, -, +$. The two C^1 limits used above preserve these signs for $F'(t)$, since $\theta, S_\epsilon, \delta_\epsilon > 0$. Equation (13.17) and $U' > 0$ then give forward return multipliers less than one, greater than one, and less than one. All are positive. Increasing t moves inward in the period annulus; reversing the order leaves this alternating pattern unchanged. □

A The base annulus and its three coefficient functions

The collar in Proposition 2.2 must lie beyond an energy threshold where three coefficient functions form an extended complete Chebyshev system. We obtain this threshold from the base field (2.1). Put $y = z - 1$ and denote time by θ. The first integral is

$$H_0(x, z) = \frac{x^2}{2} + z - 1 - \log z. \tag{A.1}$$

As shown in Section 2, every level $H_0 = h > 0$ is a regular oval with two simple turning points. Its lower intersection with $x = 0$ has height $s(h) \in (0, 1)$, where $s(h) - 1 - \log s(h) = h$.

The coefficient functions are derivatives of the energy displacement, so we first fix its parameter coordinates. In this appendix $t = B + m$ is a coefficient coordinate, whereas θ denotes time. Put $p = (\tau_0, \ell, a, t, B)$ and consider

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}\theta} &= -y + Bx^2 + (t - B)y^2 + ax, \\ \frac{\mathrm{d}z}{\mathrm{d}\theta} &= xz + (\tau_0 - a + \ell)y^2 + (\tau_0 - a)y. \end{aligned} \tag{A.2}$$

Thus $c = \tau_0 - a$, $d = c + \ell$ and $m = t - B$. On a compact energy interval $J \subset (0, \infty)$, the base flights are compact in $z > 0$ and their two $x = 0$ crossings are transverse, with the other crossing above $z = 1$. The finite-flight argument in Lemma 2.4, now on these compact ovals, gives a positive first lower return $P(h, p)$, jointly analytic near $J \times \{0\}$ after a finite covering of J.

Define the energy displacement and its three coefficients by

$$\begin{gathered} \Delta(h, p) = P(h, p) - 1 - \log P(h, p) - h, \\ \mathfrak{a}(h) = \partial_{\tau_0}\Delta(h, 0), \qquad \mathfrak{b}(h) = \partial_\ell \Delta(h, 0), \qquad \mathfrak{c}(h) = -\partial_a \partial_t \Delta(h, 0). \end{gathered} \tag{A.3}$$

Uniqueness of the local return identifies these definitions on overlapping compact intervals. Consequently the three functions are real analytic on $(0, \infty)$.

For functions $f_1, \dots, f_k$, we use the Wronskian convention

$$W[f_1, \dots, f_k] = \det\bigl(f_j^{(i-1)}\bigr)_{1 \le i, j \le k}.$$

An ordered family is an extended complete Chebyshev system on an interval if every nonzero linear combination of its first k members has at most $k-1$ zeros there, counted with multiplicity, for each k.

Proposition A.1 (An independent Chebyshev threshold)**.** *Put $u = \log z$ and $g(u) = e^u - 1 - u$. For $h > 0$, let $u_-(h) < 0 < u_+(h)$ be the two roots of $g(u) = h$, and set $X_h(u) = \sqrt{2(h - g(u))}$ between them. The coefficients in (A.3) satisfy*

$$\begin{aligned} \mathfrak{a}(h) &= 2\int_{u_-(h)}^{u_+(h)} X_h(u)\,\mathrm{d}u, \\ \mathfrak{c}(h) &= 2\int_{u_-(h)}^{u_+(h)} X_h(u)(e^u - 1)^2\,\mathrm{d}u, \\ \mathfrak{b}(h) &= 2\int_{u_-(h)}^{u_+(h)} X_h(u)(e^u - e^{-u})\,\mathrm{d}u. \end{aligned} \tag{A.4}$$

There is a constant $H_{\mathrm{ECT}} > 1$, depending only on the base field, such that on $(H_{\mathrm{ECT}}, \infty)$

$$W[\mathfrak{a}] > 0, \qquad W[\mathfrak{a}, \mathfrak{c}] > 0, \qquad W[\mathfrak{a}, \mathfrak{c}, \mathfrak{b}] < 0. \tag{A.5}$$

In particular $(\mathfrak{a}, \mathfrak{c}, \mathfrak{b})$ is an extended complete Chebyshev system on that interval.

Proof. We first express the three variations as integrals over the base oval. Two exact differential identities then express $\mathfrak{c}$ and $\mathfrak{b}$ in terms of the area integral $\mathfrak{a}$. The Wronskian signs follow from the large-energy estimates for $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$.

First and mixed variations. Along (A.2), differentiation of (A.1) gives the exact identity

$$\frac{\mathrm{d}H_0}{\mathrm{d}\theta} = \tau_0 y^2 + \ell\frac{y^3}{z} + a(x^2 - y^2) + txy^2 + B(x^3 - xy^2). \tag{A.6}$$

Its integral between successive lower intersections is Δ. The right-hand side vanishes identically at $p = 0$. Hence its first parameter derivatives are obtained by integrating the displayed coefficients along the base oval: derivatives of the flight and its endpoints give no first-order term. On the base field,

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(xy + \frac{x^3}{3}\right) = x^2 - y^2, \qquad \frac{\mathrm{d}u}{\mathrm{d}\theta} = x.$$

It follows that $\partial_a\Delta(h, 0) = 0$ and

$$\mathfrak{a} = \oint y^2\,\mathrm{d}\theta = \oint x^2\,\mathrm{d}\theta = 2\int_{u_-}^{u_+} X_h\,\mathrm{d}u.$$

The first t and B derivatives vanish as well: after $x\,\mathrm{d}\theta = \mathrm{d}u$, the integrands y^2 and $x^2 - y^2$ have equal values on the two branches, which are traversed in opposite u directions. For the ℓ derivative, use $(X_h)_u = -(e^u - 1)/X_h$ to obtain

$$\begin{aligned}\mathfrak{b} &= 2\int_{u_-}^{u_+} \frac{(e^u - 1)^3 e^{-u}}{X_h}\,\mathrm{d}u \\ &= -2\int_{u_-}^{u_+} (e^u - 1)^2 e^{-u}(X_h)_u\,\mathrm{d}u = 2\int_{u_-}^{u_+} (e^u - e^{-u})X_h\,\mathrm{d}u.\end{aligned}$$

The boundary terms vanish because X_h vanishes at both endpoints.

To compute the mixed derivative, restrict to $\tau_0 = \ell = B = 0$. At $a = 0$ the field has the first integral

$$H_t(x, z) = H_0(x, z) - t\kappa(z - 1), \qquad \kappa(y) = \int_0^y \frac{v^2}{1 + v}\,\mathrm{d}v.$$

In (x, u) coordinates it is the Hamiltonian field $\dot{x} = -\partial_u H_t$, $\dot{u} = \partial_x H_t$, and the a perturbation is $(x, -y)$. For t sufficiently small on a fixed compact energy interval, the continued regular ovals $H_t = k$ therefore have the first a-variation

$$\begin{aligned}M(k, t) &= \oint_{H_t = k} (x\,\mathrm{d}u + y\,\mathrm{d}x) \\ &= 2\int_{u_-(k,t)}^{u_+(k,t)} X_t(u, k)(1 - e^u)\,\mathrm{d}u, \qquad X_t(u, k) = \sqrt{2\big(k - g(u) + t\kappa(e^u - 1)\big)}.\end{aligned} \tag{A.7}$$

Here the dot refers to θ, and $u_\pm(k, t)$ are the continued simple turning points. The first equality follows directly by integrating $\partial_a(\mathrm{d}H_t/\mathrm{d}\theta)$ on the unperturbed $a = 0$ oval. In the second equality we integrated $y\,\mathrm{d}x$ by parts.

The mixed variation is defined using the original energy H_0. We must therefore convert both the incoming label and the outgoing energy in this Hamiltonian calculation back to H_0. On the lower section,

$$H_t(0, s(h)) = h - t\kappa(s(h) - 1), \qquad \frac{\partial_z H_0(0, s)}{\partial_z H_t(0, s)} = \frac{1}{1 - t(s - 1)}.$$

Since at $a = 0$ the lower return is the identity, these formulas give

$$\partial_a \Delta(h,p)|_{\tau_0=\ell=a=B=0} = \frac{M(h - t\kappa(s(h)-1), t)}{1 - t(s(h)-1)}. \tag{A.8}$$

At $t = 0$, $M(k,0) = -2\int X_k g' \,\mathrm{d}u = 0$, because $X_k g' = -(X_k^3)_u/3$. Thus $M(k,0)$ and its k derivative vanish for every $k > 0$. Differentiating (A.8) at $t = 0$ leaves only $M_t(h,0)$. It follows that

$$\begin{aligned}\mathfrak{c}(h) = -M_t(h,0) &= 2\int_{u_-}^{u_+} \frac{\kappa(e^u-1)g'(u)}{X_h(u)}\,\mathrm{d}u \\ &= -2\int_{u_-}^{u_+} \kappa(e^u-1)(X_h)_u\,\mathrm{d}u = 2\int_{u_-}^{u_+} X_h(u)(e^u-1)^2\,\mathrm{d}u.\end{aligned}$$

The differentiations of (A.7) are legitimate on every fixed compact energy interval: a square-root coordinate at each simple turning point removes the moving endpoint, and the first derivative in the original coordinate has an integrable inverse-square-root singularity. This proves all three formulas in (A.4).

Two Abel differential identities. The same endpoint argument permits one energy derivative in (A.4). Since $g'' = 1 + g'$,

$$\mathfrak{c}' = 2\int_{u_-}^{u_+} \frac{(g')^2}{X_h}\,\mathrm{d}u = 2\int_{u_-}^{u_+} g'' X_h\,\mathrm{d}u = \mathfrak{a}, \qquad \int_{u_-}^{u_+} g' X_h\,\mathrm{d}u = 0.$$

In particular $2\int e^u X_h\,\mathrm{d}u = \mathfrak{a}$. Set $J_-(h) = 2\int_{u_-}^{u_+} e^{-u}X_h\,\mathrm{d}u$, so that $\mathfrak{b} = \mathfrak{a} - J_-$. Integrating

$$(e^{-u}X_h)_u = -e^{-u}X_h - \frac{1-e^{-u}}{X_h}$$

between the turning points gives $J_-' = J_- + \mathfrak{a}'$. We have therefore proved

$$\mathfrak{c}' = \mathfrak{a}, \qquad \mathfrak{b}' = \mathfrak{b} - \mathfrak{a}. \tag{A.9}$$

As $h \downarrow 0$, the ovals shrink to the nondegenerate minimum of g and all three integrals tend to zero. Integrating (A.9) with these initial values yields

$$\mathfrak{c}(h) = \int_0^h \mathfrak{a}(v)\,\mathrm{d}v, \qquad \mathfrak{b}(h) = -e^h\int_0^h e^{-v}\mathfrak{a}(v)\,\mathrm{d}v. \tag{A.10}$$

The area integral through two derivatives. Let $\omega(v) = u_+'(v) - u_-'(v)$ for $v > 0$. Changing variables $v = g(u)$ on the two branches gives

$$\mathfrak{a}(h) = 2\sqrt{2}\int_0^h (h-v)^{1/2}\omega(v)\,\mathrm{d}v. \tag{A.11}$$

Write $E_\pm(v) = e^{u_\pm(v)}$. The defining equations imply

$$u_- = -v - 1 + E_-, \qquad E_+ - \log E_+ = v + 1.$$

For large v they give $E_- = O(e^{-v})$ and $E_+ = v + O(\log v)$. For the latter estimate, one first bounds $v < E_+ < 2v$ and then substitutes in $E_+ = v + 1 + \log E_+$. The exact inverse derivative formulas

$$u_\pm' = (E_\pm - 1)^{-1}, \qquad u_\pm'' = -E_\pm(E_\pm - 1)^{-3}$$

consequently imply

$$\omega(v) = 1 + v^{-1} + O(v^{-2}\log v), \qquad \omega'(v) = -v^{-2} + O(v^{-3}\log v). \tag{A.12}$$

Near zero, $q = \operatorname{sgn}(u)\sqrt{2g(u)}$ is an analytic coordinate with derivative one. Its analytic inverse gives $\omega(v) = \sqrt{2}\, v^{-1/2} + O(v^{1/2})$.

Choose once a smooth cutoff equal to one near zero and zero for $v \geq V$, with a fixed $V > 1$. Decompose $\omega - 1 = r_0 + r_\infty$, where r_0 is the cutoff part. Then $r_0 \in L^1(0,\infty)$ has support in $[0, V]$, whereas r_∞ is smooth, vanishes near zero, and satisfies

$$|r_\infty(v)| \leq \frac{C}{1+v}, \qquad |r'_\infty(v)| \leq \frac{C}{(1+v)^2}.$$

For $h > 2V$, the compactly supported part contributes $O(h^{1/2-j})$ to the jth derivative, $0 \leq j \leq 2$. For r_∞, a split at $v = h/2$ gives

$$\left| \frac{\mathrm{d}^j}{\mathrm{d}h^j} \int_0^h (h-v)^{1/2} r_\infty(v)\, \mathrm{d}v \right| \leq C h^{1/2-j} \log h, \qquad j = 0, 1.$$

To obtain the second derivative, first transfer a derivative to r_∞ by integration by parts:

$$\int_0^h (h-v)^{-1/2} r_\infty(v)\, \mathrm{d}v = 2 \int_0^h (h-v)^{1/2} r'_\infty(v)\, \mathrm{d}v.$$

The lower boundary term is zero since r_∞ vanishes near zero. Differentiating this identity now gives

$$\frac{\mathrm{d}^2}{\mathrm{d}h^2} \left(2\sqrt{2} \int_0^h (h-v)^{1/2} r_\infty(v)\, \mathrm{d}v \right) = \sqrt{2} \int_0^h (h-v)^{-1/2} r'_\infty(v)\, \mathrm{d}v.$$

Subtracting the constant kernel $h^{-1/2}$ on the right writes it as

$$\sqrt{2}\, h^{-1/2} r_\infty(h) + \sqrt{2} \int_0^h \big((h-v)^{-1/2} - h^{-1/2} \big) r'_\infty(v)\, \mathrm{d}v.$$

On $[0, h/2]$ the kernel difference is at most $Cvh^{-3/2}$, which gives $Ch^{-3/2} \log h$. On $[h/2, h]$ the derivative r'_∞ is bounded by Ch^{-2}, which gives $Ch^{-3/2}$; the boundary term has that same bound. Adding the explicit convolution with the constant density one proves

$$\begin{aligned} \mathfrak{a}(h) &= \frac{4\sqrt{2}}{3} h^{3/2} + O(h^{1/2} \log h), \\ \mathfrak{a}'(h) &= 2\sqrt{2}\, h^{1/2} + O(h^{-1/2} \log h), \\ \mathfrak{a}''(h) &= \sqrt{2}\, h^{-1/2} + O(h^{-3/2} \log h). \end{aligned} \tag{A.13}$$

The convolution estimates establish each derivative bound directly, with constants fixed by g and the chosen cutoff.

The three initial Wronskians. The positive area integral gives $W[\mathfrak{a}] = \mathfrak{a} > 0$ for all $h > 0$. Equations (A.10) and (A.13) give

$$\mathfrak{c}(h) = \frac{8\sqrt{2}}{15} h^{5/2} + O(h^{3/2} \log h), \qquad \mathfrak{c}' = \mathfrak{a}, \qquad \mathfrak{c}'' = \mathfrak{a}'.$$

It follows that

$$W[\mathfrak{a}, \mathfrak{c}] = \mathfrak{a}^2 - \mathfrak{a}'\mathfrak{c} = \frac{64}{45} h^3 + O(h^2 \log h). \tag{A.14}$$

To estimate the third column, evaluate the positive Laplace moment of $\mathfrak{a}$. Changing the order of integration is justified by nonnegativity, and gives

$$\begin{aligned} K_0 := \int_0^\infty e^{-h} \mathfrak{a}(h)\, \mathrm{d}h &= 2\sqrt{2} \int_{-\infty}^\infty e^{-g(u)}\, \mathrm{d}u \int_0^\infty e^{-v} v^{1/2}\, \mathrm{d}v \\ &= \sqrt{2\pi} \int_{-\infty}^\infty e^{1+u-e^u}\, \mathrm{d}u = e\sqrt{2\pi}. \end{aligned} \tag{A.15}$$

The last integral uses $z = e^u$. Thus

$$\mathfrak{b}(h) = -K_0 e^h + e^h \int_h^\infty e^{-v}\mathfrak{a}(v)\,\mathrm{d}v = -K_0 e^h + O(h^{3/2}).$$

The exact identities $\mathfrak{b}' = \mathfrak{b} - \mathfrak{a}$ and $\mathfrak{b}'' = \mathfrak{b} - \mathfrak{a} - \mathfrak{a}'$ now show that

$$\mathfrak{b}^{(j)}(h) = -K_0 e^h + O(h^{3/2}), \qquad j = 0, 1, 2. \tag{A.16}$$

In the expansion along the third column, the $\mathfrak{b}''$ minor is (A.14), while the other two minors are $O(h^2)$ and $O(h)$. Equation (A.16) therefore yields

$$W[\mathfrak{a}, \mathfrak{c}, \mathfrak{b}] = -\frac{64e\sqrt{2\pi}}{45} e^h h^3 \left(1 + O\left(\frac{\log h}{h}\right)\right). \tag{A.17}$$

Choose $H_{\mathrm{ECT}} > \max\{1, 2V\}$ so large that the relative errors in (A.14) and (A.17) are at most $1/2$ for every $h > H_{\mathrm{ECT}}$. This choice involves only the base integral. It proves all signs in (A.5).

For completeness, the initial-Wronskian criterion [1, Lemma 2.3(b)] reduces here to the following two quotient identities. Put $r = \mathfrak{c}/\mathfrak{a}$ and $v = \mathfrak{b}/\mathfrak{a}$. Then

$$r' = \frac{W[\mathfrak{a}, \mathfrak{c}]}{\mathfrak{a}^2} > 0, \qquad \left(\frac{v'}{r'}\right)' = \frac{W[\mathfrak{a}, \mathfrak{c}, \mathfrak{b}]}{\mathfrak{a}^3 (r')^2} < 0.$$

After division by $\mathfrak{a} > 0$, a three-function combination is $G = A + Cr + Dv$. If $D \neq 0$, then $G'/r' = C + Dv'/r'$ is strictly monotone; otherwise $G = A + Cr$. Rolle's inequality with multiplicity (Lemma 3.1) gives the bounds two, one and zero for the three initial subfamilies. This argument also applies to finite selections on the open tail and to included regular endpoints. □

Corollary A.2 (Stability on a fixed compact energy interval)**.** *Let $J = [h_1, h_2] \subset (H_{\mathrm{ECT}}, \infty)$ with $h_1 < h_2$. There is a number $\delta_J > 0$ such that any three real analytic functions (a_*, c_*, b_*) defined on a neighborhood of J and satisfying*

$$\|a_* - \mathfrak{a}\|_{C^2(J)} + \|c_* - \mathfrak{c}\|_{C^2(J)} + \|b_* - \mathfrak{b}\|_{C^2(J)} < \delta_J$$

form an extended complete Chebyshev system on J. The permitted size δ_J depends on the fixed interval.

Proof. Compactness gives nonzero margins for all three Wronskians on J. Their signs persist under a small $C^2(J)$ perturbation, and the same quotient argument gives the count, including the endpoint germs. □

B Common domains and parameter-dependent coordinates

Coefficient division compares a local map with its center value while holding a common input fixed. A coefficient-dependent input change therefore enters the coefficient derivative. We record this rule, then check that phase changes preserve the whole passage and regular connections preserve flat outputs. The physical domain is the intersection of the scalar whole-path conditions, regular connection domains, positive central heights and the chosen collar. The integral comparisons use larger signed domains before this physical restriction.

Lemma B.1 (A fixed input under coefficient division)**.** *Let $b = (b_0, \eta) \in B_0 \times \mathbb{R}^k$, and let a parameterized input change $x = \phi(s, b)$ be defined on a common product $J \times B$. Assume its inverse exists on a slightly larger common interval and both maps have bounded derivatives through total order $N+1$. Suppose $F(x, b)$ has those bounds on the entire image and $F(x, b_0, 0) = 0$ for every represented center input. Define $G(s, b) = F(\phi(s, b), b)$ and suppose $(s, b_0, t\eta)$ remains in the common domain for $0 \le t \le 1$. Then the actual divided columns are*

$$G_i(s, b) = \int_0^1 \partial_{\eta_i} G(s, b_0, t\eta)\, \mathrm{d}t, \qquad G = \sum_i \eta_i G_i.$$

They have bounded total order N on $J \times B$. In general these columns are not obtained by composing previously divided columns of F with $\phi(s, b)$.

Proof. Apply the fundamental theorem of calculus to $t \mapsto G(s, b_0, t\eta)$ with s fixed. Its value at zero vanishes by the center identity on the entire represented interval. At each coefficient point the chain rule reads

$$\partial_{\eta_i} G = F_{\eta_i}(\phi, b) + F_x(\phi, b)\phi_{\eta_i}(s, b).$$

Even though $F_x = 0$ at the center, the second term need not vanish along the coefficient segment. It must be retained in the divided column. Repeated chain and product rules through order N use only the bounded $N + 1$ jets of F and ϕ on the stated image. Integration proves the bounds, and the prescribed inverse margins make this a common coordinate on every open overlap. □

In Section 9, form the separated columns only after fixing the actual common input. Two input derivatives of these columns use three input and one coefficient derivatives of the nonconstant critical logarithm. Their coefficient Lipschitz bounds use two of each. Both requirements have total order four and follow from the order-five primitive estimate. Counting only derivatives at a fixed coefficient would miss the second chain-rule term.

Lemma B.2 (Changing a scalar phase)**.** *Suppose $x' = qg_1(x, b)$ and $y = \Phi(x, b)$ is an increasing regular phase change, with positive bounded derivative and inverse derivative, on a common entry-to-exit interval. Allow also a time change $d\widetilde{t}/dt = a(x, b) > 0$, with bounded positive unit a. The exact new scalar factor is*

$$g_2(\Phi(x, b), b) = \frac{\Phi_x(x, b)}{a(x, b)} g_1(x, b).$$

The whole-path passing conditions are equivalent in the two phases. For a normally contracting equation $Z' = -Z\beta(x, Z, b)$, the linear normal action and the nonlinear relative integral are unchanged when their time and phase factors are transformed together.

Proof. Differentiate $y = \Phi(x, b)$. Its positive multiplier and increasing Φ preserve positivity on the whole closed entry-to-exit interval; corresponding scalar zeros cannot be crossed in finite time. The normal rate becomes $\widetilde{\beta} = \beta/a$, so change of variables gives

$$\int_{x_e}^{x_R} \frac{\beta(x, 0, b)}{qg_1(x, b)}\, \mathrm{d}x = \int_{\Phi(x_e)}^{\Phi(x_R)} \frac{\widetilde{\beta}(y, 0, b)}{qg_2(y, b)}\, \mathrm{d}y.$$

The same substitution applies to the relative integral along the actual orbit. At a separated lost exit, the normal domination and integrable-tail bound of Appendix D justify taking the infinite-time limit. □

The positive phase factors in Section 5 identify the entire passages and their contractions. For a quadratic factor a nonpositive gap retains only entries above the positive root, toward the fixed right exit. For a weak factor $u(\gamma + u)$ it retains $u > \max(0, -\gamma)$. For the cubic raw factor $xF_I(x)$ one uses the positive component of F_I adjoining the fixed exit. Each condition must hold on the whole entry-to-exit path; two endpoint values do not determine passage.

Lemma B.3 (Regular maps and flat outputs)**.** *Suppose a normally contracting transition has the form $G(r, b) = L(b)re^{j(r,b)}$, with bounded selected mixed derivatives of j and of the absolute factor L, while derivatives of $\log L$ may be unbounded. Let two regular maps fixing zero have positive bounded units $\Phi_{\text{in}}(r, b) = ra(r, b)$ and $\Phi_{\text{out}}(z, b) = zc(z, b)$. Their composition has the exact form*

$$\Phi_{\text{out}} \circ G \circ \Phi_{\text{in}} = La(0, b)c(0, b)re^{\psi(r,b)},$$

where the selected jets of ψ are bounded on the common domain. They extend to a lost passage whenever the relative unit and absolute factor have the corresponding extensions. This statement supplies no uniform inverse derivative for the composite map.

Proof. The exact relative logarithm is

$$\psi = \log \frac{a(r, b)}{a(0, b)} + j(ra(r, b), b) + \log \frac{c(Lra(r, b)e^{j(ra(r,b),b)}, b)}{c(0, b)}.$$

Every denominator has a fixed positive margin. The mixed derivatives use absolute derivatives of L, so no factor L^{-1} or coefficient derivative of $\log L$ occurs. The chain rule gives the bounds and extensions. At $L = 0$ the composite and its input derivative vanish, so the assertion gives no uniform inverse bound. □

For the contact and weak-center arguments, a displacement such as $\delta \mapsto \delta + 3(|\tau| + |\zeta| + |\nu|)$ chooses a center with the original coefficient vector held fixed. Apply Taylor's formula on that prescribed segment in the larger domain; the absolute values are not differentiated. The center identity holds on an open passing neighborhood of each retained input, so its passive center-gap derivative vanishes. This gives the linear transverse term and the quadratic remainder used in those arguments.

C Endpoint calculations

The first calculation determines the resonant sign used at an upper saddle. The second computes the exact linear return and its transverse columns on the closed radial interval needed at the joint center.

C.1 The resonant sign from the original upper field

The incoming lower-loss chart uses the sign of the first orbital resonant coefficient of the original upper field. We compute it from (4.32), beginning on $\nu = 0$. Its equilibrium satisfies

$$X_s = -\bar{c}Y_s, \qquad Y_s = \frac{1 + \epsilon\bar{a}\bar{c}}{\bar{m} + \epsilon\bar{B}\bar{c}^2}.$$

The trace is zero precisely when

$$0 = \bar{c}[1 + \epsilon(\bar{m} - 2\bar{B})] + \epsilon\bar{a}\bar{m} + \epsilon\bar{a}\bar{c}^2(1 - \epsilon\bar{B}) + \epsilon^2\bar{B}\bar{c}^3. \tag{C.1}$$

Indeed the trace, multiplied by $Y_s + \epsilon$, is $\bar{c}Y_s(1 - 2\epsilon\bar{B}) + \epsilon(\bar{a} + \bar{c})$; substitution of Y_s gives the displayed polynomial. The implicit function theorem yields $\bar{c} = -\epsilon\bar{a}\bar{m} + O(\epsilon^2)$. Thus $X_s = \epsilon\bar{a} + O(\epsilon^2)$ and $Y_s = 1/\bar{m} + O(\epsilon^2)$.

Put $x = X - X_s$, $y = Y - Y_s$. To first order in ϵ, expansion of $1/(Y + \epsilon)$ in the first component gives the quadratic and cubic terms

$$\epsilon P_2 = \epsilon(\bar{B}\bar{m}x^2 - \bar{a}\bar{m}^2xy + \bar{m}^3y^2),$$
$$\epsilon P_3 = \epsilon(-\bar{B}\bar{m}^2x^2y + \bar{a}\bar{m}^3xy^2 - \bar{m}^4y^3).$$

The second component is affine, and the omitted coefficients are $O(\epsilon^2)$ in the reserved finite coefficient jets. At zero scale use $x = \sqrt{\bar{m}}(u - v)$, $y = u + v$. The linear part is $(\sqrt{\bar{m}}u, -\sqrt{\bar{m}}v)$. The coefficients of u^2v and uv^2 in P_3 are respectively

$$\bar{B}\bar{m}^3 + \bar{a}\bar{m}^{7/2} - 3\bar{m}^4, \qquad \bar{B}\bar{m}^3 - \bar{a}\bar{m}^{7/2} - 3\bar{m}^4.$$

Since $\dot{u} = (\dot{y} + \dot{x}/\sqrt{\bar{m}})/2$ and $\dot{v} = (\dot{y} - \dot{x}/\sqrt{\bar{m}})/2$, the sum of the two resonant vector-field coefficients is $\epsilon\bar{a}\bar{m}^3 + O(\epsilon^2)$. The quadratic terms affect this coefficient only at $O(\epsilon^2)$: their coefficients and normalizing changes are $O(\epsilon)$, since the nonresonant denominators have fixed margins. The $O(\epsilon)$ change of linear eigenvectors applied to P_3 has the same order of effect.

Dividing the two components to form the orbital equation changes this sum by the factor $-1/\lambda_u$. Thus its V^2 coefficient in the notation of (4.33) is $-\epsilon\bar{a}\bar{m}^{5/2} + O(\epsilon^2)$. The analytic construction of Section 4.4 vanishes exactly both at $\epsilon = 0$ and on the reversible face. Restricting to $\kappa = \nu = 0$ and applying these two divisions therefore strengthens the remainder to

$$\sigma = -\chi_{\mathrm{loc}}^2\epsilon\bar{a}[\bar{m}^{5/2} + O(\epsilon)]. \tag{C.2}$$

Here $\chi_{\mathrm{loc}} > 0$ is the common local section scale. The two axis shears have no quadratic terms affecting this constant coefficient; the common section rescaling multiplies it by χ_{loc}^2. The sign is unchanged by the common reversal $U = -u, V = -v$ that selects the returning quadrant. On a compact block $\bar{a} > a_* > 0$, analyticity of σ/ϵ then gives $\sigma < -c\epsilon$ for small $|\kappa| + |\nu| + \epsilon$. This supplies the sign for the prepared original field on that compact block.

C.2 The exact linear return and the radial joint columns

In this subsection B, m, a, c denote the barred coefficients and $M = -m > 0$. At zero primary scale and zero line breaking the upper orbital field is

$$X' = -1 - MY, \qquad Y' = X + cY, \qquad c > 0. \tag{C.3}$$

Put $g = M - c^2/4 > 0$, $\Omega = 2\sqrt{g}$ and

$$F_U(p) = 1 - cp + Mp^2, \qquad F_L(p) = 1 + ap + Bp^2.$$

For the first upper return $(p, 0) \mapsto (-q, 0)$, the reciprocal ratio $\eta = p/q$ becomes exponentially small as the repeated limit is approached. We factor out that exponential to retain mixed gap derivatives on the closed radial interval. An exact determinant identity then gives the principal rate without differentiating the divergent transit time.

Lemma C.1. *Fix $\gamma > 0$ and $0 < d < \gamma$. On $0 \le s \le \gamma - d$, with coefficients close to $(c, M) = (2\gamma, \gamma^2)$ and $g > 0$, the reciprocal return ratio $\eta = p/q$, $p = 1/s$, has the form*

$$\eta(s, c, g) = \Lambda\, \mathcal{B}(s, c, g, \Lambda), \qquad \Lambda = e^{-\pi c/\Omega}. \tag{C.4}$$

The function $\mathcal{B}$ is a positive analytic unit on a common neighborhood including $s = g = \Lambda = 0$, and $\eta(0) = \Lambda$. For each fixed N there are $a_N, C_N > 0$ such that

$$|\partial_s^i \partial_c^j \partial_g^k \eta| \le C_N e^{-a_N/\sqrt{g}}, \qquad i + j + k \le N.$$

Zero extension for $g \le 0$ has these finite jets; it is an auxiliary extension, not a return through a missing upper passage. At every actual finite return,

$$\frac{q_p}{q} = \frac{p}{F_U(p)} \left(M + \frac{c}{q} + \frac{1}{q^2} \right). \tag{C.5}$$

Proof. Set $Z = Y + 1/M$ and $V = Y' = X + cY$. The quadratic expression $E = V^2 - cZV + MZ^2$ satisfies $E' = cE$. Thus at the return time T,

$$1 + cq + Mq^2 = e^{cT} F_U(p).$$

The transverse determinant formula gives $q_p = pe^{cT}/q$. Eliminate e^{cT} between these two identities to obtain (C.5).

For the uniform formula put

$$a_0 = 2M - cs, \quad F_s = s^2 - cs + M, \quad d_0 = (a_0^2 + \Omega^2 s^2)^{1/2} = 2\sqrt{MF_s},$$

$$R = \exp\left[\frac{c}{\Omega} \arctan \frac{\Omega s}{a_0} \right], \qquad k = \Lambda R.$$

The first three quantities have positive margins on the chosen interval. The even power series of $\arctan z/z$ shows that R and d_0 are analytic in (s, c, g) on a common neighborhood of $g = 0$. In the explicit solution, with $z = \Omega T/2$, the return equation is

$$e^{cz/\Omega} \left(\cos z + \frac{2Mp - c}{\Omega} \sin z \right) = 1.$$

Its homogeneous zero is $z_0 = \pi - \arctan(\Omega s/a_0)$. Write $z = z_0 - \Omega h$ and $h = skW$. The equation becomes

$$d_0 W \operatorname{sinc}(\Omega skW) = e^{cskW}. \tag{C.6}$$

Here $\operatorname{sinc}(0) = 1$. At $k = 0$, $W = 1/d_0$, and the W derivative has the positive margin d_0. The analytic implicit function theorem therefore gives a common W. Only even powers of Ω occur in this equation.

It remains to verify that this implicit solution is the first positive upper return. On the preceding interval $0 < z < z_0$, the bracket in the return equation is positive. The derivative of its product with the exponential has the sign of

$$2Mp \cos z + \frac{2M(cp - 2)}{\Omega} \sin z.$$

Both coefficients are positive in the present separated block: $cp > 2$ after a fixed shrink. It changes sign exactly once before z_0. The product first increases from one and then decreases to zero, so it meets one once more, at the solution just constructed. Along the preceding segment $Y > 0$.

Differentiating the explicit solution in time and using (C.6) gives

$$\eta = \frac{2Mke^{ch}}{d_0\cos(\Omega h) - cske^{ch}}.$$

The denominator has a positive margin. This proves (C.4); at $s = 0$, $R = 1$, $h = 0$ and $d_0 = 2M$, giving its exact endpoint value. Every fixed number of c, g derivatives of Λ costs only finite inverse powers of g, absorbed by its exponential. The chain rule proves the stated mixed bounds and the finite-order zero extension. □

The lower principal actions satisfy

$$A'_+(p) = p/F_L(p), \qquad A'_-(q) = q/(1 - aq + Bq^2).$$

The joint-center comparison uses the derivative of $K^0 = A_+(p) - A_-(q(p))$. Differentiation removes the input-independent divergent actions. To express the remaining rate in the radial input, define

$$L_s = s^2 + as + B, \qquad U_s = s^2 - cs + M, \qquad D_s = B - as\eta + s^2\eta^2, \qquad \tau = a + c, \quad \varsigma = B - M.$$

They have fixed positive margins on the separated interval and its slightly larger coefficient box.

Lemma C.2 (Exact principal columns on the closed radial interval)**.** *On the entire interval of Lemma C.1,*

$$\partial_s K^0 = \frac{M\tau + \varsigma(c - s)}{BL_sU_s} + \frac{(M\tau + c\varsigma)\eta + \varsigma s\eta^2}{BU_sD_s}. \tag{C.7}$$

The right side, with the auxiliary extension of η, has bounded mixed jets of each prescribed finite order on the common coefficient box. At the center $a = -c$, $M = B$, put $F = s^2 - cs + B$ and $D = B + cs\eta + s^2\eta^2$. Its transverse columns in (τ, ς) are exactly

$$\mathsf{A} = \frac{1}{F^2} + \frac{\eta}{FD}, \qquad \mathsf{V} = \frac{c - s}{BF^2} + \frac{c\eta + s\eta^2}{BFD}. \tag{C.8}$$

In the actual gap coordinates at zero scale, with B, δ fixed, the center columns are

$$U_g = \mathsf{A} - \frac{c}{2}\mathsf{V}, \qquad V_g = -2\mathsf{V}.$$

As $B \to \gamma^2$ and the common gap $g = \delta \downarrow 0$, they converge in every prescribed finite C^k norm on the entire closed interval to

$$U_0(s) = -\frac{1}{\gamma(\gamma - s)^3}, \qquad V_0(s) = -\frac{2(2\gamma - s)}{\gamma^2(\gamma - s)^4}. \tag{C.9}$$

Proof. Insert (C.5) in $(K^0)_p = p/F_L - qq_p/(1 - aq + Bq^2)$, substitute $p = 1/s$, $q = 1/(s\eta)$, and use $\partial_s = -p^2\partial_p$. Clearing the three displayed positive denominators gives (C.7). It has no pole at $s = 0$; indeed its value there is

$$(1 + \Lambda)\frac{M\tau + c\varsigma}{B^2M}.$$

Thus the reciprocal-return term in (C.7) contributes at the radial endpoint even when the center gap is fixed and positive.

Both numerators vanish at the center through their exact factors τ, ς. Derivatives of η therefore contribute no additional center column. Differentiating the remaining factors gives (C.8). The zero-scale form of (5.18) gives $d\varsigma = -(c/2)d\tau - 2d\zeta$ at fixed B, δ. A passive c variation makes no contribution because the whole center function is zero. Finally $c = 2\sqrt{B-g}$ on that center, and Lemma C.1 absorbs the reciprocal-return terms with their required derivatives. Direct substitution gives (C.9). For fixed $B = \gamma^2$ the error is $O(g) + O(e^{-a/\sqrt{g}})$; varying B adds $O(|B - \gamma^2|)$. All these estimates include $s = 0$. □

D Uniform estimates for relative transition maps

D.1 Mixed variations and a disappearing exit

The proof of Proposition 4.3 has two parts. Normal variational equations give decay of mixed jets at fixed time; the fixed transverse exit then controls the hitting-time jets. Normal decay dominates their growth, so differentiated endpoint terms vanish at a lost exit and the relative integral extends to infinite time. We work on one common real domain and choose the derivative order before reducing it. Retaining $N+2$ joint derivatives of the triangular field, initial sections and coefficient changes gives relative jets through total order N. Any two derivatives spent later on a logarithmic rate must already be included in N here.

Let $w(t;a,\mathbf{b})$ be the scalar solution starting at a and let $Z(t;z,a,\mathbf{b})$ be its normal solution. A selected derivative means a product of ordinary $a,z,\mathbf{b}$ derivatives and, in the lower case, E_ϵ. Euler derivatives obey the same product rule as ordinary derivatives; when commuting them through the vector field, derivatives of $q=\epsilon$ retain its factor ϵ. In the upper case $q=1$. Differentiation in frozen parameters gives inhomogeneous variational equations and hence polynomial factors in time. We estimate these equations directly, using the scalar-flow bounds.

For a normal jet of positive total order, differentiating $\dot Z=-g(w,Z)$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}D^\alpha Z=-g_Z(w,Z)D^\alpha Z+\mathcal{R}_\alpha(t). \tag{D.1}$$

Terms in $\mathcal{R}_\alpha$ containing only scalar and coefficient variations still contain Z: all derivatives of $g(w,0)$ in those variables vanish. Every other term contains a lower normal variation. The scalar variations satisfy (4.9). At order one, variation of constants in (D.1) gives a polynomial time factor times $e^{-(b_*-c_1\varpi)t}$. At order k, products of previously bounded variations have the same form with a larger polynomial and a larger constant multiplying ϖ. The convolution estimate

$$\int_0^t e^{-b_*(t-s)}(1+s)^m e^{-(b_*-c\varpi)s}\,\mathrm{d}s\le C(1+t)^{m+1}e^{-(b_*-c\varpi)t}$$

closes the induction. Include the finitely many constants needed through order $N+1$ in c_N before choosing ϖ. We obtain

$$|D^\alpha Z(t)|\le C_N(1+t)^{M_N}e^{-(b_*-c_N\varpi)t},\qquad |\alpha|\le N+1. \tag{D.2}$$

For $\alpha=0$, the stronger estimate $|Z(t)|\le|z|e^{-b_*t}$ holds. At normal input zero, normal-input variations in (D.2) need not vanish; they still decay. Their decay makes the relative integral differentiable at zero normal input as well.

The identity

$$\beta(w,Z)-\beta(w,0)=Z\int_0^1\beta_Z(w,\theta Z)\,\mathrm{d}\theta$$

and (D.2) now bound every derivative of its left-hand side by (4.14). Replacing β by g_Z proves the corresponding estimate for the logarithm of the normal derivative. The estimate is uniform through $z=0$.

Write the fixed-exit equation as $H(T,\theta)=\phi_T(a(\theta);\mathbf{b}(\theta))-v_R(\theta)=0$. Its time derivative has magnitude at least cq. At order k, the chain rule separates $H_tD^\alpha T$ from products $(\partial_t^mD^\gamma H)(T)\prod_{\ell=1}^m D^{\beta_\ell}T$, where $|\gamma|+\sum|\beta_\ell|=k$ and every remaining time jet has order below k. Mixed time derivatives of the flow follow from its equation with the same finite-order slow-growth bounds. Dividing by the fixed-exit velocity proves (4.10) by induction. Regular parameter dependence of either section is included in H. The identity $\partial_a\phi_T=f(v_R)/f(a)$ already places any reciprocal entry speed in the scalar variational bound; no entry-speed margin is required.

For a quadratic passage from $-v_R$ to v_R, the more explicit calculation is useful. With $f(v) = \mu + v^2$ and $\mu > 0$,

$$T = q^{-1} \int_{-v_R}^{v_R} \frac{\mathrm{d}v}{\mu + v^2}, \qquad |\partial_\mu^j T| \le C_j q^{-1} \mu^{-j-1/2} \le C_j q^{2j} (1+T)^{2j+1}.$$

Thus a vanishing interior gap costs only a fixed power of the transit time in this case. At a split root, the preceding implicit-flow argument supplies the appropriate slow exponential bound instead.

At total order k, differentiating an integral with endpoint T gives its fixed-time differentiated integral and a finite sum of terms

$$(\partial_t^{m-1} D^\gamma \mathcal{I})(T) \prod_{\ell=1}^{m} D^{\beta_\ell} T, \qquad m \ge 1, \quad |\beta_\ell| \ge 1, \quad |\gamma| + \sum_\ell |\beta_\ell| = k. \tag{D.3}$$

The integrand in each product uses total order at most $k-1$. Its normal factor, the time-jet bounds, and the combined slow exponent reserved before choosing ϖ bound every term by $C_N q^{-M_N} (1+T)^{M_N} e^{-(b_* - 2c_N \varpi)T}$. When $q \to 0$, separated sections give $q^{-1} \le T/c$; the remaining polynomial is absorbed by the positive normal margin. Thus all endpoint terms vanish uniformly at a disappearing exit.

For the absolute factor $L = e^{-I_0}$, differentiate $I_0 = \int_0^T \beta(w(t), 0)\,\mathrm{d}t$ only at finite T. Its derivatives have polynomial and slow-exponential bounds from the same flow and time equations. The chain rule for e^{-I_0} then multiplies finite products of these derivatives by $L \le e^{-b_* T}$. This proves absolute exponential bounds for all selected derivatives of L. The action I_0 itself may diverge.

At the separated-section slow-root boundaries of Proposition 4.3, the transit time tends to infinity. On the complete auxiliary flow define $j_\infty = -\int_0^\infty \mathcal{I}(t)\,\mathrm{d}t$. The integrable majorant permits differentiation through the integral. The difference $j_T - j_\infty$ is a tail, and its selected derivatives, including (D.3), tend to zero. Define $j = j_T$ on the passing side and $j = j_\infty$ on the nonpassing side. These derivatives agree at the lost-exit boundary; induction on derivative order, or the one-variable fundamental theorem along coordinate segments, proves the resulting C^N extension in the selected ordinary variables, together with the selected Euler scale bounds. Define the absolute factor to be zero on the nonpassing side. Its jets tend to zero by the absolute estimate, giving a separate extension in the same derivative class. Both extensions are defined on the prescribed auxiliary domain. Their restriction to real passing orbits gives the transition maps used in the comparison.

D.2 The central kernel on a real coefficient domain

The central connection requires a different estimate: the reference is controlled with a Gaussian weight rather than by ordinary high-order state derivatives. We prove the mixed bounds asserted in Proposition 4.7. All integrals may be rescaled by $X = Ry$ to the fixed interval $0 \le y \le 1$. Write $F_\epsilon = F_+ + \epsilon \bar{m}$ and split the exact graph field as

$$\mathfrak{f}(X, z) = a_0(X) z + \frac{\nu}{F_\epsilon(X)} + \mathcal{N}(X, z), \qquad a_0 = \frac{X}{\epsilon F_\epsilon} - \frac{\bar{c}}{F_\epsilon}.$$

Direct differentiation of (4.21) on its signed tube gives

$$|\mathcal{N}_z(X, z)| \le C[(X/\epsilon + 1)|z| + |\nu|], \qquad \mathcal{N}(X, 0) = 0. \tag{D.4}$$

The backward kernel $K(X, S) = \exp[-\int_X^S a_0(u)\,\mathrm{d}u]$ satisfies

$$|K(X, S)| \le C e^{-c(S^2 - X^2)/\epsilon}, \qquad \int_X^R |K(X, S)|\,\mathrm{d}S \le \frac{C\epsilon}{X + \sqrt{\epsilon}}. \tag{D.5}$$

For the last inequality, split at $X = \sqrt{\epsilon}$ and use $S^2 - X^2 \geq (S - X)^2$ in the first case and $S^2 - X^2 \geq X(S - X)$ in the second. The terminal term $K(X, R)\epsilon\nu h_b$ has the same weight, since $R \geq R_0$.

In the norm $\|z\|_* := \sup_X (X + \sqrt{\epsilon})|z(X)|/\epsilon$, the integral of the forcing in the backward equation has size $C|\nu|$. On $\|z\|_* \leq C_0|\nu|$, $(X/\epsilon)|z| \leq C_0|\nu|$; hence (D.4) and (D.5) make its nonlinear part a contraction after $|\nu|$ and the tube have been reduced. This proves existence, uniqueness and the first bound in (4.19) for the continuation of the given terminal graph. Its uniqueness identifies it with that graph on the overlap with the lower phase construction.

For each selected coefficient or Euler derivative, differentiating the kernel introduces at most a fixed polynomial in $(S^2 - X^2)/\epsilon$; the differentiated R endpoint is treated in the fixed y coordinate. Such polynomials are absorbed by a smaller positive Gaussian exponent. The terminal data retain their ϵ weight, as do their Euler derivatives. Differentiate the contracting equation. Its highest unknown derivative occurs through the same integral operator of norm less than $1/2$, while its other terms involve lower reference jets and the differentiated kernels. Induction yields the second bound in (4.19). The identity $z_b = 0$ at $\nu = 0$ and one additional ν derivative restore its ν factor whenever no ν derivative was prescribed. Accordingly, retain one extra coefficient order when that divided bound is needed. All of these arguments use real differentiation and the weighted integral norm.

The linear coefficient about this reference is $a = \mathfrak{f}_z(X, z_b)$. The exact identity

$$D - zD_z = F_\epsilon - \epsilon\bar{m}z^2$$

shows that $a - X/(\epsilon F_+)$ and its selected coefficient and Euler jets are bounded pointwise. The term $F_\epsilon - F_+ = \epsilon\bar{m}$ removes the apparent inverse scale in that difference; every remaining dangerous term contains $(X/\epsilon)z_b$, which the weighted bound controls. Taylor division of the nonlinear difference gives

$$B_c(X, v) = \int_0^1 (1 - \theta)\mathfrak{f}_{zz}(X, z_b + \theta v)\,\mathrm{d}\theta.$$

Its ordinary v derivatives and selected coefficient derivatives have bounds $C_N(X/\epsilon + 1)$ on the fixed signed input tube. Retaining $N + 2$ derivatives of the graph field supplies these N orders; the reference construction may retain $N + 3$ when its divided ν bound is also used.

The exact kernel $E(X) = \exp[-\int_X^R a]$ has the same Gaussian bound. Its selected derivatives have that bound times a fixed polynomial in $(R^2 - X^2)/\epsilon$. Consequently,

$$\int_0^R (X/\epsilon + 1)|D^\alpha E(X)|\,\mathrm{d}X \leq C_N. \tag{D.6}$$

For the principal term, the substitution $u = (R^2 - X^2)/\epsilon$ evaluates the integral with weight X/ϵ directly; the remaining term follows from the same Gaussian estimate and $R \geq R_0$.

Equation (4.22) is therefore a contraction for Q in a fixed small ball and for W in a fixed interval independent of the scale. Differentiate it successively in W, the coefficients and E_ϵ. The highest Q jet always appears with its original contracting linearization. All other terms are integrals of products of lower jets, derivatives of B_c, and at least one kernel factor E. Equation (D.6) bounds them. Solving at each order by the inverse of $I - \mathcal{T}_Q$, whose norm is at most two, proves the required mixed bounds. This argument works at $W = 0$ as well as at nonzero input. Finally a_r has bounded selected jets, so its integral supplies the bounds for $\log a_c$ and its positive upper and lower margins.

These estimates hold at any finite order chosen in advance, with constants that may depend on that order. They control ordinary section and coefficient derivatives and Euler scale derivatives of the integrated relative functions. At the Gaussian center the reference has the stated weighted bound; uniform high-order X derivatives and ordinary ϵ derivatives of the final relative maps remain outside these estimates.

D.3 Estimates for the independent source maps

Fix every derivative order before the coefficient box and scale bound. Use the selected jets of Section 6.1, retaining two extra coefficient orders for divided coefficients. We construct the core, tail inverse and upper return on independent common domains before imposing physical scale relations; constants may depend on the finite order.

Core estimate in Proposition 6.1. Put $H = \vartheta^{-2}$ at a real base point. Complexify the independent parameters in the product

$$|\widetilde{\vartheta}/\vartheta - 1| < 1/32, \quad |\widetilde{\beta}_i| < 2b_0, \quad |\widetilde{\widehat{\ell}}| < 4, \quad |\widetilde{s}| < \tfrac{3}{2}e^{-H}.$$

The integration variable X remains real. With tildes suppressed in the functions, the elementary estimates

$$|F - 1| < 5b_0, \quad |(\vartheta/\widetilde{\vartheta})^2/F - 1| < 1/10, \quad |\widetilde{\beta}_c/F| < 3b_0$$

hold after fixing b_0 small. Integration of the real part of the exponent of J gives

$$\frac{9}{20}HX^2 - 3b_0X \le \log|J(X)| \le \frac{11}{20}HX^2 + 3b_0X. \tag{D.7}$$

In particular $|J| \le e^{3H/5}$, whereas $|J^{-1}| \le e^{3b_0}e^{-9HX^2/20}$. Consequently

$$|I(X)| \le \frac{32e^{3b_0}}{31(1-5b_0)}\int_0^\infty e^{-9t^2/20}\,dt < 2.$$

Rescaling the reciprocal Gaussian by $\widetilde{\vartheta}$ therefore bounds I uniformly as the scale vanishes.

The normalized equation can be continued through signed zeros of U because its remainder contains no division by U. Write $m_* = \widetilde{\vartheta}^2\widetilde{\beta}_m$, $c_* = \widetilde{\vartheta}\widetilde{\beta}_c$, $A_X = X/\widetilde{\vartheta} - c_*$ and $H_z = 1 + 2m_* - m_*z$. Direct substitution into (6.4) gives

$$U_X = \frac{\widetilde{\widehat{\ell}}}{\widetilde{\vartheta}FJ} + \mathcal{R}_c, \qquad U(0) = 1,$$

where

$$\mathcal{R}_c = \frac{\widetilde{s}}{\widetilde{\vartheta}}\left\{\frac{\widetilde{\widehat{\ell}}UH_z}{DF} - \frac{2\widetilde{\widehat{\ell}}U}{D} + JU^2\left(\frac{A_XH_z}{DF} + \frac{c_* + \widetilde{s}\widetilde{\widehat{\ell}}}{D}\right)\right\}.$$

This equation is holomorphic also at $\widetilde{s} = 0$. On a provisional tube $|U| \le 16$ one has $|z| \le 24e^{-2H/5} < 1/8$, $|F| > 3/4$ and $|D| > 1/2$. It follows, with an absolute constant C, that

$$|\mathcal{R}_c| \le C|\widetilde{s}|H(1 + |J|), \qquad \sup_{0\le X\le 1}\left|\int_0^X \mathcal{R}_c\,dt\right| \le CHe^{-2H/5}.$$

For large H this last quantity is less than $1/48$. The integral equation now improves the tube to $|U| \le 1 + 4\cdot 2 + 1/48 < 10$. Continuation on the compact integration interval proves existence on the entire product and the same bound for $R = U - 1 - \widetilde{\widehat{\ell}}I$.

At a real point, Cauchy radii $\vartheta/64, s/4, b_0/2, 1/2$ lie in this product. They bound every weighted ordinary derivative of R by $C_NHe^{-2H/5}$, and hence every selected Euler word by $C_Ne^{-H/3}$. The same argument applied to I bounds its selected jets. The independent neighborhoods around individual real points also cover the boundary $s = e^{-H}$. On the real slice $D \ge 7/8 - 3b_0 > 27/32$. On $0 \le X \le \vartheta$ one has $J < 2$ and $F < 9/8$, so $I(1) > 4/9$. Thus $U_{\widehat{\ell}}(1) > 1/4$, after another increase of H. At $\widehat{\ell} = -3$ the endpoint is negative, whereas at zero it is positive. Strict monotonicity gives the unique simple threshold; in fact

$$\widehat{\ell}_c = -I(1)^{-1} + O(e^{-H/3}), \qquad U(1) = (\widehat{\ell} - \widehat{\ell}_c)G_c, \quad 1/4 < G_c < 3.$$

The threshold and the factor inherit the finite parameter jets by the implicit function theorem and its mean-value formula.

Finally, at any real zero of z its derivative has the sign of $\widehat{\ell}$. If $\widehat{\ell} < 0$, all such zeros cross downwards, so a second zero is impossible; its endpoint sign determines passage. If $\widehat{\ell} > 0$, a positive solution cannot cross downwards; if $\widehat{\ell} = 0$, uniqueness prevents a zero. These observations give the whole-path threshold assertion, including its equality case. □

Independent tail and inverse estimates. Keep the preceding complex neighborhoods and take, around each real $v \in [7/16, 9/16]$, a complex disc of radius $1/32$. Decrease the scale radius if necessary so that

$$H/3 < \operatorname{Re}(\widetilde{v}/\widetilde{\vartheta}^2) < 2H/3, \qquad |\widetilde{v}| < 3/5.$$

Hence the exponential in (6.6) satisfies

$$|z| \le e^{-(1-r)H/3}, \quad |z^{-1}| \le e^{2(1-r)H/3}, \quad \int_0^1 |z|\,dr \le 3/H.$$

Writing its denominator as $X + \eta_d$, one obtains

$$|\eta_d| \le Cb_0\vartheta^2 + C\vartheta e^{-H/3}.$$

The second term uses $|s/z| \le (3/2)e^{-H/3}$ and therefore holds before any physical entry is inserted.

Use $E = X^2/2$. On a provisional tube $\operatorname{Re} X > 3/4$, $|X| < 2$, its equation has denominator bounded away from zero. Integration of the equation and the preceding integral of $|z|$ give

$$|E(r) - (1/2 + r\widetilde{v})| \le C(b_0 + \vartheta^2), \qquad 0 \le r \le 1.$$

Indeed the contributions of $\beta_B X^2 + \beta_a X$ are $O(b_0)$; those of $z, \vartheta^2\beta_m$ and η_d/X have integrals $O(\vartheta^2) + O(b_0\vartheta^2) + O(\vartheta e^{-H/3})$. For small fixed b_0, then small ϑ_0, the right side improves the tube, for example to $\operatorname{Re} X > 9/10$, $|X| < 3/2$. The branch of the square root containing $X(0) = 1$ therefore continues to $r = 1$.

Cauchy estimates on smaller fixed entry discs bound all entry jets, and those on the parameter product bound the selected jets. In particular $|T_v - 1| < 1/64$ after the same fixed choices. The endpoint values at $v = 7/16, 9/16$ enclose $[31/32, 33/32]$ strictly, since their limits are $15/16, 17/16$. The implicit function theorem on this fixed interval, with $T_v > 63/64$, supplies the inverse and its finite mixed jets. On the real slice the original time satisfies $r' = \vartheta(X + \eta_d)/v > 0$; this is positive on the closed r-interval for every fixed field and scale, so the passage takes finite positive time.

At the same state, the difference of the tail vector fields for $\widehat{\ell}$ and zero is $\widehat{\ell}$ times $O(\vartheta e^{-H/3})$. Their state derivatives are bounded on the tube. Variation of constants and analytic division in $\widehat{\ell}$ therefore give the divided estimate (6.7), including its selected jets by the same complex neighborhoods. To divide the inverses, use

$$T_{\widehat{\ell}}^{-1}(e) - T_0^{-1}(e) = -\frac{T_{\widehat{\ell}}(T_0^{-1}(e)) - e}{\displaystyle\int_0^1 T'_{\widehat{\ell}}(T_0^{-1}(e) + t[T_{\widehat{\ell}}^{-1}(e) - T_0^{-1}(e)])\,dt}.$$

The real denominator is uniformly positive; differentiation of this identity gives the same finite jet bounds for the divided inverse. □

The upper passage and its endpoint layers. Set $g(Y) = \beta_m Y - 1$, $f(X) = \beta_B X^2 + \beta_a X$, $G(X, Y) = X + \beta_c Y$ and

$$Q = \frac{\lambda}{Y + \lambda}, \qquad r_u = \frac{Y^2}{Y + \lambda}.$$

On either endpoint layer the exact graph equation is

$$\frac{dX}{dY} = \frac{(1-Q)g + Qf}{G + \alpha r_u}. \tag{D.8}$$

Fix the layer height $d = 1/8$. On the right tube $1 < X < 2$, and the left tube $-2 < X < -1$, $0 \le Y \le d$, the denominator has absolute value at least $7/8$. For $E = X^2/2 + Y$ one has

$$\frac{dE}{dY} = \frac{Y(\beta_c + \beta_m X) + QX(f-g) + \alpha r_u}{G + \alpha r_u}.$$

Since $\int_0^d Q\, dY = \lambda \log(1 + d/\lambda)$, the energy variation on each layer is made less than $1/512$ by the fixed choices. The resulting energy bounds improve the assumed X tubes: at the most restrictive endpoint, $X^2/2 \ge 3/4 - d - 1/512 > 1/2$.

In the middle $Y \ge d$, divide the field by Y to obtain

$$\dot{X} = g + \lambda f/Y < -7/8, \qquad \dot{Y} = (1 + \lambda/Y)G + \alpha Y.$$

On $|X| \le 2$, $d \le Y \le 3/2$, the energy derivative is bounded by $C(|\beta| + \lambda + |\alpha|)$ and the travel time by a fixed constant. The total energy variation is less than $1/128$ after shrinking the parameters. As long as $|X| \le 1$, the energy lower bound implies $Y > d$; on $X < -1$ the vertical derivative is strictly negative. The arc therefore reaches the left layer transversely, before leaving this tube. On that layer Y decreases to zero and the original vertical velocity there is $\lambda X < 0$. The analogous right-layer velocity starts positive. These signs prove that the constructed passage is the first upper passage and has positive time.

Use (D.8) on the two fixed Y-intervals. Every E_λ derivative of Q is Q times a polynomial in Q; also $E_\lambda r_u = -Q r_u$, with the same property after iteration. Thus all selected graph-field derivatives are bounded on the tube. For a mixed jet of order k, the variational equation has the form

$$Z_k' = AZ_k + P_k, \qquad \|A\| \le C, \tag{D.9}$$

where P_k contains bounded lower jets. Variation of constants on the fixed layer intervals, followed by the transverse middle exit, proves the mixed bounds by induction. Subtracting two such equations bounds their jet differences by the integrated field-error carrier and lower jet differences. This also proves a bounded analytic divided difference $(\mathcal{U}_\alpha - \mathcal{U}_0)/\alpha$. □

D.4 Principal maps and their differentiated errors

We compare the three unforced maps with their limits in the reserved mixed jets. This gives the divided source coefficients and their $O(r_s)$ errors on the full contraction cube. At $\widehat{\ell} = 0$ the core expansion yields

$$v_{\pm,0} = 1 - \int_0^1 \frac{t}{1 + \beta_{a,\pm} t + \beta_B t^2}\, dt + O(\vartheta^2)$$

in every prescribed selected jet after the physical $s(\vartheta)$ is substituted. The exponential remainder absorbs the polynomial factors introduced by $E_\vartheta s = 2hs/(1-s)$. For the unforced tail, its limiting energy equation is

$$E_r = v(1 + 2\beta_B E + \beta_a \sqrt{2E}), \qquad E(0) = 1/2.$$

The integral of the difference from the exact equation is $O(\vartheta^2)$ on the common tube. This follows term by term from $\int |z|\, dr = O(\vartheta^2)$ and the explicit $\vartheta^2\beta_m, \vartheta^2\beta_c$ terms. Selected Euler derivatives of z have the form $P((1-r)/\vartheta^2)z$, whose integrals have the same bound. Use (D.9) on the fixed r-interval with this $O(\vartheta^2)$ carrier and bounded coefficient and entry jets. Induction

gives the comparison in every reserved mixed derivative. In the limit, $dS_\beta(X)/dr = v$, so inserting the principal core entry gives $S_\beta(\xi_\pm) = 1$.

For the upper unforced passage the limiting field is $\dot X = -1 + \beta_m Y$, $\dot Y = X + \beta_c Y$. On its endpoint layers the difference of the exact graph equation from the limiting equation is exactly $Q(f-g)/G$. Every selected Euler derivative of this difference is bounded by $C_N Q$ after coefficient and state derivatives. Thus the integral carrier is $O(\lambda(1+|\log\lambda|))$. In the middle the field error and its jets are $O(\lambda)$. Equation (D.9), with these carriers on the two fixed layers and the compact transverse middle arc, gives

$$|\mathcal{U}_0 - \mathcal{U}_{\mathrm{pr}}|_N \le C_N \vartheta^2 (1 + |\log\vartheta|).$$

The implicit middle exit is differentiated only at its uniformly transverse section. Combining the three comparisons with the common tail inverse gives the claimed $O(r_s)$ error for the unforced shooting function on the entire contraction cube. The specified Hadamard integrals consume two coefficient derivatives and preserve that error. Hence they prove the error assertion for A, C without changing their normalization.

At $\beta = 0$ the principal tail is $1/2 + v$ and the principal upper action is the identity; consequently $\mathcal{M}_{\mathrm{pr}} = \mathrm{id}$. On a small coefficient box analyticity and the differentiated error therefore give, at an independent entry,

$$|\partial_v \mathcal{M} - 1| + |\partial_v^2 \mathcal{M}| + |\partial_v^3 \mathcal{M}| + h|D_{L,\mathrm{ind}}\partial_v \mathcal{M}| + h|D_{L,\mathrm{ind}}\partial_v^2 \mathcal{M}| + h^2|D_{L,\mathrm{ind}}^2 \partial_v \mathcal{M}| \le C(|\beta| + r_s).$$

The unforced entry has the corresponding bounds $h|D_L v_{+,0}| + h^2|D_L^2 v_{+,0}| \le C(|\beta| + r_s)$. Here $D_{L,\mathrm{ind}}$ is the fixed-original-coefficient operator (6.18), with the entry independent until its composition with the logarithmic entries below.

D.5 Moving logarithmic entries and relative second derivatives

Write $D = D_L$, $\zeta = \zeta_+$ and $w_t = v_{+,0} - t\zeta/h$. Let ε_1 bound the preceding small principal-map and unforced-entry quantities. Since $Dh = 1 - s$, $D^2 h = s$ and $s \le 1/h$,

$$|D(\zeta/h)| \le |D\zeta|/h + |\zeta|/h^2, \qquad |D^2(\zeta/h)| \le |D^2\zeta|/h + 2|D\zeta|/h^2 + 3|\zeta|/h^3.$$

Uniformly in $0 \le t \le 1$, it follows that

$$|Dw_t| \le \varepsilon_1/h + |D\zeta|/h + |\zeta|/h^2 =: W_1,$$
$$|D^2 w_t| \le \varepsilon_1/h^2 + |D^2\zeta|/h + 2|D\zeta|/h^2 + 3|\zeta|/h^3 =: W_2.$$

Differentiating $S = \int_0^1 \mathcal{M}_v(w_t)\,dt$ gives

$$|S - 1| \le \varepsilon_1, \quad |DS| \le \varepsilon_1/h + \varepsilon_1 W_1, \quad |D^2 S| \le \varepsilon_1/h^2 + 2\varepsilon_1 W_1/h + \varepsilon_1 W_2 + \varepsilon_1 W_1^2.$$

The last term includes the full $\mathcal{M}_{vvv}(Dw_t)^2$ contribution. The product rule and (6.20), together with $|\zeta|/h \le 1$, imply

$$|D^j\{\zeta(S-1)\}| \le C\varepsilon_1 |D^j\zeta|, \qquad j = 0, 1, 2.$$

For example, at order two the terms containing $(D\zeta)^2$ are bounded by $C|D^2\zeta|$; factors $|\zeta|/h$ are at most one; the remaining terms reduce to the inequalities $|\zeta| \le C|D\zeta| \le C^2|D^2\zeta|$. Thus the relative estimate remains uniform as either normalized core approaches zero within its retained domain.

Finally, the independent forced error $\mathcal{E}$ has all the reserved jets bounded by $C_N e^{-h/3}$. Two physical derivatives after insertion of $v_\pm$ introduce at most two inverse core factors, since each differentiation of a logarithm adds one such factor. On the retained positive-core domain their product is at most $s^{-4\delta_0}$. All other losses are fixed powers of h from physical substitution and the leading factor $h\widehat{\ell}$. Because $s = e^{-h-1+s}$ and $1/3 - 4\delta_0 > 1/4$, these losses are absorbed in $C\widehat{\ell} e^{-h/4}$. Analytic division by $\widehat{\ell}$ before differentiation preserves the explicit factor $\widehat{\ell}$ even at zero. This proves (6.22) and completes the estimates needed for the source curvature.

D.6 Finite-order estimates for the exact long passage

The exact saddle power accounts for the nonintegrable part of the long flight. Its remaining correction is integrable, and differentiating that correction proves Proposition 10.1, including the ordinary terminal-height jets at zero. Fix the total order N first. The constants may depend on N and the fixed heights and sections, but are independent of the long time.

Use the terminal variables of (10.6). The rectangle is chosen with $W < -c$ and $Q > 1/2$, so $0 \le w(t) \le w_e e^{-t/2}$. The exact subtraction of the saddle ratio gives

$$|Q - \beta_\epsilon| \le C\left(|\gamma|e^{-t} + |J|e^{-2t} + w + \epsilon(|\bar a| + |\nu|)\varphi w + \epsilon\varphi^2 w\right). \tag{D.10}$$

A coefficient derivative may remove the small coefficient multiplying a term, but leaves its exponential or weighted factor. These factors control the differentiated integral: for every fixed integer i,

$$\int_0^L (1+t)^i w(t)\,\mathrm{d}t \le C_i, \qquad \int_0^L (1+t)^i(\varphi+\varphi^2)w(t)\,\mathrm{d}t \le C_i H^{1/2}(1+L)^i.$$

Indeed $\varphi = He^t$ and $e^L = \varphi_0/H$; integration of $He^{t/2}$ and $H^2e^{3t/2}$ proves the second estimate.

Write

$$w = w_e e^{-\beta_\epsilon t} e^{R(t)}, \qquad R(0) = 0, \qquad R_t = -(Q - \beta_\epsilon).$$

The zeroth-order bound follows by integrating (D.10). At each derivative order, the highest variation of R solves a linear inhomogeneous equation with coefficient $-wQ_w$. Its absolute integral is bounded independently of L, so variation of constants has a common bound. A γ or J derivative introduces e^{-t} or e^{-2t}; an $\bar a$ or ν derivative introduces $\epsilon\varphi w$. An $H\partial_H$ derivative of an explicit coefficient retains φw or $\varphi^2 w$. A θ derivative of the removed power introduces a factor t, always accompanied by one of these integrable carriers.

The same variation-of-constants induction, with these carriers and finite polynomial factors in t, bounds all lower-jet products through the fixed order N. At the moving endpoint, $H\partial_H L = -1$; the boundary terms retain $e^{-L/2}$, e^{-L}, or $(\varphi_0 + \varphi_0^2)w(L)$, so the same induction includes them.

For ordinary terminal-height derivatives, use the equation for R rather than $\log w_e$. Its explicit height dependence is $w_e e^{-\beta_\epsilon t} e^R$, smooth also at $w_e = 0$. Each resulting source contains $e^{-\beta_\epsilon t}$ times controlled lower variations. Thus every reserved ordinary height derivative is bounded jointly with all auxiliary coefficient derivatives. The original map keeps its exact front factor w_e.

To construct the limiting unit, set $H = 0$, hence $\varphi = 0$, in the equation for R and integrate on $[0,\infty)$. The convergent integral defines $\Psi_0(\gamma, J, w_e; \theta)$. Comparing its solution on $[0, L]$ with the positive-H solution uses only the last two carriers in (D.10); the omitted infinite tail is $O(e^{-L/2})$. Variation of constants proves an $O(H^{1/2})$ difference at order zero. The differentiated comparisons add finitely many powers of $|\log H|$. Choose one $q \in (0, 1/2)$ to absorb all of them after N has been fixed. This proves the stated mixed $O(H^q)$ limits.

For $x \in \{\bar a, \nu\}$ put $Y_x = \partial_x R$. Since β_ϵ is independent of x, the exact coefficient variations satisfy

$$(Y_x)_t + wQ_w Y_x = \epsilon F_x, \qquad Y_x(0) = 0, \qquad F_{\bar a} = \frac{G\varphi w}{W^2}, \quad F_\nu = \frac{\varphi w(1-e)(W+G)}{W^2}.$$

Dividing this equation by ϵ gives

$$(Z_x)_t + wQ_w Z_x = F_x, \qquad Z_x(0) = 0.$$

This equation defines Z_x also at $\epsilon = 0$, and uniqueness gives $Y_x = \epsilon Z_x$. Both forcing terms have the carrier φw. Their auxiliary derivatives are bounded by $C(1+t)^{M_N}\varphi e^{-t/2}$; ordinary

height derivatives replace the factor w_e by bounded variations. The other differentiated sources contain a lower variation of Z_x multiplied by an integrable variation of wQ_w. The same variation-of-constants induction through total order $N-1$, including ordinary ϵ derivatives and the moving endpoint, therefore gives

$$S_x := Z_x(L), \qquad \|S_x\|_{N-1}^{\mathrm{aux}} \le CH^{1/2}(1+L)^{M_N} \le CH^q.$$

Thus the coefficient variations have the stated factorization with the full auxiliary derivative budget.

The terminal-height limit is nonconstant. At $\gamma = J = 0$ the limiting equation has $Q = (k+2w)/(b+w)$; direct separation gives

$$t = \frac{b}{k}\log\frac{w_e}{w} + \frac{1}{2k}\log\frac{k+2w}{k+2w_e}, \qquad \Psi_0(0,0,w_e;\theta) = -\frac{1}{2b}\log(1+2w_e/k).$$

This also verifies its ordinary smoothness at zero height. No assertion here concerns an ordinary C^2 extension in H at zero.

D.7 Arrival and confinement for returning upper arcs

To connect the original axis input to the long map, use $t = 1/X$, $v = Z/X$ and divide the field by $X > 0$. The exact equations are

$$\begin{aligned} t'/t = {} & -\theta - \bar{m}v^2 + t[(1+2\epsilon\bar{m})v - \epsilon\bar{a}] - \epsilon(1+\epsilon\bar{m})t^2, \\ v' = {} & bv + \bar{d}v^2 - \bar{m}v^3 + t[(1+2\epsilon\bar{m})v^2 - \epsilon(\bar{d}+\nu+\bar{a})v] \\ & + t^2[-\epsilon(1+\epsilon\bar{m})v + \epsilon^2\nu]. \end{aligned}$$

The upper half-plane has $v \ge \epsilon t$. Fix $V = 1/\varphi_0$. For $0 < v \le V$ and sufficiently small entering $t, \epsilon, \bar{d}, \bar{m}, \nu$, these equations give $v' \ge v/2$. The apparent constant forcing, divided by v, is bounded by $\epsilon|\nu|t$. In the first equation discard the favorable $-\theta$ and use $\epsilon|\bar{a}|t/v \le |\bar{a}|$. Then

$$\frac{\mathrm{d}\log t}{\mathrm{d}v} \le C(1+|\bar{m}|v+t).$$

Integrate on a small fixed t strip and use first exit: an entry $t = 1/p$ has $t \le C_V/p$ up to $v = V$. It cannot exit through $Y = 0$, since $Y_\xi = \epsilon X > 0$ there, and cannot stay forever below V because $v' \ge v/2$. Thus it reaches the fixed section with $w = vt \to 0$ as $p \to \infty$. The flight time is finite for each entry and need not be uniformly bounded. Reflection gives the incoming assertion.

For the middle passage choose $w_1 = w_0 + 12$ and then $L \ge 10$, large enough for the reflected adjoining signs on $0 \le w \le w_1$. Only after these choices reduce the coefficient and scale bounds. Put

$$R_0 = L^2 w_0, \quad K_0 = L+2, \quad A_0 = 3(L+2), \quad R_* = 4(L^2+L+2), \quad R_{\max} = R_0 + 2A_0L + R_* + 2.$$

Then $R_{\max}/L^2 = w_0 + 10 + 16/L + 10/L^2 < w_1$. On $-L \le \hat{\varphi} \le L$, $0 \le R \le R_{\max}$ the exact projective field (10.20) converges to

$$\hat{\varphi}' = M - \hat{\varphi}^2 - C\hat{\varphi} - R, \qquad R' = -R(\hat{\varphi} + C).$$

For the normalized faces $|C|, |M| \le 1$, a common scale restriction therefore gives

$$R' \le K_0 R, \qquad \hat{\varphi}' \le -R/2 \quad (R \ge R_*).$$

For $\mathcal{V} = R + A_0(\hat{\varphi} + L)$ it follows that $\mathcal{V}' \le -K_0R/2 < 0$ when $R \ge R_*$. At a first crossing of $\mathcal{V} = R_0 + 2A_0L + R_* + 1$, one would have $R > R_*$ and a negative crossing derivative, a

contradiction. This barrier lies strictly below the previously fixed radial boundary. The field points inward at $\hat{\varphi} = L$, and $R = 0$ is invariant. Hence an arc stays in this compact rectangle until its first meeting with $\hat{\varphi} = -L$.

This is a spatial statement. A given returning upper arc must make that meeting, because its later incoming endpoint has $\hat{\varphi} = -q/(\epsilon\lambda) < -L$. It is not asserted that every initial point in the rectangle returns. At the meeting, $w = R/L^2 < w_1$. The reflected adjoining signs then continue the same arc outward to the fixed horizontal section with decreasing w; uniqueness excludes a second middle excursion. No upper bound for the intervening time is used.

D.8 Relative repeated-root limits at reciprocal input zero

The repeated upper action may diverge, while the input derivatives of its logarithmic comparison retain ordinary limits in the actual reciprocal tag $t = 1/P$. We separate this constant action from the relative units. In the horizontal matching $H = \lambda L$ is fixed and the terminal height varies, so the ordinary height estimates above apply at reciprocal zero.

Apply Corollary 4.4 to (10.23), with $q = 1$, $f(v) = \mu_u + v^2$ and the normal margins of Section 4.5. Fix all needed orders before choosing the strip so that $C_N\varpi < \beta_*/8$. At its fixed positive exit v_R the scalar velocity has a positive margin. The full strip between $\pm v_R$ passes only for $\mu_u > 0$; its time tends to infinity as the gap vanishes. Its time derivatives have polynomial bounds in $1 + T$, and moving entries satisfy the same fixed-exit jet estimate, by Appendix D.1. The prepared field, sections and reserved flow jets converge on bounded time intervals. The corollary therefore gives convergence of the relative amplitudes and their ordinary radial derivatives through normal input zero and a lost exit. The absolute action remains in its constant multiplier. For $C = 2g$, $M = -g^2$, put

$$V = \hat{z} + g^{-2}, \qquad Z_1 = \hat{x} + g\hat{z} - g^{-1}, \qquad r = Z_1/V.$$

The limiting affine equations give $V' = gV + Z_1$, $Z_1' = gZ_1$, $r' = -r^2$, and the first integral $\log(Vr) - g/r$ on the relevant positive branch. At $(P, 0)$ it is $\log(P - P_c) - P_c/(P - P_c)$, $P_c = 1/g$. The incoming and outgoing relative logarithmic derivatives are $-P/(P - P_c)^2$ and $-1/P$. These are the limits used in (10.26). To extend the derivative limit to reciprocal zero we need an ordinary input bound. In the two-infinite repeated row, $\mu_u > 0$ and the slow sections are fixed. Reserve four ordinary input orders in the tag and fixed connecting maps. Lemma 10.3 gives a regular change between the actual radial section coordinate and $t = 1/P$, through zero. The connections to the fixed slow sections stay away from the angular root. Their normal entry is therefore t times a positive unit with three bounded ordinary derivatives. Proposition 4.3, including normal input zero, gives the same relative bounds across the long slow strip. At the horizontal connections use Proposition 10.1 with $H = \lambda L$ fixed while the terminal height varies. Its ordinary height jets through zero supply the remaining relative units. Thus the inputs in this row are

$$u_+ = c_+ tV_+(t), \qquad u_- = c_- tV_-(t), \qquad c_+ = \Lambda, \quad c_- = \Lambda e^{-A_u}, \quad 0 < c_- \le c_+ \le 1,$$

where $V_\pm$ have common ordinary C^3 bounds and positive lower margins. These bounds remain uniform as $\mu_u \downarrow 0$. Put $l_\pm = 1 + tV_\pm'/V_\pm \ge 1/2$ on a common small interval. Apply Lemma 4.13 with $N = 2$ on the fixed coefficient fibre, using $s = tV_+(t)$, $s' = V_+ l_+ > c$, $k = (V_-/V_+) \circ t(s)$ and $1 + sk'/k = l_-/l_+ > c$. The two lower inputs are $c_+ s$ and $c_- sk(s)$. Absorb $c_\pm^{h_I/\epsilon}$ into their constant multipliers; $P_\pm(c_\pm s)$ have the required ordinary C^3 bounds, and $H_\pm = h_I + uP_\pm' > c$. The lemma bounds $\mathcal{K} - [\epsilon \log(K_-/K_+) - h_I A_u]$ in ordinary C^2, so it bounds $\mathcal{K}(t) - \mathcal{K}(0)$ in the same norm. Only input derivatives are taken: A_u and the divergent lower actions remain in the constants. The estimate holds on the passing side, then restricts to the lower whole-path and positive-central interval; the varying terminal height requires no ordinary C^2 extension in

H. The common C^2 bound makes the first derivatives at $t = 0$ and at a fixed small $t = \eta$ differ by at most $C\eta$. Convergence at $\eta > 0$, followed by $\eta \downarrow 0$, gives the endpoint derivative limit.

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